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1 Further Lebesgue integration theory

1.1 Notation and the space C_c^k

Notation:

Let $U \subseteq \mathbb{R}^n$ be open. Then $C^k(U)$ is the space of all k -times continuously differentiable complex valued functions on U and $C^\infty(U) = \bigcap_{k=0}^\infty C^k(U)$. These are complex vector spaces.

Now let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n) \in (\mathbb{Z}_{\geq 0})^n$. Then define

$$|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_n \text{ and}$$

$$\frac{\partial^{|\alpha|} f}{\partial x^\alpha} = \left(\frac{\partial}{\partial x_1} \right)^{\alpha_1} \left(\frac{\partial}{\partial x_2} \right)^{\alpha_2} \dots \left(\frac{\partial}{\partial x_n} \right)^{\alpha_n} f$$

We can also write this as $D^\alpha f$.

We will also write $x^\alpha = (x_1)^{\alpha_1} (x_2)^{\alpha_2} \dots (x_n)^{\alpha_n}$ and $\alpha! = \alpha_1! \alpha_2! \dots \alpha_n!$

Note that in this course, most theorems work in the real or complex numbers unless stated otherwise. Notable examples where we need one and not the other are the Riesz Representation Theorem only giving a natural isometric isomorphism in the reals since it is anti-isometric in the complex numbers, and Fourier transforms only making sense in the complex numbers.

Definition. Let $\phi \in C(U)$ for some open set U , then its support is the closure (relative to U , not $\mathbb{R}^n$) of the set $\{x \in U : \phi(x) \neq 0\}$.

For $0 \leq k \leq \infty$ define $C_c^k(U)$ to be the subset of $C^k(U)$ consisting of functions with compact support. Note that this is a vector subspace of $C^k(U)$.

Note: The relative condition is important: The function that is identically 1 on $B_1(0)$ does not have compact support because $\overline{B_1(0)}$ is the closure in $\mathbb{R}^n$, not in $B_1(0)$. In fact, if the closure in $\mathbb{R}^n$ is not a subset of U then there will be an accumulation point of $\{x \in U : \phi(x) \neq 0\}$ missing from the support so there will not be compact support. We therefore may say that compact support means the support is compact and a subset of U .

Proposition. There exists a non-negative function $\psi : C_c^\infty(\mathbb{R}^n)$ such that its support is $\overline{B_1(0)}$ and $\psi(0) \neq 0$ and its integral over $\mathbb{R}^n$ is 1.

Proof. Let $\chi(t) = \begin{cases} 0 & t \leq 0 \\ e^{-1/t} & t > 0 \end{cases}$. Then this is C^∞ . Now define $\phi(x) = \chi(1 - 2|x|^2)$, then this satisfies all conditions of the proposition except possibly the integral (recall from Analysis that compositions of C^∞ functions are C^∞), which we solve by dividing it by its total integral.

□

Corollary. For any open set U , $C_c^\infty(U)$ is not the trivial subspace consisting of the 0 function.

Let $U \subseteq U' \subseteq \mathbb{R}^n$ and U, U' be open. Note that if $f : U \rightarrow \mathbb{C}$ has compact support, say the support is K , then the continuous function on K equaling the distance from that point to the boundary of U attains a non-zero minimum on K , so as you approach the boundary of U , f will approach 0 and in fact f will be identically 0 if you are sufficiently close to the boundary. Therefore, we can extend f to a function with compact support on U' by setting it to 0 on $U' \setminus U$ and it is guaranteed to still be smooth. This means that $C_c^k(U) \subseteq C_c^k(U')$.

1.2 Convolutions of functions

Let U be open and let $\phi \in C_c^k(U)$, then extend ϕ to $\mathbb{R}^n$ by $\phi = 0$ on U^C . Define $T_x\phi : U \rightarrow \mathbb{C}$ by $y \mapsto \phi(y - x)$ for some x . Then by the above discussion, if $x \in B_\varepsilon(0)$ for ε sufficiently small then $T_x\phi$ will have compact support.

Note that we may sometimes write $\tau_x\phi$ for $T_x\phi$ and it means exactly the same thing.

Proposition. Suppose $p \in [1, \infty)$ and $g \in L^p(\mathbb{R}^n)$. Let (z_j) be a sequence in $\mathbb{R}^n$ such that $z_j \rightarrow 0$ as $j \rightarrow \infty$. Then if $T_{z_j}g$ is defined as above then $\|T_{z_j}g - g\| \rightarrow 0$ in the L^p norm.

Proof. Step 1. Suppose g is the indicator function of a rectangle defined by

$$R = (a_1, b_1] \times (a_2, b_2] \times \dots \times (a_n, b_n]$$

Now when we do T_{z_j} to g each side is translated by a distance of at most $|z_j|$. Without loss of generality assume j is large enough such that $|z_j|$ is less than the minimum side length of the rectangle. Then we have that when we translate by a component of the z_j vector, imagining doing this one at a time, at each step the integral of the absolute value of the difference is at most $2|z_j|I_{\max}^{n-1}$ where $I_{\max}$ is the maximum side length of the rectangle. Therefore we have that

$$\|T_{z_j}g - g\|_p^p \leq 2n|z_j|I_{\max}^{n-1} \rightarrow 0$$

Since $p < \infty$ it follows that the theorem is true for each such g

Step 2. Now suppose g is the indicator function of a measurable set of finite measure and fix $\varepsilon > 0$. By properties of the Lebesgue measure there exists compact K and open U such that $K \subseteq A \subseteq U$ and $|U \setminus K| < (\frac{\varepsilon}{3})^p$. Since U is open we can write U as the union of open rectangles by $U = \bigcup_\alpha R_\alpha$. Since K is compact it is covered by a finite subset of these, so write $K \subseteq \bigcup_{i=1}^N R_i := B$.

Note that $K \subseteq B \subseteq U$ so $A \triangle B \subseteq U \setminus K$ so it follows that we have

$$\|1_A - 1_B\|_p = \|A \triangle B\|_p^{\frac{1}{p}} < \frac{\varepsilon}{3}$$

Also by step 1 we know there exists J such that $j \geq J$ implies

$$\|T_{z_j} 1_B - 1_B\|_p < \frac{\varepsilon}{3}$$

Therefore

$$\begin{aligned} \|T_{z_j} 1_A - 1_A\|_p &= \|T_{z_j} 1_A - T_{z_j} 1_B + T_{z_j} 1_B - 1_B + 1_B - 1_A\|_p \\ &\leq \|T_{z_j} 1_A - T_{z_j} 1_B\|_p + \|T_{z_j} 1_B - 1_B\|_p + \|1_B - 1_A\|_p \\ &= \|T_{z_j} 1_B - 1_B\|_p + 2\|1_B - 1_A\|_p < \varepsilon \end{aligned}$$

Note that if g can be written as $\sum_{i=1}^N g_i 1_{A_i}$ where A_i each have finite measure, then we have

$$\|T_{z_j} g - g\|_p \leq \sum_{i=1}^N |g_i| \|T_{z_j} 1_{A_i} - 1_{A_i}\|_p \rightarrow 0$$

Step 3. Now suppose that $g \in L^p(\mathbb{R}^n)$.

Fix $\varepsilon > 0$. Then by definition of the Lebesgue integral there exists a simple function $\tilde{g} = \sum_{i=1}^n \tilde{g}_i 1_{A_i}$ where $\tilde{g}_i \in \mathbb{C}$ and $|A_i| < \infty$ such that (after considering the real and imaginary parts separately then positive and negative parts separately) $\|g - \tilde{g}\|_p < \frac{\varepsilon}{3}$. By step 2 we can find J such that $j \geq J$ implies

$$\|T_{z_j} \tilde{g} - \tilde{g}\|_p < \frac{\varepsilon}{3}$$

Now we have

$$\begin{aligned} \|T_{z_j} g - g\|_p &= \|T_{z_j} g - T_{z_j} \tilde{g} + T_{z_j} \tilde{g} - \tilde{g} + \tilde{g} - g\|_p \\ &\leq \|T_{z_j} g - T_{z_j} \tilde{g}\|_p + \|T_{z_j} \tilde{g} - \tilde{g}\|_p + \|\tilde{g} - g\|_p \\ &= \|T_{z_j} \tilde{g} - \tilde{g}\|_p + 2\|\tilde{g} - g\|_p < \varepsilon \end{aligned}$$

This completes the proof of the proposition. □

Definition. Let f, g be functions $\mathbb{R}^n \rightarrow \mathbb{C}$. Then we define their convolution as follows, provided the integral in question exists:

$$(f \star g)(x) = \int_{\mathbb{R}^n} f(y) g(x - y) dy$$

Remark. If f is the density of a probability measure ν then this is equal to the convolution $\nu \star g$ as defined in the probability and measure course.

Proposition. Let $f, g, h \in C_c(\mathbb{R}^n)$. Then $f \star g = g \star f$ and $f \star (g \star h) = (f \star g) \star h$ and

$$\int_{\mathbb{R}^n} (f \star g)(x) dx = \int_{\mathbb{R}^n} f(x) dx \int_{\mathbb{R}^n} g(x) dx$$

Proof. Note that it is not enough to just prove the product formula then use commutativity/associativity of multiplication because functions having the same integral does not make them equal. So using the change of variables $y = x - z$ we have

$$(f \star g)(x) = \int_{\mathbb{R}^n} f(y) g(x - y) dy = \int_{\mathbb{R}^n} f(x - z) g(z) dz = (g \star f)(x)$$

We use the fact that there is compact support to use Fubini's theorem. Then we have (using the substitution $w = y + z$)

$$\begin{aligned} [f \star (g \star h)](x) &= \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(z) h(x - y - z) dz \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(w - y) h(x - w) dw \right) dy \end{aligned}$$

$$= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) g(w-y) dy \right) h(x-w) dw = [(f \star g) \star h](x)$$

Now we calculate

$$\begin{aligned} \int_{\mathbb{R}^n} (f \star g)(x) dx &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) g(x-y) dy \right) dx \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) g(x-y) dx \right) dy \\ &= \int_{\mathbb{R}^n} \left(f(y) \int_{\mathbb{R}^n} g(x-y) dx \right) dy \\ &= \int_{\mathbb{R}^n} \left(f(y) \int_{\mathbb{R}^n} g(z) dz \right) dy \\ &= \int_{\mathbb{R}^n} f(x) dx \int_{\mathbb{R}^n} g(x) dx \end{aligned}$$

as required. □

Lemma. Let $f \in C_c(\mathbb{R}^n)$ and let (z_i) be a sequence in $\mathbb{R}^n$ with $z_i \rightarrow 0$. Then for any $x \in \mathbb{R}^n$ we have $T_{z_j} f(x) \rightarrow f(x)$ as $j \rightarrow \infty$

Proof. Since $z_j \rightarrow 0$ this follows by continuity of f and the definition of $T_{z_j} f$ □

Lemma. Let $f \in C_c(\mathbb{R}^n)$ and let (z_i) be a sequence in $\mathbb{R}^n$ with $z_i \rightarrow 0$. Then

$$|T_{z_j} f(x)| \leq \left(\sup_{\mathbb{R}^n} |f| \right) 1_{B_R(0)}(x)$$

for some $R > 0$ and for all j

Proof. Since $z_j \rightarrow 0$ there exists some $P > 0$ such that $z_j \in B_P(0)$ for all j . Then note that $\text{supp}(f)$ is also bounded which means that there exists $R > 0$ such that $\text{supp}(T_{z_j} f) \subseteq B_R(0)$.

This implies that $T_{z_j} f = T_{z_j} f 1_{B_R(0)}(x)$ so we have

$$|T_{z_j} f(x)| = |T_{z_j} f(x)| 1_{B_R(0)}(x) \leq \left(\sup_{\mathbb{R}^n} |f| \right) 1_{B_R(0)}(x)$$

□

Definition. Given a function f and $h \neq 0$ define

$$\Delta_i^h f(x) = \frac{f(x + h e_i) - f(x)}{h}$$

Lemma. Let $f \in C_c^1(\mathbb{R}^n)$ and let (h_j) be a sequence in $\mathbb{R}$ with $h_j \rightarrow 0$. Then for any $x \in \mathbb{R}^n$ we have

$$\left| \Delta_i^{h_j} f(x) \right| \leq \left(\sup_{\mathbb{R}^n} |D_i f| \right) 1_{B_R(0)}(x)$$

for some $R > 0$ and all j

Proof. As above there is a P with $|h_j| \leq P$ for all j . For similar reasoning to the proof above there exists R such that $\text{supp} \Delta_i^{h_j} f \subseteq B_R(0)$. Therefore $\Delta_i^{h_j} f(x) = \Delta_i^{h_j} f(x) 1_{B_R(0)}(x)$. Therefore

$$\left| \Delta_i^{h_j} f(x) \right| = \left| \Delta_i^{h_j} f(x) 1_{B_R(0)}(x) \right| \leq \left(\sup_{\mathbb{R}^n} |D_i f| \right) 1_{B_R(0)}(x)$$

Note: By the mean value theorem $\Delta_i^{h_j} f(x)$ always equals $D_i f(x)$ for some x so the last inequality is justified.

□

Definition. $L_{\text{loc}}^1(\mathbb{R}^n)$ is the set of functions that are Lebesgue integrable on every compact subset of $\mathbb{R}^n$, quotiented out by functions that are equal almost everywhere. Note that this contains all bounded measurable functions, for example.

Note that this means “locally integrable” - the definitions are equivalent as if it is integrable on every compact set then it is integrable on a neighbourhood about every point. If it is integrable on a neighbourhood about every point then by the finite subcover property it is integrable on every compact set.

Theorem. Let α be a multi-index as introduced at the beginning and let $f \in L_{\text{loc}}^1(\mathbb{R}^n)$ and $g \in C_c^k(\mathbb{R}^n)$ for some $k \geq 0$. Then $f \star g \in C^k(\mathbb{R}^n)$ and

$$D^\alpha (f \star g) = f \star (D^\alpha g)$$

for any multi-index α with $|\alpha| \leq k$.

Proof. Step 1. We would like to prove this for $k = 0$. It suffices to prove that $f \star g$ is continuous. To do this it is enough to prove the result that $f \star g(x - z_j) \rightarrow f \star g(x)$ for any $x \in \mathbb{R}^n$ and sequence $z_j \rightarrow 0$. Note that

$$f \star g(x - z_j) = \int_{\mathbb{R}^n} f(y) g(x - z_j - y) dy = \int_{\mathbb{R}^n} f(y) T_{z_j} g(x - y) dy$$

Hypothetically if we could swap the limit and the integral this would complete the proof. Indeed we can because by an earlier lemma and $g \in C_c(\mathbb{R}^n)$ we have

$$|T_{z_j} g(x - y)| \leq \left(\sup_{\mathbb{R}^n} |g| \right) 1_{B_R(0)}(x - y)$$

Therefore

$$|f(y) T_{z_j} g(x - y)| \leq |f(y)| \left(\sup_{\mathbb{R}^n} |g| \right) 1_{B_R(0)}(x - y)$$

This is therefore a suitable dominating function for the dominated convergence theorem because $f \in L_{\text{loc}}^1(\mathbb{R}^n)$.

Step 2. Now suppose that $k = 1$. Then $f \star D_i g$ is continuous by step 1. To show that $f \star g$ is indeed C^1 we will show that for any $x \in \mathbb{R}^n$ and any sequence $(h_j) \subseteq \mathbb{R}$ with $h_j \rightarrow 0$ we have

$$\lim_{j \rightarrow \infty} \Delta_i^{h_j} f \star g(x) = f \star D_i g(x)$$

It will then follow from step 1 that this partial derivative is indeed continuous which then implies that $f \star g$ is C^1 and the theorem is true for $k = 1$, so it remains to do this limit calculation.

$$\begin{aligned} \Delta_i^{h_j} f \star g(x) &= \frac{f \star g(x + h_j e_i) - f \star g(x)}{h_j} \\ &= \int_{\mathbb{R}^n} f(y) \left(\frac{g(x + h_j e_i - y) - g(x - y)}{h_j} \right) dy \\ &= \int_{\mathbb{R}^n} f(y) \Delta_i^{h_j} g(x - y) dy \end{aligned}$$

An argument analogous to the previous case where we use a lemma above and apply dominated convergence implies the limit.

Step 3. The general case is now an easy induction. □

Lemma. Let $K \subseteq \mathbb{R}^n$ be compact and let $F \subseteq \mathbb{R}^n$ be closed. Then their Minkowski sum

$$K + F = \{x + y \in \mathbb{R}^n : x \in K, y \in F\}$$

is closed

Proof. Let z_m be a sequence in $K + F$ with $z_m \rightarrow z$, then we want to show that z is in $K + F$. For each z_m write it as $x_m + y_m$ with x_m in K and y_m in F . Then take a subsequence x_{m_k} that is convergent and consider $x_{m_k} + y_{m_k}$. Note that $y_{m_k} = z_{m_k} - x_{m_k}$ is also convergent, and the limit point of $x_{m_k} + y_{m_k}$ is in $K + F$ since they are both closed, as required. □

Definition. For a general measurable function its support is the set of all closed sets where it is 0 almost everywhere outside that closed set.

Proposition. Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $g \in C^k_c(\mathbb{R}^n)$ for some $k \geq 0$. Then

$$\text{supp}(f \star g) \subseteq \text{supp}(f) + \text{supp}(g)$$

Proof. If $f \star g(x) \neq 0$ then there exists $y \in \mathbb{R}^n$ such that $y \in \text{supp}(f)$ and $x - y = z \in \text{supp}(g)$, by definition of $f \star g$. Therefore, $x = y + z$ with $y \in \text{supp}(f)$ and $z \in \text{supp}(g)$. This implies that

$$\{x \in \mathbb{R}^n : f \star g(x) \neq 0\} \subseteq \text{supp}(f) + \text{supp}(g)$$

But we know $\text{supp}(f) + \text{supp}(g)$ is closed since $\text{supp}(f)$ is closed and $\text{supp}(g)$ is compact. Therefore, the closure of $\{x \in \mathbb{R}^n : f \star g(x) \neq 0\}$ is in $\text{supp}(f) + \text{supp}(g)$ as required. □

Lemma. Suppose $1 \leq p \leq \infty$ and let q satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then for a measurable function $f : \mathbb{R}^n \rightarrow \mathbb{C}$, we have

$$\|f\|_p = \sup \left\{ \int_{\mathbb{R}^n} |f(x)g(x)| dx : g \in L^q(\mathbb{R}^n), \|g\|_{L^q} \leq 1 \right\}$$

Proof. Assume $\|g\|_q \leq 1$. By Holder's inequality we have

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_p \|g\|_q \leq \|f\|_p$$

Taking the supremum over all such g implies

$$\|f\|_p \geq \sup \left\{ \int_{\mathbb{R}^n} |f(x)g(x)| dx : g \in L^q(\mathbb{R}^n), \|g\|_{L^q} \leq 1 \right\}$$

To show that this bound is attained we will exhibit a function g that achieves this bound. We split this into 3 cases.

Case 1. $1 < p, q < \infty$

In this case set $g(x) = \frac{|f(x)|^{p-1}}{\|f\|_p^{p/q}}$

Then

$$\|g\|_q = \int_{\mathbb{R}^n} \frac{|f(x)|^{(p-1)q}}{\|f\|_p^p} dx = \int_{\mathbb{R}^n} \frac{|f(x)|^p}{\|f\|_p^p} dx = 1$$

Now evaluate the integral

$$\int_{\mathbb{R}^n} |f(x) g(x)| dx = \int_{\mathbb{R}^n} \frac{|f(x)|^p}{\|f\|_p^{p/q}} dx = \frac{\|f\|_p^p}{\|f\|_p^{p-1}} = \|f\|_p$$

Therefore the supremum is attained.

Case 2. $p = 1, q = \infty$

Setting $g = 1$ everywhere gets the job done.

Case 3. $p = \infty, q = 1$

For any $\varepsilon > 0$ the set $A_\varepsilon = \{x : |f(x)| > \|f\|_\infty - \varepsilon\}$ has positive measure by definition of $\|f\|_\infty$. If necessary, choose a measurable subset $B \subseteq A_\varepsilon$ such that B has finite measure. Define $g(x) = \frac{1}{\mu(B)} 1_B(x)$, then $\|g\|_{L^1} = 1$. Then we have

$$\int_{\mathbb{R}^n} |f(x) g(x)| dx = \frac{1}{\mu(B)} \int_B |f(x)| dx \geq \frac{1}{\mu(B)} \int_B (\|f\|_\infty - \varepsilon) dx = \|f\|_\infty - \varepsilon$$

Then the supremum is at least $\|f\|_\infty$ since ε was arbitrary.

If $\|f\|_\infty$ is infinite then f is infinite on a non-measure 0 set, so setting g to be 1 on some non-measure-0 finite-measure subset of this set proves that the lemma is true in the extended reals i.e. $[0, \infty]$.

□

Theorem. (Minkowski's integral inequality) Suppose $1 \leq p < \infty$ and $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$ is integrable. Set $G(y) = \int_{\mathbb{R}^n} F(x, y) dx$.

$$\left[\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dx \right|^p dy \right]^{\frac{1}{p}} \leq \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} |F(x, y)|^p dy \right]^{\frac{1}{p}} dx$$

Proof. Let $\|g\|_{L^q} \leq 1$ where $\frac{1}{p} + \frac{1}{q} = 1$. Then

We have

$$\begin{aligned} \int_{\mathbb{R}^n} |G(y) g(y)| dy &= \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dx \right| |g(y)| dy \\ &\leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| |g(y)| dx dy \end{aligned}$$

Since the integrand is non-negative we have

$$= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)| |g(y)| dy \right) dx$$

By Holder's inequality we get

$$\begin{aligned} &\leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p dy \right)^{\frac{1}{p}} \left(\int_{\mathbb{R}^n} |g(y)|^q dy \right)^{\frac{1}{q}} dx \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p dy \right)^{\frac{1}{p}} dx \end{aligned}$$

Now to deduce the theorem we want to change the left hand side of this chain of inequalities to

$$\left[\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dx \right|^p dy \right]^{\frac{1}{p}}$$

By the preceding lemma we have

$$\|G\|_p = \sup_{\|g\|_q \leq 1} \int_{\mathbb{R}^n} |G(y) g(y)| dy$$

But $\|G\|_p$ is exactly

$$\left[\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dx \right|^p dy \right]^{\frac{1}{p}}$$

Now since the bound holds for $\int_{\mathbb{R}^n} |G(y) g(y)| dy$ every such g , it also holds for the supremum, so done. $\square$

Proposition. Suppose $\phi : C_c^\infty(\mathbb{R}^n)$ is non-negative, has support contained in $B_1(0)$, and integrates to 1 over $\mathbb{R}^n$ and define $\phi_\varepsilon(y) = \frac{1}{\varepsilon^n} \phi\left(\frac{y}{\varepsilon}\right)$. Then if $f \in C_c^k(\mathbb{R}^n)$ then $\phi_\varepsilon \star f$ is C_∞ and we have

$$D^\alpha(\phi_\varepsilon \star f) \rightarrow D^\alpha f$$

as $\varepsilon > 0$ and the convergence is uniform on $\mathbb{R}^n$ for any multi-index with $|\alpha| \leq k$.

Proof. Note that it is easy to see that $\int_{\mathbb{R}^n} \phi_\varepsilon(y) dy = 1$

By the earlier general theorem on the derivative of convolutions, we have that $\phi_\varepsilon \star f$ is C_∞ and that $D^\alpha(\phi_\varepsilon \star f) = \phi_\varepsilon \star D^\alpha f$ for any $|\alpha| \leq k$. We now calculate

$$\begin{aligned} D^\alpha(\phi_\varepsilon \star f)(x) - D^\alpha f(x) &= \int_{\mathbb{R}^n} \phi_\varepsilon(y) D^\alpha f(x-y) dy - D^\alpha f(x) \int_{\mathbb{R}^n} \phi_\varepsilon(y) dy \\ &= \int_{\mathbb{R}^n} \phi_\varepsilon(y) [D^\alpha f(x-y) - D^\alpha f(x)] dy \\ &= \int_{B_1(0)} \phi(z) [D^\alpha f(x-\varepsilon z) - D^\alpha f(x)] dz \end{aligned}$$

Using $y = \varepsilon z$ and the fact that ϕ has support in $B_1(0)$. Therefore

$$\begin{aligned} |D^\alpha(\phi_\varepsilon \star f)(x) - D^\alpha f(x)| &\leq \int_{B_1(0)} \phi(z) |D^\alpha f(x-\varepsilon z) - D^\alpha f(x)| dz \\ &\leq \sup_{z \in B_1(0)} |D^\alpha f(x-\varepsilon z) - D^\alpha f(x)| \int_{B_1(0)} \phi(z) dz \\ &= \sup_{z \in B_1(0)} |D^\alpha f(x-\varepsilon z) - D^\alpha f(x)| \end{aligned}$$

Since $D^\alpha f$ is continuous by assumption and has compact support it is uniformly continuous. For $\epsilon > 0$ there exists a δ such that if $v, w \in \mathbb{R}^n$ with $|v, w| < \delta$ we have

$$|D^\alpha f(v) - D^\alpha f(w)| < \epsilon$$

Now assume $\varepsilon < \delta$, then we have

$$\sup_{z \in B_1(0)} |D^\alpha f(x-\varepsilon z) - D^\alpha f(x)| < \epsilon$$

This is a bound for $|D^\alpha(\phi_\varepsilon \star f)(x) - D^\alpha f(x)|$ that is x -independent so we have convergence and it is uniform as required. $\square$

Lemma. $L^p(\mathbb{R}^n) \subseteq L_{\text{loc}}^1(\mathbb{R}^n)$

Proof. Let $\frac{1}{p} + \frac{1}{q} = 1$ and let E be a compact subset of $\mathbb{R}^n$. By Holder's inequality with $f = f1_E$ and $g = 1$ we have

$$\int_E |f| \leq \mu(E)^{1-\frac{1}{p}} \|f\|_p$$

Then the result follows. $\square$

Proposition. Suppose $\phi : C_c^\infty(\mathbb{R}^n)$ is non-negative, has support contained in $B_1(0)$, and integrates to 1 over $\mathbb{R}^n$ and define $\phi_\varepsilon(y) = \frac{1}{\varepsilon^n} \phi\left(\frac{y}{\varepsilon}\right)$. Then if $g \in L^p(\mathbb{R}^n)$ with $1 \leq p < \infty$, then $\phi_\varepsilon \star g$ is C_∞ and $\phi_\varepsilon \star g \rightarrow g$ in $L^p(\mathbb{R}^n)$ as $\varepsilon \rightarrow 0$

Note that we expect something convolved with something like ϕ_ε to approximately equal the function - since convolution on probability densities gives the density of the sum of the distribution and adding a probability distribution concentrated near 0 will not change the density much.

Proof. Because $L^p(\mathbb{R}^n) \subseteq L^1_{\text{loc}}(\mathbb{R}^n)$, an earlier theorem establishes the smoothness of $\phi_\varepsilon \star g$. By the calculation from the previous proposition, we have

$$|(\phi_\varepsilon \star g)(x) - g(x)| \leq \int_{\mathbb{R}^n} \phi(z) |g(x - \varepsilon z) - g(x)| dz$$

Therefore by Minkowski's integral inequality we have

$$\begin{aligned} \|(\phi_\varepsilon \star g)(x) - g(x)\|_p &= \left[\int_{\mathbb{R}^n} |(\phi_\varepsilon \star g)(x) - g(x)|^p dx \right]^{\frac{1}{p}} \\ &\leq \left[\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} \phi(z) |g(x - \varepsilon z) - g(x)| dz \right|^p dx \right]^{\frac{1}{p}} \\ &\leq \int_{\mathbb{R}^n} \left[\int_{\mathbb{R}^n} \phi(z)^p |g(x - \varepsilon z) - g(x)|^p dx \right]^{\frac{1}{p}} dz \\ &= \int_{\mathbb{R}^n} \phi(z) \|T_{\varepsilon z} g - g\|_p dz \end{aligned}$$

We will set ε_j to be a real sequence tending to 0 and we will try to show that $\|(\phi_{\varepsilon_j} \star g)(x) - g(x)\|_p \rightarrow 0$. Since $\|T_{\varepsilon_j z} g\|_p = \|g\|_p$ we have

$$\phi(z) \|T_{\varepsilon_j z} g - g\|_p \leq 2\phi(z) \|g\|_p$$

Since $\phi(z)$ is continuous and supported on a compact set, it is bounded, so the dominated convergence theorem may be applied to $\phi(z) \|T_{\varepsilon_j z} g - g\|_p$. By an earlier proposition we have that this tends pointwise to 0, so the dominated convergence theorem implies the result. □

Proposition. Suppose $\phi : C_c^\infty(\mathbb{R}^n)$ is non-negative, has support contained in $B_1(0)$, and integrates to 1 over $\mathbb{R}^n$. Suppose that $f \in C^k(\mathbb{R}^n)$ with $\sup_{\mathbb{R}^n} |D^\alpha f| < \infty$ for all $|\alpha| \leq k$ and $g \in L^1(\mathbb{R}^n)$ with $g \geq 0$ and $\int_{\mathbb{R}^n} g(x) dx = 1$. Set $g_\varepsilon(y) = \varepsilon^{-n} g(\varepsilon^{-1}y)$. Then $f \star g_\varepsilon \in C^k(\mathbb{R}^n)$ and as $\varepsilon \rightarrow 0$ for any multi-index with $|\alpha| \leq k$ we have

$$D^\alpha (f \star g_\varepsilon)(x) \rightarrow D^\alpha f(x)$$

Proof. By an earlier theorem we have $D^\alpha (f \star g_\varepsilon) = (D^\alpha f) \star g_\varepsilon$ for any $|\alpha| \leq k$. So we have

$$D^\alpha (f \star g_\varepsilon)(x) = \int_{\mathbb{R}^n} g(z) D^\alpha f(x - \varepsilon z) dz$$

For each fixed $x \in \mathbb{R}^n$ and $z \in \mathbb{R}^n$ we have

$$g(z) D^\alpha f(x - \varepsilon z) \rightarrow g(z) D^\alpha f(x)$$

as $\varepsilon \rightarrow 0$. Also, we have

$$|g(z) D^\alpha f(x - \varepsilon z)| \leq g(z) \sup_{\mathbb{R}^n} |D^\alpha f|$$

so the dominated convergence theorem implies that

$$D^\alpha (f \star g_\varepsilon)(x) \rightarrow D^\alpha f(x) \int_{\mathbb{R}^n} g(z) dz = D^\alpha f(x)$$

□

Proposition. Let $U \subseteq \mathbb{R}^n$ be open and let $K \subseteq U$ be compact. Then there exists $\chi \in C_c^\infty(U)$ such that $\chi = 1$ in a neighbourhood of K .

Proof. Note that by the extreme value theorem there exists $\varepsilon > 0$ such that the nearest distance from K to the boundary of U is greater than 4ε .

Define $K_\varepsilon = K + \overline{B_{2\varepsilon}(0)}$. This is compact because it is the sum of 2 compact sets and it is contained in U .

Now define ϕ_ε as earlier. Consider $\chi = \phi_\varepsilon \star 1_{K_\varepsilon}$. Then since this is an L^1 function convolved with a C_c^∞ function we know that χ is C^∞ .

By the way, this process of smoothening a function by convolving it is called mollification, and the function ϕ_ε is called a mollifier.

Given χ and using the $\text{supp}(f \star g) \subseteq \text{supp}(f) + \text{supp}(g)$ property of convolutions proved earlier we get

$$\text{supp}(\chi) \subseteq K_\varepsilon + \text{supp}(\phi_\varepsilon) \subseteq K + \overline{B_{2\varepsilon}(0)} + \overline{B_\varepsilon(0)} = K + \overline{B_{3\varepsilon}(0)} \subseteq U$$

This implies that $\chi \in C_c^\infty(U)$. Now suppose $x \in K + B_\varepsilon(0)$, then $x + B_\varepsilon(0) \subseteq K_\varepsilon$ so

$$\begin{aligned} \chi(x) &= \int_{\mathbb{R}^n} \phi_\varepsilon(y) 1_{K_\varepsilon}(x-y) dy \\ &= \int_{B_\varepsilon(0)} \phi_\varepsilon(y) 1_{K_\varepsilon}(x-y) dy = \int_{B_\varepsilon(0)} \phi_\varepsilon(y) dy = 1 \end{aligned}$$

Thus $\chi(x)$ is indeed 1 in $x + B_\varepsilon(0)$ which is a neighbourhood of K .

□

1.3 Lebesgue differentiation theorem

From now on if S is a set and I write $|S|$ it means $\mu(S)$ unless otherwise specified.

Definition. Given an integrable function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ define the Hardy-Littlewood Maximal function Mf as follows:

$$Mf(x) = \sup_{r>0} \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y)| dy$$

Lemma. (Wiener's covering Lemma) Suppose $B_1, B_2, \dots, B_N$ is a finite collection of open balls in $\mathbb{R}^n$. Then there exists a subcollection $B_{i_1}, B_{i_2}, \dots, B_{i_k}$ of disjoint balls such that

$$\bigcup_{i=1}^N B_i \subseteq \bigcup_{j=1}^k 3B_{i_j}$$

It then follows that

$$\text{vol}\left(\bigcup_{i=1}^N B_i\right) \leq 3^n \sum_{j=1}^k \text{vol}(B_{i_j})$$

Proof. Without loss of generality assume $B_1, B_2, \dots, B_N$ are ordered largest to smallest. Start by picking B_1 , then it is easy to see that the next ball not fully covered by $3B_1$ is disjoint from B_1 since it is smaller; draw a picture if you need to, so we pick the next one not fully covered. Repeatedly pick the first not-fully-covered one until you are done; it is always disjoint from the previous ones because it is not covered by 3 times any of the previous ones.

□

Lemma. Let $f \in L^1(\mathbb{R}^n)$ for $n \geq 1$. Then Mf is measurable, finite almost everywhere, and for all $\lambda > 0$ we have

$$\mu(\{x : Mf(x) > \lambda\}) \leq \frac{3^n}{\lambda} \|f\|_{L^1}$$

Proof. Define $A_\lambda := \{x : Mf(x) > \lambda\}$. Then for each $x \in A_\lambda$ there exists a radius r_x such that

$$\frac{1}{\text{vol}(B_{r_x}(x))} \int_{B_{r_x}(x)} |f(y)| dy > \lambda$$

Now we first want to show that A_λ is open. Suppose for a contradiction that $x \in A_\lambda$, $x_m \rightarrow x$ and $x_m \notin A_\lambda$. Then by the dominated convergence theorem we have

$$\frac{1}{\text{vol}(B_{r_x}(x))} \int_{B_{r_x}(x_m)} |f(y)| dy \rightarrow \frac{1}{\text{vol}(B_{r_x}(x))} \int_{B_{r_x}(x)} |f(y)| dy$$

however, by assumption, the RHS is $> \lambda$, and also by assumption and definition of Mf the LHS is $\leq \lambda$ for all m , so this is a contradiction.

Since A_λ is open it follows that Mf is measurable.

Now let $K \subseteq A_\lambda$ be compact and cover it by $B_{r_x}(x)$ for each x and take a finite subcover $B_1, B_2, \dots, B_N$ by compactness. By Wiener's covering lemma there exists a disjoint subcollection $B_{i_1}, B_{i_2}, \dots, B_{i_k}$ with

$$|K| \leq 3^n \sum_{j=1}^k |B_{i_j}|$$

Therefore we have

$$|K| \leq \frac{3^n}{\lambda} \sum_{j=1}^k \int_{B_{i_j}} |f(y)| dy$$

since the volume of something is at least the integral of something at least λ over that volume divided by λ .

Therefore we have

$$|K| \leq \frac{3^n}{\lambda} \|f\|_{L^1}$$

The result follows since this holds for all $K \subseteq A_\lambda$ and the Lebesgue measure equals the supremum over such K .

Finally note that as $\lambda \rightarrow \infty$, $\mu(A_\lambda)$ tends to 0 so indeed Mf is finite almost everywhere. □

Theorem. (Lebesgue Differentiation Theorem) Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be integrable. Then for almost every $x \in \mathbb{R}^n$ we have

$$\lim_{r \rightarrow 0} \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| dy = 0$$

Note that this holds if f is defined on an open set $U \subseteq \mathbb{R}^n$ by considering $f1_U$ defined on $\mathbb{R}^n$.

Proof. For each $\lambda > 0$ define

$$A_\lambda = \left\{ x \in \mathbb{R}^n : \limsup_{r \rightarrow 0} \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| dy > 2\lambda \right\}$$

If we can show that $|A_\lambda| = 0$ for all λ then we will be done because the set of points that fail the proposition is $\bigcup_{n=1}^{\infty} A_{\frac{1}{n}}$.

To do this fix $\varepsilon > 0$. We can find $g \in C_c^\infty(\mathbb{R}^n)$ such that $\|f - g\|_{L^1} < \varepsilon$. (I proved this in my probability and measure notes). Now we have

$$\begin{aligned} & \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| dy \\ & \leq \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y) - g(y)| dy + \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |g(y) - g(x)| dy + |g(x) - f(x)| \end{aligned}$$

By the definition of the Hardy-Littlewood Maximal function we have the following bound for the first term:

$$\frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y) - g(y)| dy \leq M[f - g](x)$$

Since g is continuous we have

$$\limsup_{r \rightarrow 0} \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |g(y) - g(x)| dy = 0$$

Therefore

$$\limsup_{r \rightarrow 0} \frac{1}{\text{vol}(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| dy \leq M[f - g](x) + |f(x) - g(x)|$$

If $x \in A_\lambda$ that means that either $M[f - g](x) > \lambda$ or $|f(x) - g(x)| > \lambda$. By the previous lemma we know

$$\mu\{x : M[f - g](x) > \lambda\} \leq \frac{3^n}{\lambda} \varepsilon$$

By Markov's inequality (although this is not probability, it is the same principle, we have)

$$\mu\{x : |f(x) - g(x)| > \lambda\} \leq \frac{\|f - g\|_{L^1}}{\lambda} \leq \frac{\varepsilon}{\lambda}$$

Therefore we have

$$|A_\lambda| \leq \frac{1 + 3^n}{\lambda} \varepsilon$$

Thus it is 0 since ε was arbitrary.

□

Definition. We say that an x where the theorem above holds is a Lebesgue point of f .

1.4 Other measure theory stuff

Theorem. Suppose $A \subseteq \mathbb{R}^n$ is measurable with finite measure. Then for all $\varepsilon > 0$ there exists a set B which is a finite union of rectangles such that the symmetric difference $A \triangle B$ has measure less than ε

Proof. This follows from the fact that by general measure theory (i.e. the outer measure from the proof of the Caratheodory extension theorem) a countable union of rectangles covering the set can get to a measure within $\frac{\varepsilon}{2}$ (since an uncountable set of rectangles cannot have finite total measure) and some finite subcollection of the rectangles can have a symmetric difference of less than $\frac{\varepsilon}{2}$ to that countable collection of rectangles and therefore less than ε to the original set.

□

Theorem. (Egorov's theorem) Let $E \subseteq \mathbb{R}^n$ be measurable with finite measure and let $f_k \rightarrow f$ a.e. on E . Then given $\varepsilon > 0$ there exists a closed set $A_\varepsilon \subseteq E$ such that $\mu(E \setminus A_\varepsilon) \leq \varepsilon$ and $f_k \rightarrow f$ uniformly on A_ε .

Proof. By discarding a set of measure 0 if necessary we can assume $f_k \rightarrow f$ everywhere. For each $n, k \in \mathbb{N}$ let

$$E_k^n = \left\{ x \in E : |f_j(x) - f(x)| < \frac{1}{n} \text{ for all } j > k \right\}$$

Fixing n we note that $E_k^n \subseteq E_{k+1}^n$ and that $\bigcup_{k=1}^{\infty} E_k^n = E$ by pointwise convergence. By countable additivity it follows that $\mu(E_k^n) \xrightarrow{k \rightarrow \infty} \mu(E)$. Pick k_n such that $\mu(E \setminus E_{k_n}^n) < 2^{-n}$, then we have that for all $j > k_n$ and $x \in E_{k_n}^n$ that $|f_j(x) - f(x)| < \frac{1}{n}$. Now pick N with $\sum_{n=N}^{\infty} 2^{-n} < \frac{\varepsilon}{2}$ and let

$$A'_\varepsilon = \bigcap_{n=N}^{\infty} E_{k_n}^n$$

Now observe

$$\mu(E \setminus A'_\varepsilon) \leq \sum_{n=N}^{\infty} \mu(E \setminus E_{k_n}^n) < \frac{\varepsilon}{2}$$

Now let $\delta > 0$. Pick $n > \max(N, \frac{1}{\delta})$ and note that $x \in A'_\varepsilon$ implies that $x \in E_{k_n}^n$. We deduce that $|f_j(x) - f(x)| < \delta$ for all $j > k_n$ and therefore $f_j \rightarrow f$ uniformly on A'_ε . Also, we can pick a closed set $A_\varepsilon \subseteq A'_\varepsilon$ such that $\mu(A'_\varepsilon \setminus A_\varepsilon) < \frac{\varepsilon}{2}$ and thus $\mu(E - A_\varepsilon) < \varepsilon$.

□

Theorem. (Lusin's theorem) Let $E \subseteq \mathbb{R}^n$ be measurable with finite measure and let f be measurable and finite valued on E . Then for all $\varepsilon > 0$ there exists a closed set $F_\varepsilon \subseteq E$ with $\mu(E \setminus A_\varepsilon) \leq \varepsilon$ and f restricted to F_ε being continuous.

Example. If $f = 1_{\mathbb{Q}}$ on $[0, 1]$ then it is nowhere continuous but it would be if we consider it as a function of the irrationals between 0 and 1 which is “almost all” of its domain.

Proof. First suppose that f is a simple function so $f = \sum_{k=1}^m a_k 1_{A_k}$ for some measurable sets A_k with finite measure. We may suppose that the A_k are disjoint and they have union E (we may add a 0 term to achieve this). For all $\varepsilon > 0$ we can pick compact sets $K_k \subseteq A_k$ with

$$|A_k \setminus K_k| < \frac{\varepsilon}{m}$$

Let $B = \bigcup_{k=1}^m K_k$. Then $|E \setminus B| < \varepsilon$. Since the sets K_k are compact and disjoint (so the minimum distance between any 2 of them is strictly positive) and f is constant on each of them, it follows that f is continuous on B .

Now let f_n be a sequence of simple functions such that $f_n \rightarrow f$ almost everywhere (possible by measurability). Then we can find C_n with measure less than 2^{-n} with f_n continuous outside of them and by Egorov's theorem we can find a set $A_{\frac{\varepsilon}{3}}$ such that $\mu(E \setminus A_{\frac{\varepsilon}{3}}) < \frac{\varepsilon}{3}$ and $f_n \rightarrow f$ uniformly on $A_{\frac{\varepsilon}{3}}$. Let N be sufficiently large such that $\sum_{n=N}^{\infty} 2^{-n} < \frac{\varepsilon}{3}$. Now define

$$F'_\varepsilon = A_{\frac{\varepsilon}{3}} \setminus \bigcup_{n=N}^{\infty} C_n$$

This means that $\mu(E \setminus F'_\varepsilon) < \frac{2\varepsilon}{3}$ and f_n is continuous on $E \setminus F'_\varepsilon$ for each $n > N$. Since they converge uniformly to f it follows that f is continuous on F'_ε . Therefore, picking $F_\varepsilon \subseteq F'_\varepsilon$ closed with $\mu(E \setminus F_\varepsilon) < \varepsilon$ gives the result.

□

Definition. Let $(E, \mathcal{E}, \mu)$ be a measure space. It is said to be complete if it contains all subsets of measure 0 sets.

Proposition. Let $(E, \mathcal{M}, \mu)$ be a measure space obtained from the Caratheodory extension theorem. Then it is complete.

Proof. Let $A \in \mathcal{M}$ be a measurable subset and let N be a null set i.e. a subset of a measure 0 set. Then $A \cup N$ has the same outer measure as A and satisfies the criterion to be in $\mathcal{M}$, since for all $B \subseteq E$ we indeed have

$$\mu^*(B) = \mu^*(B \cap (A \cup N)) + \mu^*(B \cap (A \cup N)^C)$$

which is the criterion to be in $\mathcal{M}$, as adding subsets of N to a set affects its outer measures by an arbitrarily small amount and therefore by 0.

□

Proposition. Let $(E, \mathcal{M}, \mu)$ be a complete measure space and let f be a measurable function on E . If $g : E \rightarrow \mathbb{R}$ is such that $f = g$ almost everywhere then g is measurable.

Proof. Follows from the fact that $A = \{f < a\}$ is measurable and $\{g < a\}$ differs from that on a null set and thus is also measurable.

□

Proposition. Let $(E, \mathcal{E}, \mu)$ be a measure space and let $1 \leq p < \infty$ and let g_n be a sequence in $L^p(E, \mu)$ such that

$$\sum_{n=1}^{\infty} \|g_n\|_p < \infty$$

Then there exists $f \in L^p(E, \mu)$ such that

$$\sum_{n=1}^{\infty} g_n = f$$

where the sum converges pointwise almost everywhere and in $L^p(E, \mu)$

Proof. Fix representative functions from each equivalence class in L^p , i.e. for each $g_n \in L^p$ pick some $\tilde{g}_n$ in its equivalence class.

Define $h_n, h : E \rightarrow [0, \infty]$ by

$$h_n = \sum_{k=1}^n |\tilde{g}_k|$$

$$h = \sum_{k=1}^{\infty} |\tilde{g}_k|$$

The monotone convergence theorem implies that

$$\int_E h_n^p d\mu \rightarrow \int_E h^p d\mu$$

Define $K := \sum_{k=1}^{\infty} \|g_k\|_p$. By Minkowski's inequality we have

$$\|h_n\|_p \leq \sum_{k=1}^n \|g_k\|_p \leq K$$

It follows that $h \in L^p$ with $\|h\|_p \leq K$ which implies that h is finite almost everywhere. At each x where h is finite we have that $\sum_{k=1}^{\infty} \tilde{g}_k$ converges absolutely and therefore converges. This means the sum in the statement of the proposition converges almost everywhere to a function f that we will define to be 0 in places where the sum does not converge. In particular, this f satisfies $|f| \leq h$ so $\|f\|_p \leq \|h\|_p \leq K$ (note $\|f\|_p \leq \|h\|_p$ by the Minkowski inequality and the definitions of f and h). Also, we have

$$\left| f - \sum_{k=1}^n \tilde{g}_k \right|^p \leq \left(|f| + \sum_{k=1}^n |\tilde{g}_k| \right)^p \leq (2h)^p$$

So the dominated convergence theorem implies that $\sum_{n=1}^{\infty} \tilde{g}_n$ converges to f in L^p . Note that different choices of representatives result in everything differing on a countable union of measure 0 sets which is still a measure 0 set.

□

Theorem. Let f_n be a Cauchy sequence in $L^p(E, \mu)$ for $1 \leq p \leq \infty$. Then it has a subsequence that converges pointwise almost everywhere.

Proof. **Step 1.** First suppose $p < \infty$.

By the Cauchy property we can find a subsequence such that $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$. Then if $g_k = f_{n_{k+1}} - f_{n_k}$ then $\sum_{n=1}^{\infty} \|g_n\|_p < \infty$ so by the previous proposition there exists $g \in L^p$ such that $\sum g_k = g$ almost everywhere. The result therefore follows.

Step 2. Now suppose $p = \infty$.

For each $m \in \mathbb{N}$ there exists an n such that for all $j, k \geq n$ we have $|f_j(x) - f_k(x)| < \frac{1}{m}$ for all x except on a measure 0 set which we will call $N_{j,k,m}$. Then by countable additivity and properties of countable sets the union of all $N_{j,k,m}$ over all triples of integers j, k, m is a measure 0 set which we will call N .

By completeness of $\mathbb{R}$ or $\mathbb{C}$ we have that for each $x \notin N$ we have $f_j(x) \rightarrow f(x)$ for some $f(x) \in \mathbb{C}$ where we define $f(x)$ to be 0 in N . Then note that for $j, k \geq n$ we have

$$\sup_{x \notin N} |f_j(x) - f_k(x)| < \frac{1}{m}$$

Sending $k \rightarrow \infty$ we get

$$\sup_{x \notin N} |f_j(x) - f(x)| < \frac{1}{m}$$

So we conclude $f_j \rightarrow f$ in L^∞ and we actually didn't have to take a subsequence.

□

Theorem. Let E be a measure space and let $1 \leq p < \infty$ and let S be the set of all complex, measurable simple functions on E such that

$$\mu(\{x : s(x) \neq 0\}) < \infty$$

Then S is dense in $L^p(E, \mu)$

Proof. First let f be real and non-negative.

Let $f \geq 0$ with $f \in L^p(E, \mu)$ and define

$$f_n(x) = \min \{2^n \lfloor 2^{-n} f(x) \rfloor, n\}$$

Then $f_n \in S$ and $0 \leq f_n \leq f$ and $f_n, f \in L^p(E, \mu)$ and $f_n \rightarrow f$ and $|f - f_n|^p \leq |f|^p$ so the dominated convergence implies that $f_n \rightarrow f$ in L^p .

If f is general complex split it into its positive real, negative real, positive imaginary and negative imaginary parts and add the results.

□

2 Further Banach and Hilbert space analysis

2.1 Topological vector spaces

First we look at some basic properties of topological spaces.

Notation: Let E be any subset of a topological space. Then we write E° to be the interior of E which is the union of all open sets contained in E .

Definition. Recall that a base or basis for a topological space is a collection of open sets such that every open set is a union of some open sets in this collection. Define a local base to be a collection γ of open neighbourhoods of a point p such that every neighbourhood contains a member of γ .

Proposition. Let S be a Hausdorff topological space and let $E \subseteq S$. Then p is a limit point of E if and only if every neighbourhood of p contains infinitely many elements of E .

Proof. If every neighbourhood of p contains infinitely many elements of E then p is obviously a limit point.

Conversely, let p be a limit point and let U be a neighbourhood intersecting E in only finitely many points, say $\{x_1, x_2, \dots, x_N\}$. Then by the Hausdorff property we know there exists disjoint open sets U_i, V_i with $x_i \in U_i$ and $p \in V_i$ since the neighbourhood must intersect E in a point other than p itself by definition of a limit point.

Now consider $\bigcap_{i=1}^N V_i \cap U$. Then this contains p and does not contain any other points of E contradicting the limit point assumption. □

Proposition. Let $U \subseteq \mathbb{R}^n$ be open. Then there exists a sequence of compact sets K_n such that K_n is in the interior of K_{n+1} for all n and their union is U , and any compact set $K \subseteq U$ is contained in some K_n . We call this sequence an *exhaustion* of U .

Proof. It clearly holds if we let $d(x)$ be the nearest distance from x to the boundary of U and observe that this is a continuous function, then define

$$K_n = \overline{B_n(0)} \cup \left\{ x \in U : d(x) \geq \frac{1}{n} \right\}$$

Then each arbitrary K compact is contained in some K_n by boundedness and the fact that it has a minimum distance that is greater than $\frac{1}{n}$ for some n by the extreme value theorem. □

In this section a topological vector space has the same definition as in linear analysis. In this course, we add the additional condition that sets consisting of single points are closed, which we will later show implies (in a topological vector space) that the space is Hausdorff, and is therefore equivalent as we know from the topological spaces course that Hausdorff implies sets consisting of single points are closed.

Definition. A subset E of a topological vector space is bounded if for every neighbourhood V of 0 there exists $s > 0$ such that $E \subseteq tV$ whenever $t > s$.

Theorem. A normed space with the metric topology is a topological vector space. Furthermore, in a normed space E is bounded in the above sense if and only if it is bounded in the norm sense.

Proof. Note that the metric topology is Hausdorff so one-point sets are indeed closed.

Step 1. Check that addition is continuous

Let U be an open set in X and let $x, y \in X$ such that $z = x + y \in U$. By openness of U in the norm topology there exists $r > 0$ such that $B_r(z) \subseteq U$. If we let $W = B_{r/2}(x) \times B_{r/2}(y)$ then $W \subseteq (+)^{-1}(U)$ so every point in $(+)^{-1}(U)$ has a product of open balls around it, so W is open in the product topology. This implies that pre-images of open sets are open so addition is continuous.

Step 2. Check that scalar multiplication is continuous

Let ϕ be the underlying field for the vector space. Let $x \in X$ and $a \in \phi$ such that $z = ax \in U$. By openness of U in the norm topology there exists $r > 0$ such that the set $B_r(z) \subseteq U$.

Let $0 < \varepsilon < \frac{\min\{r,1\}}{1+\|x\|+|a|}$. Let $W = B_\varepsilon(a) \times B_\varepsilon(x)$ and let $(a', x') \in W$. Then we have

$$\begin{aligned} \|a'x' - z\| &= \|(a' - a)x' - a(x - x')\| \leq \|(a' - a)x'\| + \|a(x - x')\| < \varepsilon(\|x\| + \varepsilon) + |a|\varepsilon \\ &\leq \varepsilon(\|x\| + 1 + |a|) < r \end{aligned}$$

Therefore we have $W \subseteq (\cdot)^{-1}(U) \subseteq \phi \times X$. Since $(a, x) \in (\cdot)^{-1}(U)$ was arbitrary and there is a product of open balls W around it in $(\cdot)^{-1}(U)$, it follows that $(\cdot)^{-1}(U)$ is open so multiplication is continuous.

Step 3. Check the thing about boundedness

Suppose $E \subseteq X$ is bounded in the sense defined above. Then $E \subseteq tB_1(0)$ for some $t > 0$ which implies that E is bounded in the norm sense.

Conversely, if E is bounded in the norm sense then it is contained in $B_M(0)$ and every open neighbourhood of 0 contains some $B_r(0)$ by definition so for this open neighbourhood $s = \frac{M}{r}$ works to make E bounded in the sense defined above.

□

Remark. Although the theorem above establishes equivalence in a specific case, notions of boundedness can be not the same in general.

Proposition. Let S_1, S_2, S_3 be topological spaces and let $f : S_1 \times S_2 \rightarrow S_3$ be continuous with respect to the product topology. Then for each $a \in S_1$ and $b \in S_2$ the maps $f_a : S_2 \rightarrow S_3$ sending y to $f(a, y)$ and $f_b : S_1 \rightarrow S_3$ sending x to $f(x, b)$ are continuous.

Note that the converse is false: Continuity with each variable constant does not generally imply continuity.

Proof. For $a \in S_1$ define $i_a : S_2 \rightarrow S_1 \times S_2$ by $i_a(y) = (a, y)$. Then this is continuous since pre-images of unions of products of open base sets must be unions of open base sets in S_2 and thus open in S_2 . Then since f is continuous and $f_a = f \circ i_a$, f_a is also continuous since compositions of continuous functions are continuous. The symmetric argument proves the f_b case.

□

Corollary. Addition and scalar multiplication by non-zero values are homeomorphisms from a vector space X to itself since they are continuous and have continuous inverses. This principle tells us that subsets E of a topological vector space are open if and only if all translates $a + E$ and non-zero multiples λE are open. In particular, we have the following:

Proposition. Let X be a topological vector space and let β' be a local base at 0. Then the collection

$$\beta = \{a + B : a \in X, B \in \beta'\}$$

is a base for the topology of X .

Proof. Note that β consists of open sets since translations of open sets are open in topological vector spaces. Let U be open and pick $x \in U$, then we have that $-x + U$ is a neighbourhood of the origin so there exists $B \in \beta'$ such that $x + B \subseteq U$. Therefore for any U open we have an element of β contained in U containing every point so β is a base for the topology of X .

□

Proposition. Let C be a convex subset of a topological vector space X . Then C° is convex.

Proof. Given $t \in (0, 1)$ and $x, y \in C^\circ$ we need to prove $tx + (1 - t)y \in C^\circ$. Since $x \in C^\circ$ there exists an open neighbourhood U of x such that $U \subseteq C$. Now define $\phi(v) = tv + (1 - t)y$, then this is continuous since we are in a topological vector space, and in fact it is a homeomorphism, so $V = \phi(U) = tU + (1 - t)y$ is an open set containing $\phi(x)$. Since $U \subseteq C$ and $y \in C$ and C is convex it follows that any point in V is a convex combination of two points in C so we must have $V \subseteq C$ and therefore $tx + (1 - t)y \in V \subseteq C^\circ$. □

Definition. Let U be a subset of a vector space V . We say U is balanced if $\lambda U \subseteq U$ for all $\lambda \in \phi$ with $|\lambda| \leq 1$.

Proposition. Let X be a topological vector space. Then if $U \subseteq X$ is an open neighbourhood of 0 then U contains a balanced open neighbourhood of 0.

Proof. Since scalar multiplication is continuous there exists a $\delta > 0$ and V open such that $\alpha V \subseteq U$ for all $|\alpha| < \delta$. So the set of all αv with $|\alpha| < \delta$ and $v \in V$ is the desired balanced open neighbourhood. □

Proposition. Let X be a topological vector space. Then if $U \subseteq X$ is an open convex neighbourhood of 0 then U contains a convex balanced open neighbourhood of 0.

Proof. Let $A = \bigcap_{|\alpha|=1} \alpha U$ and let W be the balanced neighbourhood from the proof of the previous proposition. Since W is balanced we have $\alpha^{-1}W = W$ whenever $|\alpha| = 1$ so $W \subseteq \alpha U$ for all $|\alpha| = 1$ so $W \subseteq A$. This means that A° is a neighbourhood of the origin and $W \subseteq A^\circ \subseteq A \subseteq U$ so A° is a non-empty open neighbourhood of 0. Since U is convex, αU is for all α so A is the intersection of convex sets so A is also convex. Therefore, A° is also convex as the interior of a convex set in a topological vector space.

It remains to prove that A° is balanced.

Let $0 \leq r \leq 1$ and let $|\beta| = 1$. To show A is balanced it suffices to show that $r\beta A \subseteq A$. Note that

$$r\beta A = \bigcap_{|\alpha|=1} r\beta\alpha U = \bigcap_{|\alpha|=1} r\alpha U$$

Since αU is convex and contains 0 we have $r\alpha U \subseteq \alpha U$ (since αU contains all points on the line segment from 0 to αu for every $u \in U$) so it follows that A is balanced.

To show that A° is also balanced note that for $\alpha = 0$ we have $0A^\circ \subseteq A^\circ$ and if $0 < \alpha \leq 1$ then $x \mapsto \alpha x$ is a map sending open subsets of A to open subsets of A since it is a homeomorphism and A is balanced, so it also sends A° to open subsets of A° which means $\alpha A^\circ \subseteq A^\circ$. □

Theorem. A topological vector space as defined with the “sets of single points are closed” property is Hausdorff.

Proof. We first claim that every neighbourhood W of 0 contains a balanced open set U such that $U + U \subseteq W$. Indeed, note that $0 + 0 = 0$ so by continuity of $+$ there exists neighbourhoods U_1, U_2 of 0 such that $U_1 + U_2 \subseteq W$, so $U' = U_1 \cap U_2$ has $U' + U' \subseteq W$. Then taking a balanced open neighbourhood of 0 gives the desired U .

Now let $x, y \in X$ with $x \neq y$. Since $x \in \{y\}^C$ there exists an open neighbourhood W of x with $y \notin W$ so $W - x$ is a neighbourhood of 0 so there exists balanced U with $U + U \subseteq W - x$. Then it remains to check that $x + U$ and $y + U$ have empty intersection as this will imply Hausdorffness as x, y were chosen arbitrarily. If there exists $a, b \in U$ with $x + a = y + b$ then $y = x + a - b$ and $a, -b \in U$ by balancedness so $y \in x + U + U$ which is a contradiction. □

Therefore in this course we will assume Hausdorff is part of the definition of a topological vector space.

Theorem. Sets consisting of single points are bounded in a topological vector space X .

Proof. Fix $x \in X$ and let $f_x(\lambda) = \lambda x$. Then this is continuous by an earlier proposition so $f_x^{-1}(W)$ is open in $\mathbb{R}$ for all W an open neighbourhood of 0 in X . Since $0 \in W$ we have that there is some ε with $(-\varepsilon, \varepsilon) \in f_x^{-1}(W)$. In other words, $\lambda x \in W$ for $\lambda \in (0, \varepsilon)$ which means that $x \in tW$ for $t > \varepsilon^{-1}$.

□

Theorem. If E_1, E_2 are bounded sets in a topological vector space then $E_1 + E_2$ is bounded.

Proof. Let W be an open neighbourhood of 0, then by earlier work there exists an open neighbourhood U of 0 such that $U + U \subseteq W$. By boundedness of E_1, E_2 there exists $s \in \mathbb{R}$ such that if $t > s$ then $t^{-1}E_1 \subseteq U$ and $t^{-1}E_2 \subseteq U$. Then we have that for each such t

$$t^{-1}(E_1 + E_2) \subseteq U + U \subseteq W$$

This implies that $E_1 + E_2 \subseteq tW$ for t sufficiently large so $E_1 + E_2$ is bounded.

□

Corollary. If E is bounded in a topological vector space and x is a point then $x + E$ is bounded by the previous 2 theorems.

Theorem. If x_n is a sequence in a topological vector space X such that the set $\{x_n : n \in \mathbb{N}\}$ is bounded and a_n is a sequence of real or complex numbers with $a_n \rightarrow 0$ then $a_n x_n \rightarrow 0$.

Proof. Let W be an open neighbourhood of 0 in X . Then we know there exists a balanced and open U with $U \subseteq W$. Then there exists $s > 0$ such that $x_n \in tU$ for all $t \geq s$ and all $n \in \mathbb{N}$. Since $a_n \rightarrow 0$ there exists N such that $|a_n| < s^{-1}$ for all $n \geq N$. We have $x_n \in sU$ and $|sa_n| < 1$ for all $n \geq N$, and $|sa_n| < 1$ implies (since U is balanced) that $sa_n x_n \in sU$ so $a_n x_n \in U \subseteq W$ for n sufficiently large.

Since $a_n x_n$ eventually goes into every open neighbourhood of 0 we conclude that it converges to 0.

□

Definition. Let X be a vector space with a metric d . Then d is said to be invariant if for all $x, y, z \in X$ we have $d(x + z, y + z) = d(x, y)$.

Proposition. Let X be a vector space with an invariant metric and induced topology. Then given a sequence $x_n \subseteq X$ with $x_n \rightarrow 0$ there exist scalars $a_n \rightarrow \infty$ with $a_n x_n \rightarrow 0$

Proof. First we claim that if d is invariant then $d(nx, 0) \leq nd(x, 0)$ for all integers n . It is true for $n = 1$ and by induction we have (if it is true for $1, 2, 3, \dots, k-1$) that

$$d(kx, 0) \leq d(kx, x) + d(x, 0) \leq (k-1)d(x, 0) + d(x, 0) = kd(x, 0)$$

Now since $x_n \rightarrow 0$ for any $m \in \mathbb{N}$ there exists N_m such that $n > N_m$ implies $d(x_n, 0) < m^{-2}$. Now assume $N_m < N_{m+1}$ and define $a_n = m$ for $N_m \leq n < N_{m+1}$. Then for any n , where we let m be such that $N_m \leq n < N_{m+1}$, we have $d(a_n x_n, 0) \leq md(x_n, 0) < m^{-1} \rightarrow 0$ as required.

□

Definition. A Cauchy sequence in a topological vector space is one such that for every neighbourhood U of 0 there exists an integer N such that $x_n - x_m \in U$ whenever $n, m \geq N$.

Note that if X is a vector space with an invariant metric d and the topology induced by the metric, then a sequence is Cauchy with respect to the metric d if and only if it is Cauchy with respect to the topology of X .

In the following definitions, X is a topological vector space.

Definition. X is a locally convex topological vector space if it has a local base consisting of convex members.

Definition. X is locally bounded if 0 has a bounded neighbourhood

Definition. X is locally compact if 0 has a compact neighbourhood

Definition. X is metrizable if there exists a metric d on X such that its topology is the metric topology induced by d

Definition. X is an F-space if its topology is induced by a complete invariant metric

Definition. X is a Frechet space if it is a locally convex F-space

Definition. X is normable if it is metrizable and we can put a norm on it with that corresponding metric.

Definition. X has the Heine-Borel property if every closed and bounded (in the topological vector space sense) subset of X is compact. Note that bounded in both sense agrees in $\mathbb{R}^n$ with, say, the weak topology trivially by definition.

Easy exercise: Show that $\mathbb{R}^n$ belongs to all classes above. We've done most of the parts in earlier levels.

Example. We showed in Linear analysis that an infinite dimensional space with the weak topology is not normable.

Note that the converse to the Heine-Borel property is always true:

Proposition. Let X be a topological vector space and let $K \subseteq X$ be compact. Then K is closed and bounded.

Proof. It is closed because it is a standard fact in topology that compact sets in Hausdorff spaces are closed. Therefore we just need to show that compact $\implies$ bounded.

Let U be a neighbourhood of 0 . Let $W \subseteq U$ be an open, balanced neighbourhood of 0 - we proved earlier that this exists.

Next we want to show that

$$\bigcup_{n=1}^{\infty} nW = X$$

To do this, fix $x \in X$. Since the map $a \mapsto ax$ is continuous, the set of all a with $ax \in W$ is open and contains 0 and therefore contains n^{-1} for sufficiently large n . Therefore $n^{-1}x \in W$ so $x \in nW$ for sufficiently large n . Since W is balanced, we have $sW \subseteq tW$ whenever $0 < s < t$.

Now since $U = \{nW : n \in \mathbb{N}\}$ is an open cover of X , it is also an open cover of K . Therefore by compactness there exists a finite collection $n_1, n_2, \dots, n_N$ (assume without loss of generality these are in ascending order) such that

$$K \subseteq \bigcup_{i=1}^N n_i W = n_N W \subseteq n_N U$$

Since U was arbitrary it follows that K is bounded.

□

We will now talk about locally convex topological vector spaces. Recall from Linear analysis that a seminorm has all the properties of a norm except for the “ 0 if and only if x is 0 ” property. We say that the $\lambda p(x) = p(\lambda x)$ for $\lambda > 0$ property is called being positive homogeneous.

If p is a seminorm then a standard triangle inequality argument gives $|p(x) - p(y)| \leq p(x - y)$ (hint: to show $|x| \leq y$ show that $x \leq y$ and $-x \leq y$). We have that $p(x) \geq 0$ for all x and $p(0) = 0$.

Proposition. Let p be a seminorm on a vector space X , then the set $\{x : p(x) = 0\}$ is a vector subspace of X .

Proof. Suppose $p(x) = p(y) = 0$. By the triangle inequality.

$$0 \leq p(\lambda x + \mu y) \leq |\lambda|p(x) + |\mu|p(y) = 0$$

so $p(\lambda x + \mu y) = 0$.

□

Proposition. Let p be a seminorm on a vector space X , then the set $B = \{x : p(x) < 1\}$ is convex and balanced

Proof. B is balanced by the property $p(\lambda x) = |\lambda|p(x)$. To see that B is convex let $x, y \in B$ and let $0 < t < 1$. Then

$$p(tx + (1 - t)y) \leq tp(x) + (1 - t)p(y) < 1$$

so $tx + (1 - t)y \in B$ which implies B is convex as t, x, y were arbitrary.

□

Lemma. Suppose we have a set S and a collection β of subsets satisfying

1. If $x \in S$ there exists some $B \in \beta$ with $x \in B$.
2. If $B_1, B_2 \in \beta$ then for all $x \in B_1 \cap B_2$ there exists $B \in \beta$ with $x \in B$ and $B \subseteq B_1 \cap B_2$

Then the set T of unions of elements of β is a topological space.

Proof. It contains an empty set by the nullary union, and by property 1 the set S itself is in T . It is clearly closed under arbitrary unions, and by property 2 we see that it is closed under finite intersections, by picking a B for every $x \in B_1 \cap B_2$ and taking the union of each such B to get $B_1 \cap B_2 \in \beta$ so that we can extend to finite intersections by induction.

□

Definition. A family P of seminorms is said to be separating if for all $x \in X$ with $x \neq 0$ there exists $p \in P$ with $p(x) \neq 0$.

Theorem. Let P be a separating family of seminorms on a vector space X . Associate to each $p \in P$ and $n \in \mathbb{N}$ with $n > 0$ the set

$$V(p, n) = \left\{ x \in X : p(x) < \frac{1}{n} \right\}$$

Let β' be the collection of all finite intersections of the sets $V(p, n)$. Then β' is a convex, balanced, local base for a topology T on X which turns X into a locally convex topological vector space that makes every $p \in P$ continuous and with the property that $E \subseteq X$ is bounded if and only if every $p \in P$ is bounded on E .

Proof. Define

$$\beta := \{x + B, x \in X, B \in \beta'\}$$

Step 1. We will prove that β is a base for a topology on X which we will call T .

Since $0 \in B$ for any $B \in \beta'$ we have that for any $x \in X$ there is an element of β containing x . Now let $B_1, B_2 \in \beta$. We will write

$$B_1 = y + \bigcap_{i=1}^N V(p_i, n_i)$$

$$B_2 = z + \bigcap_{j=1}^M V(q_j, m_j)$$

for some $y, z \in X$ and $p_i, q_j \in P$ and $N, M, n_i, m_j \in \mathbb{N}$.

Fix $x \in B_1 \cap B_2$. Then $x \in B$ for some $B \in \beta$. We claim that for n'_i, m'_j sufficiently large, B defined as follows satisfies $B \subseteq B_1 \cap B_2$

$$B = x + \left(\bigcap_{i=1}^N V(p_i, n'_i) \right) \cap \left(\bigcap_{j=1}^M V(q_j, m'_j) \right)$$

To prove this, note that by definition of the V 's and the fact that $x \in B_1 \cap B_2$ we have

$$p_i(x - y) < \frac{1}{n_i} \text{ and } q_j(x - z) < \frac{1}{m_j} \text{ for all } i = 1, 2, \dots, N, j = 1, 2, \dots, M$$

Therefore there exists n'_i and m'_j sufficiently large so that

$$p_i(x - y) + \frac{1}{n'_i} < \frac{1}{n_i} \text{ and } q_j(x - z) + \frac{1}{m'_j} < \frac{1}{m_j} \text{ for all } i = 1, 2, \dots, N, j = 1, 2, \dots, M$$

Note also that if $w \in B$ then

$$p_i(w - x) < \frac{1}{n'_i} \text{ and } q_j(w - x) < \frac{1}{m'_j} \text{ for all } i = 1, 2, \dots, N, j = 1, 2, \dots, M$$

By the triangle inequality for seminorms we have

$$p_i(w - y) \leq p_i(w - x) + p_i(x - y) < \frac{1}{n'_i} + p_i(x - y) < \frac{1}{n_i}$$

For the same reason, $q_j(w - z) < \frac{1}{m_j}$ so indeed we have $B \subseteq B_1 \cap B_2$ since $w \in B$ implies $w \in B_1 \cap B_2$.

Therefore we have the conditions of the earlier lemma so β is indeed a base for a topology on X .

Step 2. We will prove that β' is a local base that is convex and balanced.

To establish that it is a local base, let U be any open neighbourhood of 0. Then U contains some basis element $y + B, B \in \beta'$ containing 0. Note that $-y \in B$. It suffices to prove that $y + B$ contains an element of β' . Write $B = \bigcap_{i=1}^N V(p_i, n_i)$, then for every $i \in \{1, 2, \dots, N\}$ we have $p_i(-y) < \frac{1}{n_i}$. Define $\varepsilon_i = \frac{1}{n_i} - p_i(-y)$ and as earlier pick m_i sufficiently large so that $\frac{1}{m_i} < \varepsilon_i$. Now define $B' = \bigcap_{i=1}^N V(p_i, m_i)$. To show that $B' \subseteq y + B$ let $w \in B'$. By the triangle inequality we have

$$p_i(w - y) \leq p_i(w) + p_i(-y) < \frac{1}{m_i} + p_i(-y) < \frac{1}{n_i}$$

Since this is true for all i we have $w - y \in B$ so $w \in y + B$ as required.

To prove that each element of β' is convex and balanced it is sufficient to show that each $V(p, n)$ is convex and balanced because intersections of convex sets are convex and balanced. The good news is that we recently proved that $\{x : B(x) < 1\}$ is convex and balanced for any seminorm, so just replace 1 with $\frac{1}{n}$ then we are done.

Step 3. We aim to prove that we have a topological vector space. The first step in doing this will be to show that singletons $\{x\}$ are closed.

Let $x, y \in X$ with $x \neq y$. Then since $x - y \neq 0$ and P is separating, there exists $p \in P$ such that $p(x - y) > 0$. Thus, there exists $n \in \mathbb{N}$ such that $np(x - y) > 1$. For this n we have that $x \notin (y + V(p, n))$ so we may write $\{x\}^C$ as a union of sets like this which are open in T which implies that $\{x\}$ is closed.

Step 4. The second step in showing that we have a topological vector space will be to show that addition is continuous.

Let U be open in X and pick $z \in U$. Then there exists $p_i \in P$ and $n_i, N \in \mathbb{N}$ such that

$$\bigcap_{i=1}^N V(p_i, n_i) \subseteq -z + U$$

Now let $(x, y) \in +^{-1}(z)$ which is a fancy way of saying $x + y = z$. Define

$$V_1 = x + \bigcap_{i=1}^N V(p_i, 2n_i) \text{ and } V_2 = y + \bigcap_{i=1}^N V(p_i, 2n_i)$$

and suppose that $(w_1, w_2) \in V_1 \times V_2$. Then for all i we have

$$p_i(w_1 + w_2 - z) = p_i(w_1 - x + w_2 - y) < \frac{1}{2n_i} + \frac{1}{2n_i} = \frac{1}{n_i}$$

This means that $V_1 + V_2 \subseteq U$ so $V_1 \times V_2 \subseteq (+)^{-1}(U)$ which means that everything in $(+)^{-1}(U)$ has a product topology basis set around it so is open so addition is continuous.

Step 5. The final step in showing that we have a topological vector space will be to show that scalar multiplication is continuous.

Let U be open in X and pick $z \in U$. Then there exists $p_i \in P, n_i, N \in \mathbb{N}$ such that

$$\bigcap_{i=1}^N V(p_i, n_i) \subseteq -z + U$$

Suppose that $(a, x) \in (\cdot)^{-1}(z)$ which is a fancy way of saying $ax = z$.

Let ϕ be the underlying field (from now on we will work in $\mathbb{C}$ unless stated otherwise) and take $0 < \varepsilon < \min_i (2n_i p_i(x))^{-1}$ where the min is over i with $p_i \neq 0$ and $n'_i > 2n_i (|\alpha| + (2n_i p_i(x))^{-1})$. (If) Define

$$V = x + \bigcap_{i=1}^N V(p_i, n'_i) \text{ and } D = \{\beta \in \phi : |\alpha - \beta| < \varepsilon\}$$

Suppose $(b, y) \in D \times V$, then it suffices to prove that $by \in U$ as then $(\cdot)^{-1}(U)$ will contain an open base set about each point, so let's now verify this.

First assume $p_i(x)$ is not 0 for any x .

We have

$$p_i(by - ax) = p_i(b(y - x) - (a - b)x) \leq p_i(b(y - x)) + p_i((a - b)x) \leq \frac{|b|}{n'_i} + \varepsilon p_i(x)$$

The last inequality is by definition of V, D and the fact that we are working with seminorms. Then we have

$$\leq \frac{|\alpha| + \varepsilon}{n'_i} + \varepsilon p_i(x)$$

Now substituting in our definitions for ε and n'_i we have

$$< \frac{|\alpha| + (2n_i p_i(x))^{-1}}{2n_i (|\alpha| + (2n_i p_i(x))^{-1})} + (2n_i p_i(x))^{-1} p_i(x) = \frac{1}{n_i}$$

This implies that $(by - ax) \in \bigcap_{i=1}^N V(p_i, n_i)$ so an open set about each point is in the pre-image of an arbitrary open set.

If $p_i(x) = 0$ for some x then we can define $n'_i > n_i (|a| + \varepsilon)$ and do the bounding as follows:

$$p_i(by - ax) \leq |b| p_i(y) \leq \frac{|\alpha| + \varepsilon}{n'_i} < \frac{1}{n_i}$$

so we still get the desired result.

Step 6. We have a topological vector space. The next thing to check is that it makes p continuous.

If $a < b$ we want to show that $p^{-1}(a, b)$ is open (it is enough to do this for the base sets as if pre-images of those are open then so are pre-images of their unions which are unions of their pre-images). Let $x \in p^{-1}(a, b)$ i.e. $a < p(x) < b$. Consider $U = x + V(p, n)$, then if $y \in U$ then $|p(y) - p(x)| \leq p(y - x) < \frac{1}{n}$ by the triangle inequality so for n sufficiently large we have $p(y) \in (a, b)$ so that $x \in U \subseteq p^{-1}(a, b)$ and U is one of the open base sets.

Step 7. It remains to show that $E \subseteq X$ is bounded if and only if every $p \in P$ is bounded on E .

Suppose E is bounded and fix $p \in P$. Since $V(p, 1)$ is a neighbourhood of the origin, we have that $E \subseteq kV(p, 1)$ for some $k < \infty$ by boundedness, which implies $p(x) < k$ for all $x \in E$ so p is bounded on E .

Conversely, suppose that every $p \in P$ is bounded on E . Let U be a neighbourhood of the origin. Then there exists $p_i \in P$ and $n_i, N \in \mathbb{N}$ such that

$$\bigcap_{i=1}^N V(p_i, n_i) \subseteq U$$

Also by assumption there exist $M_i < \infty$ such that $p_i < M_i$ on E for each $i \in \{1, 2, \dots, N\}$. It now follows straightforwardly that if $n > M_i n_i$ for all i then $E \subseteq \bigcap_{i=1}^N V\left(p_i, \frac{1}{M_i}\right) \subseteq nU$, which establishes boundedness.

□

Note: The topology induced by P as described in the theorem above is just like in linear analysis where it is the coarsest topology that makes all elements of P continuous. The reason why this is true is that any topology making each p continuous must contain each ε -open ball with respect to p , and thus any finite intersection of these sets i.e. for each fixed x any such topology must contain all

$$U = \bigcap_{j=1}^m B_{p_j}(x, \varepsilon_j) = x + \bigcap_{j=1}^m \{y \in X : p_j(y) < \varepsilon_j\}$$

Theorem. Let $P = \{p_i : i \in \mathbb{N}\}$ be a countable separating family of seminorms on a vector space X and let T be the topology induced by P as described in the theorem above. Then the locally convex topological vector space is metrizable and the topology agrees with the one induced by the invariant metric

$$d(x, y) = \sum_{i=1}^{\infty} 2^{-i} \frac{p_i(x - y)}{1 + p_i(x - y)}$$

Proof. Without loss of generality assume all p_i 's are non-zero since the zero ones won't contribute anything to the metric and being separating only requires the condition on non-zero seminorms.

Step 1. To see that d is an invariant metric, note that it is symmetric since $p(-x) = p(x)$ for a seminorm and if it were to be a metric it would be invariant due to depending only on $x - y$.

Note that $F : t \mapsto \frac{t}{1+t}$ is smooth, increasing, concave and bijects $[0, \infty)$ into $[0, 1)$. Thus, $d(x, y)$ is a sum of non-negative terms and thus also non-negative, and it is non-zero if $x \neq y$ since the family is separating so at least one non-zero p_i must be non-zero when evaluated at $x - y$.

It remains to check that d satisfies the triangle inequality. Note that F satisfies $F(x) + F(y) \geq F(x + y)$ since it is concave and $F(0) = 0$. Now since p is a seminorm and F is increasing and subadditive we have

$$F(p(x - y)) \leq F(p(x - z) + p(z - y)) \leq F(p(x - z)) + F(p(z - y))$$

Thus we conclude

$$d(x, y) = \sum 2^{-i} F(p_i(x - y)) \leq \sum (2^{-i} F(p_i(x - z)) + F(p_i(z - y))) = d(x, z) + d(z - y)$$

Step 2. Now we need to show that the topology induced by d agrees with the topology T induced by the family of seminorms P . The open sets of T_d are precisely those which can be written as a union of the open balls $B_r(x) = \{y \in X : d(x, y) < r\}$.

Step 2A. We will show that $T_d \subseteq T$.

We know that each p_i is T -continuous on X by the previous theorem. Since $|F(t)| < 1$ we conclude by the Weierstrass M -test that the sum in $d(x, y)$ converges uniformly. Therefore d is continuous as a real-valued function on $X \times X$ with the product topology coming from T since it is a uniform limit of continuous functions (to see this, d is continuous and $x, y \mapsto x - y$ is continuous and we are composing these). Therefore the map $d_x : X \rightarrow \mathbb{R}$ given by $d_x(y) = d(x, y)$ is continuous, so $d_x^{-1}(-r, r) = B_r(x)$ is an open set in the T topology so $T_d \subseteq T$.

Step 2B. We will show that $T \subseteq T_d$.

Let W be an open set of T so that $x \in W$. By definition of T there exists B with $x + B \subseteq W$ where there exists $p_{i_k} \in P$ and $n_k, N \in \mathbb{N}$ such that

$$B = \bigcap_{k=1}^N V(p_{i_k}, n_k)$$

Now suppose $d(x, y) < \varepsilon * 2^{-M}$ where $M \geq i_k$ and $0 < \varepsilon < \frac{1}{2}n_k^{-1}$ for all $k = 1, 2, \dots, N$. Note that $\varepsilon < \frac{1}{2}$. Then for $1 \leq i \leq M$ we have

$$\frac{p_i(x - y)}{1 + p_i(x - y)} \leq \varepsilon$$

so we have

$$p_i(x - y) \leq \frac{\varepsilon}{1 - \varepsilon} \leq 2\varepsilon$$

Therefore if $y \in B_{\varepsilon 2^{-M}}(x)$ then $y - x \in B$ so $y \in W$. Since x was arbitrary we can write W as a union of open balls for the metric d so $T \subseteq T_d$.

□

Note that the balls $B_r(0)$ need not be convex even though the topology is, and even though $d(x, y) < 1$ for all $x, y \in X$ it is not true that all subsets of X are bounded in the sense introduced above for a topological space.

Theorem. Let X and Y be a topological vector spaces whose topologies are generated by the families of seminorms P and Q respectively. A linear map $T : X \rightarrow Y$ is continuous if and only if for every seminorm $q \in Q$, there exist $p_1, \dots, p_n \in P$ and a constant $C > 0$ such that for all $x \in X$:

$$q(Tx) \leq C \max_{1 \leq i \leq n} p_i(x)$$

Proof. ($\Rightarrow$) Assume T is continuous. Let $q \in Q$. The set

$$V = \{y \in Y : q(y) < 1\}$$

is an open neighbourhood of the origin in Y . Because T is continuous at the origin, the preimage

$$T^{-1}(V) = \{x \in X : q(Tx) < 1\}$$

is an open neighbourhood of the origin in X . By the definition of the topology on X generated by P , this preimage must contain a basic open neighbourhood of the origin. Therefore, there exist an $\varepsilon > 0$ and a finite subset of seminorms $p_1, \dots, p_n \in P$ such that the set

$$U = \{x \in X : p_i(x) < \varepsilon \text{ for all } i = 1, \dots, n\}$$

satisfies $U \subseteq T^{-1}(V)$. Define the continuous seminorm $p(x) = \max_{1 \leq i \leq n} p_i(x)$. We now have that $p(x) < \varepsilon \implies q(Tx) < 1$. For any $x \in X$, consider two cases: If $p(x) > 0$, scale x by choosing $x' = \frac{\delta}{p(x)}x$ for any $0 < \delta < \varepsilon$. Then $p(x') = \delta < \varepsilon$, which guarantees $x' \in U$. Consequently, $q(Tx') < 1$. By the linearity of T and the absolute homogeneity of seminorms, we can factor out the scalar to get $\frac{\delta}{p(x)}q(Tx) < 1$, meaning $q(Tx) < \frac{1}{\delta}p(x)$. Taking the limit as $\delta \rightarrow \varepsilon$ yields $q(Tx) \leq \frac{1}{\varepsilon}p(x)$. If $p(x) = 0$, then for any $\alpha > 0$, we have $p(\alpha x) = \alpha p(x) = 0 < \varepsilon$. This implies $\alpha x \in U$, so $q(T(\alpha x)) < 1$, leading to $q(Tx) < 1/\alpha$. Since this holds for all $\alpha > 0$, we must have $q(Tx) = 0$, which trivially satisfies $q(Tx) \leq \frac{1}{\varepsilon}p(x)$. Setting $C = 1/\varepsilon$ completes the forward implication.

($\Leftarrow$) Assume the seminorm inequality holds. Because T is linear, it suffices to prove continuity at the origin. The topology on Y is generated by finite intersections of subbasic open sets of the form $V_{q,\varepsilon} = \{y \in Y : q(y) < \varepsilon\}$ for $q \in Q$ and $\varepsilon > 0$. It is therefore sufficient to show that the preimage of any such $V_{q,\varepsilon}$ contains an open neighbourhood of the origin in X . By assumption, for our chosen q , there exists $C > 0$ and $p_1, \dots, p_n \in P$ such that

$$q(Tx) \leq C \max_{1 \leq i \leq n} p_i(x)$$

Define the open set

$$U = \{x \in X : p_i(x) < \frac{\varepsilon}{C} \text{ for all } i = 1, \dots, n\}$$

For any $x \in U$, we have $\max_{1 \leq i \leq n} p_i(x) < \frac{\varepsilon}{C}$. Applying the assumed inequality gives $q(Tx) \leq C \max_{1 \leq i \leq n} p_i(x) < C \left(\frac{\varepsilon}{C}\right) = \varepsilon$. Thus, $T(U) \subseteq V_{q,\varepsilon}$, meaning $T^{-1}(V_{q,\varepsilon})$ is open in X , completing the proof. □

2.2 Building on everything from Linear Analysis

Theorem. Let K be a non-empty, closed, convex set in a Hilbert space H . Then K contains a unique element of least norm.

Proof. Let $d = \inf \{\|x\| : x \in K\}$. Then given $x, y \in K$ the parallelogram law gives

$$\frac{1}{4} \|x - y\|^2 = \frac{1}{2} \|x\|^2 + \frac{1}{2} \|y\|^2 - \left\| \frac{x + y}{2} \right\|^2$$

Then convexity of K implies $\frac{x+y}{2} \in K$ so we have

$$\|x - y\|^2 \leq 2\|x\|^2 + 2\|y\|^2 - 4d^2$$

Then this implies that if $\|x\| = \|y\| = d$ then $x = y$ so the infimum can be attained at at most one point. Now suppose x_n is a sequence in K with $\|x_n\| \rightarrow d$. Then we have

$$\|x_n - x_m\|^2 \leq 2\|x_n\|^2 + 2\|x_m\|^2 - 4d^2$$

This implies that x_n is Cauchy so $x_n \rightarrow x$ for some $x \in H$. Since K is closed, $x \in K$ and since the norm is continuous we must have $\|x\| = d$ so the infimum is attained. □

Recall that Hilbert spaces have orthogonal decompositions and that the Riesz representation theorem says that bounded linear operators are exactly the functions that send $x \mapsto (z, x)^1$. Also recall that if K is any set we have $K^{\perp\perp} = \overline{\text{span}(K)}$.

Proposition. A hilbert space H is separable if and only if it has a complete countable orthonormal system.

Proof. Recall theory from Linear analysis. Then argue that since it has a complete countable orthonormal system with dense linear span then we proved in linear analysis that this means it is separable, and that Gram-Schmidt turns this into an orthonormal system. □

Also from linear analysis recall Bessel's inequality.

Proposition. Let $U \subseteq \mathbb{R}^n$ be measurable. Then $L^2(U)$ has a complete countable orthonormal basis.

¹Annoyingly, my reference text now seems to use the first slot as the conjugate linear one, unlike linear analysis where it used the second. I told you in level 11.2 that this was annoying as nobody can agree on a convention.

Proof. It is enough to check that $L^2(U)$ is separable. To do this, it is enough to check that $L^2(\mathbb{R}^n)$ is separable by considering $P : L^2(\mathbb{R}^n) \rightarrow L^2(U)$ defined by $f \mapsto f1_U$. The dense countable subset of $L^2(\mathbb{R}^n)$ is dense in $L^2(U)$ after we do this restriction since the square integral difference only becomes smaller.

To do this, note that the set of simple functions over rational rectangles with rational complex coefficients, which is countable, is dense. To see this, fix $\varepsilon > 0$ and $f \in L^2(\mathbb{R}^n)$, then we want to try to approximate f .

Note that complex valued simple functions with bounded support are dense in L^2 by an earlier proposition so there exists one with $\|f - s\|_2 < \frac{\varepsilon}{3}$.

By an earlier theorem, for each measurable set A_k in the simple function s there exists a finite union of rectangles B_k such that $|A_k \triangle B_k|$ is sufficiently small so that we end up with s_1 being a linear combination of these finite unions of rectangles satisfying $\|s - s_1\|_2 < \frac{\varepsilon}{3}$.

Of course, we can approximate real rectangles by rational rectangles then approximate the coefficients by rational numbers to get an s_2 with $\|s_1 - s_2\| < \frac{\varepsilon}{3}$.

Thus s_2 is the desired function by the triangle inequality.

□

The Riesz representation theorem can be used to prove a certain theorem in measure theory. However, we need some definitions first.

Definition. Let $(E, \mathcal{E})$ be a measurable space with measures μ and ν . Then we say ν is absolutely continuous with respect to μ , written $\nu \ll \mu$, if for any measurable set A we have $\mu(A) = 0 \implies \nu(A) = 0$.

Definition. μ and ν are said to be mutually singular if there exists a measurable set A such that $\mu(A) = 0 = \nu(A^c)$.

Theorem. (Radon-Nikodym Theorem) Let $(E, \mathcal{E})$ be a measurable space with finite measures μ, ν and suppose $\nu \ll \mu$. Then there exists a non-negative $w \in L^1(E, \mu)$ such that for all $A \in \mathcal{E}$ we have

$$\nu(A) = \int_A w d\mu$$

Furthermore, for any non-negative measurable F we have

$$\int_E F(x) d\nu = \int_E F(x) w(x) d\mu(x)$$

Proof. Let $\alpha = \mu + 2\nu$ and $\beta = 2\mu + \nu$. Then these are finite measures on $(E, \mathcal{E})$. Now we will consider the Hilbert space $L^2(E, \alpha) = L^2(E, \mu) \cap L^2(E, \nu)$. Define $\Lambda : H \rightarrow \mathbb{C}$ by

$$\Lambda(f) = \int_E f d\beta$$

Then we have

$$|\Lambda(f)| = \int_E |f| d\beta \leq 2 \int_E |f| d\alpha \leq 2\sqrt{\alpha(E)} \|f\|_{L^2(E, \alpha)}$$

The last inequality is by Cauchy-Schwarz applied to $|f|$ and 1.

Therefore it follows that Λ is bounded on H and linear so by the Riesz representation theorem there exists a $g \in H$ such that for all $f \in H$ we have

$$\int_E f(x) d\beta(x) = \int_E f(x) g(x) d\alpha(x)$$

By definition of α and β we now get

$$2 \int_E f(x) d\mu + \int_E f(x) d\nu = 2 \int_E f(x) g(x) d\nu + \int_E f(x) g(x) d\mu$$

Rearranging gives

$$\int_E f * (2g - 1) d\nu = \int_E f * (2 - g) d\mu$$

And there exists $g \in H$ such that this is true for all $f \in H$.

Now take $f = 1_A$ for $A = \{x \in E : g(x) < \frac{1}{2}\}$. Then we must have $\int_E f * (2g - 1) d\nu = 0$ which implies $\int_E f * (2 - g) d\mu = 0$ but this is the integral of something $> 3/2$ so the only way this can happen is if $\mu(A) = 0$ which implies $\nu(A) = 0$ which means $g \geq \frac{1}{2}$ μ -almost everywhere and ν -almost everywhere. Now by setting $A = \{x \in E : g(x) > 2\}$ and $f = 1_A$ we have that $\int_E f * (2 - g) d\mu = 0$ which means that $g \leq 2$ μ -almost everywhere and thus also ν -almost everywhere. By redefining g on a measure 0 set we may assume that $\frac{1}{2} \leq g(x) \leq 2$ for all $x \in E$. By the monotone convergence theorem we have that $\int_E f * (2g - 1) d\nu = \int_E f * (2 - g) d\mu$ holds for any non-negative measurable f .

Let $Z = \{x : g(x) = \frac{1}{2}\}$. Then we have $\mu(Z) = 0$ since the integral $\int_E f * (2g - 1) d\nu$ gives 0 for $f = 1_Z$ but $\int_E f * (2 - g) d\mu$ has a integrand bounded below by $\frac{3}{2}$ on Z so we must have $\mu(Z) = 0$ and since $\nu \ll \mu$ we must have $\nu(Z) = 0$. Given non-negative measurable F define, for all $x \in Z^c$

$$f(x) = \frac{F(x)}{2g(x) - 1} \text{ and } w(x) = \frac{2 - g(x)}{2g(x) - 1}$$

and set $f(x) = w(x) = 0$ in Z . Then using $\mu(Z) = \nu(Z) = 0$ we deduce

$$\int_E F(x) d\nu = \int_E f * (2 - g) d\mu = \int_E F(x) w(x) d\mu$$

Setting $F(x) = 1$ gives $w \in L^1(E, \mu)$ and that w has the desired property.

□

Recall from linear analysis that dual spaces separate points, and are complete.

Lemma. Let $1 \leq p < \infty$ and let $\frac{1}{p} + \frac{1}{q} = 1$. Let $u : L^p(\mathbb{R}^n; \mathbb{R}) \rightarrow \mathbb{R}$ be a bounded, positive, real-linear map. Then there exists non-negative $g \in L^q(\mathbb{R}^n; \mathbb{R})$ with $\|g\|_{L^q} = \|u\|_{(L^p)^*}$ such that $u(f) = \int_{\mathbb{R}^n} f(x) g(x) dx$ for all $f \in L^p(\mathbb{R}^n; \mathbb{R})$

Remark. The $p = 2$ case follows from the Riesz representation theorem.

Proof. For $E \subseteq \mathbb{R}^n$ Lebesgue measurable let $\mu(E) = \int_E e^{-|x|^2} dx$ and observe that $\mu(\mathbb{R}^n)$ is finite. For a measurable set A define

$$\nu(A) := u\left(e^{-\frac{|x|^2}{p}} 1_A\right)$$

Then this is a measure on $\mathbb{R}^n$. To see countable additivity, if $B = \cup_{n=1}^{\infty} A_n$ for disjoint measurable A_n then setting $B_k = \cup_{n=1}^k A_n$ we have

$$\left\| e^{-\frac{|x|^2}{p}} 1_B - e^{-\frac{|x|^2}{p}} 1_{B_k} \right\|_p = [\mu(B \setminus B_k)]^{\frac{1}{p}} \rightarrow 0$$

This is where we need $p < \infty$ - if p were infinite then $[\mu(B \setminus B_k)]^{\frac{1}{p}}$ would be identically 1.

Therefore ν is indeed a finite measure on $\mathbb{R}^n$. Note also that $\nu \ll \mu$ since if $\mu(A) = 0$ then

$$\left\| e^{-\frac{|x|^2}{p}} 1_A \right\|_p = [\mu(A)]^{\frac{1}{p}} = 0$$

which implies that $u\left(e^{-\frac{|x|^2}{p}} 1_A\right)$ is also 0.

Therefore, the Radon-Nikodym theorem implies that there exists a non-negative $G \in L^1(\mathbb{R}^n, \mu)$ such that

$$\nu(A) = \int_A G(x) d\mu = \int_A G(x) e^{-|x|^2} dx$$

By linearity, note that if $f = e^{-\frac{|x|^2}{p}} F$ for some simple function $F = \sum a_i 1_{F_i}$ then we have

$$\begin{aligned} u(f) &= \sum_{i=1}^N a_i u \left(e^{-\frac{|x|^2}{p}} 1_{F_i} \right) = \sum_{i=1}^N a_i \nu(F_i) = \sum_{i=1}^N a_i \int_{F_i} G(x) e^{-|x|^2} dx \\ &= \sum_{i=1}^N a_i \int_{\mathbb{R}^n} 1_{F_i} e^{-\frac{|x|^2}{q}} e^{-\frac{|x|^2}{p}} dx = \int_{\mathbb{R}^n} f(x) G(x) e^{-\frac{|x|^2}{q}} dx \end{aligned}$$

Now set $g(x) := G(x) e^{-\frac{|x|^2}{q}}$ so we have $u(f) = \int_{\mathbb{R}^n} f(x) g(x) dx$ for some g and all functions of the form $f = e^{-\frac{|x|^2}{p}} F$ for some simple function F .

We claim such functions are dense in $L^p(\mathbb{R}^n; \mathbb{R})$. Indeed, define $\phi : L^p(\mathbb{R}^n, \mu) \rightarrow L^p(\mathbb{R}^n, dx)$ by $\phi(H)(x) = e^{-|x|^2/p} H(x)$. This preserves the L^p norm because

$$\|\phi(H)\|_{L^p(dx)}^p = \int_{\mathbb{R}^n} \left| e^{-|x|^2/p} H(x) \right|^p dx = \int_{\mathbb{R}^n} e^{-|x|^2} |H(x)|^p dx = \|H\|_{L^p(\mu)}^p$$

Therefore, simple functions being dense in any measure space implies the claim as we have an isomorphism from simple functions in $L^p(\mu(\mathbb{R}^n))$ to our “ $f = e^{-\frac{|x|^2}{p}} F$ for some simple function F ” functions in $L^p(dx(\mathbb{R}^n))$. Note that again we have used here that $p < \infty$, as this was assumed in my theorem about density in L^p earlier.

By boundedness of u and density of the f functions above where $u(f) = \int_{\mathbb{R}^n} f(x) g(x) dx$ is known to hold in L^p we have, for all $f \in L^p$ that

$$\int_{\mathbb{R}^n} |f(x) g(x)| dx = \int_{\mathbb{R}^n} |f(x)| g(x) dx = u(|f|) \leq \|u\|_{(L^p)^*} \|f\|_p$$

This implies that

$$\sup \left\{ \int_{\mathbb{R}^n} |f(x) g(x)| dx : f \in L^p(\mathbb{R}^n; \mathbb{R}), \|f\|_p \leq 1 \right\} \leq \|u\|_{(L^p)^*}$$

To check that $\|g\|_q \geq \|u\|_{(L^p)^*}$, note that by definition of the operator norm, we have

$$\sup \left\{ \left| \int_{\mathbb{R}^n} f(x) g(x) dx \right| : f \in L^p(\mathbb{R}^n; \mathbb{R}), \|f\|_p \leq 1 \right\} = \|u\|_{(L^p)^*}$$

Note that it is weird, as this is the supremum of stuff that is *smaller* than the stuff the supremum above, by the integral triangle inequality, which forces the two suprema to be exactly equal to $\|u\|_{(L^p)^*}$.

Now $\|g\|_q = \|u\|_{(L^p)^*}$ follows from the lemma we proved before proving the Minkowski integral inequality. □

Theorem. Let $1 \leq p < \infty$ and let q satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then $L^q(\mathbb{R}^n)$ is isometrically isomorphic to $L^p(\mathbb{R}^n)^*$ for real-valued functions, where again we note that this is true for $p = 2$ by the Riesz representation theorem.

Proof. Define $\Lambda_g(f) = \int_{\mathbb{R}^n} g(x) f(x) dx$ as a map $L^p(\mathbb{R}^n) \rightarrow \mathbb{R}$. Note that Holder's inequality implies that $|\int f g| \leq \|g\|_q \|f\|_p$ so if $f \in L^p$ and $g \in L^q$ then $\Lambda_g \in (L^p)^*$. Now the map from $L^p \rightarrow (L^p)^*$ defined by $g \mapsto \Lambda_g$ is linear and injective. The fact that it is surjective and isometric as well follows from the lemma above - any linear map corresponds to a $g \in L^q$ and the norms match the way we want them to. □

Note that the 'Gaussian' measure used in the proof above is outer regular, since given a measurable set there exists a covering of it by open sets with the measure difference $< \varepsilon$ for any ε in the real number case, and this measure difference only gets smaller when we switch to the Gaussian world.

Now suppose we are given a σ -algebra $\mathcal{M}$ on $\mathbb{R}^n$ containing $\mathcal{B}(\mathbb{R}^n)$ and a measure μ on M such that $\mu(\mathbb{R}^n)$ is finite and μ is both outer regular and inner regular. Then we note that if $f : (\mathbb{R}^n, \mathcal{M}) \rightarrow (\mathbb{R}^n, \mathcal{B})$ is continuous then it is measurable. We define a map $\Lambda : C_c^0(\mathbb{R}^n; \mathbb{R}) \rightarrow \mathbb{R}$ by

$$\Lambda(f) = \int_{\mathbb{R}^n} f(x) d\mu$$

This is now a positive, bounded linear map, in the sense that if we put the supremum norm on C_c^0 then the operator norm of this map is $\mu(\mathbb{R}^n)$ (this is by considering a function that is 1 on arbitrarily large subsets of $\mathbb{R}^n$).

We now want to know if given a map Λ we can recover the measure μ . Although 1_A is not continuous we could recover the measure if only we were allowed to put in 1_A . To solve this problem, we try to approximate 1_A . More precisely, let $O \subseteq \mathbb{R}^n$ be open (it suffices to compute the measure for open sets by outer regularity). Then define $O_k = O \cap B_k(0)$ and

$$\chi_k(x) := \begin{cases} 1 & x \in O_k, d(x, \partial O_k) \geq \frac{1}{k} \\ 0 & x \notin O_k \text{ or } \frac{1}{2k} > O_k, d(x, \partial O_k) \\ 2kd(x, \partial O_k) - 1 & x \in \frac{1}{2k} \leq O_k, d(x, \partial O_k) < \frac{1}{k} \end{cases}$$

Then these each have compact support and support in O and increases monotonically to 1_O and is continuous with compact support so the monotone convergence theorem on these, which can each be measured, lets us recover the measure of O in the sense that this measure becomes uniquely determined. In fact, if O is open then we have

$$\mu(O) = \sup \{ \Lambda(g) : g \in C_c^0(\mathbb{R}^n; \mathbb{R}), 0 \leq g \leq 1_O, \text{supp}(g) \subseteq O \}$$

We now want to prove the converse i.e. that given any bounded linear operator $\Lambda : C_c^0(\mathbb{R}^n; \mathbb{R}) \rightarrow \mathbb{R}$ it corresponds to a measure μ . Funnily enough, and perhaps annoyingly, this is also called the Riesz Representation Theorem.

Recall that a Radon measure on a Hausdorff space is one which is inner regular, outer regular and finite on compact sets.

Theorem. (Riesz Representation Theorem, but different) Let X be a locally compact Hausdorff topological space and let Λ be a positive (i.e. sends non-negative real functions to non-negative reals) linear functional on $C_c(X)$. Then there exists a σ -algebra $\mathcal{M}$ on X which contains all borel sets of X and there exists a unique finite outer regular measure μ that is finite on compact sets and inner regular on open sets on $\mathcal{M}$ with finite measure such that $\Lambda f = \int_X f d\mu$ for every $f \in C_c(X)$. Note that once we prove this, we can take the completion of $\mathcal{M}$ by adding subsets of measure 0 sets.

Proof. In this proof V denotes an open set and K denotes a compact set unless specified otherwise.

Step 1. We show that if such a μ exists then it is unique.

Indeed, such a measure is determined on open sets by its values on compact sets and on arbitrary sets by its value on open sets, so it suffices to prove that if K is compact then $\mu_1(K) = \mu_2(K)$ whenever μ_1 and μ_2 are measures for which the theorem holds. Fix K and $\varepsilon > 0$. Then by regularity and properties of the measure there exists $V \supseteq K$ such that $\mu_2(V) < \mu_2(K) + \varepsilon$. By Urysohn's lemma (see linear analysis) there exists f continuous such that f is 1 on K and 0 outside V . Now we have

$$\mu_1(K) = \int_X 1_K d\mu_1 \leq \int_X f d\mu_1 = \Lambda f = \int_X f d\mu_2 \leq \int_X 1_V d\mu_2 = \mu_2(V) < \mu_2(K) + \varepsilon$$

By interchanging the roles of μ_1 and μ_2 and applying the same argument again we note that $\mu_1(K) \leq \mu_2(K) \leq \mu_1(K)$ so they are equal on compact sets. Therefore we just need to show that there exists a measure μ satisfying the formula $\Lambda f = \int_X f d\mu$ with the desired properties.

Step 2. We will construct μ and $\mathcal{M}$ and in later steps prove that our construction has the desired properties.

For every open set V in X define

$$\mu(V) = \sup \{ \Lambda f : f < 1_V, \text{supp}(f) \subseteq V \}$$

Then note that this is an increasing set function on open sets. For arbitrary E let $\mu(E)$ be the outer measure.

Let $\mathcal{E}$ be the class of all $E \subseteq X$ with finite outer measure such that

$$\mu(E) = \sup \{ \mu(K) : K \subseteq E, K \text{ is compact} \}$$

Let $\mathcal{M}$ be the class of all $E \subseteq X$ such that $E \cap K \in \mathcal{E}$ for every compact K . Then it is clear that this is complete (i.e. contains subsets of measure 0 sets) since the set function μ is increasing and $\mu(E) = 0$ implies that $E \in \mathcal{E}$ and $E \in \mathcal{M}$. It remains to prove that $\mathcal{M}$ is a σ -algebra containing the Borel sets and μ is a measure on $\mathcal{M}$ with the properties we want - this is what we will do now. In fact, if we can show that μ is a measure at all on $\mathcal{M}$ and $\mathcal{M}$ is a σ -algebra that contains the Borel sets then from how it was constructed we know it is outer regular. Inner regularity and finiteness on compact sets are not immediate though, as the supremum of finite things need not be finite and while the condition for $\mathcal{M}$ is related to inner regularity it is not exactly that.

Step 3. We will extend Urysohn's lemma (which immediately implies $K \subseteq V$ implies there is a continuous function with $1_K < f < 1_V$ since the space is Hausdorff and V^C is closed) to say that not only can we make the continuous function 0 outside the open set V but can also make one such that its support is compact and contained in V . This extension requires the space to be locally compact, it is here that we actually use local compactness.

To do this, note that each point x in K has an open neighbourhood U_x with compact closure, and the U_x have a finite subcover of K by compactness. Let U be the union of this finite subcover, then the closure of U is a finite union of compact sets so it is also compact. Now consider the compact set $C = \overline{U} \setminus V \subseteq X \setminus V$.

Then since X is Hausdorff, for each point $p \in C$ there exists disjoint open sets separating p and K . To see this, note that if $q \in K$ then there is U_q, V_q disjoint with $p \in U_q$ and $q \in V_q$. Then we play the "cover everything in K take finite subcover then finite intersection of the U_q 's" game. Let W be an open set covering K disjoint from p , then its closure does not contain p as there exists a closed set containing K disjoint from p . Since p was an arbitrary point in $C = \overline{U} \setminus V$, we have $K \subseteq W \subseteq \overline{W} \subseteq V$ with $\overline{W}$ compact (since it can be constructed to be a closed subset of $\overline{U}$ by insisting every V_q is in U and by Hausdorffness closed subsets of compact sets are compact).

Here is what happens now: $Y = \overline{W}$ is compact so $Y \setminus W$ is a closed subset of Y disjoint from K and the standard Urysohn's lemma on $Y \setminus W$ and K gives one with compact support in V .

From now on I will say $f \prec V$ if $f < 1_V$ and $\text{supp}(f) \subseteq V$ and f has compact support and is continuous. I will also say $K \prec f$ if f is continuous, has compact support, and is 1 on K . Note that $K \prec f \prec V$ is equivalent to $1_K < f \prec V$.

Step 4. We prove that if $V_1, V_2, \dots, V_n$ are open in a locally compact Hausdorff space X and K is compact and contained in their union then there exist functions $h_i \prec V_i$ for $i = 1, 2, \dots, n$ such that $h_1(x) + h_2(x) + \dots + h_n(x) = 1$ whenever $x \in K$.

Each $x \in K$ has an open neighbourhood W_x with compact closure contained in V_i for some i depending on x (by the same principle as step 3, with $K = \{x\}$ and $V = V_i$ for some V_i containing x).

By compactness there are point $x_1, x_2, \dots, x_m$ with $K \subseteq W_{x_1} \cup W_{x_2} \cup \dots \cup W_{x_m}$. If $1 \leq i \leq n$ then let H_i be the union of the $\overline{W_{x_j}}$ that lie in V_i . Then by Urysohn's lemma there are continuous functions such that $1_{H_i} < g_i \prec V_i$. Now define

$$\begin{aligned} h_1 &= g_1 \\ h_2 &= (1 - g_1) g_2 \\ &\vdots \\ h_n &= (1 - g_1) (1 - g_2) \dots (1 - g_{n-1}) g_n \end{aligned}$$

Then each $h_i \prec V_i$ and by an easy induction we have

$$h_1 + h_2 + \dots + h_n = 1 - (1 - g_1) (1 - g_2) \dots (1 - g_n)$$

Since $K \subseteq H_1 \cup H_2 \dots H_n$ at least one of $g_i(x)$ is 1 at each point $x \in K$ so $h_1 + h_2 + \dots + h_n$ is 1 in K as it is 1 minus a product that collapses to 0.

Step 5. We will show that μ is countably subadditive.

Let $E_1, E_2, \dots$ be arbitrary subsets of X . Then we want to show that $\mu(\cup E_i) \leq \sum \mu(E_i)$.

We first show that μ is finitely subadditive on open sets. Let V_1, V_2 be open. Then choose $g \prec V_1 \cup V_2$. It is here we use the fact that $g \prec V_1 \cup V_2$ and not merely $g < 1_{V_1 \cup V_2}$ - we use step 4 to find functions $h_1 \prec V_1$ and $h_2 \prec V_2$ such that

$h_1 + h_2 = 1$ in the support of g . Hence, $h_i g \prec V_i$ (Note: the LHS is a product of things between 0 and 1 and therefore the RHS is as well) and $g = h_1 g + h_2 g$. Now we have, by linearity,

$$\Lambda g = \Lambda(h_1 g) + \Lambda(h_2 g) \leq \mu(V_1) + \mu(V_2)$$

Since this holds for every $g \prec V_1 \cup V_2$ we have that $\mu(V_1 \cup V_2) \leq \mu(V_1) + \mu(V_2)$.

If $\mu(E_i)$ is infinite for some i then we trivially have $\mu(\cup E_i) \leq \sum \mu(E_i)$. Suppose therefore that $\mu(E_i)$ is finite for each i . Then for each $\varepsilon > 0$ there exists open sets $V_i \supseteq E_i$ with $\mu(V_i) < \mu(E_i) + 2^{-i}\varepsilon$. Then let $V = \cup V_i$ and let $f \prec V$. Since f has compact support we see that $f \prec V_1 \cup \dots \cup V_n$ for some n . Now for each n , and induction on the $\mu(V_1 \cup V_2) \leq \mu(V_1) + \mu(V_2)$ condition we have

$$\Lambda f \leq \mu(V_1 \cup \dots \cup V_n) \leq \mu(V_1) + \dots + \mu(V_n) \leq \sum_{i=1}^{\infty} \mu(E_i) + \varepsilon$$

Since this holds for every $f \prec V$ and $\cup E_i \subseteq V$ (by how the V_i 's were defined in the previous paragraph) it follows that

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \mu(V) \leq \sum_{i=1}^{\infty} \mu(E_i) + \varepsilon$$

This proves countable subadditivity as ε was arbitrary.

Step 6. We will prove that all compact sets are in $\mathcal{E}$ and that if K is compact then $\mu(K) = \inf \{\Lambda f : K \prec f\}$. This will imply that μ is finite on compact sets.

If $K \prec f$ and $0 < a < 1$ let $V_a = \{x : f(x) > a\}$. Then V_a is open and $K \subseteq V_a$ and $ag \leq f$ whenever $g \prec V_a$. Therefore, by how we defined μ on open sets,

$$\mu(K) \leq \mu(V_a) = \sup \{\Lambda g : g \prec V_a\} \leq a^{-1} \Lambda f$$

Therefore letting $a \rightarrow 1$ we have $\mu(K) \leq \Lambda f$. Thus $\mu(K)$ is finite since Λf is finite. Now clearly, we have $K \in \mathcal{E}$.

Note that if $\varepsilon > 0$ there exists an open set V such that $\mu(V) < \mu(K) + \varepsilon$. By step 3 there exists f such that $1_K < f \prec V$. Thus, by definition of μ on open sets,

$$\Lambda f \leq \mu(V) < \mu(K) + \varepsilon$$

Since there is an f such that this is true for all $\varepsilon > 0$ and since we have from a few lines ago that $\mu(K) \leq \Lambda f$ for all f with $K \prec f$ it follows that $\mu(K) = \inf \{\Lambda f : K \prec f\}$, and that the set $\{\Lambda f : K \prec f\}$ is indeed non-empty.

Step 7. We will prove that $\mathcal{E}$ contains every open set V with $\mu(V) < \infty$.

Let a be a real number with $a < \mu(V)$. Then there exists $f \prec V$ with $a < \Lambda f$ by definition of μ on open sets. If W is any open set which contains the support K of f then $f \prec W$ so $\Lambda f \leq \mu(W)$. Therefore, since μ is the outer measure, we have that $\Lambda f \leq \mu(K)$ by taking infimum on the right. This exhibits a compact $K \subseteq V$ with $a < \mu(K)$ and since a was arbitrary we have that the supremum of $\mu(K)$ over compact sets in V must be $\mu(V)$, since it's clearly not greater.

Step 8. We will show that if $E = \bigcup_{i=1}^{\infty} E_i$ where E_i is a family of pairwise disjoint members of $\mathcal{E}$ then $\mu(E) = \sum_{i=1}^{\infty} \mu(E_i)$. Also, if $\mu(E) < \infty$ then $E \in \mathcal{E}$.

First, let K_1, K_2 be disjoint compact sets. Then I will prove that $\mu(K_1 \cup K_2) = \mu(K_1) + \mu(K_2)$. Urysohn's lemma guarantees the existence of $f \in C_c(X)$ such that $f(x) = 1$ on K_1 and $f(x) = 0$ on K_2 and $0 \leq f \leq 1$. For all $\varepsilon > 0$, by step 6 there exists g such that $K_1 \cup K_2 \prec g$ and $\Lambda g < \mu(K_1 \cup K_2) + \varepsilon$. It follows that $K_1 \prec fg$ and $K_2 \prec (1-f)g$ and by linearity of Λ we have

$$\mu(K_1) + \mu(K_2) \leq \Lambda(fg) + \Lambda((1-f)g) = \Lambda g < \mu(K_1 \cup K_2) + \varepsilon$$

Since ε was arbitrary and μ is subadditive, finite additivity on compact sets follows from step 5.

Now if $\mu(E)$ is infinite then the formula $\mu(E) = \sum_{i=1}^{\infty} \mu(E_i)$ follows from step 5, since we must have that $\sum_{i=1}^{\infty} \mu(E_i)$ is infinite. Therefore, assume that $\mu(E) < \infty$ and choose $\varepsilon > 0$. Since $E_i \in \mathcal{E}$ there exists compact sets $H_i \subseteq E_i$ with

$\mu(H_i) > \mu(E_i) - 2^{-i}\varepsilon$. Setting $K_n = H_1 \cup H_2 \cup \dots \cup H_n$, since the H_i 's are disjoint as the E_i 's are disjoint, we have by finite additivity on compact sets that

$$\mu(E) \geq \mu(K_n) = \sum_{i=1}^n \mu(H_i) > \sum_{i=1}^n \mu(E_i) - \varepsilon$$

Since this holds for every n and $\varepsilon > 0$, we have $\mu(E) \geq \sum_{i=1}^{\infty} \mu(E_i)$, with equality because step 5 implies the other inequality direction. But then if $\mu(E) < \infty$ and $\varepsilon > 0$ then this implies the existence of an N such that

$$\mu(E) \leq \sum_{i=1}^n \mu(E_i) + \varepsilon$$

Therefore, it follows that

$$\mu(E) \leq \sum_{i=1}^n \mu(E_i) + \varepsilon < \sum_{i=1}^n \mu(H_i) + 2\varepsilon = \mu(K_N) + 2\varepsilon$$

This shows that, since ε was arbitrary, $\mu(E)$ is the supremum of the measure of compact sets contained inside it, so $E \in \mathcal{E}$.

Step 9. We want to prove that if $E \in \mathcal{E}$ and $\varepsilon > 0$ then there is a compact K and open V such that $\mu(V - K) < \varepsilon$.

Note that by definition there exists $K \subseteq E$ and $E \subseteq V$ such that

$$\mu(V) - \frac{\varepsilon}{2} < \mu(E) < \mu(K) + \frac{\varepsilon}{2}$$

Since $V - K$ is open with finite measure it is in $\mathcal{E}$ by step 7. Therefore, step 8 implies that

$$\mu(K) + \mu(V - K) = \mu(V) > \mu(K) + \varepsilon$$

so the result follows.

Step 10. We want to prove that if $\mathcal{E}$ is closed under finite unions and intersections.

Fix sets $A, B \in \mathcal{E}$ so we want to show that their union and intersection is in $\mathcal{E}$. If $\varepsilon > 0$ then step 9 proved the existence of sets K_i, V_i such that $K_1 \subseteq A \subseteq V_1$ and $K_2 \subseteq B \subseteq V_2$ and $\mu(V_i - K_i) < \varepsilon$ for $i = 1, 2$. Now we have

$$A \setminus B \subseteq V_1 \setminus K_2 \subseteq (V_1 \setminus K_1) \cup (K_1 \setminus V_2) \cup (V_2 \setminus K_2)$$

Step 5 (subadditivity) implies that

$$\mu(A \setminus B) \leq 2\varepsilon + \mu(K_1 \setminus V_2)$$

Since $K_1 \setminus V_2$ is a compact subset of $A \setminus B$ we know that $A \setminus B \in \mathcal{E}$ by definition since ε was arbitrary.

Also, since $A \setminus B$ and B are disjoint elements of $\mathcal{E}$ and they are 2, each with finite measure, step 8 guarantees $A \cup B \in \mathcal{E}$. Also, $A \cap B = A \setminus (A \setminus B)$ so $A \cap B \in \mathcal{E}$ as well.

Step 11. We want to prove that $\mathcal{M}$ is a σ -algebra that contains all Borel sets.

Let K be an arbitrary compact subset of X . If $A \in \mathcal{M}$ then $A^C \cap K = K \setminus (A \cap K)$ so $A^C \cap K$ is the difference between 2 elements of $\mathcal{E}$ (since $\mathcal{E}$ contains compact sets by an earlier step and contains $A \cap K$ by definition of $\mathcal{M}$ in terms of $\mathcal{E}$) so also in $\mathcal{E}$ by the proof in step 10, which means that if $A \in \mathcal{M}$ then $A^C \in \mathcal{M}$.

Next, let A_i be a countable family of elements of $\mathcal{M}$ and let A be their union. Put $B_1 = A_1 \cap K$ and for $n > 1$ define

$$B_n = (A_n \cap K) \setminus (B_1 \cup B_2 \cup \dots \cup B_{n-1})$$

Then this is a disjoint sequence of members of $\mathcal{E}$ (symmetric difference between stuff in $\mathcal{E}$ by definition with finite unions of stuff in $\mathcal{E}$ by induction). Also, the union of all B_n is $A \cap K$. By step 8, we therefore know that $A \cap K$ is in $\mathcal{E}$ so by definition $A \in \mathcal{M}$. Finally, if C is closed, then $C \cap K$ is compact so $C \cap K \in \mathcal{E}$ so $C \in \mathcal{M}$ which implies that $X \in \mathcal{M}$. Therefore, $\mathcal{M}$ is indeed a σ -algebra on X containing all closed, and hence all Borel, subsets of X .

Step 12. We want to prove that $\mathcal{E}$ consists of precisely the sets $E \in \mathcal{M}$ with finite measure. Note that this implies that μ is countably additive and thus a measure on $\mathcal{M}$ as it is countably additive on $\mathcal{E}$, so μ is a measure on $\mathcal{M}$.

If $E \in \mathcal{E}$ then steps 6 and 10 imply that $E \cap K \in \mathcal{E}$ for every compact K so $E \in \mathcal{M}$.

Therefore we just need to check that if $E \in \mathcal{M}$ and $\mu(E) < \infty$ then $E \in \mathcal{E}$. Therefore, fix $\varepsilon > 0$. Then there exists an open set V with $E \subseteq V$ and $\mu(V) < \infty$ and a compact set $K \subseteq V$ with $\mu(V - K) < \varepsilon$ (by steps 7 and 9). Since $E \cap K \in \mathcal{E}$ there is a compact set $H \subseteq E \cap K$ with

$$\mu(E \cap K) < \mu(H) + \varepsilon$$

Since $E \subseteq (E \cap K) \cup (V \setminus K)$ we have, by subadditivity of μ ,

$$\mu(E) \leq \mu(E \cap K) + \mu(V \setminus K) < \mu(H) + 2\varepsilon$$

This implies that $E \in \mathcal{E}$.

Step 13. We want to prove that for every $f \in C_c(X)$, $\Lambda f = \int_X f d\mu$.

Note that in fact, if this is true for real f then it is also true for complex f . Although we have to be careful with complex numbers because the Hilbert space version of the Riesz representation theorem only gives an anti-isometry in the complex case, we didn't actually use that theorem in the proof of this one, so this one actually does work for complex f .

Also, if we can prove that $\Lambda f \leq \int_X f d\mu$ for all real f we will be done, by considering $-f$ and using linearity of Λ and the integral.

Now, let K be the support of a real $f \in C_c(X)$ and let $[a, b]$ be an interval which contains the range of f (by the topological version of the extreme value theorem). Fix $\varepsilon > 0$ and choose y_i for $i = 0, 1, 2, \dots, n$ such that $y_i - y_{i-1} < \varepsilon$ and $y_0 < a < y_1 < \dots < y_n = b$ (note I did indeed mean $y_0 < a$ and not $y_0 = a$ - we will see why shortly). Put

$$E_i = \{x : y_{i-1} < f(x) \leq y_i\} \cap K$$

Since f is continuous, f is Borel measurable, and the sets E_i are therefore disjoint Borel sets whose union is K . For each $i = 1, 2, \dots, n$ there are open sets $V_i \supseteq E_i$ such that

$$\mu(V_i) < \mu(E_i) + \frac{\varepsilon}{n}$$

and such that $f(x) < y_i + \varepsilon$ for all $x \in V_i$. By step 4, there exist functions $h_i \prec V_i$ such that $\sum h_i = 1$ on K . Therefore, $f = \sum h_i f$ and step 6 and linearity of Λ shows that

$$\mu(K) \leq \Lambda \left(\sum h_i \right) = \sum \Lambda h_i$$

Since $h_i f \leq (y_i + \varepsilon) h_i$ and since $y_i - \varepsilon < f(x)$ on E_i we have (noting that Λ has $f < g \implies \Lambda f \leq \Lambda g$ as it is linear and positive and by considering $g - f$)

$$\begin{aligned} \Lambda f &= \sum_{i=1}^n \Lambda(h_i f) \leq \sum_{i=1}^n (y_i + \varepsilon) \Lambda h_i \\ &= \sum_{i=1}^n (|a| + y_i + \varepsilon) \Lambda h_i - |a| \sum_{i=1}^n \Lambda h_i \\ &\leq \sum_{i=1}^n (|a| + y_i + \varepsilon) \left[\mu(E_i) + \frac{\varepsilon}{n} \right] - |a| \mu(K) \end{aligned}$$

The last line follows from the definition of V_i and the fact $h_i \prec V_i$. Noting that $\sum \mu(E_i) = \mu(K)$ we get

$$\begin{aligned} &= \sum_{i=1}^n (|a| + y_i + \varepsilon) [\mu(E_i)] + \frac{\varepsilon}{n} \sum_{i=1}^n (|a| + y_i + \varepsilon) - |a| \mu(K) \\ &= \sum_{i=1}^n (y_i - \varepsilon) [\mu(E_i)] + \frac{\varepsilon}{n} \sum_{i=1}^n (|a| + y_i + \varepsilon) + 2\varepsilon \mu(K) \end{aligned}$$

$$\leq \int f d\mu + \varepsilon \left[\frac{1}{n} \sum_{i=1}^n (|a| + y_i + \varepsilon) + 2\mu(K) \right]$$

Since the bracketed stuff is a fixed constant depending only on f and ε was arbitrary, the inequality we wanted to prove follows. □

Definition. Let $0 < \lambda \leq 1$. Then a function $f : [a, b] \rightarrow \mathbb{C}$ is γ -Holder continuous if, roughly speaking, for ε we have that the δ that works has δ^γ is on the order of ε . More precisely, we say the condition is if

$$\|f\|_{C^{0,\gamma}} := \sup_{x \in I} |f(x)| + \sup_{x, y \in I, x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\gamma} < \infty$$

The set $C^{0,\gamma}(I)$ of γ -Holder continuous functions is a Banach space with this norm.

Proof. Easy check is that the second term is a seminorm and satisfies linearity and the triangle inequality and the first term in the sum is a norm, so we actually have a normed space. The only hard part is completeness. Let f_k be Cauchy in $C^{0,\gamma}([a, b])$. Then since $\|f_k - f_m\|_{C^0} \leq \|f_k - f_m\|_{C^{0,\gamma}}$ we have that f_k is Cauchy in C^0 so it converges uniformly. Easy check is that the uniform limit is also a limit in the C^γ seminorm. Explicitly, if m, n are large enough for a given ε then it follows by fixing m and letting $n \rightarrow \infty$ and taking supremum over x, y in

$$\frac{|(f_m(x) - f_n(x)) - (f_m(y) - f_n(y))|}{|x - y|^\gamma} \leq \|f_m - f_n\|_{C^{0,\gamma}} < \varepsilon$$

To check that the limit is indeed in $C^{0,\gamma}$, we have for N large enough that

$$\|f\|_{C^{0,\gamma}} \leq \|f - f_N\|_{C^{0,\gamma}} + \|f_N\|_{C^{0,\gamma}} \leq \varepsilon + \|f - f_N\|_{C^0} + \|f_N\|_{C^{0,\gamma}} < \infty$$

Therefore f is indeed in $C^{0,\gamma}$ and $f_N \rightarrow f$ in $C^{0,\gamma}$ by combining the 2 convergence results. □

Proposition. The closed unit ball in $C^{0,\gamma}([a, b])$ is compact in the C^0 -topology.

Proof. Compactness in a metric space is equivalent to sequential compactness which we have by Arzela-Ascoli since the family is uniformly bounded and equicontinuous by the Holder continuous thing. □

Recall from linear analysis the Banach-Alaoglu theorem: that closed unit balls in dual spaces are compact in the weak-star topology, which is coarser than the weak topology which is coarser still than the strong topology.

Proposition. Let X be a separable Banach space and let D be a countable dense subset $\{x_k : k \in \mathbb{N}\}$. Let $P = \{p_k : k \in \mathbb{N}\}$ be the family of seminorms on X^* defined by $p_k : \Lambda \mapsto |\Lambda(x_k)|$ and let T_P be the associated topology as defined in Linear analysis i.e. the coarsest topology that makes every p_k continuous. Let $\overline{B}^*$ be the closed unit ball in X^* . Then $T_P|_{\overline{B}^*} = T_{w^*}|_{\overline{B}^*}$ and these are both metric topologies.

Proof. Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$ i.e. the field we are working with. We first want to show P is separating. To do this, let $\Lambda \in X^*$ satisfy $p_k(\Lambda) = 0$ for all $k \in \mathbb{N}$ i.e. $\Lambda(x_k) = 0$ for all $x_k \in D$. Then $\Lambda : X \rightarrow \mathbb{K}$ is continuous by definition of the dual space. A continuous function that vanishes on D must vanish everywhere on X since D is dense, so if Λ is non-zero then P must be separating. If $\Lambda = 0$ then the definition of separating seminorms specifies non-zero Λ (It said x in the definition but here we are considering X^* so Λ is an arbitrary element) so separating becomes Vacuous.

By an earlier theorem, it follows that T_P is a metric topology (see the thing about countable separating family of seminorms). It remains, now, to check that $T_P|_{\overline{B}^*} = T_{w^*}|_{\overline{B}^*}$.

Note that since $P \subseteq X^{**}$ we have that $T_P|_{\overline{B}^*} \subseteq T_{w^*}|_{\overline{B}^*}$ so it remains to check the inclusion in the other direction.

To do this, let U be open in $T_{w^*}|_{\overline{B}^*}$ and let $\Lambda_0 \in U$ be arbitrary, then we want to show that Λ_0 is an interior point of U with respect to the $T_P|_{\overline{B}^*}$ topology.

By definition of the subspace topology there exists O open in the weak-star topology on X^* such that $U = O \cap \overline{B}^*$. Since the weak-star topology is a topological vector space its topology is translation-invariant so there exists a weak-star open neighbourhood of the origin in X^* , say W , such that $\Lambda_0 + W_{w^*} \subseteq O$. Now W_{w^*} must contain a basic weak-star open neighbourhood of 0, which is given by $\bigcap_{j=1}^N V(x^{(j)}, n_j)$ where $V(x, n) = \{\Lambda \in X^* : |\Lambda(x)| < \frac{1}{n}\}$ are the subbasic neighbourhoods about 0 in the weak-star topology of X^* .

Therefore, it suffices to prove that every subbasic open neighbourhood of 0 in the weak-star topology contains a subbasic open neighbourhood of 0 in the $T_P|_{\overline{B}^*}$ topology, as we can change subbasic to basic by taking finite intersections and note that if we can prove this then U has a basic open subset of $T_P|_{\overline{B}^*}$ about every point contained in U , which implies that anything in $T_{w^*}|_{\overline{B}^*}$ is a union of things in $T_P|_{\overline{B}^*}$ which implies the other inclusion.

In other words, we want to prove that for every $x \in X$ and $n \in \mathbb{N}$ there exists $x_k \in D$ and $m \in \mathbb{N}$ such that $V(x_k, m) \subseteq V(x, n)$.

Fix $x \in X$, then by density, for all $\varepsilon > 0$ there exists $x_i \in D$ such that $\|x - x_i\| < \varepsilon$. If $\Lambda \in V(x_i, m)$ then

$$|\Lambda(x)| \leq \|\Lambda\| \|x - x_i\| + |\Lambda(x_i)| < \varepsilon + \frac{1}{m}$$

Taking $\varepsilon < \frac{1}{2n}$ and $m > 2n$ we have $\Lambda \in V(x, n)$. It is here we use the fact that we are in the closed ball, as we must have $\|\Lambda\| \leq 1$.

□

Corollary. If x_k is a dense sequence in X then a sequence Λ_j in $\overline{B}^*$ converges to Λ if and only if as $j \rightarrow \infty$ we have $\Lambda_j(x_k) \rightarrow \Lambda(x_k)$ for every k .

Corollary. Compactness in the weak-star topology is equivalent to sequential compactness.

Note that this gives an alternative proof of Banach-Alaoglu from the one given in linear analysis, for the case the space is separable. The idea is to use Bolzano-Weierstrass and diagonalization to get sequential compactness which is equivalent to compactness because the topology is metric.

Corollary. Let $1 < p \leq \infty$ and let (f_j) be a sequence of functions in $L^p(\mathbb{R}^n)$ each satisfying $\|f_j\|_p \leq 1$. Also let g satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then there exists $f \in L^p$ and a subsequence f_{j_k} such that $\|f\|_p \leq 1$ and for every $g \in L^q$ we have

$$\int_{\mathbb{R}^n} g(x) f_{j_k}(x) dx \rightarrow \int_{\mathbb{R}^n} g(x) f(x) dx$$

In fact we can replace the bound 1 by any K by rescaling appropriately.

Proof. This is because we are looking at the closed unit ball in the weak-star topology in a dual space which is compact by Banach-Alaoglu and is a metric topology and therefore also sequentially compact.

□

Recall from linear analysis that the Hahn-Banach theorem says

Theorem. Let X be a real or complex vector space and $p : X \rightarrow \mathbb{R}$ be a seminorm and $M \subseteq X$ be a subspace. Suppose $l : M \rightarrow \mathbb{R}$ is linear and satisfies $l(y) \leq p(y)$ for all $y \in M$. Then there exists a linear operator $l' : X \rightarrow \mathbb{R}$ such that $l = l'$ on M and $l' \leq p$ on X .

Now we have the following cool results, which relate to the geometrically obvious draw-a-picture fact that in $\mathbb{R}^n$ any 2 convex sets have a plane between them:

Lemma. Let X be a Banach space and suppose $A \subseteq X$ is a convex open neighbourhood of 0. For $x \in X$ define the Minkowski functional of A by $\mu_A(x) = \inf \{t > 0 : t^{-1}x \in A\}$. Then μ_A is sublinear (i.e. satisfies the triangle inequality) and satisfies $\mu_A(x) \leq k\|x\|$ for some $k > 0$. Also, $\mu_A(y) < 1$ whenever $y \in A$.

Proof. The fact that $\mu < 1$ is because for each $y \in A$ there is a ε such that $(1 + \varepsilon)y \in A$ by openness.

Note that μ_A is positively homogenous because if $\lambda > 0$ then $\inf \{t > 0 : t^{-1}\lambda x \in A\} = \lambda \inf \{t > 0 : t^{-1}x \in A\}$.

The fact that μ_A satisfies the triangle inequality is because: By definition of the infimum for all $\varepsilon > 0$ there exists $s, t > 0$ such that $s < \mu_A(x) + \varepsilon$ and $\frac{x}{s} \in A$ and $t < \mu_A(y) + \varepsilon$ and $\frac{y}{t} \in A$. Then consider the convex combination of $\frac{x}{s}$ and $\frac{y}{t}$ with weights $\frac{s}{s+t}$ and $\frac{t}{s+t}$. Then this is $\frac{x+y}{s+t}$. Since A is convex we must have that $\frac{x+y}{s+t}$ is in A which means that $\mu_A(x+y) \leq s+t < \mu_A(x) + \mu_A(y) + 2\varepsilon$, so letting $\varepsilon \rightarrow 0$ the result follows.

The fact that we have $\mu_A(x) \leq k\|x\|$ is because there is an open ball around 0 in A by openness, say of radius ε , and $\mu_A(x) < 1$ inside this open ball, and so by linearity we can take $k = \varepsilon^{-1}$. □

Theorem. Let A, B be disjoint non-empty convex sets in a real or complex Banach space X and suppose A is open. Then there exists $\Lambda \in X^*$ and $\gamma \in \mathbb{R}$ such that for all $x \in A$ and $y \in B$ we have

$$\operatorname{Re}(\Lambda x) < \gamma \leq \operatorname{Re}(\Lambda y)$$

If B is also open the second inequality may be taken to be strict.

Proof. Note that it suffices to prove the result for real Banach spaces because then if X is complex we can find a real-linear $l : X \rightarrow \mathbb{R}$ which separates A, B as stated in the theorem and then set $\Lambda = l(x) - il(ix)$ which is complex-linear.

Pick $a_0 \in A$ and $b_0 \in B$ and let $x_0 = b_0 - a_0$. Let $C = A - B + x_0$. This is a convex open neighbourhood of 0 in X since A is open, and since A, B are disjoint we have $x_0 \notin C$. Let $p(x) = \inf \{t > 0 : t^{-1}x \in C\}$, then by the lemma this is a sublinear functional satisfying $p(x) \leq k\|x\|$ for some $k > 0$ and $p(y) < 1$ for some $y \in C$. Since $x_0 \notin C$ we have $p(x_0) \geq 1$.

Let M be the subspace generated by x_0 and on this space define the linear functional $f(tx_0) = t$. If $t \geq 0$ then $f(tx_0) = t \leq tp(x_0) = p(tx_0)$ and if $t < 0$ then $f(tx_0) = t < 0 \leq p(tx_0)$ so $f \leq p$ on M so by Hahn-Banach we can extend f to a linear functional Λ satisfying

$$-k\|x\| \leq -p(-x) \leq \Lambda(x) \leq p(x) \leq k\|x\|$$

Therefore $\Lambda \in X^*$. Now let $a \in A$ and $b \in B$, then

$$\Lambda a - \Lambda b + 1 = \Lambda(a - b + x_0) \leq p(a - b + x_0) < 1$$

the last inequality is since $a - b + x_0 \in C$ so we have $\Lambda a < \Lambda b$. This implies that $\Lambda(A)$ and $\Lambda(B)$ are disjoint convex subsets of $\mathbb{R}$ and $\Lambda(A)$ is open by the open mapping theorem from linear analysis. The result therefore follows. If B is open then so is $\Lambda(B)$ so we indeed can get a strict inequality. □

Theorem. Let A, B be disjoint non-empty convex sets in a real or complex Banach space X and suppose A is compact and B is closed. Then there exists $\Lambda \in X^*$ and $\gamma_1, \gamma_2 \in \mathbb{R}$ such that for all $x \in A$ and $y \in B$ we have

$$\operatorname{Re}(\Lambda x) < \gamma_1 < \gamma_2 < \operatorname{Re}(\Lambda y)$$

i.e. such that there is a gap.

Proof. As in the previous theorem we may assume X is a real Banach space.

Since A is compact and B is closed we have that $\inf \{\|a - b\| : a \in A, b \in B\} = d > 0$. Let $V = B_{\frac{d}{2}}(0)$ and consider $A + V$. Then this is open, convex and disjoint from B so applying part *a* with $A + V$ in place of A shows that there exists $\Lambda \in X^*$ such that $\Lambda(A + V)$ and $\Lambda(B)$ are disjoint convex subsets of $\mathbb{R}$ with $\Lambda(A + V)$ to the left of $\Lambda(B)$ (by taking $-\Lambda$ if necessary). Since $\Lambda(A + V)$ is open and $\Lambda(A)$ is a compact subset of $\Lambda(A + V)$ the result follows. $\square$

Corollary. By applying the previous theorem with $A = \{x_0\}$ and $B = \overline{M}$ and rescaling the resulting Λ appropriately, where M is a subspace of a Banach space X and $x_0 \in X \setminus \overline{M}$, we get the result from linear analysis that there exists $\Lambda \in X^*$ such that $\Lambda x_0 = 1$ and $\Lambda x = 0$ for every $x \in M$.

3 Test functions and distributions

3.1 The test function spaces

Let $\Omega \subseteq \mathbb{R}^n$ be open and consider the vector space $C^\infty(\Omega)$.

Recall that a Frechet space is a locally convex topological vector space with a topology induced by a complete invariant metric. We want to put a topology on $C^\infty(\Omega)$ that makes it a Frechet space with the Heine-Borel property (which says closed + bounded implies compact).

Alright so here is how we do this. First note that there is a sequence of compact sets such that each is in the interior of the next and their union is Ω as proved earlier. Call this K_n and define, for α a tuple of integers,

$$p_n(f) = \max \{|D^\alpha f(x)|, x \in K_n, |\alpha| \leq n\}$$

Note that this definition includes the zero tuple $\alpha = \emptyset$. Then this family is separating because if $f \neq 0$ then $f(x) \neq 0$ at some point $x \in \Omega$ so for sufficiently large n we have $p_n(f) > 0$. Thus, the family of seminorms $P = \{p_n : n \in \mathbb{N}\}$ induces a topology T on $C^\infty(\Omega)$ that is locally convex and metrizable and forms a topological vector space and is translation and scale invariant, by an earlier theorem.

Note, also, that p_n is increasing as n increases. Therefore, the local base for this topology can actually be simplified to sets of the form

$$V_N = \left\{ f \in C^\infty(\Omega) : p_N(f) < \frac{1}{N} \right\}$$

We will call this topology $\mathcal{E}(\Omega)$.

Proposition. A sequence f_n in $\mathcal{E}(\Omega)$ converges to f if and only if $D^\alpha f_n \rightarrow D^\alpha f$ uniformly on compact sets for each multi-index α

Proof. By the translation invariance of the topology we can assume that $f = 0$. The sequence f_n tends to 0 in $\mathcal{E}(\Omega)$ if and only if for each N there exists m_N such that $f_n \in V_N$ for all $n \geq m_N$.

First, suppose that $f_n \rightarrow 0$ in $\mathcal{E}(\Omega)$. Fix α and let $K \subseteq \Omega$ be any compact subset. For any ε there exists N such that $N > \max\{|\alpha|, \varepsilon^{-1}\}$ and $K \subseteq K_N$. If $n \geq m_N$ then $f_n \in V_N$ which implies that

$$\sup_K |D^\alpha f_n| \leq \sup_{K_N} |D^\alpha f_n| \leq p_N(f_n) \leq \frac{1}{N} < \varepsilon$$

Thus $D^\alpha f_n \rightarrow 0$ uniformly on K .

Conversely, suppose that for each multi-index α and compact set K we have that $D^\alpha f_n \rightarrow 0$ uniformly on K . Then given some N there for each α with $|\alpha| \leq N$ we have $D^\alpha f_n \rightarrow 0$ uniformly on K_N which implies there exists m_α such that if $n \geq m_\alpha$ we have

$$\sup_{K_N} |D^\alpha f_n| \leq \frac{1}{N}$$

Thus if $m = \max_{|\alpha| \leq N} m_\alpha$ then for all $n > m$ we have $f_n \in V_N$ and $f_n \rightarrow 0$ in $\mathcal{E}$. $\square$

The above theorem is basically the principle that convergence in a topology induced by some norms is equivalent to convergence in all these norms.

Theorem. The topological vector space $\mathcal{E}(\Omega)$ is a Frechet space with the Heine-Borel property.

Proof. Step 1. To check that it is a Frechet space we just need to prove it is complete since we know it has the topology induced by an invariant metric and is locally convex.

A sequence f_n in $\mathcal{E}(\Omega)$ is Cauchy if for all N there exists an M such that for all $i, j \geq M$ we have

$$(\forall \alpha : |\alpha| < N) \sup_{K_N} |D^\alpha f_i - D^\alpha f_j| < \frac{1}{N}$$

Since the K_N exhaust Ω this implies that there exists continuous functions g^α such that $D^\alpha f_n \rightarrow g^\alpha$ uniformly on compact subsets of Ω , by the Cauchy criterion for uniform convergence of functions.

Now recall from basic analysis that on $[a, b]$, uniform convergence of derivatives and pointwise convergence of functions at a point implies that the functions converge uniformly to a differentiable function with the limit derivative, and if the derivatives are continuous then the limit of the derivatives is continuous as the uniform limit of continuous functions. Also, this result extends trivially to directional derivatives and can be extended to higher indices by simple induction.

Therefore, there exists a C^∞ function f such that $f_n \rightarrow f$ and $D^\alpha f_n \rightarrow D^\alpha f$ on every K_N by this result and thus globally.

This proves completeness.

Step 2. We want to show that closed and bounded implies compact, for which it suffices to prove that any bounded sequence has a convergent subsequence since we are in a metric space. Let $E \subseteq \mathcal{E}(\Omega)$ be closed and bounded. By an earlier theorem, boundedness is equivalent to existence of M_N such that $p_N(f) < M_N$ for all $f \in E$. For some M_N 's we have that $|D^\alpha f| < M_N$ for all $|\alpha| = N$ holds for all $f \in E$. By the mean value inequality this implies that for each β with $|\beta| < N - 1$ the set $\{D^\beta f : f \in E\}$ is equicontinuous on K_{N-1} and it is trivially uniformly bounded by the condition $p_N(f) < M_N$.

Thus, Arzela-Ascoli implies there is a subsequence that converges uniformly on K_0 . Call this $f_{n_k^1}$. Suppose now that an increasing sequence of integers n_k^N are given with the property that $(D^\beta f_{n_k^N})$ converges uniformly on K_{N-1} for all $|\beta| \leq N - 1$. Then we know that for each β with $|\beta| \leq N$ the set $\{D^\beta f_{n_k^N} : f \in E, k \in \mathbb{N}\}$ is equicontinuous and pointwise bounded on K_N . We can therefore extract a further sequence $f_{n_k^{N+1}}$ that converges uniformly on K_N for all $|\beta| \leq N$. By induction we can find a subsequence with the required property for all n and the usual diagonal sequence $F_k = f_{n_k^k}$ has that $D^\alpha F_k$ converges on compact subsets for any α .

Note that the limit function is actually in E because E is closed. Therefore E is sequentially compact. □

Definition. Let $K \subseteq \mathbb{R}^n$ be compact. Denote by $\mathcal{D}_K$ the space of all $f \in C^\infty(\mathbb{R}^n)$ with support in K . If $K \subseteq \Omega$ then $\mathcal{D}_K$ may be identified with a vector subspace of $C^\infty(\Omega)$ that we will shortly show is closed with respect to the $\mathcal{E}(\Omega)$ topology.

Lemma. The map $\delta_x : \mathcal{E}(\Omega) \rightarrow \mathbb{C}$ given by $f \mapsto f(x)$ is continuous.

Proof. If $\phi_j \rightarrow 0$ in $\mathcal{E}(\Omega)$ then $D^\alpha \phi_j \rightarrow 0$ uniformly on every compact subset of Ω for every multi-index α . In particular, taking $\alpha = 0$ and the compact set $\{x\}$ we have $\delta_x \phi_j \rightarrow 0$. Since sequential continuity coincides with continuity on metric spaces and thus Frechet spaces, the result follows. □

Proposition. $\mathcal{D}_K$ is closed in $C^\infty(\Omega)$

Proof. We know that the set $\delta_x^{-1}(\{0\})$ is closed by the lemma. Since we can write

$$\mathcal{D}_K = \bigcap_{x \in \Omega \setminus K} \delta_x^{-1}(\{0\})$$

it follows that this is an intersection of closed sets, thus also closed.

□

Therefore we have that $\mathcal{D}_K$ is a Frechet space with the subspace topology.

Definition. Let $\Omega \subseteq \mathbb{R}^n$ be open. Then the set $\mathcal{D}(\Omega)$ is the union over the sets $\mathcal{D}_K$ with $K \subseteq \Omega$ compact.

Note that this is just $C_c^\infty(\Omega)$. However, $\mathcal{D}(\Omega)$ refers to this set with the topology we will shortly define.

Note that this is a vector space over $\mathbb{R}$ or $\mathbb{C}$.

Our goal is to put a topology on $\mathcal{D}(\Omega)$ that makes it so that it is a complete locally convex topological vector space such that the subspace topology induced on $\mathcal{D}_K$ agrees with the subspace Frechet topology from $\mathcal{E}(\Omega)$ for each compact $K \subseteq \Omega$.

Here is how we will do this.

We will define

$$\|f\|_k := \max \{|D^\alpha f(x)| : x \in \Omega, |\alpha| \leq k\}$$

For each compact $K \subseteq \Omega$ we will let T_K be the Frechet subspace topology on $\mathcal{D}_K$ induced by $\mathcal{E}(\Omega)$.

We will let β be the collection of all convex balanced sets $W \subseteq \mathcal{D}(\Omega)$ such that $\mathcal{D}_K \cap W \in T_K$ for every compact $K \subseteq \Omega$.

We will let T be the collection of all (possibly empty) unions of sets of the form $\phi + W$ with $\phi \in \mathcal{D}(\Omega)$ and $W \in \beta$.

Theorem. T is a topology for $\mathcal{D}(\Omega)$ with local base β and it makes $\mathcal{D}(\Omega)$ into a locally convex topological vector space.

Proof. Step 1. We want to prove that we have a topological space with local base β .

It is clear from the definition that T has the empty set and the whole space and is closed under arbitrary unions. It remains to check closure under finite intersections. Therefore, let $V_1, V_2 \in T$ and let $\phi \in V_1 \cap V_2$, then we want to prove that $\phi + W \subseteq V_1 \cap V_2$ for some $W \in \beta$. Setting $\phi = 0$ and $V_2 = \mathcal{D}(\Omega)$ implies that any neighbourhood of 0 contains an element of β . Therefore it is enough to prove $\phi + W \subseteq V_1 \cap V_2$.

From the definition of T , there exist $\phi_i \in \mathcal{D}(\Omega)$ and $W_i \in \beta$ such that $\phi \in \phi_i + W_i \subseteq V_i$ for $i = 1, 2$.

Choose a compact $K \subseteq \Omega$ such that $\phi, \phi_i \in \mathcal{D}_K$. Since $\mathcal{D}_K \cap W_i$ is open in $\mathcal{D}_K$ we claim that we have

$$\phi - \phi_i \in (1 - \delta_i) W_i$$

for some $\delta_i > 0$. This is because $\mathcal{D}_K$ is a topological vector space for each K with the topology T_K that is a topological vector space by properties of $\mathcal{E}$ so scalar multiplication is continuous so for any $\psi \in \mathcal{D}_K$ the map $F_\psi : \mathbb{R} \rightarrow \mathcal{D}_K$ given by $t \mapsto t\psi$ is continuous. Therefore the set $(F_{\phi - \phi_i})^{-1}[W_i \cap \mathcal{D}_K]$ is open in $\mathbb{R}$. In particular there exist ε_i with $(1 - 2\varepsilon_i, 1 + 2\varepsilon_i) \subseteq A$ which is equivalent to $t(\phi - \phi_i) \in W_i \cap \mathcal{D}_K$ for $t \in (1 - 2\varepsilon_i, 1 + 2\varepsilon_i)$. This implies that $\phi - \phi_i \in (1 - \delta_i) W_i$ holds for some $\delta_i > 0$.

Since W_i is convex we have

$$\phi - \phi_i + \delta_i W_i \subseteq (1 - \delta_i) W_i + \delta_i W_i = W_i$$

The last equality is something I proved in linear analysis. We now deduce that

$$\phi + \delta_i W_i \subseteq \phi_i + W_i \subseteq V_i$$

Therefore, taking $W = \delta_1 W_1 \cap \delta_2 W_2$ we have $\phi + W \subseteq V_1 \cap V_2$ as required.

Step 2. We aim to prove we have a topological vector space which we know would be locally convex by step 1.

Step 2A. We want to show that singletons are closed.

Let $\phi_1, \phi_2 \in \mathcal{D}(\Omega)$ be distinct and consider the set

$$W = \{\phi \in \mathcal{D}(\Omega) : \|\phi\|_0 \leq \|\phi_1 - \phi_2\|_0\}$$

It is easy to see that W is balanced and it is also convex because it is a metric ball in a normed space (since $\|\lambda y + (1 - \lambda)z\| \leq \max(\|y\|, \|z\|)$ by the triangle inequality). Also, the sets $\{\phi \in \mathcal{D}_K : \|\phi\|_0 < r\}$ and their translations are open in T_K for any r and all compact $K \subseteq \Omega$ by definition of T_K which is in turn by definition of $\mathcal{E}$ which has to make all p_n 's which are equal to the $\|\cdot\|_n$'s on $\mathcal{D}_K$ continuous, we conclude that $W \in \beta$. Also, $\phi_1 \notin \phi_2 + W$ and since ϕ_2 was arbitrary we must have that $\{\phi_1\}$ is closed.

Step 2B. We want to show that addition is T -continuous.

Let $U \in T$ be open and suppose $\phi_1, \phi_2 \in \mathcal{D}(\Omega)$ with $\phi_1 + \phi_2 \in U$. Since β is a local base, $\phi_1 + \phi_2 + W \subseteq U$ for some $W \in \beta$. Then by convexity of W ,

$$\left(\phi_1 + \frac{1}{2}W\right) \times \left(\phi_2 + \frac{1}{2}W\right) \subseteq (+)^{-1}(U)$$

Therefore $(+)^{-1}(U)$ is open in the product topology so addition is T -continuous.

Step 2C. We want to show that scalar multiplication is T -continuous.

Let $U \in T$ be open and suppose $a\phi \in U$ with $a \in \mathbb{K}$ and $\phi \in \mathcal{D}(\Omega)$. Since β is a local base we have $a\phi + W \subseteq U$ for some $W \in \beta$. Note that

$$a'\phi' - a\phi = a'(\phi' - \phi) + (a' - a)\phi$$

Since $\mathcal{D}_K$ is a topological vector space, continuity of multiplication restricted to $\mathcal{D}_K$ for a compact K which contains the support of ϕ ensures we can choose $\delta > 0$ such that $a\phi \in \frac{1}{2}W$ if $|a| < \delta$. Now set $\varepsilon = (2(|a| + \delta))^{-1}$. By the fact that W is balanced and convex we have that $|a' - a| < \delta$ and $\phi' - \phi \in \varepsilon W$ implies that

$$a'\phi' - a\phi \in \frac{1}{2}W + \frac{1}{2}W = W$$

so indeed scalar multiplication is continuous.

□

Proposition. A convex balanced subset V of $\mathcal{D}(\Omega)$ is open if and only if $V \in \beta$

Proof. Since β is a local base we know if $V \in \beta$ then V is open.

If V is a convex, balanced open set we want to check that $\mathcal{D}_K \cap V \in T_K$ for every compact $K \subseteq \Omega$.

Let K be compact and pick $\phi \in \mathcal{D}_K \cap V$. Since β is a local base, we have $\phi + W \subseteq V$ for some $W \in \beta$. Therefore,

$$\phi + (\mathcal{D}_K \cap W) \subseteq \mathcal{D}_K \cap V$$

To see this note the LHS is in V by $\phi + W \subseteq V$ and in $\mathcal{D}_K$ as the sum of 2 things in the vector space $\mathcal{D}_K$.

From the definition of β we know $\mathcal{D}_K \cap W \in T_K$ so $\mathcal{D}_K \cap V$ is open in $\mathcal{D}_K$, and thus also in T_K as required.

□

Proposition. The subspace topology $\mathcal{D}_K$ inherits from $\mathcal{D}(\Omega)$ for any K is the same as the one it inherits from $\mathcal{E}(\Omega)$.

Proof. First we claim that any element of $T|_{\mathcal{D}_K}$ is an element of T_K . To see this, note that every set that is open in T is of the form $\bigcup_i (x_i + V_i)$ for $V_i \in \beta$ convex, balanced and open in T since T is a topological vector space. Therefore, the

sets open in $T|_{\mathcal{D}_K}$ are sets of the form $\bigcup_i (\mathcal{D}_K \cap (x_i + V_i))$. If $\mathcal{D}_K \cap (x_i + V_i)$ contains a point y then $y = x_i + v$ for some $v \in V_i$. Because V_i is a convex balanced open neighbourhood of 0, the set $x_i + V_i$ can be recentered around $y \in \mathcal{D}_K$ with a convex balanced neighbourhood $W \in \beta$ such that $y + W \subseteq x_i + V_i$. By the same reason as in the proof of the previous proposition we have $y + (\mathcal{D}_K \cap W) \subseteq \mathcal{D}_K \cap (x_i + V_i)$. By the proof of the previous proposition and the fact that we have a topological vector space, $y + (\mathcal{D}_K \cap W)$ is open in T_K . It now follows that, since y was arbitrary, every $y \in \mathcal{D}_K \cap U$ has a T_K -neighbourhood contained in $\mathcal{D}_K \cap U$, so $\mathcal{D}_K \cap U$ is open in T_K .

Therefore it remains to show that if $E \in T_K$ then E is open in $T|_{\mathcal{D}_K}$.

We want to show that $E = \mathcal{D}_K \cap U$ for some $U \in T$. Suppose $\phi \in E$. Then by definition of T_K there exists N, δ such that

$$\{\psi \in \mathcal{D}_K : \|\psi - \phi\|_N < \delta\} \subseteq E$$

Define

$$W_\phi := \{\psi \in \mathcal{D}(\Omega) : \|\psi\|_N < \delta\}$$

Then $W_\phi \in \beta$ and

$$\mathcal{D}_K \cap (\phi + W_\phi) = \phi + \mathcal{D}_K \cap W_\phi \subseteq E$$

The last inclusion is by the definition of W_ϕ, N, δ . Now we have that $E = \mathcal{D}_K \cap U$ for some $U \in T$ because we can take

$$U = \bigcup_{\phi \in E} (\phi + W_\phi)$$

□

Proposition. If E is a bounded (in TVS sense) subset of $\mathcal{D}(\Omega)$ then $E \subseteq \mathcal{D}_K$ for some $K \subseteq \Omega$, and there are real numbers $M_N < \infty$ such that every $\phi \in E$ satisfies the inequalities $\|\phi\|_N \leq M_N$ for each $N \in \mathbb{N}$.

Proof. First suppose that $E \subseteq \mathcal{D}(\Omega)$ is not in any $\mathcal{D}_K$, then we want to show that it is not bounded. Let K_m be an exhaustion of Ω . Since $E \not\subseteq \mathcal{D}_{K_m}$ for each m we can find $\phi_m \in E$ such that $\text{supp}(\phi_m) \not\subseteq K_m$. In particular there exists $x_m \in \Omega \setminus K_m$ with $\phi_m(x_m) \neq 0$. Let

$$W = \{\phi \in \mathcal{D}(\Omega) : \forall m \in \mathbb{N} |\phi(x_m)| < m^{-1} |\phi_m(x_m)|\}$$

Then W is convex and balanced. Suppose $K \subseteq \Omega$ is compact, then there exists M such that $K \subseteq K_M$. In particular this implies that $x_m \notin K$ for $m \geq M$, so $x_m \notin K$ for $m \geq M$. Therefore, $\mathcal{D}_K \cap W$ is an intersection of finitely many open sets (as for $m \geq M$ the condition does not restrict $\mathcal{D}_K \cap W$ any further) so $W \in \beta$. However, $\phi_m \notin mW$ for any m so E is not bounded.

Thus, any bounded set in $\mathcal{D}(\Omega)$ belongs to $\mathcal{D}_K$ for some K . By the previous proposition it follows that E is bounded in $\mathcal{D}_K$. The final part of the theorem follows from the fact that a set is bounded if and only if every norm in the set of seminorms forming the space is bounded on that set from a theorem we proved earlier.

□

Corollary. Since $\mathcal{D}_K$ has the Heine-Borel property inherited from $\mathcal{E}$ as shown earlier, the previous proposition implies that so does $\mathcal{D}(\Omega)$.

Proposition. Cauchy sequences in topological vector spaces are bounded.

Proof. Let U be an arbitrary neighbourhood of 0 in X . By continuity of addition in a topological vector space there exists a neighbourhood W of 0 such that $W + W \subseteq U$ and we know from earlier every neighbourhood of 0 contains a balanced neighbourhood of 0, so let $V \subseteq W$ be balanced, then $V + V \subseteq U$.

If x_n is T -Cauchy there is an integer N such that $m, n \geq N$ implies $x_n - x_m \in V$. We know finite sets in topological vector spaces are bounded so we just need to check boundedness of the tail of the sequence.

Boundedness of the tail is because $n > N$ implies $x_n \in x_N + V$. By boundedness of finite sets there exists λ such that $\lambda > 1$ and $\{x_1, x_2, \dots, x_N\} \subseteq \lambda V$. Also, V is balanced so $V \subseteq \lambda V$. Therefore we have, for $n > N$,

$$x_n \in x_N + V \subseteq \lambda V + V \subseteq \lambda(V + V) \subseteq \lambda U$$

We conclude because U was an arbitrary neighbourhood of 0.

□

Corollary. Every Cauchy sequence in $\mathcal{D}(\Omega)$ is in $\mathcal{D}_K$ for some compact K .

Corollary. Every Cauchy sequence in $\mathcal{D}(\Omega)$ is Cauchy with respect to T_K for some K

Corollary. By considering the neighbourhoods of T_K of balls of the form $\{\psi \in \mathcal{D}_K : \|\psi\|_N < N^{-1}\}$ we see that every sequence that is Cauchy in $\mathcal{D}(\Omega)$ is Cauchy with respect to each $\|\cdot\|_N$ norm.

Corollary. Since $\mathcal{D}_K$ is complete by completeness of $\mathcal{E}$, $\mathcal{D}$ is complete i.e. Cauchy sequences converge.

Proposition. A sequence tending to 0 in a topological vector space is Cauchy.

Proof. By continuity of scalar addition for every neighbourhood U there is a neighbourhood W with $W - W \subseteq U$ and there is I such that $i > I$ implies $\phi_i \in W$ so $\phi_i - \phi_j \in U$ so the sequence is Cauchy as U was arbitrary.

□

Corollary. If $\phi_i \rightarrow 0$ in $\mathcal{D}(\Omega)$ then there is a compact $K \subseteq \Omega$ containing the support of every ϕ_i such that $D^\alpha \phi_i \rightarrow 0$ uniformly as $i \rightarrow \infty$ for every multi-index α .

Lemma. Let $\Lambda : X \rightarrow Y$ be a linear map between topological vector spaces. Then Λ is continuous if and only if for every neighbourhood of 0 in Y its pre-image has an open neighbourhood of 0 in X .

Proof. For the non-trivial direction assume that Λ is continuous at 0. Then if $\phi_0 \in X$ then for any neighbourhood U of $\Lambda(\phi_0)$ the set $W = U - \Lambda(\phi_0)$ is a neighbourhood of 0 in Y and by linearity, the pre-images satisfy

$$\Lambda^{-1}(U) = \Lambda^{-1}(W + \Lambda(\phi_0))$$

Now let $x \in \Lambda^{-1}(W + \Lambda(\phi_0))$, then $\Lambda(x) \in W + \Lambda(\phi_0)$ so by linearity of Λ we have $\Lambda(x - \phi_0) \in W$. Therefore, $x = \phi_0 + (x - \phi_0) \in \phi_0 + \Lambda^{-1}(W)$. On the other hand, if $x \in \phi_0 + \Lambda^{-1}(W)$ then $x = \phi_0 + v$ where $\Lambda(v) \in W$. By linearity,

$$\Lambda(x) = \Lambda(\phi_0 + v) = \Lambda(\phi_0) + \Lambda(v) \in \Lambda(\phi_0) + W$$

Therefore

$$\Lambda^{-1}(U) = \Lambda^{-1}(W + \Lambda(\phi_0)) = \phi_0 + \Lambda^{-1}(W)$$

Thus, $\Lambda^{-1}(U)$ contains an open neighbourhood around ϕ_0 , as $\Lambda^{-1}(W)$ is open by continuity at 0.

□

Definition. Let X, Y be topological vector spaces. Then a linear map $\Lambda : X \rightarrow Y$ is said to be bounded if it maps bounded sets to bounded sets.

Theorem. Let Y be a locally convex topological vector space and let $\Lambda : \mathcal{D}(\Omega) \rightarrow Y$ be a linear map. Then the following are equivalent:

- a) Λ is continuous
- b) Λ is bounded
- c) If $\phi_i \rightarrow 0$ in $\mathcal{D}(\Omega)$ then $\Lambda \phi_i \rightarrow 0$ in Y

d) The restrictions of Λ to every $\mathcal{D}_K \subseteq \mathcal{D}(\Omega)$ are continuous.

e) For each compact $K \subseteq \Omega$ there exists $N \in \mathbb{N}$ and a constant C such that for all $\phi \in C_c^\infty(K)$ we have

$$|\Lambda[\phi]| \leq C \sup_{x \in K} \sum_{|\alpha| \leq N} |D^\alpha \phi(x)|$$

Proof. $a \implies b$:

Let E be a bounded set in $\mathcal{D}(\Omega)$ and let W be a neighbourhood of 0 in Y . Since Λ is continuous there exists a neighbourhood V of 0 in $\mathcal{D}(\Omega)$ such that $\Lambda(V) \subseteq W$.

Since E is bounded there exists $s > 0$ such that $E \subseteq tV$ for all $t > s$. Since Λ is linear we have

$$\Lambda(E) \subseteq \Lambda(tV) = t\Lambda(V) \subseteq tW$$

therefore $\Lambda(E)$ is bounded.

$b \implies c$:

We know that if $\phi_i \rightarrow 0$ in $\mathcal{D}(\Omega)$ then there exists $K \subseteq \Omega$ such that $\phi_i \rightarrow 0$ in $\mathcal{D}_K$ and ϕ_i is contained within $\mathcal{D}_K$ by the corollaries above. Since $\mathcal{D}_K$ is metrizable with an invariant metric there exist scalars $a_i \rightarrow \infty$ such that $a_i \phi_i \rightarrow 0$ in $\mathcal{D}_K$ (by an earlier proposition), and hence in $\mathcal{D}(\Omega)$ by the subspace topology property. By linearity of Λ we have $\Lambda(\phi_i) = a_i^{-1} \Lambda(a_i \phi_i)$.

Now $a_i \phi_i$ is Cauchy, hence bounded, so $\Lambda(a_i \phi_i)$ is bounded by assumption, and as $a_i^{-1} \rightarrow 0$ and $\Lambda(a_i \phi_i)$ is bounded, an earlier proposition also implies that $\Lambda(\phi_i) = a_i^{-1} \Lambda(a_i \phi_i) \rightarrow 0$.

$c \implies d$:

Again by the subspace topology and (c) we have that if $\phi_i \rightarrow 0$ in $\mathcal{D}_K$ then $\Lambda \phi_i \rightarrow 0$. Suppose that the restriction of Λ to $\mathcal{D}_K$ is not continuous. Then by the lemma above, there exists a neighbourhood W of 0 in Y such that $\Lambda^{-1}(W)$ contains no neighbourhood of 0 in $\mathcal{D}_K$. Since $\mathcal{D}_K$ is metrisable, pick a metric d which generates T_K and construct a sequence $x_n \rightarrow 0$ such that $x_n \notin \Lambda^{-1}(W)$. Then $x_n \rightarrow 0$ in $\mathcal{D}_K$ and hence also in $\mathcal{D}(\Omega)$, but $\Lambda(x_n) \not\rightarrow 0$, contradicting (c).

$d \implies a$:

Suppose that U is a convex, balanced open neighbourhood of the origin in Y and set $V = \Lambda^{-1}(U)$. Then V is convex and balanced by the linearity of Λ . Then by an earlier proposition V is open if and only if it is in β i.e. if $\mathcal{D}_K \cap V$ is open in $\mathcal{D}_K$ for every K compact in Ω . But then this is the case as Λ is continuous when restricted to each $\mathcal{D}_K$. Thus, V is open.

Now suppose that W is any open set in Y and let $\phi \in \Lambda^{-1}(W)$. Since Y is locally convex there exists a convex balanced neighbourhood of 0 in U such that $\Lambda \phi + U \subseteq W$. Since Λ is linear, we have $\phi + \Lambda^{-1}(U) \subseteq \Lambda^{-1}(W)$ (see the proof of the previous lemma) so $\phi + \Lambda^{-1}(U)$ is open so there is an open neighbourhood about every point in the pre-image so $\Lambda^{-1}(W)$ is open so Λ is continuous.

Therefore $a - d$ are equivalent. It remains to relate this to condition e .

$d \implies e$:

Fix K . By continuity, $\Lambda^{-1}(B_1(0))$ contains a basic open neighbourhood of 0, say U . Then there exists N, C such that

$$\left\{ \phi \in \mathcal{D}_K(\Omega) : p_N(\phi) < \frac{1}{C} \right\} \subseteq U \subseteq \{ \phi \in \mathcal{D}_K(\Omega) : |\Lambda(\phi)| < 1 \}$$

If $p_N(\phi) = 0$ then $p_N(\lambda\phi) = 0$ for every λ so we must have $\Lambda(\phi) = 0$.

If $p_N(\phi) > 0$ then for any $\delta > 0$ consider

$$\psi = \frac{\varepsilon}{p_N(\phi) + \delta} \phi$$

Then $p_N(\psi) < \varepsilon$ so $\Lambda(\psi) < 1$, so by linearity,

$$|\Lambda(\phi)| \leq \frac{1}{\varepsilon} (p_N(\phi) + \delta)$$

Taking $\delta \rightarrow 0$ gives the result where $C = \frac{1}{\varepsilon}$ is the appropriate constant.

$e \implies d$:

Let N, C be as in the statement of the theorem.

Given $\varepsilon > 0$ the set $V = \{\phi \in \mathcal{D}_K(\Omega) : p_N(\phi) < \frac{\varepsilon}{C}\}$ is a basic open neighbourhood of 0 in $\mathcal{D}_K(\Omega)$ by definition of the topology. For any $\phi \in V$ we therefore have $|\Lambda(\phi)| < \varepsilon$ so the restriction of Λ to K is continuous. □

To summarize, we have shown that if Ω is open in $\mathbb{R}^n$ we can equip $C_c^\infty(\Omega)$ with a topology such that

1. The vector space operations of addition and scalar multiplication are continuous
2. A sequence ϕ_j converges to 0 if there exists a compact $K \subseteq \Omega$ such that the support of every function in the sequence is in K and for each multi-index α we have $\sup_{x \in K} |D^\alpha \phi_j| \rightarrow 0$, and $\phi_j \rightarrow \phi$ if $\phi_j - \phi \rightarrow 0$.
3. Linear maps from $\mathcal{D}(\Omega)$ into a locally convex vector space are continuous if and only if they are sequentially continuous.

Now recall that $T_x \phi(y) = \phi(y - x)$. Then we have

Proposition. Let $\phi \in \mathcal{D}(\Omega)$. Let $\delta > 0$ be such that $T_x \phi \in \mathcal{D}(\Omega)$ for $|x| < \delta$. Then if $x_n \subseteq \mathbb{R}^n$ is a sequence tending to 0 contained in $B_\delta(0)$, then $T_{x_n} \phi \rightarrow \phi$ in $\mathcal{D}(\Omega)$.

Proof. Note that there exists $\delta > 0$ such that $\text{supp}(\phi) + B_{2\delta}(0) \subseteq \Omega$. Then for $|x| < \delta$ we have

$$\text{supp} T_x \phi \subseteq \overline{\text{supp} \phi + B_\delta(0)} \subseteq \text{supp} \phi + B_{2\delta}(0) \subseteq \Omega$$

Then if i is large enough, $\text{supp}(T_{x_i} \phi) \subseteq \overline{\text{supp} \phi + B_\delta(0)}$. For each α , $D^\alpha \phi$ is continuous on a compact set so it is uniformly continuous so

$$\sup_K |T_{x_n} D^\alpha \phi - D^\alpha \phi| \rightarrow 0$$

□

Proposition. Let $\phi \in \mathcal{D}(\Omega)$. For h sufficiently small define the forward difference quotient as

$$\Delta_i^h \phi = \frac{1}{h} (T_{-he_i} \phi - \phi)$$

where e_i is the standard basis for $\mathbb{R}^n$. Then $\Delta_i^h \phi \rightarrow D_i \phi$ as $h \rightarrow 0$ in the $\mathcal{D}$ topology.

Proof. As in the argument above there exists a compact $K \subseteq \Omega$ such that $\text{supp}(\Delta_i^h \phi) \subseteq K$ for h sufficiently small. Therefore by the mean value theorem, for each $x \in K$ there exists $t_x \in (0, h)$ such that

$$D^\alpha \Delta_i^h \phi(x) = \frac{D^\alpha \phi(x + he_i) - D^\alpha \phi(x)}{h} = D_i D^\alpha \phi(x + t_x e_i)$$

Given $\varepsilon > 0$, by uniform continuity as in the argument above if t_x is small enough the difference is less than ε and we conclude similarly. □

Example. If $\phi(x)$ is non-zero on $\mathbb{R}$ the sequence $\phi_j(x) = \frac{1}{j} \phi(x - j)$ does not converge in $\mathcal{D}(\mathbb{R})$ because its support is not contained in a compact set.

Note that the inclusion map $i : \mathcal{D}(\Omega) \rightarrow \mathcal{E}(\Omega)$ is continuous, since convergence in $\mathcal{D}$ is strictly stronger than convergence in $\mathcal{E}$.

Note that the subspace topology on C_c^∞ of $\mathcal{E}$ is NOT $\mathcal{D}$. To see this, if ϕ is non-zero but has compact support, then as above the sequence $\phi_j(x) = \frac{1}{j}\phi(x-j)$ does not converge in $\mathcal{D}(\mathbb{R})$, but it does converge in $\mathcal{E}(\mathbb{R})$ since in every compact set it is eventually identically 0.

Definition. A function $\phi \in C^\infty(\mathbb{R}^n)$ is said to be rapidly decreasing if for all multi-indices α and $N \in \mathbb{N}$ we have

$$\sup_{x \in \mathbb{R}^n} \left| (1 + |x|)^N D^\alpha \phi(x) \right| < \infty$$

Example. Functions with compact support are rapidly decreasing. So is the function $x \mapsto e^{-|x|^2}$.

We will define the family of seminorms $\mathcal{P} = p_N$ where

$$p_N(\phi) = \sup_{x \in \mathbb{R}^n, |\alpha| \leq N} \left| (1 + |x|)^N D^\alpha \phi(x) \right|$$

Then this is a separating family of seminorms giving rise to a topology $\mathcal{S}$ such that the vector space operations are continuous. Since functions with compact support are rapidly decreasing we get that $\mathcal{D}(\mathbb{R}^n)$ is contained in $\mathcal{S}$.

$\mathcal{S}$ is called the Schwartz space.

Using the same proof as the analogous theorem above, or the principle that convergence in a topology induced by some norms is equivalent to convergence in all these norms, we see that convergence in $\mathcal{S}$ is equivalent to uniform convergence of $p_N(\phi_j)$ for every $N \in \mathbb{N}$.

Also, for the same reason as in $\mathcal{E}(\mathbb{R}^n)$ this topology on $\mathcal{S}$ is metrizable. We have that

$$\mathcal{D}(\mathbb{R}^n) \subseteq \mathcal{S} \subseteq \mathcal{E}(\mathbb{R}^n)$$

and the inclusion map is continuous in each case, again since it is easy to see that the notions of convergence are weaker. We will justify now why convergence in $\mathcal{S}$ implies convergence in $\mathcal{E}$: To do this, set $N = 0$ on the exponent and note that restricting to a subset preserves uniform convergence. Allowing $N = 0$ is not necessary since if $p_N \rightarrow 0$ for all $N \geq 1$ then

$$\sup_K |D^\alpha \phi_j| \leq \sup_{\mathbb{R}^n} |D^\alpha \phi_j| \leq \sup_{\mathbb{R}^n} \left| (1 + |x|)^N D^\alpha \phi_j \right| = p_N \rightarrow 0$$

3.2 Definition and basic properties of distributions

Consider a general linear PDE of order k where L is a differential operator:

$$Lu = \sum_{|\alpha| \leq k} a_\alpha D^\alpha u = f$$

Assume that each $a_\alpha \in C^\infty(\Omega)$ for an open $\Omega \subseteq \mathbb{R}^n$.

We'll generalize solutions to the situation where u and f need not be of class C^k . We'll say that u and f belong to a space X .

We will want X to be such that the inclusion map $i : C^\infty(\Omega) \rightarrow X$ is injective, and X is a vector space over $\mathbb{C}$. We want to come up with a way to 'differentiate' elements of X that is compatible with the inclusion map and a topology on X such that the above operations are continuous.

If you took the tripos in a normal order, i.e. taking methods before this which I'm sure would make me lose my mind, you would probably have a better idea of why we want to do this and what the motivation is.

Definition. A distribution is a continuous element of $\mathcal{D}^*(\Omega)$ i.e. a continuous linear functional on the space of test functions.

Definition. Note that in condition (e) of the continuity criterion above, if there exists an N that works for every compact K (C is allowed to depend on K) then we say that Λ has finite order and the least such N is its order.

Now we will look at examples of distributions:

Example. The Dirac delta: For $x \in \Omega$ define $\delta_x \phi := \phi(x)$.

This is trivially linear. It satisfies the continuity bound with $C = 1$ and $N = 0$ and has order 0 because

$$|\phi(x_0)| \leq \sup_{x \in K} |\phi(x)|$$

Example. If $f \in L^1_{loc}$, then define the distribution T_f by

$$T_f \phi := \int_{\Omega} f(x) \phi(x) dx$$

Again, clearly linear. As for continuity and order,

$$|T_f \phi| = \int_K |f(y)| |\phi(y)| dy \leq \left(\int_K |f(y)| dy \right) \sup_{y \in K} |\phi(y)|$$

Since $f \in L^1_{loc}$ and K is compact, $(\int_K |f(y)| dy)$ is a finite constnat, so the order is 0.

Example. Let $\phi \in \mathcal{D}(\mathbb{R})$, then define

$$PV \left(\frac{1}{x} \right) [\phi] = \lim_{\varepsilon \rightarrow 0} \left[\int_{-\infty}^{-\varepsilon} \frac{\phi(x)}{x} dx + \int_{\varepsilon}^{\infty} \frac{\phi(x)}{x} dx \right]$$

This is clearly linear if the limit exists which we are about to prove is the case.

We can write

$$PV \left(\frac{1}{x} \right) [\phi] = \lim_{\varepsilon \rightarrow 0} \left[\int_{-\infty}^{-\varepsilon} \frac{\phi(x) - \phi(-x)}{x} dx \right]$$

On the other hand, we have

$$\left| \frac{\phi(x) - \phi(-x)}{x} \right| = \left| \int_{-1}^1 \phi'(xt) dt \right| \leq 2 \sup_{\mathbb{R}} |\phi'|$$

From this we conclude that the limit exists and that if $\phi \in C_c^\infty(B_R(0))$ then

$$\left| PV \left(\frac{1}{x} \right) [\phi] \right| \leq 4R \sup_{\mathbb{R}} |\phi'|$$

So we have a distribution of order at most one.

To show that the order is exactly one we must show that it is not 0.

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be smooth with $g(t) = 0$ for $t \leq 1$ and $g(t) = 1$ for $t \geq 2$ and $0 \leq g(t) \leq 1$.

Fix $\chi \in C_c^\infty(\mathbb{R})$ such that $0 \leq \chi \leq 1$ everywhere and $\chi(x) = 1$ on $[0, \frac{1}{2}]$ and $\chi(x) = 0$ for $|x| \geq 1$.

Define $\phi_n(x) = g(nx) \chi(x)$ for $x \geq 0$ and $\phi_n(x) = -g(-nx) \chi(-x)$ for $x < 0$. Define $\phi_n(0) = 0$ for all n .

Note that $\phi_n \in C_c^\infty([-1, 1])$ and $\sup_{x \in [-1, 1]} |\phi_n(x)| \leq 1$ for all n .

Then the functional is given by

$$\int_{-\infty}^{\infty} \frac{\phi_n(x)}{x} dx = 2 \int_0^1 \frac{g(nx) \chi(x)}{x} dx$$

since $\frac{g(nx)\chi(x)}{x}$ was constructed to be an even function and is 0 away from 0.

For $n > 4$ the integral can be bounded below by $2 \ln\left(\frac{n}{4}\right)$ by noting that $g(nx)\chi(x) = 1$ on $\left[\frac{2}{n}, \frac{1}{2}\right]$.

If the distribution had order 0 there would exist a constant C such that for all n we have

$$PV\left(\frac{1}{x}\right)[\phi_n] \leq C \sup_{\mathbb{R}} |\phi_n| \leq C$$

Therefore the order is 1.

Now define

$$i : C^\infty(\Omega) \rightarrow \mathcal{D}^*(\Omega)$$

by

$$f \mapsto T_f$$

Note that $C^\infty \subseteq C^0 \subseteq L^1_{loc}$ by the extreme value theorem. We hope to be able to identify distributions with smooth functions in this way, and then we will consider distributions that are not representable in this way as 'generalized' functions, like the dirac delta, that we will hopefully see in the methods course why we actually care about them.

To achieve this goal we want to verify that i is injective. The idea is to recover f at each point by multiplying it by scaled bump functions.

Given $\phi : \mathbb{R}^n \rightarrow \mathbb{C}$ and $x \in \mathbb{R}^n$ define $T_x\phi(y) = \phi(y - x)$. Define $\check{\phi}(y) = \phi(-y)$. By convention, define $T_x\check{\phi}(y) = \phi(x - y)$ so that the translator operator acts before inversion. Then we have the following:

Theorem. Let $f \in C^k(\Omega)$ and as earlier let ϕ_ε be a function that integrates to 1 and is smooth and supported on a ε open ball about 0. Define for any x with $d(x, \partial\Omega) \geq \varepsilon$

$$f_\varepsilon(x) := T_f[T_x\check{\phi}_\varepsilon]$$

Then for any compact subset $K \subseteq \Omega$ and any $|\alpha| \leq k$ we have

$$\sup_{x \in K} |D^\alpha f_\varepsilon(x) - D^\alpha f(x)| \rightarrow 0$$

as $\varepsilon \rightarrow 0$

In particular we can recover $D^\alpha f(x)$ using only T_f .

Proof. Fix $K \subseteq \Omega$ compact. Recall that there exists $\delta > 0$ and $\chi \in C_c^\infty(\Omega)$ such that $\chi = 1$ on $K + B_\delta(0)$. For $\varepsilon < \delta$, since $\text{supp}(\phi_\varepsilon) \subseteq B_\varepsilon(0)$ we have for $x \in K$:

$$f_\varepsilon(x) = \int_{\Omega} f(y) T_x\check{\phi}_\varepsilon(y) dy = \int_{\mathbb{R}^n} \chi(y) f(y) \phi_\varepsilon(x - y) dy = \phi_\varepsilon \star (\chi f)$$

Since χ is smooth we have $\chi f \in C_c^k(\mathbb{R}^n)$ so as $\varepsilon \rightarrow 0$ we have for any $|\alpha| \leq k$ by an earlier theorem about convolving with ϕ_ε that

$$D^\alpha(\phi_\varepsilon \star (\chi f)) \rightarrow D^\alpha(\chi f)$$

and the convergence is uniform on $\mathbb{R}^n$ as $\varepsilon \rightarrow 0$. In particular we have uniform convergence in K as required. The fact that χ is 1 on $K + B_\varepsilon$ justifies this.

□

Corollary. If $f, g \in C^0$ then $T_f = T_g$ implies $f = g$ since the above theorem gives a way to recover f by picking a clever family of test functions.

3.3 Differentiating distributions

We want our definition of the derivative of a distribution to coincide with the derivative of a function.

If f is C^1 then we know that $T_{D_i f} \phi = \int_{\Omega} [D_i f](x) \phi(x) dx$. Since $\phi \in C_c^\infty$, then by Fubini's theorem we can integrate by parts without picking up any boundary terms to get

$$T_{D_i f} \phi = - \int_{\Omega} f(x) [D_i \phi](x) dx = -T_f [D_i \phi]$$

We now define the D^α derivative of a distribution u to be the distribution $D^\alpha u$ which acts on a test function $\phi \in C_c^\infty(\Omega)$ as

$$(D^\alpha u)[\phi] = (-1)^{|\alpha|} u[D^\alpha \phi]$$

Since a distribution acts on smooth functions, a derivative is defined for any distribution.

Example. The derivative of the dirac delta function on $\mathbb{R}$ is a function that sends $\phi \rightarrow -\frac{d\phi}{dx}|_0$.

Example. Consider the Heaviside step function on $\mathbb{R}$ (Recall from the differential equations course that this is $1_{x \geq 0}$). We can define the associated distribution T_H as

$$T_H \phi = \int_{\mathbb{R}} H(x) \phi(x) dx$$

for $\phi \in C_c^\infty(\mathbb{R})$. Easy check is this is actually a distribution (Hint: it has order 0 check the bound for an arbitrary compact set). Then its derivative is given by

$$-T_H [D_x \phi] = - \int_{\mathbb{R}} H(x) \phi'(x) dx = - \int_0^\infty \phi'(x) dx = \phi(0)$$

Therefore its derivative is the dirac delta as we would 'expect' if we naively treat the fundamental theorem of calculus as a general principle as we are meant to in applied contexts.

Definition. Multiplying a distribution u by a smooth function f is defined by $f u(\phi) = u(f \phi)$

Proposition. T is linear over C^∞ i.e. Let $f_1, f_2 \in C_0(\Omega)$ and let $a \in C^\infty(\Omega)$. Then

$$a T_{f_1} + T_{f_2} = T_{a f_1 + f_2}$$

Proof. Take a test function $\phi \in \mathcal{D}(\Omega)$. Then by linearity, and the definition of multiplication of distributions we have

$$\begin{aligned} (a T_{f_1} + T_{f_2}) \phi &= (a T_{f_1}) \phi + (T_{f_2}) \phi = T_{f_1} (a \phi) + T_{f_2} (\phi) \\ &= \int_{\Omega} (f_1(x) a(x) + f_2(x)) \phi(x) dx = T_{a f_1 + f_2} (\phi) \end{aligned}$$

We conclude as this holds for every test function. □

Proposition. If $f \in C^k(\Omega)$ then $D^\alpha T_f = T_{D^\alpha f}$ for $|\alpha| \leq k$. Also, $i \circ D^\alpha = D^\alpha \circ i$ on functions of class C^k whenever $|\alpha| \leq k$ so derivatives work as we expect.

Proof. Let $\phi \in \mathcal{D}(\Omega)$ be a test function. Then we have

$$(D^\alpha T_f)[\phi] = (-1)^{|\alpha|} T_f [D^\alpha \phi] = (-1)^{|\alpha|} \int_{\Omega} f(x) D^\alpha \phi(x) dx$$

Doing integration by parts $|\alpha|$ times and noting that there are no boundary terms because ϕ has compact support, we get

$$= (-1)^{|\alpha|} (-1)^{|\alpha|} \int_{\Omega} D^\alpha f(x) \phi(x) dx = (T_{D^\alpha f})[\phi]$$

We therefore have

$$(D^\alpha \circ i) f = D^\alpha (T_f) = T_{D^\alpha f} = (i \circ D^\alpha) f$$

□

Corollary. By linearity of $f \mapsto T_f$ in C^∞ and the proposition above, we have: Let $f \in C^k(\Omega)$ and let L be a differential operator so that $L = \sum_{|\alpha| \leq k} a_\alpha D^\alpha$. Then $T_{Lf} = LT_f$

Theorem. Let $f \in C^0(\Omega)$ define the distribution T_f and suppose that for any multi-index α with $|\alpha| \leq k$ there exists $g^\alpha \in C^0(\Omega)$ such that

$$D^\alpha T_f = T_{g^\alpha}$$

Then $f \in C^k$ and $D^\alpha f = g^\alpha$ in the sense of classical derivatives.

Proof. First we show the result for $k = 1$. Given a compact subset K of Ω and ϕ_ε a bump function as earlier for ε sufficiently small, for $x \in K$ define $f_\varepsilon(x) = \int_\Omega f(y) \phi_\varepsilon(x-y) dy$ as earlier. We know that $f_\varepsilon \rightarrow f$ uniformly on K as $\varepsilon \rightarrow 0$ by a recent theorem. We have

$$D_i f_\varepsilon(x) = \int_\Omega f(y) \frac{\partial}{\partial x_i} \phi_\varepsilon(x-y) dy = - \int_\Omega f(y) \frac{\partial}{\partial y_i} \phi_\varepsilon(x-y) dy = (D_i T_f) [T_x \check{\phi}_\varepsilon] = T_{g_i} [T_x \check{\phi}_\varepsilon]$$

As $\varepsilon \rightarrow 0$ we have $T_{g_i} [T_x \check{\phi}_\varepsilon] = g_i \star \phi_\varepsilon \rightarrow g_i$ uniformly on K . Therefore we have $f_\varepsilon \rightarrow f$ and $D_i f_\varepsilon \rightarrow g_i$ uniformly on K as $\varepsilon \rightarrow 0$. This implies that $f \in C^1(K)$ with $D_i f = g_i$.

We have this for all K by repeated application of the $k = 1$ result.

□

Example. As cursed and non-rigorous as this example is, it's good intuition. If we say that the derivative of $H(x)$ is $\delta(x)$ naively and are like, what if we can just be silly like that, we might say that the derivative of $\delta(x)$ is ∞^2 for 0^- and $-\infty^2$ for 0^+ , and you might intuitively expect that integrating this against f gives $-f'(0)$, as it would give the difference between f at 0^- and 0^+ divided by ∞ .

The reason it works intuitively this way is because we have defined it directly using familiar properties like integration by parts.

3.4 Convergence of distributions

Consider the weak- $\star$ topology on $\mathcal{D}^*(\Omega)$. Specifically, we will index seminorms by $\phi \in \mathcal{D}(\Omega)$ so that each seminorm is given by $p_\phi(u) = |u(\phi)|$. Then consider the topology induced by these. This is separating since if $u \neq 0$ then there exists a ϕ with $p_\phi(u) \neq 0$ so it is separating. Therefore, by an earlier theorem, we get a locally convex topological vector space.

This topology has the property that a sequence $u_j \subseteq \mathcal{D}^*$ converges to 0 in the weak- $\star$ topology if and only if $u_j[\phi] \rightarrow 0$ for all $\phi \in \mathcal{D}(\Omega)$.

Example. Let $f_j \subseteq C^0(\Omega)$ be a sequence of functions such that $f_j \rightarrow 0$ uniformly on any compact set $K \subseteq \Omega$. Then $T_{f_j} \rightarrow 0$ in $\mathcal{D}^*(\Omega)$. This is because for any $\phi \in \mathcal{D}(\Omega)$ its support is in some compact K and

$$|T_{f_j}[\phi]| = \left| \int_K f_j(y) \phi(y) dy \right| \leq \mu(K) \sup_K |f_j \phi|$$

The right hand side tends to 0 by uniform convergence so $T_{f_j}[\phi] \rightarrow 0$ for any ϕ .

Example. By a similar argument if $f_j \subseteq L^1_{loc}(\Omega)$ then if $\|f_j\|_{L^1(K)} \rightarrow 0$ for every compact $K \subseteq \Omega$ then $T_{f_j} \rightarrow 0$ in $\mathcal{D}^*(\Omega)$.

Example. For $\phi \in \mathcal{D}(\mathbb{R})$ set

$$u[\phi] = \sum_{m=-\infty}^{\infty} \phi^{(|m|)}(m)$$

where the superscript denotes the $|m|$ 'th derivative. For each ϕ this has finitely many terms so it is trivial that this is linear, and it is an easy check that this is a distribution as it satisfies the bound for some C, N for each K (pick N with

$K \subseteq [-N, N]$). However, it is also easy to check that it has infinite order. To do this, suppose there is an integer N and constant C_K for each K such that for all $\phi \in \mathcal{D}(\mathbb{R})$ supported in K we have

$$|T[\phi]| \leq C_K \sup_{x \in K, 0 \leq j \leq N} |\phi^{(j)}(x)|$$

Note: this is clearly equivalent to the inequality with the sum.

To get a contradiction (and hence prove that this example has infinite order) let $m > N$ and let $K = [m - 0.5, m + 0.5]$. Define the sequence, for $k \in \mathbb{N}$

$$\phi_k(x) = \frac{1}{k^{N+0.5}} \chi(x - m) e^{ik(x-m)}$$

where χ is a standard bump function with $\chi(0) = 1$ supported in $[-0.5, 0.5]$.

Then for the semi-norm of the ϕ_k 's, the maximum derivative up to order N approaches 0, as splitting it into $\chi(x - m)$ and $\frac{1}{k^{N+0.5}} e^{ik(x-m)}$ factors and then applying the Leibniz rule means the derivatives up to order N are $o(1)$ as $k \rightarrow \infty$ as the derivatives of χ are bounded by a constant.

Also, the infinite sum in the norm collapses to $\phi_k^{(m)}(m)$ which by the Leibniz rule has magnitude at 0 equal to

$$\frac{1}{k^{N+0.5}} \sum_{l=0}^m \binom{m}{l} \chi^{(m-l)}(0) (ik)^l$$

As k grows, the dominant term corresponds to $l = m$ and this is $i^m \chi(0) k^{m-N-0.5}$, which is unbounded, which contradicts the inequality we assumed at the start.

Now we have a distribution with infinite order. Now consider the sequence of distributions

$$u_M = \sum_{m=-M}^M (-1)^m D^{|m|} \delta_m$$

Let $\phi \in \mathcal{D}(\Omega)$ be any test function, then there exists R such that it is supported in a subset of $B_R(0)$ and for $M \geq R$ we have

$$u_M[\phi] = \sum_{m=-M}^M \phi^{(|m|)}(m) = u[\phi]$$

Note that the last equality is an exact equality because $M \geq R$. Also, we can write

$$u = \sum_{m=-\infty}^{\infty} D^{|m|} \delta_m$$

where the sum converges in $\mathcal{D}(\Omega)$, trivially so because for every K it is eventually constant.

Example. Consider the bump functions $T_x \check{\phi}_\varepsilon(y) = \phi_\varepsilon(x - y)$. Then we want to show that as $\varepsilon \rightarrow 0$ we have $T_{T_x \check{\phi}_\varepsilon} \rightarrow \delta_x$ in $\mathcal{D}^*$. To do this, recall that $T_{T_x \check{\phi}_\varepsilon}[\psi] = \phi_\varepsilon \star \psi(x)$. By properties of convergence of convolutions with ϕ_ε we therefore have $T_{T_x \check{\phi}_\varepsilon}[\psi] \rightarrow \psi(x) = \delta_x[\psi]$. The result follows since ψ was arbitrary.

3.5 Convolutions and PDE examples

Taking together some of the propositions above, we have

Theorem. Suppose that $f \in C^0(\Omega)$ and that $T \in \mathcal{D}^*(\Omega)$ satisfies the distributional equation

$$\sum_{|\alpha| \leq k} a_\alpha D^\alpha T = T_f$$

for $a_\alpha \in C^\infty(\Omega)$ and that there exist functions $u_\alpha \in C^0(\Omega)$ such that $D^\alpha T = T_{u_\alpha}$. Then $u := u_0 \in C^k$ is a classical solution of the equation

$$\sum_{|\alpha| \leq k} a_\alpha D^\alpha u = f$$

This means we could try to solve a PDE by trying to find a distributional solution before worrying about whether it is in fact a classical solution.

Take L to be the partial differential operator $L := \sum_{|\alpha| \leq k} a_\alpha D^\alpha$ and suppose now that each a_α is constant and that $\Omega = \mathbb{R}^n$. We seek solutions to the distributional equation $Lu = T_f$ and we hope that this will lead to classical solutions in $C^k(\Omega)$ of the analogous PDE.

Definition. A distribution $u \in \mathcal{D}'(\Omega)$ is supported in the closed set $K \subseteq \Omega$ if $u[\phi] = 0$ for all $\phi \in C_c^\infty(\Omega \setminus K)$ i.e. when ϕ is 0 in K . The support of a distribution is the intersection of all closed sets with this property. If this resulting set is compact we say that u has compact support.

Some easy checks: Note that the support of a distribution corresponding to a continuous function f is equal to the support of f . Also, if u is supported in K then so is $D^\alpha u$ for any multi-index α . Also, since $\text{supp}(\delta) = \{0\}$ there cannot be any continuous function that dirac delta corresponds to as we kind of already know.

Now let f be continuous on $\mathbb{R}^n$ and $\phi \in C_c^\infty(\mathbb{R}^n)$. Then recall that the convolution of the two functions is defined to be $(f \star \phi)(x) = T_f[T_x \check{\phi}]$. Define the convolution of a distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ with a test function $\phi \in \mathcal{D}(\mathbb{R}^n)$ by the formula

$$(u \star \phi)(x) = u[T_x \check{\phi}]$$

Proposition. Let $u_1, u_2 \in \mathcal{D}'(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$. Then $u_1 \star \phi = u_2 \star \phi$ for all ϕ implies $u_1 = u_2$.

Proof. Since $u[\phi] = (u \star \check{\phi})(0)$ we have $u_1 \phi = u_1 \star \check{\phi}(0) = u_2 \star \check{\phi}(0) = u_2 \phi$.

□

Proposition. Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$. Then $u \star \phi \in C^\infty(\mathbb{R}^n)$ and

$$D^\alpha (u \star \phi) = u \star D^\alpha \phi = D^\alpha u \star \phi$$

for every multi-index α

Proof. We calculate

$$\begin{aligned} \Delta_i^h [u \star \phi](x) &= \frac{(u \star \phi)(x + he_i) - (u \star \phi)(x)}{h} = \frac{1}{h} (u[T_{(x+he_i)} \check{\phi}] - u[T_x \check{\phi}]) \\ &= u \left[\frac{1}{h} (T_{(x+he_i)} \check{\phi} - T_x \check{\phi}) \right] = u[\Delta_i^h T_x \check{\phi}] \end{aligned}$$

Now if ϕ is a test function we have, if this limit exists,

$$\lim_{h \rightarrow 0} \Delta_i^h T_x \check{\phi}(y) = \lim_{h \rightarrow 0} \frac{\phi(x + he_i - y) - \phi(x - y)}{h} = D_i \phi(x - y) = (T_x(D_i \phi))(y)$$

By an earlier proposition about forward difference quotients (immediately generalizes even if we drop the 'forward' condition), the limit exists and convergence is also in the $\mathcal{D}$ topology. Therefore, using the characterization of continuity of distributions we may write

$$\lim_{h \rightarrow 0} \Delta_i^h [u \star \phi](x) = u[T_x(D_i \phi)] = u \star D_i \phi(x)$$

Therefore by repeating the argument for higher derivatives we conclude that the first equality holds.

To get the second equality, we calculate

$$D^\alpha [T_x \check{\phi}](y) = \frac{\partial^{|\alpha|}}{\partial y^\alpha} [\phi(x - y)] = (-1)^{|\alpha|} (D^\alpha \phi)(x - y) = (-1)^{|\alpha|} [T_x(D^\alpha \phi)](y)$$

Using the definition of the derivative and convolution of a distribution we have

$$(D^\alpha u \star \phi)(x) = D^\alpha u[T_x \check{\phi}] = (-1)^{|\alpha|} u[D^\alpha (T_x \check{\phi})]$$

Therefore by the line of working above that this is

$$= u [T_x (\check{D}^\alpha \phi)] = (u \star D^\alpha \phi)(x)$$

□

Proposition. Let $u \in \mathcal{D}^*(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$. Then

$$\text{supp}(u \star \phi) \subseteq \text{supp}(u) + \text{supp}(\phi)$$

In particular if u has compact support then $u \star \phi$ has compact support and is in $\mathcal{D}(\mathbb{R}^n)$

Proof. Suppose for $x \in \mathbb{R}^n$ that $u \star \phi(x) \neq 0$. Then we must have

$$\text{supp}(u) \cap \text{supp}(T_x \check{\phi}) \neq \emptyset$$

by the definition of the convolution. In particular, there exists $z \in \text{supp}(u)$ such that $T_x \check{\phi}(z) = \phi(x - z) \neq 0$. Thus $x - z \in \text{supp}(\phi)$ for $z \in \text{supp}(u)$ so we have

$$\{x : u \star \phi(x) \neq 0\} \subseteq \text{supp}(u) + \text{supp}(\phi)$$

We may take the closure on the left hand side since $\text{supp}(u) + \text{supp}(\phi)$ is closed, so done.

□

Noting that the convolution is associative for ordinary continuous functions as we proved earlier, if u_2 has compact support we define the convolution of 2 distributions $u_1 \star u_2$ to be the unique distribution that satisfies, for all test functions ϕ , the identity

$$(u_1 \star u_2) \star \phi = u_1 \star (u_2 \star \phi)$$

Theorem. If u_2 has compact support then there actually exists a distribution w satisfying $w \star \phi = u_1 \star (u_2 \star \phi)$

This is surprisingly difficult to actually prove. We need the following lemma:

Lemma. Let $\chi \in \mathcal{D}(\mathbb{R}^n)$ and let u_1 be any distribution, then the map $\chi \mapsto u_1 \star \chi$ is continuous $\mathcal{D}(\mathbb{R}^n) \rightarrow \mathcal{E}(\mathbb{R}^n)$

Proof. Recall that by definition of the distribution convolution on test functions,

$$(u_1 \star \chi_j)(x) = u_1[\tau_x \check{\chi}_j].$$

To see why $(u_1 \star \chi_j) \rightarrow 0$ in $\mathcal{E}(\mathbb{R}^n)$ (i.e., uniform convergence of all derivatives on every compact $K \subset \mathbb{R}^n$): Using the derivative identity $D^\alpha(u_1 \star \chi_j) = u_1 \star (D^\alpha \chi_j)$, we have for any multi-index α :

$$D^\alpha(u_1 \star \chi_j)(x) = u_1[\tau_x(D^\alpha \check{\chi}_j)].$$

Because u_1 is a distribution, for any compact set $L \subseteq \mathbb{R}^n$, there exist $C_L > 0$ and $N_L \in \mathbb{N}$ such that for all $\theta \in \mathcal{D}_L(\mathbb{R}^n)$:

$$|u_1[\theta]| \leq C_L \sum_{|\beta| \leq N_L} \sup_{y \in L} |D^\beta \theta(y)|$$

Since $\chi_j \rightarrow 0$ in $\mathcal{D}(\mathbb{R}^n)$, there is a fixed compact $S \subseteq \mathbb{R}^n$ containing $\text{supp}(\chi_j)$ for all j . If $x \in K$ (a fixed compact set), then

$$\text{supp}(\tau_x(D^\alpha \check{\chi}_j)) = x - \text{supp}(D^\alpha \chi_j) \subseteq K - S := L,$$

where L is a fixed compact set independent of j and $x \in K$.

Define $D_y^\beta f$ to mean the derivative with respect to y .

Applying the estimate for u_1 on the compact set L :

$$\sup_{x \in K} |D^\alpha(u_1 \star \chi_j)(x)| \leq C_L \sum_{|\beta| \leq N_L} \sup_{x \in K} \sup_{y \in L} |D_y^\beta[\tau_x(D^\alpha \check{\chi}_j)](y)|.$$

Since $D_y^\beta[\tau_x(D^\alpha \check{\chi}_j)](y) = (-1)^{|\alpha|+|\beta|} D^{\alpha+\beta} \chi_j(x-y)$, this yields:

$$\sup_{x \in K} |D^\alpha(u_1 \star \chi_j)(x)| \leq C_L \sum_{|\beta| \leq N_L} \sup_{z \in \mathbb{R}^n} |D^{\alpha+\beta} \chi_j(z)|.$$

Because $\chi_j \rightarrow 0$ in $\mathcal{D}(\mathbb{R}^n)$, every derivative $D^{\alpha+\beta} \chi_j \rightarrow 0$ uniformly on $\mathbb{R}^n$. The right-hand side converges to 0, proving $\sup_{x \in K} |D^\alpha(u_1 \star \chi_j)(x)| \rightarrow 0$.

□

Now we can finally prove the existence of convolution of distributions.

Proof. To prove the existence of the convolution $u_1 \star u_2 \in \mathcal{D}^*(\mathbb{R}^n)$ satisfying the definition, we can construct the distribution explicitly.

Define the functional $w : \mathcal{D}(\mathbb{R}^n) \rightarrow \mathbb{C}$ by:

$$w[\phi] := (u_1 \star (u_2 \star \check{\phi}))(0)$$

Because u_2 has compact support, the proposition above guarantees that $u_2 \star \check{\phi} \in \mathcal{D}(\mathbb{R}^n)$. Consequently, the proposition above that ensures that $u_1 \star (u_2 \star \check{\phi}) \in C^\infty(\mathbb{R}^n)$, making its evaluation at 0 well-defined. Since u_1 and u_2 are continuous linear functionals, w inherits linearity, confirming $w \in \mathcal{D}^*(\mathbb{R}^n)$ if we can prove continuity.

To prove w is continuous: w is the composition of four functions and we will show that they are each continuous:

Function 1: $\mathcal{D} \rightarrow \mathcal{D}$ sending $\phi \mapsto \check{\phi}$. The pre-images of the open sets are exactly the open sets as the topology, as weird as it is, is not asymmetric in that way, so it is continuous.

Function 2: $\mathcal{D} \rightarrow \mathcal{D}$ sending $\psi \mapsto u_2 \star \psi$. This is continuous because if $\psi_j \rightarrow 0$ in $\mathcal{D}$ then we know from the lemma above that $u_2 \star \psi_j \rightarrow 0$ in $\mathcal{E}$. However, the convergence must be in a fixed compact set, which is supplied by the proposition above the lemma above.

Function 3: $\mathcal{D} \rightarrow \mathcal{E}$ sending $\chi \mapsto u_1 \star \chi$. This is continuous by the lemma above.

Function 4: Evaluation at 0. Let $E : \mathcal{E}(\mathbb{R}^n) \rightarrow \mathbb{C}$ be given by $E(f) = f(0)$. Since E is linear, it suffices to prove continuity at 0: given $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that $0 \in K_N$ and $N > \varepsilon^{-1}$. Then for any $f \in V_N$,

$$|E(f)| = |f(0)| \leq p_N(f) < \frac{1}{N} < \varepsilon$$

Thus $V_N \subseteq E^{-1}(B_\varepsilon(0))$, so E is continuous at 0, and hence continuous everywhere on $\mathcal{E}(\mathbb{R}^n)$.

We must show that $w \star \phi = u_1 \star (u_2 \star \phi)$ for all $\phi \in \mathcal{D}(\mathbb{R}^n)$. Using the definition of convolutions we evaluate

$$(w \star \phi)(x) = w[\tau_x \check{\phi}]$$

Substituting the definition of w , this becomes:

$$(w \star \phi)(x) = (u_1 \star (u_2 \star (\tau_x \check{\phi}')))(0)$$

For any $y \in \mathbb{R}^n$, the inner term simplifies algebraically:

$$(\tau_x \check{\phi})(y) = \tau_x \check{\phi}(-y) = \phi(x+y) = \tau_{-x} \phi(y)$$

Thus, $(\tau_x \check{\phi}) = \tau_{-x} \phi$, yielding:

$$(w \star \phi)(x) = (u_1 \star (u_2 \star \tau_{-x} \phi))(0)$$

For any $T \in \mathcal{D}^*(\mathbb{R}^n)$ and $\psi \in \mathcal{D}(\mathbb{R}^n)$:

$$(T \star \tau_{-x} \psi)(y) = T[\tau_y(\tau_{-x} \psi)]$$

We will now change to the sensible convention but only for the remainder of this proof. $\tau_y \check{\phi}(x) = \phi(y-x)$ and $\tau_y \check{\phi}(x) = \phi(-y-x)$ from now on but only until the end of this proof.

Expanding the argument of T :

$$(\tau_{-x}^{\check{}}\psi)(z) = (\tau_{-x}\psi)(-z) = \psi(-z+x) = \check{\psi}(z-x) = \tau_x\check{\psi}(z)$$

This implies $\tau_y(\tau_{-x}^{\check{}}\psi) = \tau_y\tau_x\check{\psi} = \tau_{x+y}\check{\psi}$. Therefore:

$$(T \star \tau_{-x}\psi)(y) = T[\tau_{x+y}\check{\psi}] = (T \star \psi)(x+y) = \tau_{-x}(T \star \psi)(y)$$

This confirms the identity $T \star \tau_{-x}\psi = \tau_{-x}(T \star \psi)$. Applying this relation first to $T = u_2$ and $\psi = \phi$, we get $u_2 \star \tau_{-x}\phi = \tau_{-x}(u_2 \star \phi)$. Because $u_2 \star \phi \in \mathcal{D}(\mathbb{R}^n)$, we can apply the identity again to $T = u_1$ and $\psi = u_2 \star \phi$:

$$u_1 \star (u_2 \star \tau_{-x}\phi) = u_1 \star \tau_{-x}(u_2 \star \phi) = \tau_{-x}(u_1 \star (u_2 \star \phi))$$

Evaluating this at 0:

$$(w \star \phi)(x) = \tau_{-x}(u_1 \star (u_2 \star \phi))(0) = (u_1 \star (u_2 \star \phi))(x)$$

We conclude that $w \star \phi = u_1 \star (u_2 \star \phi)$ for all $\phi \in \mathcal{D}(\mathbb{R}^n)$. The distribution therefore exists and is therefore unique by a few propositions ago. □

Proposition. Suppose that $u_1, u_2 \in \mathcal{D}^*(\mathbb{R}^n)$ are distributions and that u_2 has compact support. Then

$$D^\alpha(u_1 \star u_2) = u_1 \star D^\alpha u_2 = D^\alpha u_1 \star u_2$$

Proof. Consider the convolution of $D^\alpha(u_1 \star u_2)$ with an arbitrary test function, then a previous proposition lets us shuffle the derivatives around, so we have

$$D^\alpha(u_1 \star u_2) \star \phi = (u_1 \star u_2) \star D^\alpha \phi = u_1 \star (u_2 \star D^\alpha \phi) = u_1 \star (D^\alpha u_2 \star \phi) = (u_1 \star D^\alpha u_2) \star \phi$$

This establishes the first equality. For the second equality note

$$(u_1 \star D^\alpha u_2) \star \phi = u_1 \star D^\alpha(u_2 \star \phi) = D^\alpha u_1 \star (u_2 \star \phi) = (D^\alpha u_1 \star u_2) \star \phi$$

Then done since ϕ was an arbitrary test function. □

Lemma. Let $\phi(x)$ be infinitely differentiable. Then for all k , $\phi(x)$ can be written as $\sum_{j=0}^{k-1} \frac{\phi^{(j)}(0)}{j!} x^j + x^k \chi(x)$ where $\chi(x)$ is infinitely differentiable.

Proof. Taylor's theorem with the integral form of the remainder gives:

$$\phi(x) = \sum_{j=0}^{k-1} \frac{\phi^{(j)}(0)}{j!} x^j + R_k(x)$$

where the remainder $R_k(x)$ is:

$$R_k(x) = \frac{1}{(k-1)!} \int_0^x (x-t)^{k-1} \phi^{(k)}(t) dt$$

Substituting $t = sx$ (so $dt = xds$):

$$R_k(x) = \frac{1}{(k-1)!} \int_0^1 (x-sx)^{k-1} \phi^{(k)}(sx) \cdot xds = x^k \cdot \frac{1}{(k-1)!} \int_0^1 (1-s)^{k-1} \phi^{(k)}(sx) ds$$

Define:

$$\chi(x) = \frac{1}{(k-1)!} \int_0^1 (1-s)^{k-1} \phi^{(k)}(sx) ds$$

Since $\phi \in C^\infty(\mathbb{R})$, the integrand $(s, x) \mapsto (1-s)^{k-1}\phi^{(k)}(sx)$ is smooth on $[0, 1] \times [x-1, x+1]$ for each fixed x value. By differentiation under the integral sign, $\chi(x)$ is infinitely differentiable on $\mathbb{R}$, with:

$$\chi^{(m)}(x) = \frac{1}{(k-1)!} \int_0^1 s^m (1-s)^{k-1} \phi^{(k+m)}(sx) ds \text{ for all } m \geq 0$$

This yields the desired decomposition with $\chi \in C^\infty$.

□

In the following proposition, xu denotes multiplying the distribution u by the smooth function x .

Proposition. Let $u \in \mathcal{D}^*(\mathbb{R})$ and suppose that $x^k u = 0$ for $k \in \mathbb{N}$. Then $u = \sum_{j=0}^{k-1} c_j D^j \delta_0$ for some complex constants c_j .

Proof. By the definition of multiplication by a smooth function in $\mathcal{D}^*(\mathbb{R})$, $x^k u = 0$ means that for all test functions $\phi \in \mathcal{D}(\mathbb{R})$:

$$(x^k u)[\phi] = u[x^k \phi] = 0$$

Fix a cut-off function $\psi \in \mathcal{D}(\mathbb{R})$ such that $\psi(x) = 1$ in a neighbourhood of 0. Using Taylor's theorem, any $\phi \in \mathcal{D}(\mathbb{R})$ can be expanded around $x = 0$ to order $k-1$:

$$\phi(x) = \left(\sum_{j=0}^{k-1} \frac{\phi^{(j)}(0)}{j!} x^j \right) \psi(x) + x^k \chi(x)$$

This works with χ smooth by the lemma above. After multiplying the Taylor polynomial by the smooth function ψ , there is still a χ that works since that multiplication only affects things away from 0.

Applying the distribution u to ϕ and using linearity gives:

$$\begin{aligned} u[\phi] &= u \left[\sum_{j=0}^{k-1} \frac{\phi^{(j)}(0)}{j!} x^j \psi(x) + x^k \chi(x) \right] \\ u[\phi] &= \sum_{j=0}^{k-1} \frac{\phi^{(j)}(0)}{j!} u[x^j \psi] + u[x^k \chi] \end{aligned}$$

By our initial hypothesis, $u[x^k \chi] = (x^k u)[\chi] = 0$. The evaluation $u[x^j \psi]$ is simply some complex constant for each j . We are left with:

$$u[\phi] = \sum_{j=0}^{k-1} \frac{u[x^j \psi]}{j!} \phi^{(j)}(0)$$

Recall that the j -th distributional derivative of the Dirac delta acts as $\delta_0^{(j)}[\phi] = (-1)^j \phi^{(j)}(0)$. Substituting this in, we can rewrite the summation:

$$u[\phi] = \sum_{j=0}^{k-1} c_j \delta_0^{(j)}[\phi]$$

where $c_j = \frac{(-1)^j u[x^j \psi]}{j!} \in \mathbb{C}$. Since this holds for all $\phi \in \mathcal{D}(\mathbb{R})$, the most general distribution satisfying $x^k u = 0$ is a linear combination of the Dirac delta and its first $k-1$ derivatives:

$$u = \sum_{j=0}^{k-1} c_j \delta_0^{(j)}$$

□

Definition. Given a linear Partial Differential Operator with constant coefficients $L := \sum_{|\alpha| \leq k} a_\alpha D^\alpha$, we say a distribution G (often called a Green's function) is a fundamental solution of the partial differential operator if it satisfies the distributional equation $LG = \delta_0$.

Lemma. Let $u \in \mathcal{D}^*(\mathbb{R}^n)$ be a distribution with compact support. Then $\delta_0 \star u = u$.

Proof. Let $\psi \in \mathcal{D}(\mathbb{R}^n)$ be an arbitrary test function. By the convolution definition for a distribution and a test function, we evaluate $\delta_0 \star \psi$ at a point $x \in \mathbb{R}^n$:

$$(\delta_0 \star \psi)(x) = \delta_0[\tau_x \check{\psi}]$$

By the definition of the Dirac delta, evaluating δ_0 on the test function $\tau_x \check{\psi}$ simply returns the value of that test function at 0:

$$(\delta_0 \star \psi)(x) = (\tau_x \check{\psi})(0)$$

Using the identity $\tau_x \check{\psi}(y) = \psi(x - y)$ evaluated at $y = 0$:

$$(\tau_x \check{\psi})(0) = \psi(x - 0) = \psi(x)$$

Thus, $(\delta_0 \star \psi)(x) = \psi(x)$ for all x , meaning $\delta_0 \star \psi = \psi$. This makes intuitive sense, as if ϕ_ε is a thin tall mollifier as defined earlier and we convolve f with this we have seen that we get an approximation to f , and the dirac delta, crudely speaking, can be intuitively thought of as $\lim_{\varepsilon \rightarrow 0} \phi_\varepsilon$.

Now we let $u_1 = \delta_0$ and $u_2 = u$. Since u has compact support, convolution is valid. For any test function $\phi \in \mathcal{D}(\mathbb{R}^n)$, the defining property gives:

$$(\delta_0 \star u) \star \phi = \delta_0 \star (u \star \phi)$$

By properties of convolutions proved earlier, since u and ϕ both have compact support, the resulting convolution $\psi = u \star \phi$ also has compact support, meaning $u \star \phi \in \mathcal{D}(\mathbb{R}^n)$.

Therefore, we can substitute $\psi = u \star \phi$ into the result from earlier:

$$\delta_0 \star (u \star \phi) = u \star \phi$$

Which leaves us with:

$$(\delta_0 \star u) \star \phi = u \star \phi$$

Since we have shown that $(\delta_0 \star u) \star \phi = u \star \phi$ holds for all $\phi \in \mathcal{D}(\mathbb{R}^n)$, it immediately follows that:

$$\delta_0 \star u = u$$

□

Proposition. Suppose that $G \in \mathcal{D}^*(\mathbb{R}^n)$ is a fundamental solution of L and let $u_0 \in \mathcal{D}^*(\mathbb{R}^n)$ be a distribution with compact support. Then the distribution $u := G \star u_0$ solves the distributional equation

$$\sum_{|\alpha| \leq k} a_\alpha D^\alpha u = u_0$$

Proof. Linearity and differentiability properties of the convolution imply that

$$L(G \star u_0) = \sum_{|\alpha| \leq k} a_\alpha D^\alpha (G \star u_0) = \sum_{|\alpha| \leq k} a_\alpha (D^\alpha G \star u_0) = \left(\sum_{|\alpha| \leq k} a_\alpha D^\alpha G \right) \star u_0 = (LG) \star u_0$$

Now by definition of the fundamental solution we have $LG = \delta_0$ so we have

$$L(G \star u_0) = (LG) \star u_0 = \delta_0 \star u_0$$

Then the previous lemma implies the result.

□

Therefore, loosely speaking, we can think of $G\star$ as a sort of inverse to the operator L .

We now revisit the Laplace/Poisson equation from Vector calculus for the case of $f \in C_c^2(\mathbb{R}^3)$ and $w \in C^2(\mathbb{R}^3)$. A lot of this will be similar to what we did in Vector calculus. We want to consider the problem of $\nabla^2 w = f$ where ∇^2 is the differential operator defined by

$$\nabla^2 := \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Theorem. The Laplacian is rotation invariant.

Proof. Work in $\mathbb{R}^n$.

Let $R \in O(n)$ be an orthogonal matrix representing a rotation. For the Laplacian, let $y = Rx$ so that $y_j = R_{ji}x_i$. By the chain rule, $\frac{\partial}{\partial x_i} = \sum_j R_{ji} \frac{\partial}{\partial y_j}$. Applying this twice gives:

$$\nabla_x^2 f(Rx) = \sum_i \frac{\partial^2}{\partial x_i^2} f(Rx) = \sum_i \sum_{j,k} R_{ji} R_{ki} \frac{\partial^2 f}{\partial y_j \partial y_k}$$

Since $RR^T = I$, we have $\sum_i R_{ji} R_{ki} = \delta_{jk}$. Substituting this yields:

$$\nabla_x^2 f(Rx) = \sum_{j,k} \delta_{jk} \frac{\partial^2 f}{\partial y_j \partial y_k} = \sum_j \frac{\partial^2 f}{\partial y_j^2} = (\Delta f)(Rx)$$

Thus, ∇^2 commutes with rotations and is rotationally invariant. □

Based on this fact, we guess that we can write $G = T_g$ for a function g that we will assume is smooth and radial and defined for $r \neq 0$.

Now if this is the case, g satisfies the Laplace equation away from 0. Since $\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$, the chain rule gives:

$$\frac{\partial g}{\partial x_i} = g'(r) \frac{x_i}{r}$$

Differentiating a second time:

$$\frac{\partial^2 g}{\partial x_i^2} = \frac{\partial}{\partial x_i} \left(g'(r) \frac{x_i}{r} \right) = g''(r) \frac{x_i^2}{r^2} + g'(r) \left(\frac{1}{r} - \frac{x_i^2}{r^3} \right)$$

Summing over $i = 1, 2, 3$, noting that $\sum_{i=1}^3 x_i^2 = r^2$ and $\sum_{i=1}^3 1 = 3$:

$$\nabla^2 g = g''(r) \frac{r^2}{r^2} + g'(r) \left(\frac{3}{r} - \frac{r^2}{r^3} \right) = g''(r) + \frac{2}{r} g'(r)$$

Multiplying the expression by r :

$$r \nabla^2 g = r g''(r) + 2 g'(r)$$

By the product rule:

$$\frac{d^2}{dr^2} (rg(r)) = \frac{d}{dr} (r g'(r) + g(r)) = r g''(r) + g'(r) + g'(r) = r g''(r) + 2 g'(r)$$

Therefore:

$$\nabla^2 g = \frac{1}{r} \frac{d^2}{dr^2} (rg)$$

For $r > 0$, $\nabla^2 g = 0$ is equivalent to:

$$\frac{d^2}{dr^2} (rg) = 0$$

Thus, $g = \frac{A}{r} + B$.

Since a constant satisfies the laplace equation we may assume $B = 0$. Since an easy bounding argument shows that $g \in L^1_{loc}(\mathbb{R}^3)$ it defines a distribution. We want to prove that for a suitable choice of A , $G = T_g$ satisfies $\nabla^2 G = \delta_0$. We calculate

$$\nabla^2 T_g(\phi) = T_g(\nabla^2 \phi) = \int_{\mathbb{R}^3} g(x) \nabla^2 \phi(x) dx$$

By the dominated convergence theorem (since g is in L^1_{loc} and ϕ has compact support) we have, for $R > 0$ large enough to contain $\text{supp}(\phi)$ that

$$= \lim_{\varepsilon \rightarrow 0} \int_{B_R(0) \setminus B_\varepsilon(0)} g(x) \nabla^2 \phi(x) dx$$

By the Green's second identity (from Vector calculus) we have

$$\int_{B_R(0) \setminus B_\varepsilon(0)} g(x) \nabla^2 \phi(x) - \phi(x) \nabla^2 g(x) dx = \int_{\partial(B_R(0) \setminus B_\varepsilon(0))} \left[g \frac{\partial \phi}{\partial n} - \phi \frac{\partial g}{\partial n} \right] dA$$

Since $\nabla^2 g$ is 0 away from the origin and $\phi = 0$ in a neighbourhood of $\partial B_R(0)$ we have

$$\int_{B_R(0) \setminus B_\varepsilon(0)} g(x) \nabla^2 \phi(x) dx = \int_{\partial(B_\varepsilon(0))} \left[\phi \frac{\partial g}{\partial n} - g \frac{\partial \phi}{\partial n} \right] dA$$

The sign change is because we are now working on the inner boundary. Now we estimate

$$\sup_{y \in \partial B_\varepsilon(0)} \left| g(y) \frac{\partial \phi}{\partial n}(y) \right| = \sup_{y \in \partial B_\varepsilon(0)} \frac{|A|}{\varepsilon} \left| \frac{\partial \phi}{\partial n}(y) \right| \leq \frac{|A|}{\varepsilon} \sup_{y \in \mathbb{R}^3} |\nabla^2 \phi|$$

Therefore

$$\left| \int_{\partial(B_\varepsilon(0))} g \frac{\partial \phi}{\partial n} dA \right| \leq 4\pi\varepsilon^2 \times \frac{|A|}{\varepsilon} \sup_{y \in \mathbb{R}^3} |\nabla^2 \phi| = C\varepsilon \rightarrow 0$$

For the $\phi \frac{\partial g}{\partial n}$ term note that on $\partial(B_\varepsilon(0))$ we have $\frac{\partial}{\partial n} = \frac{\partial}{\partial r}$ so that for any $y \in B_\varepsilon(0)$ we have

$$\phi(y) \frac{\partial g}{\partial n}(y) = \frac{-A}{\varepsilon^2} \phi(y)$$

We therefore have

$$\int_{\partial(B_\varepsilon(0))} \phi \frac{\partial g}{\partial n} dA = (-4\pi A) \times \frac{1}{\text{Surface Area}(\partial B_\varepsilon(0))} \int_{\partial(B_\varepsilon(0))} \phi(y) dA(y)$$

For any $\delta > 0$ since ϕ is continuous there exists $\varepsilon > 0$ such that $|\phi(y) - \phi(0)| < \delta$ for all $y \in B_\varepsilon(0)$. We estimate

$$\begin{aligned} & \left| \frac{1}{\text{Surface Area}(\partial B_\varepsilon(0))} \int_{\partial(B_\varepsilon(0))} \phi(y) dA(y) - \phi(0) \right| \\ & \leq \frac{1}{\text{Surface Area}(\partial B_\varepsilon(0))} \int_{\partial(B_\varepsilon(0))} |\phi(y) - \phi(0)| dA(y) \\ & < \frac{\delta}{\text{Surface Area}(\partial B_\varepsilon(0))} \int_{\partial(B_\varepsilon(0))} 1 dA(y) = \delta \end{aligned}$$

Therefore

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\text{Surface Area}(\partial B_\varepsilon(0))} \int_{\partial(B_\varepsilon(0))} \phi(y) dA(y) = \phi(0)$$

Putting this all together gives

$$\Delta T_g[\phi] = (-4\pi A) \times \phi(0) = -4\pi A (\delta_0[\phi])$$

Therefore we have $\nabla^2 T_g = \delta_0$ if $A = -\frac{1}{4\pi}$ so this is our fundamental solution. Therefore, if $f \in C_c^0(\mathbb{R}^3)$ then a solution is given by $u = T_g \star T_f$.

Proposition. For $f \in C_c^0(\mathbb{R}^n)$ and $g \in L^1_{loc}(\mathbb{R}^n)$ we have $T_{f \star g} = T_f \star T_g$

Proof. Since f has compact support, the distribution T_f also has compact support. Therefore, the convolution of T_g and T_f is well-defined. The convolution $T_g \star T_f$ is the unique distribution satisfying $(T_g \star T_f) \star \phi = T_g \star (T_f \star \phi)$ for all test functions $\phi \in \mathcal{D}(\mathbb{R}^n)$. For any distribution u , its action on a test function ψ can be recovered by evaluating the convolution at the origin: $u[\psi] = (u \star \check{\psi})(0)$. Applying this identity to $T_g \star T_f$ with the test function ϕ yields

$$(T_g \star T_f)[\phi] = ((T_g \star T_f) \star \check{\phi})(0) = (T_g \star (T_f \star \check{\phi}))(0) = T_g[T_f \star \check{\phi}]$$

By the definition of the convolution of a distribution and a test function, we have

$$(T_f \star \check{\phi})(x) = T_f[\tau_x \check{\phi}] = T_f[\tau_x \phi]$$

Evaluating $T_f[\tau_x \phi]$ gives $\int_{\mathbb{R}^n} f(y) \phi(y - x) dy$. Applying the spatial inversion operator $\check{h}(x) = h(-x)$ to this result yields

$$T_f \star \check{\phi}(x) = \int_{\mathbb{R}^n} f(y) \phi(x + y) dy$$

Evaluating the action of T_g on this new test function gives

$$T_g[T_f \star \check{\phi}] = \int_{\mathbb{R}^n} g(x) \left(\int_{\mathbb{R}^n} f(y) \phi(x + y) dy \right) dx$$

Because f has compact support and $\phi \in \mathcal{D}(\mathbb{R}^n)$, the variables $y \in \text{supp } f$ and $x + y \in \text{supp } \phi$ are restricted to bounded sets, which forces x to be bounded as well. Given $g \in L^1_{loc}(\mathbb{R}^n)$ (which accommodates the singularity at 0), the function $g(x) f(y) \phi(x + y)$ is absolutely integrable over $\mathbb{R}^n \times \mathbb{R}^n$. This permits the use of Fubini's theorem to interchange the order of integration, resulting in $\int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(x) \phi(x + y) dx \right) dy$. Substituting $z = x + y$ (so $dx = dz$) transforms the expression into

$$\int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(z - y) \phi(z) dz \right) dy$$

Rearranging the terms gives

$$\int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} g(z - y) f(y) dy \right) \phi(z) dz$$

The inner integral is exactly the convolution $(g \star f)(z)$. Substituting this back leaves $\int_{\mathbb{R}^n} (g \star f)(z) \phi(z) dz$, which is the definition of $T_{g \star f}[\phi]$. Since $\phi \in \mathcal{D}(\mathbb{R}^n)$ was arbitrary, $T_g \star T_f = T_{g \star f}$. □

Given this, if $f \in C^2_c(\mathbb{R}^3)$ then by properties of convolutions from earlier, we know $g \star f$ is smooth and that $w = g \star f$ is a solution of the classical equation. Thus, the solution we seek is

$$w(x) = \frac{-1}{4\pi} \int_{\mathbb{R}^3} \frac{f(y)}{|x - y|} dy$$

This is a solution for general $f \in C^2_c(\mathbb{R}^3)$ however we don't know that it is the general solution.

3.6 The dual space of $\mathcal{E}(\Omega)$

Consider the dual space of $\mathcal{E}(\Omega)$, denoted $\mathcal{E}^*(\Omega)$, the set of continuous linear maps $\mathcal{E}(\Omega) \rightarrow \mathbb{C}$ with the convergence is with respect to the topology of $\mathcal{E}(\Omega)$ which is a metric topology.

For a linear map $u : \mathcal{E}(\Omega) \rightarrow \mathbb{C}$ to belong to $\mathcal{E}^*(\Omega)$ it is enough that for every sequence $\phi_j \subset \mathcal{E}(\Omega)$ that tends to 0 we have $u[\phi_j] \rightarrow 0$.

Note that $\mathcal{D}$ is a subspace of $\mathcal{E}$ with the property that if a sequence tends to 0 in $\mathcal{D}(\Omega)$ it tends to 0 in $\mathcal{E}(\Omega)$. Thus, $\mathcal{E}^*(\Omega) \subseteq \mathcal{D}^*(\Omega)$ once we restrict elements of $\mathcal{E}^*$ to their action on compactly supported functions. To see that the direction is correct, convergence in $\mathcal{E}(\Omega)$ is a weaker condition so u of more sequences have to tend to 0 so being in $\mathcal{E}^*$ becomes a stronger condition.

Therefore, everything in the space $\mathcal{E}^*(\Omega)$ is a distribution when we restrict its action to stuff with compact support.

Proposition. Let $u : \mathcal{E}(\Omega) \rightarrow \mathbb{C}$ be a linear map. Then u is continuous if and only if there is some compact $K \subseteq \Omega$, $N \in \mathbb{N}_0$ and $C > 0$ such that

$$|u[\phi]| \leq C \sup_{x \in K, |\alpha| \leq N} |D^\alpha \phi(x)| \text{ for all } \phi \in \mathcal{E}(\Omega)$$

Proof. First, assume the formula is true. Pick a sequence $\phi_j \in \mathcal{E}(\Omega)$ which converges to 0 in $\mathcal{E}(\Omega)$, then we want to show that $u(\phi_j) \rightarrow 0$. But because it converges to 0 we have that for all α and any compact $K' \subset \Omega$ we have $\sup_{x \in K', |\alpha| \leq N} |D^\alpha \phi_j(x)| \rightarrow 0$ so therefore $|u[\phi_j]| \rightarrow 0$ as it satisfies the inequality above. Therefore, u is continuous.

To show the opposite implication we will assume the formula is not true for any K, N, C . Let $K_1, K_2, \dots$ be an exhaustion of Ω . We will prove not-continuity by constructing a clever sequence.

If the formula does not hold for $K = K_j, N = j, C = j$ for any j , then for every j there exists $\phi_j \in \mathcal{E}(\Omega)$ such that

$$|u[\phi_j]| > j \sup_{x \in K_j, |\alpha| \leq j} |D^\alpha \phi_j(x)|$$

Define $\psi_j(x) = \frac{\phi_j(x)}{|u[\phi_j]|}$, then $\psi_j \in \mathcal{E}(\Omega)$. It remains to check that $\psi_j \rightarrow 0$ in $\mathcal{E}(\Omega)$ as then we will be done since $|u[\psi_j]| = 1$ for all j by construction.

Fix a compact $K \subset \Omega$ and a multi-index α . For sufficiently large j , $K \subseteq K_j$ and $j \geq |\alpha|$. Thus:

$$\sup_{x \in K} |D^\alpha \phi_j(x)| \leq \sup_{x \in K_j, |\beta| \leq j} |D^\beta \phi_j(x)|$$

Therefore we have

$$\sup_{x \in K} |D^\alpha \psi_j(x)| = \frac{1}{|u[\phi_j]|} \sup_{x \in K} |D^\alpha \phi_j(x)| < \frac{\sup_{x \in K} |D^\alpha \phi_j(x)|}{j \sup_{x \in K_j, |\beta| \leq j} |D^\beta \phi_j(x)|} \leq \frac{1}{j}$$

Therefore, $D^\alpha \psi_j$ tends to 0 on K so done since α and K were arbitrary.

□

Example. For $f \in C_c^0(\Omega)$ define T_f in the usual way. Then $T_f \in \mathcal{E}^*(\Omega)$ because

$$|T_f[\phi]| \leq \int_{\Omega} |f(x)| dx \cdot \sup_{y \in \text{supp } f} |\phi(y)|$$

Example. If $x \in \Omega$ then $\delta_x \in \mathcal{E}^*(\Omega)$ because if K is any compact set containing x then

$$|\delta_x[\phi]| \leq \sup_{y \in K} |\phi(y)|$$

Example. The map $u : \mathcal{D}(\mathbb{R}) \rightarrow \mathbb{C}$ defined by

$$u[\phi] = \sum_{m=-\infty}^{\infty} \phi(m)$$

is a distribution (check the continuity bound thingy) but it is not a distribution in $\mathcal{E}^*(\mathbb{R})$ because the infinite sum does not converge for general elements of $\mathcal{E}$, such as the constant function 1.

We note that in all 3 examples, the distributions are in $\mathcal{E}^*$ if and only if they have compact support. Of course, only 3 examples doesn't really mean anything, but we have the following theorem in this case (which we will state and prove after the following 3 lemmas).

Theorem. (Generalized Leibniz rule) Let $\alpha = (\alpha_1, \dots, \alpha_d)$ be a multi-index, and let $f, g : \mathbb{R}^d \rightarrow \mathbb{C}$ (or $\mathbb{R}$) be sufficiently smooth functions (Specifically, have continuous $|\alpha|$ -th partials). Then

$$D^\alpha(fg) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (D^\beta f)(D^{\alpha-\beta} g)$$

where $\beta \leq \alpha$ means $\beta_i \leq \alpha_i$ for all $1 \leq i \leq d$, and the multi-index binomial coefficient is defined as the product of the standard 1D coefficients:

$$\binom{\alpha}{\beta} = \prod_{i=1}^d \binom{\alpha_i}{\beta_i}$$

Proof. Because partial derivatives commute for smooth functions, the d -dimensional rule is just the 1-dimensional Leibniz rule applied iteratively and independently across each of the d coordinates. □

Lemma. Let ϕ_j be a sequence in $\mathcal{E}(\Omega)$ such that $\phi_j \rightarrow \phi$ in $\mathcal{E}(\Omega)$ and $\chi \in \mathcal{D}(\Omega)$. Then $\chi\phi_j \rightarrow \chi\phi$ in $\mathcal{D}(\Omega)$

Proof. A sequence converging in $\mathcal{D}(\Omega)$ must have its support contained within a single compact set $K \subset \Omega$. Since $\chi \in \mathcal{D}(\Omega)$, it inherently has compact support, which we can define as K , and the support of $\chi\phi_j$ is contained in K for all j .

To check uniform convergence of derivatives, we use the Leibniz rule,

$$D^\alpha(\chi(\phi_j - \phi)) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^{\alpha-\beta} \chi D^\beta(\phi_j - \phi)$$

Because χ is smooth, its derivatives are continuous and bounded on the compact set K . We can bound this by a constant $M_\beta = \sup_K |D^{\alpha-\beta} \chi|$. Convergence in $\mathcal{E}(\Omega)$ requires that for any compact set (including K) and any multi-index β , $\sup_{x \in K} |D^\beta(\phi_j - \phi)| \rightarrow 0$ as $j \rightarrow \infty$. Applying the triangle inequality yields:

$$\sup_{x \in K} |D^\alpha(\chi\phi_j - \chi\phi)| \leq \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} M_\beta \sup_{x \in K} |D^\beta(\phi_j - \phi)|$$

This is a finite sum of terms that each tend to 0 as $j \rightarrow \infty$, so the entire expression tends to 0. Both conditions for convergence in $\mathcal{D}$ are met, completing the proof. □

Lemma. Let $\psi \in \mathcal{E}(\Omega)$, then there exists a sequence $\phi_j \in \mathcal{D}(\Omega)$ such that $\phi_j \rightarrow \psi$ in $\mathcal{E}(\Omega)$

Proof. Let K be a compact set and K_n an exhaustion of Ω , then $K \in K_j$ for j large enough. This means that, hypothetically, if ϕ_j were to equal ψ on K_j , that would be good news, as the derivatives would certainly converge uniformly on K as they would eventually be constant. Well the good news is that this is not a hypothetical: Recall that there exists a function that is 1 on an open set around K_j but has compact support contained in Ω for each j by something we proved earlier. Call this function R_j , then $\phi_j := R_j\psi$ gets the job done. □

Here's the promised theorem about distributions in $\mathcal{E}^*$ and distributions of compact support.

Theorem. Let u be a distribution and suppose $u \in \mathcal{E}^*(\Omega)$, then u has compact support. Also, if $u \in \mathcal{D}^*(\Omega)$ has compact support then there exists a unique $u' \in \mathcal{E}^*(\Omega)$ such that $u'[\phi] = u[\phi]$ for all $\phi \in \mathcal{D}(\Omega)$. We say u' is the extension of u as a linear map on $\mathcal{E}(\Omega)$

Proof. Suppose $u \in \mathcal{E}^*(\Omega)$. Then we have already argued that $u \in \mathcal{D}^*(\Omega)$ after restriction, and it remains to show that $\text{supp } u$ is compact. By the criterion for continuity in $\mathcal{E}^*$ there exists some compact set $K \subseteq \Omega$, $N \in \mathbb{N}$ and $C > 0$ such that

$$|u[\phi]| \leq C \sup_{x \in K, |\alpha| \leq N} |D^\alpha \phi(x)| \text{ for all } \phi \in \mathcal{E}(\Omega)$$

Now suppose that $\text{supp } \phi \subset \Omega \setminus K$. From the estimate above we have that $u[\phi] = 0$ so $\text{supp } u \subset K$ (by definition of support of a distribution for it is 0 on all functions supported on a set outside K) so we must have that $\text{supp } u$ is compact.

Now suppose that $u \in \mathcal{D}^*(\Omega)$ has compact support. Then we know there exists $\chi \in C_c^\infty(\Omega)$ such that $\chi = 1$ on $\text{supp } u$. For $\phi \in \mathcal{E}(\Omega)$ we define

$$u'[\phi] = u[\chi\phi]$$

Then $u[\chi\phi]$ makes sense since $\chi\phi$ has compact support. If $\phi_j \rightarrow 0$ in $\mathcal{E}(\Omega)$ then $\chi\phi_j \rightarrow 0$ in $\mathcal{D}(\Omega)$ by a lemma above. Thus, $u' \in \mathcal{E}^*(\Omega)$ since u is continuous. Also note that if $\phi \in \mathcal{D}(\Omega)$ then $\chi\phi - \phi$ has support in $\Omega \setminus \text{supp } u$. Therefore,

$$0 = u[\chi\phi - \phi] = u'[\phi] - u[\phi]$$

Therefore, u' and u agree on $\mathcal{D}(\Omega)$ so it remains to show that u' is unique.

Suppose that $v' \in \mathcal{E}^*(\Omega)$ satisfies $u'[\phi] = v'[\phi]$ for all $\phi \in \mathcal{D}(\Omega)$. Let $\psi \in \mathcal{E}(\Omega)$ be arbitrary. Then by the lemma above we can find $\phi_j \in \mathcal{D}(\Omega)$ such that $\phi_j \rightarrow \psi$ in $\mathcal{E}(\Omega)$. Thus, by continuity of u' and v' we have

$$u'[\psi] = \lim_{j \rightarrow \infty} u'[\phi_j] = \lim_{j \rightarrow \infty} v'[\phi_j] = v'[\psi]$$

Thus $u' = v'$ since ψ was arbitrary. □

3.7 Tempered distributions

The space of Tempered distributions is the continuous dual of $\mathcal{S}$, the Schwartz space of rapidly decreasing functions. As a reminder, $\phi \in \mathcal{S}$ if $\phi \in C^\infty(\mathbb{R}^n)$ and for any multi-index α and any $N \in \mathbb{N}$ we have

$$\sup_{x \in \mathbb{R}^n} |(1 + |x|)^N D^\alpha \phi(x)| < \infty$$

and a sequence $\phi_j \subset \mathcal{S}$ tends to 0 if for all $N \in \mathbb{N}$ and multi-indices α we have

$$\sup_{x \in \mathbb{R}^n} |(1 + |x|)^N D^\alpha \phi_j(x)| \rightarrow 0$$

Since the topology of $\mathcal{S}$ is induced by a countable separating family of seminorms, it can be induced by a metric. Thus, continuity is equivalent to sequential continuity. For a linear map $u : \mathcal{S} \rightarrow \mathbb{C}$ to belong to $\mathcal{S}^*$ it is enough that $\lim_{j \rightarrow \infty} u[\phi_j] \rightarrow 0$ for every sequence ϕ_j that tends to 0 in $\mathcal{S}$.

These are a subclass of distributions that contain all distributions from the above section and lie strictly between the 2 sets, after we restrict them to functions with compact support. This is because of the earlier assertion that convergence in $\mathcal{D}$ is stronger than convergence in $\mathcal{S}$ which is in turn stronger than convergence in $\mathcal{D}$.

Proposition. Suppose that $u : \mathcal{S} \rightarrow \mathbb{C}$ is a linear map. Then u is continuous if and only if there exist $N, k \in \mathbb{N}_0$ and $C > 0$ such that

$$|u[\phi]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} |(1 + |x|)^N D^\alpha \phi(x)| \text{ for all } \phi \in \mathcal{S}$$

Proof. The fact that continuity implies the formula happens with the same proof as in the analogous theorem above. So, we have to prove that the formula implies continuity.

Because the topology of the Schwartz space $\mathcal{S}$ is induced by a metric, sequential continuity is equivalent to continuity. Therefore, it is sufficient to prove that $\lim_{j \rightarrow \infty} u[\phi_j] = 0$ for any sequence $\{\phi_j\}_{j=1}^\infty \subset \mathcal{S}$ that converges to zero in $\mathcal{S}$.

Suppose there exist $N, k \in \mathbb{N}$ and $C > 0$ such that $|u[\phi]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} |(1 + |x|)^N D^\alpha \phi(x)|$ holds for all $\phi \in \mathcal{S}$. Let $\{\phi_j\}_{j=1}^\infty$ be an arbitrary sequence in $\mathcal{S}$ such that $\phi_j \rightarrow 0$ in $\mathcal{S}$. By the definition of convergence, for $\phi_j \rightarrow 0$ in $\mathcal{S}$, it is required that for every multi-index β and every $M \in \mathbb{N}$, the sequence satisfies $\sup_{x \in \mathbb{R}^n} |(1 + |x|)^M D^\beta \phi_j(x)| \rightarrow 0$ as $j \rightarrow \infty$. Since the limit holds for all M and β , it certainly holds for our specific N and all specific multi-indices α where $|\alpha| \leq k$. Because the condition $|\alpha| \leq k$ restricts us to a finite number of multi-indices, the maximum of these limits over this finite set also converges to zero. Therefore, $\sup_{x \in \mathbb{R}^n, |\alpha| \leq k} |(1 + |x|)^N D^\alpha \phi_j(x)| \rightarrow 0$ as $j \rightarrow \infty$. Applying the initial assumption to our sequence gives:

$$0 \leq |u[\phi_j]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} |(1 + |x|)^N D^\alpha \phi_j(x)|$$

As $j \rightarrow \infty$, the right-hand side of the inequality approaches 0. Therefore, $\lim_{j \rightarrow \infty} |u[\phi_j]| = 0$. Since sequential continuity implies continuity for this topology, the linear map u is continuous. □

Example. Suppose $f \in L^1_{loc}(\mathbb{R}^n)$ and that there exist $C > 0$ and $N \in \mathbb{N}$ such that

$$|f(x)| \leq C(1 + |x|)^N$$

Then $T_f \in \mathcal{S}^*$

The reason this is the case is because for any $\phi \in \mathcal{S}$ the functional T_f acts by $T_f[\phi] = \int_{\mathbb{R}^n} f(x) \phi(x) dx$. Using the growth condition on f , we have

$$|T_f[\phi]| \leq C \int_{\mathbb{R}^n} (1 + |x|)^N |\phi(x)| dx$$

For any integer $M \geq 0$ the quantity $(1 + |x|)^M |\phi(x)|$ has a supremum. Set $M = N + n + 1$ (where n is the spatial dimension) and rewrite the bound as

$$|T_f[\phi]| \leq C \sup_{y \in \mathbb{R}^n} \left| (1 + |y|)^{N+n+1} \phi(y) \right| \int_{\mathbb{R}^n} (1 + |x|)^{-(n+1)} dx$$

The integral is a finite constant that we will call I . Therefore T_f satisfies the bound so it is a tempered distribution.

Example. The map defined by $\phi \mapsto \int_{\mathbb{R}^n} e^{|x|^2} \phi(x) dx$ does not define a tempered distribution. This is because if we try to apply it to the function $e^{-|x|^2}$ the integral does not converge.

Example. For $\phi \in \mathcal{S}(\mathbb{R})$ the map $u[\phi] = \sum_{m=-\infty}^{\infty} m^N \phi(m)$ defines a tempered distribution.

The reason this is true is because since ϕ is rapidly decreasing, if we set $M \geq N + 2$ and set this as the exponent in the rapidly decreasing definition, the sum converges. Consequently, if the convergent sum $\sum \frac{|m|^N}{(1+|m|)^{N+2}}$ converges to K , then $|u[\phi]| \leq K \sup_x \left| (1 + |x|)^{N+2} \phi(x) \right|$ which means continuity has been confirmed so we have a tempered distribution.

To summarize: $\mathcal{D}^*$ is all distributions, $\mathcal{S}^*$ is those that don't grow too fast and $\mathcal{E}^*$ is those with compact support.

4 Fourier analysis

4.1 Properties of the Fourier transform in L^1 and L^2

Note that in this section the domain is $\mathbb{R}^n$ and not $\mathbb{C}^n$ so although we will be working with complex-valued functions, differentiable does not imply Holomorphic here. However, we will soon recall a little bit of complex analysis to help us calculate integrals.

Recall from Probability and measure: If $f \in L^1(\mathbb{R}^n)$ define its Fourier transform as $\hat{f} = \int_{\mathbb{R}^n} f(x) e^{-i(\xi \cdot x)} dx$.²

Example. If $f = 1_{(-1,1)}$ then

$$\hat{f}(\xi) = \int_{-1}^1 e^{-ix\xi} dx = \left[\frac{e^{-ix\xi}}{-i\xi} \right]_{-1}^1 = 2 \frac{\sin(\xi)}{\xi}$$

Example. Suppose $f \in L^1(\mathbb{R})$ is defined by $f(x) = e^{-|x|}$. Then we calculate

$$\begin{aligned} \hat{f}(\xi) &= \int_{-\infty}^0 e^{x(1-i\xi)} dx + \int_0^{\infty} e^{x(-1-i\xi)} dx = \left[\frac{e^{x(1-i\xi)}}{1-i\xi} \right]_{-\infty}^0 + \left[\frac{e^{x(-1-i\xi)}}{-1-i\xi} \right]_0^{\infty} \\ &= \frac{1}{1-i\xi} + \frac{1}{1+i\xi} = \frac{2}{1+\xi^2} \end{aligned}$$

Example. Let $g \in L^1(\mathbb{R})$ be given by

$$g(x) = \frac{1}{1+x^2}$$

²Yes I know this is a different convention from my probability and measure convention. Blame my reference texts as mathematicians can't seem to agree on the correct convention. You're gonna have to deal with it. There are several conventions of the Fourier transform used across various areas of maths and unfortunately I don't make the rules.

We will do this using a complex contour integral. Let γ_R be the straight path from $-R$ to R , then

Then we have

$$\hat{g}(\xi) = \lim_{R \rightarrow \infty} \int_{\gamma_R} \frac{e^{-iz\xi}}{1+z^2} dz$$

We can close the contour using a semicircle in the lower half plane. Recall a lemma we proved in the Complex Analysis course (which is called Jordan's lemma) says that if f is holomorphic on $|z| \geq r$ and $zf(z)$ is bounded for large $|z|$ then for all $A > 0$ we have that $\int_{\gamma_R} f(z) e^{iAz} dz \rightarrow 0$ as $R \rightarrow \infty$ where γ_R is the semicircular contour of radius R .

Moral of the story is if $\xi > 0$ then $\lim_{R \rightarrow \infty} \int_{\gamma_R} \frac{e^{-iz\xi}}{1+z^2} dz$ is equal to the contribution from the pole at $-i$ whenever $R > 1$. Jordan's lemma applies as although the exponent is negative, the contour is upside down, and these effects cancel out. The residue at the pole is $\frac{e^{-\xi}}{-2i}$ by a lemma from complex analysis, we multiply by $2\pi i$ and flip the sign since the contour is clockwise to get that the integral is $\pi e^{-\xi}$.

If $\xi < 0$ then by a similar argument using an upper semicircle instead we get that the integral is πe^{ξ} .

In conclusion, we have $\hat{g}(\xi) = \pi e^{-|\xi|}$.

Now consider the Gaussian $f(x) = e^{-\frac{1}{2}|x|^2}$. I did the same calculation in the probability and measure course, so I won't go through the details. However, what we end up with after a Contour integration trick is that if n is the dimension of the space then

$$\hat{f}(\xi) = (2\pi)^{\frac{n}{2}} e^{-\frac{1}{2}|\xi|^2}$$

Note that these functions are all continuous (We know that Fourier transforms in L^1 are continuous, proved in P+M) with $\sup |\hat{f}(\xi)| \leq \|f\|_1$. In all cases, we saw that the functions decayed fast. There is something more general we can say, in effect we know that if we multiply a function by faster and faster oscillations, these cancel more and more and so the resulting integral tends to 0. The precise statement is the Riemann-Lebesgue theorem that we will prove shortly. Recall that in Linear analysis we proved that Fourier series of 2π -periodic locally integrable functions converge on a 2π interval converge at all points of differentiability conditional on this theorem, well here is where we actually prove it to close that gap. As you will see, the proof isn't difficult once you have the divergence theorem.

Theorem. (Riemann-Lebesgue Lemma) let $f \in L^1(\mathbb{R}^n)$. Then $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

Proof. Note first that continuous functions with compact support are dense in L^1 . Since they can be approximated by smooth functions with compact support by convolving with ϕ_ε , we conclude that C_c^∞ is dense in L^1 . Specifically, for all $\varepsilon > 0$ there is a smooth function with compact support such that

$$\|f - f_\varepsilon\|_1 < \frac{\varepsilon}{2}$$

Now we have

$$\begin{aligned} \hat{f}_\varepsilon(\xi) &= \int_{\mathbb{R}^n} f_\varepsilon(x) e^{-ix \cdot \xi} dx = \int_{\mathbb{R}^n} f_\varepsilon(x) \nabla \cdot \left(\frac{\xi}{-i|\xi|^2} e^{-ix \cdot \xi} \right) dx \\ &= \int_{\mathbb{R}^n} \nabla \cdot \left(f_\varepsilon(x) \frac{\xi}{-i|\xi|^2} e^{-ix \cdot \xi} \right) dx - \int_{\mathbb{R}^n} Df_\varepsilon(x) \cdot \left(\frac{\xi}{-i|\xi|^2} e^{-ix \cdot \xi} \right) dx \end{aligned}$$

This line happens because $\nabla \cdot fF = \frac{\partial f F_i}{\partial x_i} = F_i \frac{\partial f}{\partial x_i} + f \frac{\partial F_i}{\partial x_i} = F \cdot \nabla f + f \nabla \cdot F$ where $f = f_\varepsilon$ and $F = \frac{\xi}{-i|\xi|^2} e^{-ix \cdot \xi}$. The left integral vanishes by the divergence theorem by restricting the integral to a large enough box since f_ε has compact support. Thus, we have

$$= - \int_{\mathbb{R}^n} \frac{\xi}{-i|\xi|^2} \cdot Df_\varepsilon(x) e^{-ix \cdot \xi} dx$$

The Cauchy-Schwarz inequality implies

$$|\hat{f}_\varepsilon(\xi)| = \left| \int_{\mathbb{R}^n} \frac{\xi}{|\xi|^2} \cdot Df_\varepsilon(x) e^{-ix \cdot \xi} dx \right| \leq \int_{\mathbb{R}^n} \frac{1}{|\xi|} |Df_\varepsilon(x)| dx = \frac{1}{|\xi|} \|Df_\varepsilon(x)\|_1$$

From this we conclude that there exists $R > 0$ such that if $|\xi| > R$ we have $|\hat{f}_\varepsilon(\xi)| < \frac{\varepsilon}{2}$. For $|\xi| > R$ we calculate

$$|\hat{f}(\xi)| \leq |\hat{f}_\varepsilon(\xi)| + |\hat{f}(\xi) - \hat{f}_\varepsilon(\xi)| \leq |\hat{f}_\varepsilon(\xi)| + \|f - f_\varepsilon\|_1 < \varepsilon$$

So done. □

Proposition. Suppose $f \in L^1(\mathbb{R}^n)$, $x \in \mathbb{R}^n$, $\lambda > 0$ and $f_\lambda(y) = \lambda^{-n} f(\lambda^{-1}y)$. Then $\hat{f}_\lambda(\xi) = \hat{f}(\lambda\xi)$. Also, define $e_y(x) = e^{i(y \cdot x)}$. Then $\widehat{e_x f}(\xi) = T_x \hat{f}(\xi)$ and $\widehat{T_x f}(\xi) = e_{-x}(\xi) \hat{f}(\xi)$.

Proof. Calculate, using the substitution $z = \lambda^{-1}x$ and the change of variables formula where the determinant is λ^n

$$\hat{f}_\lambda(\xi) = \int_{\mathbb{R}^n} f_\lambda(x) e^{-i\xi \cdot x} dx = \int_{\mathbb{R}^n} f(\lambda^{-1}x) e^{-i\xi \cdot x} \lambda^{-n} dx = \int_{\mathbb{R}^n} f(z) e^{-i\lambda\xi \cdot z} dz = \hat{f}(\lambda\xi)$$

Next, calculate

$$\widehat{e_x f}(\xi) = \int_{\mathbb{R}^n} e^{i(x \cdot y)} f(y) e^{-i(\xi \cdot y)} dy = \int_{\mathbb{R}^n} f(y) e^{-i((\xi - x) \cdot y)} dy = T_x \hat{f}(\xi)$$

Next, calculate, using substitution $z = y - x$,

$$\widehat{T_x f}(\xi) = \int_{\mathbb{R}^n} e^{-i(\xi \cdot y)} f(y - x) dy = \int_{\mathbb{R}^n} e^{-i(\xi \cdot (z+x))} f(z) dz = e^{-i\xi \cdot x} \int_{\mathbb{R}^n} e^{-i(\xi \cdot z)} f(z) dz = e_{-x}(\xi) \hat{f}(\xi)$$

□

Theorem. Suppose $f, g \in L^1(\mathbb{R}^n)$. Then $f \star g \in L^1(\mathbb{R}^n)$ and $\widehat{f \star g}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$.

Proof. To prove $f \star g \in L^1(\mathbb{R}^n)$ we calculate

$$|f \star g(x)| = \left| \int_{\mathbb{R}^n} f(y) g(x - y) dy \right| \leq \int_{\mathbb{R}^n} |f(y) g(x - y)| dy$$

Therefore (Fubini allowed because non-negative)

$$\begin{aligned} \|f \star g\|_1 &\leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(y) g(x - y)| dy \right) dx = \int_{\mathbb{R}^n} |f(y)| \left(\int_{\mathbb{R}^n} |g(x - y)| dx \right) dy \\ &= \int_{\mathbb{R}^n} |f(y)| \|g\|_1 dy = \|f\|_1 \|g\|_1 < \infty \end{aligned}$$

Now we calculate the Fourier transform:

$$\begin{aligned} \widehat{f \star g}(\xi) &= \int_{\mathbb{R}^n} f \star g(x) e^{-i\xi \cdot x} dx = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) g(x - y) dy \right) e^{-i\xi \cdot x} dx \\ &= \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} g(x - y) e^{-i\xi \cdot x} dx \right) dy = \int_{\mathbb{R}^n} f(y) \widehat{T_y g}(\xi) dy = \int_{\mathbb{R}^n} f(y) \hat{g}(\xi) e^{-i\xi \cdot y} dy = \hat{f}(\xi) \hat{g}(\xi) \end{aligned}$$

□

Lemma. Yet another one of these approximation results. Suppose $f \in C^1(\mathbb{R}^n)$ and that $f, D_j f \in L^1(\mathbb{R}^n)$ for all $j = 1, 2, \dots, n$. Then there exists $f_\varepsilon \in C_0^1(\mathbb{R}^n)$ such that $\|f_\varepsilon - f\|_1 < \varepsilon$ and $\|D_j f - D_j f_\varepsilon\|_1 < \varepsilon$

Proof. Let $\phi \in C_c^\infty(\mathbb{R}^n)$ be a standard bump function where $\phi(x) = 1$ for $|x| \leq 1$ and $\phi(x) = 0$ for $|x| \geq 2$.

Define $\phi_R(x) = \phi(x/R)$, and let $f_R(x) = f(x)\phi_R(x)$. Because $f \in C^1$ and $\phi_R \in C_c^\infty$, we immediately have $f_R \in C_0^1(\mathbb{R}^n)$.

To show this converges to f in the required norms as $R \rightarrow \infty$: For the function: $f - f_R = f(1 - \phi_R)$. Since $\phi_R \rightarrow 1$ pointwise and is bounded, $|f(1 - \phi_R)| \leq |f| \in L^1$. By the Dominated Convergence Theorem, $|f - f_R|_{L^1} \rightarrow 0$. For the derivative: Using the product rule, $D_j f_R = (D_j f)\phi_R + f(D_j \phi_R)$. The first term $(D_j f)\phi_R$ converges to $D_j f$ in L^1 by DCT, exactly as above. For the second term, the chain rule gives $D_j \phi_R(x) = \frac{1}{R}(D_j \phi)(x/R)$. If we let $C = \sup |D_j \phi|$, then $|D_j \phi_R(x)| \leq C/R$. Furthermore, $D_j \phi_R$ is only non-zero on the annulus $R \leq |x| \leq 2R$. Bounding the L^1 norm yields:

$$|f(D_j \phi_R)|_{L^1} \leq \frac{C}{R} \int_{R \leq |x| \leq 2R} |f(x)| dx$$

Since $f \in L^1$, the tail integral $\int_{|x| \geq R} |f(x)| dx$ already goes to zero as $R \rightarrow \infty$. The extra $1/R$ factor forces this term to zero even faster. Therefore, you just pick R sufficiently large such that both L^1 differences are bounded by ε .

□

Proposition. Suppose $f \in C^1(\mathbb{R}^n)$ and that $f, D_j f \in L^1(\mathbb{R}^n)$ for all $j = 1, 2, \dots, n$. Then $\widehat{D_j f}(\xi) = i\xi_j \hat{f}(\xi)$ where ξ_j is the j 'th coordinate of the vector ξ .

Proof. The principle here: Approximate it and then unapproximate it. In fact you probably will notice this has been the principle a lot in this document.

Let f_ε be as in the previous lemma. Then integrating by parts and using Fubini's theorem (allowed by continuity + compact support of everything) gives

$$\widehat{D_j f_\varepsilon}(\xi) = \int_{\mathbb{R}^n} (D_j f_\varepsilon(x)) e^{-i\xi \cdot x} dx = - \int_{\mathbb{R}^n} f_\varepsilon(x) D_j (e^{-i\xi \cdot x}) dx = i\xi_j \int_{\mathbb{R}^n} f_\varepsilon(x) e^{-i\xi \cdot x} dx$$

This implies that $\widehat{D_j f_\varepsilon}(\xi) = i\xi_j \hat{f}_\varepsilon(\xi)$. Next we calculate, using the inequality property of the Fourier transform

$$\begin{aligned} \left| \widehat{D_j f}(\xi) - i\xi_j \hat{f}(\xi) \right| &= \left| \widehat{D_j f}(\xi) - \widehat{D_j f_\varepsilon}(\xi) + i\xi_j \hat{f}_\varepsilon(\xi) - i\xi_j \hat{f}(\xi) \right| \\ &= \|D_j f(\xi) - D_j f_\varepsilon(\xi)\|_1 + |\xi| \|f - f_\varepsilon\|_1 \leq \varepsilon(1 + |\xi|) \end{aligned}$$

Therefore it equals 0 for each ξ as ε was arbitrary.

□

Proposition. Suppose $(1 + |x|)f \in L^1(\mathbb{R}^n)$. Then $\hat{f} \in C^1(\mathbb{R}^n)$ and $D_j \hat{f}(\xi) = -i\xi_j \hat{f}(\xi)$. Note that x_j here is the function sending x to its j 'th coordinate.

Proof. From the condition on f we have $x_j f \in L^1$ so $-ix_j \hat{f}$ is continuous. It suffices to prove then that for any sequence $h_k \rightarrow 0$ as $k \rightarrow \infty$ we have

$$\Delta_j^{h_k} \hat{f}(\xi) \rightarrow -i\xi_j \hat{f}(\xi)$$

To prove this we calculate

$$\Delta_j^{h_k} \hat{f}(\xi) = \frac{1}{h_k} \left(\hat{f}(\xi + h_k e_j) - \hat{f}(\xi) \right) = \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} \left(\frac{e^{-ix_j h_k} - 1}{h_k} \right) dx$$

For $x \in \mathbb{R}^n$ we have, by properties of the derivative,

$$f(x) e^{-ix \cdot \xi} \left(\frac{e^{-ix_j h_k} - 1}{h_k} \right) \rightarrow -ix_j f(x) e^{-ix \cdot \xi}$$

as $k \rightarrow \infty$. Noting that $|e^{i\theta} - 1| \leq |\theta|$ for real θ we have

$$\left| f(x) e^{-ix \cdot \xi} \left(\frac{e^{-ix_j h_k} - 1}{h_k} \right) \right| \leq |x_j f(x)|$$

The right hand side is integrable so the dominated convergence theorem implies

$$\lim_{k \rightarrow \infty} \Delta_j^{h_k} \hat{f}(\xi) = \int_{\mathbb{R}^n} -ix_j f(x) e^{-ix \cdot \xi} dx = -\widehat{ix_j f}(\xi)$$

This implies $\hat{f} \in C^1(\mathbb{R}^n)$ and $D_j \hat{f}(\xi) = -\widehat{ix_j f}(\xi)$ as required. □

Theorem. The Fourier transform is a continuous linear map from $\mathcal{S} \rightarrow \mathcal{S}$

Also, in the proof of this theorem we will see that there is a link between smoothness of the function in the real variable and fast decay of the function in the Fourier variable, as well as a link between fast decay of the function in the real variable and smoothness in the Fourier variable.

Proof. Step 1. We will prove that if $f \in C^k(\mathbb{R}^n)$ and $D^\alpha f \in L^1(\mathbb{R}^n)$ for $|\alpha| \leq k$ then there is a constant $C_k > 0$ depending only on k such that

$$\sup_{\xi \in \mathbb{R}^n} \left| (1 + |\xi|)^k \hat{f}(\xi) \right| \leq C_k \sum_{|\alpha| \leq k} \|D^\alpha f\|_1$$

Define $\xi^\alpha = \prod \xi_i^{\alpha_i}$. Then for any k there is a constant C_k (which is possible to find by expanding the brackets) such that for all $\xi \in \mathbb{R}^n$ we have

$$(1 + |\xi|)^k \leq C_k \sum_{|\alpha| \leq k} |\xi^\alpha|$$

Then repeatedly applying the proposition 2 before this one we know that

$$i^{|\alpha|} \xi^\alpha \hat{f}(\xi) = \widehat{D^\alpha f}(\xi)$$

We therefore have

$$(1 + |\xi|)^k \left| \hat{f}(\xi) \right| \leq C_k \sum_{|\alpha| \leq k} \left| i^{|\alpha|} \xi^\alpha \hat{f}(\xi) \right| = C_k \sum_{|\alpha| \leq k} \left| \widehat{D^\alpha f}(\xi) \right|$$

Taking the supremum over $\xi \in \mathbb{R}^n$ and using the $\hat{f}(\xi) \leq \|f\|_1$ principle we have

$$\sup_{\xi \in \mathbb{R}^n} \left| (1 + |\xi|)^k \hat{f}(\xi) \right| \leq C_k \sum_{|\alpha| \leq k} \|D^\alpha f\|_1$$

as required.

Step 2. We will prove that if $(1 + |x|)^k f \in L^1(\mathbb{R}^n)$ then $\hat{f} \in C^k(\mathbb{R}^n)$ and for any $|\alpha| \leq k$ we have

$$\sup_{\xi \in \mathbb{R}^n} \left| D^\alpha \hat{f}(\xi) \right| \leq \left\| (1 + |x|)^k f \right\|_1$$

By iterating the previous proposition we have $D_j \hat{f}(\xi) = (-i)^{|\alpha|} \widehat{x^\alpha f}(\xi)$. Taking the supremum of this over absolute value we have

$$\sup_{\xi \in \mathbb{R}^n} \left| D^\alpha \hat{f}(\xi) \right| \leq \|x^\alpha f\|_1 \leq \left\| (1 + |x|)^k f \right\|_1$$

The last inequality is because $1 + |x|$ is greater than all coordinates of x and so raised to the power of k it is greater than the result of taking x to the power of a multi-index with size $\leq k$.

Step 3. We will use steps 1 and 2 to prove the theorem.

Note that if $\sup_{x \in \mathbb{R}^n} (1 + |x|)^N |f(x)| < K$ and $N > n$ then

$$\|f\|_1 = \int_{\mathbb{R}^n} |f(x)| dx \leq K \int_{\mathbb{R}^n} \frac{1}{(1 + |x|)^N} dx < \infty$$

Therefore if $f \in \mathcal{S}$ then there exists a constant $C_n \left(= \int_{\mathbb{R}^n} \frac{1}{(1 + |x|)^{n+1}} dx \right)$ such that for all $M \in \mathbb{N}$ and multi-indices α we have

$$\left\| (1 + |x|)^M D^\alpha f \right\|_1 \leq C_n \sup_{x \in \mathbb{R}^n} \left((1 + |x|)^{M+n+1} |D^\alpha f(x)| \right)$$

Let $g(x) = (-i)^{|\beta|} x^\beta f(x)$. By Step 2, $\hat{g}(\xi) = D^\beta \hat{f}(\xi)$. Applying step 1 to g yields:

$$\sup_{\xi \in \mathbb{R}^n} (1 + |\xi|)^N |D^\beta \hat{f}(\xi)| \leq C_N \sum_{|\alpha| \leq N} \|D^\alpha (x^\beta f)\|_1$$

If $|\beta| \leq M$, applying the Leibniz rule expands $D^\alpha (x^\beta f)$ into a linear combination of terms bounded by a constant times $(1 + |x|)^M |D^\gamma f(x)|$, where $|\gamma| \leq |\alpha| \leq N$. Applying the L^1 bound from the previous line to each of these terms produces the following bound, absorbing the combinatorial constants from the Leibniz rule into $C_{N,M,n}$.

$$\begin{aligned} \sup_{\xi \in \mathbb{R}^n} (1 + |\xi|)^N |D^\beta \hat{f}(\xi)| &\leq C_{N,M,n} \sup_{|\alpha| \leq N} \left\| (1 + |x|)^M D^\alpha f \right\|_1 \\ &\leq C'_{N,M,n} \sup_{x \in \mathbb{R}^n, |\alpha| \leq N} \left((1 + |x|)^{M+n+1} |D^\alpha f(x)| \right) \end{aligned}$$

Thus, $\hat{f} \in \mathcal{S}$. Moreover, if f_j is a sequence in $\mathcal{S}$ with $f_j \rightarrow 0$ in $\mathcal{S}$ then $\hat{f}_j \rightarrow 0$ in $\mathcal{S}$ so the Fourier transform is continuous. □

Note that it is not true that the Fourier transform maps $\mathcal{D}(\mathbb{R}^n)$ to itself. The disproof of this in the case of $\mathbb{R}$ is implied by the following theorem, since if a real analytic function vanishes on an open set it must vanish everywhere (see Complex analysis).

Theorem. If a function is continuous with compact support on $\mathbb{R}$ then its Fourier transform is real analytic.

I include this because prior work on probability on this website actually gives a very elegant way to prove this.

Proof. If the function is 0, its Fourier transform is 0, so this is the trivial case. Otherwise, split it into positive and negative parts and use linearity to prove it for each. For the remainder of this proof assume the function has only one sign.

If you rescale the function such that it integrates to 1, you get a valid probability density function of a bounded random variable. Therefore, its Fourier transform will be its characteristic function. This is known to be analytic if the variable has a valid moment generating function on any interval around 0, see my IA probability notes. Since the random variable is bounded in absolute value, say by M , all of its k -th moments are bounded by M^k , so the resulting power series you get from the moment generating function has non-zero radius of convergence. □

Now, the Fourier inversion formula under this convention says that if $f \in L^1(\mathbb{R}^n)$ and $\hat{f} \in L^1(\mathbb{R}^n)$ then for almost every x we have

$$f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{ix \cdot \xi} d\xi$$

To see this, note that the P+M convention for $\hat{f}$ is gotten by replacing ξ with $-\xi$ in which case we recover the inversion formula we proved in that course and the value of the integral stays the same since the integral of $g(\xi)$ equals the integral of $g(-\xi)$ over $\mathbb{R}^n$ for every integrable function g .

Since the map $f(x) \mapsto \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{ix \cdot \xi} d\xi$ is continuous, we have that if f is additionally assumed to be continuous then we can upgrade a.e. convergence to convergence everywhere. Also, if $f \in L^1(\mathbb{R}^n)$ and $\hat{f} \in L^1(\mathbb{R}^n)$ then f must be almost everywhere equal to a continuous function. Also, if $\mathcal{F}$ is the fourier transform, then we have $\mathcal{F}^2 = (2\pi)^n \check{f}$. If $f \in \mathcal{S}$ this is very good news as all the conditions are satisfied so that now $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a continuous bijection with continuous inverse.

Example. Consider the differential equation $-\phi'' + \phi = f$ and suppose $f \in \mathcal{S}$. Assuming that ϕ is C^2 and taking the Fourier transform of the equation using properties above gives $-\left((i\xi)^2 \hat{\phi}(\xi)\right) + \hat{\phi}(\xi) = \hat{f}(\xi)$. Rearranging gives the equation $\hat{\phi}(\xi) = \frac{\hat{f}(\xi)}{\xi^2 + 1}$. Since the Fourier transform is a bijection on $\mathcal{S}$ there is a unique $\phi \in \mathcal{S}$ solving this equation.

Now, let $G(x) = \frac{1}{2} e^{-|x|}$. Very conveniently, we actually computed the Fourier transform of this earlier as an example (save for the trivial division by 2), and then we therefore have $\hat{\phi}(\xi) = \hat{f}(\xi) \hat{G}(\xi)$. By the convolution theorem from earlier, we

now know that $\phi = f \star G$. Now, if given an explicit f , one could compute the integral explicitly to get a solution to the ODE. I have a feeling this is probably the kind of thing we will do in IB methods.

Put on $L^2(\mathbb{R}^n)$ the inner product

$$(f, g) = \int_{\mathbb{R}^n} \overline{f(x)} g(x) dx$$

Whether this is exactly the same as conventions I've used in the past, I really don't care anymore.

Recall the following result from probability and measure:

Proposition. There exists a unique Hilbert space (multiple of an) automorphism $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ such that it agrees with the Fourier transform for all $f \in L^1 \cap L^2$.

Note that $\mathcal{F}$ is continuous because it is (up to a constant, compare to the way I stated it in my probability and measure notes) an automorphism. We call this the Fourier-Plancherel transform.

Recall also that in probability and measure we proved $f \in L^1 \cap L^2 \implies \|\hat{f}\|_2 = (2\pi)^{d/2} \|f\|_2$ (Plancherel identity). Turns out, we can generalize this a tiny bit and then we get Parseval's formula, not to be confused with the Fourier *series* version of Parseval's formula, proved in Linear analysis.

Theorem. (Parseval's formula) Let $f, g \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$. Then $\hat{f}, \hat{g} \in L^2(\mathbb{R}^n)$ and

$$(f, g) = \frac{1}{(2\pi)^n} (\hat{f}, \hat{g})$$

Proof. Start by assuming $f, g \in \mathcal{S}$. Then we have

$$\begin{aligned} (f, g) &= \int_{\mathbb{R}^n} \overline{f(x)} g(x) dx = \int_{\mathbb{R}^n} \overline{f(x)} \left(\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{g}(\xi) e^{ix \cdot \xi} d\xi \right) dx \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} \overline{f(x)} e^{ix \cdot \xi} dx \right) \hat{g}(\xi) d\xi = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \overline{\left(\int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx \right)} \hat{g}(\xi) d\xi \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \overline{\hat{f}(\xi)} \hat{g}(\xi) d\xi = \frac{1}{(2\pi)^n} (\hat{f}, \hat{g}) \end{aligned}$$

Now since $\phi_\varepsilon \star g \rightarrow g$ in L^p as $\varepsilon \rightarrow 0$ for all p , and by considering making the functions smoothly cutoff to 0 at very large values it follows that there exists a sequence f_j of smooth functions with compact support such that

$$\|f_j - f\|_1 + \|f_j - f\|_2 < \frac{1}{j}$$

and there exists an analogous sequence for g . We have

$$\sup_{\xi \in \mathbb{R}^n} |\hat{f}_j(\xi) - \hat{f}(\xi)| \leq \|f_j - f\|_1 < \frac{1}{j}$$

This implies that $\hat{f}_j \rightarrow \hat{f}$ uniformly on $\mathbb{R}^n$. By the calculation above (since f_j is in $\mathcal{S}$, and in fact this next step is also implied by the special case of this theorem proved in probability and measure) we have

$$\|\hat{f}_j - \hat{f}_k\|_2 = (2\pi)^{\frac{n}{2}} \|f_j - f_k\|_2$$

Since f_j converges in $L^2(\mathbb{R}^n)$ it is a Cauchy sequence in $L^2(\mathbb{R}^n)$ so the above line shows $\hat{f}_j$ is also a Cauchy sequence in $L^2(\mathbb{R}^n)$, hence has a limit by completeness of $L^2(\mathbb{R}^n)$, hence $\hat{f} \in L^2(\mathbb{R}^n)$ (as that is the limit since $\|\hat{f}_j - \hat{f}\|_2 \rightarrow 0$). Also, we know that the sequences $f_j, g_j, \hat{f}_j, \hat{g}_j$ all converge in L^2 so taking the limit³ gives

$$(f, g) = \frac{1}{(2\pi)^n} (\hat{f}, \hat{g})$$

□

³Since the inner product is jointly continuous in the 2 variables by the polarization identity $(x, y) = \frac{1}{4} (\|x + y\|^2 - \|x - y\|^2)$

4.2 The Fourier transform on Tempered distributions

Recall from properties and examples of Tempered distributions from earlier that since $\hat{f}$ has a supremum whenever $f \in L^1$ it follows that $T_{\hat{f}} \in \mathcal{S}^*$. For $\phi \in \mathcal{S}$ we have

$$\begin{aligned} T_{\hat{f}}[\phi] &= \int_{\mathbb{R}^n} \hat{f}(x) \phi(x) dx = \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} f(y) e^{-ix \cdot y} dy \right) \phi(x) dx \\ &= \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} \phi(x) e^{-ix \cdot y} dx \right) dy = \int_{\mathbb{R}^n} f(x) \hat{\phi}(x) dx \end{aligned}$$

In fact, for any $f \in L^1$ and $\phi \in \mathcal{S}$ we have $T_{\hat{f}}[\phi] = T_f[\hat{\phi}]$.

Motivated by this, for any tempered distribution $u \in \mathcal{S}^*$ we define its Fourier transform such that $\hat{u}[\phi] = u[\hat{\phi}]$ for all $\phi \in \mathcal{S}$.

This definition makes sense because the Fourier transform maps $\mathcal{S}$ to $\mathcal{S}$ continuously. We could not do this on $\mathcal{D}$ because the Fourier transform of a function in $\mathcal{D}$ is not in $\mathcal{D}$ (in fact this fails in $\mathbb{R}$ because it is Analytic by the theorem above so it can't vanish on an open set without vanishing everywhere).

Example. $\hat{\delta}_\xi = T_{e_{-\xi}}$. The proof of this is that by definition

$$\hat{\delta}_\xi[\phi] = \delta_\xi[\hat{\phi}] = \hat{\phi}(\xi) = \int_{\mathbb{R}^n} e^{-ix \cdot \xi} \phi(x) dx = T_{e_{-\xi}}[\phi]$$

The distributions are equal because this is true for every ϕ .

Example. For $x \in \mathbb{R}^n$ we have $\widehat{T_{e_x}} = (2\pi)^n \delta_x$. The proof of this is that if $\phi \in \mathcal{S}$ then

$$\widehat{T_{e_x}}[\phi] = T_{e_x}[\hat{\phi}] = \int_{\mathbb{R}^n} e^{ix \cdot \xi} \phi(\xi) d\xi = (2\pi)^n \phi(x) = (2\pi)^n \delta_x[\phi]$$

Example. Let α be a multi-index and denote by X^α the map $X^\alpha : x \mapsto x^\alpha$. Then for $\phi \in \mathcal{S}$ we have

$$\widehat{T_{X^\alpha}}[\phi] = T_{X^\alpha}[\hat{\phi}] = \int_{\mathbb{R}^n} \xi^\alpha \hat{\phi}(\xi) d\xi = (-i)^{|\alpha|} \int_{\mathbb{R}^n} \widehat{D^\alpha \phi}(\xi) d\xi$$

The last inequality is valid since functions in $\mathcal{S}$ are infinitely differentiable. Then by Fourier inversion formula we have

$$= (2\pi)^n (-i)^{|\alpha|} D^\alpha \phi(0) = (2\pi)^n i^{|\alpha|} (-1)^{|\alpha|} \delta_0[D^\alpha \phi] = (2\pi)^n i^{|\alpha|} D^\alpha \delta_0[\phi]$$

Therefore we have $\widehat{T_{X^\alpha}} = (2\pi)^n i^{|\alpha|} D^\alpha \delta_0$.

Definition. Let $f \in C^\infty(\mathbb{R}^n)$. Then f is said to be of tempered growth if for each α there exists a constant C_α and an integer N_α such that $|D^\alpha f(x)| \leq C_\alpha (1 + |x|)^{N_\alpha}$.

If f satisfies this then $\phi f \in \mathcal{S}$ (to be shown shortly) with $\phi \in \mathcal{S}$ and we can define $fu \in \mathcal{S}^*$ by $fu[\phi] = u[f\phi]$. To show that this is actually continuous and maps into $\mathcal{S}$ we want to show that the map $\phi \mapsto f\phi$ is continuous.

Theorem. Let f be of tempered growth, then the map $\phi \mapsto f\phi$ is a map on $\mathcal{S}$ and continuous on $\mathcal{S}$.

Proof. Continuity on $\mathcal{S}$ requires showing that the Schwartz seminorms of $f\phi$ are bounded by a finite linear combination of the Schwartz seminorms of ϕ . Using the general Leibniz rule, the derivatives of $f\phi$ are:

$$D^\alpha(f\phi) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^{\alpha-\beta} f D^\beta \phi$$

To bound a given Schwartz seminorm of $f\phi$, we multiply by $(1 + |x|)^k$ and take the supremum:

$$\sup_{x \in \mathbb{R}^n} (1 + |x|)^k |D^\alpha(f\phi)| \leq \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} \sup_{x \in \mathbb{R}^n} ((1 + |x|)^k |D^{\alpha-\beta} f(x)| |D^\beta \phi(x)|)$$

Because f is of tempered growth, we have $|D^{\alpha-\beta}f(x)| \leq C_{\alpha,\beta}(1+|x|)^{N_{\alpha,\beta}}$ for some constants $C_{\alpha,\beta}$ and integers $N_{\alpha,\beta}$. Substituting this bound gives:

$$\sup_{x \in \mathbb{R}^n} (1+|x|)^k |D^\alpha(f\phi)| \leq \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} C_{\alpha,\beta} \sup_{x \in \mathbb{R}^n} ((1+|x|)^{k+N_{\alpha,\beta}} |D^\beta \phi(x)|)$$

The terms on the right-hand side are precisely constants multiplied by the Schwartz seminorms of ϕ . This proves that $f\phi \in \mathcal{S}$ and that the map $\phi \mapsto f\phi$ is continuous. □

Define $\tau_x u[\phi] = u[\tau_{-x}\phi]^4$. Also define $\check{u}[\phi] = u[\check{\phi}]$.

We will now verify 5 basic properties of the Fourier transform on tempered distributions, including the fact that it is an invertible operation on $\mathcal{S}^*$. In all cases, remember that two distributions are equal if they agree on all test functions, I don't need to say it every time. We may use these properties in future algebra.

1.

$$\widehat{e_x u}[\phi] = e_x u[\hat{\phi}] = u[e_x \hat{\phi}] = u[\widehat{\tau_{-x}\phi}] = \hat{u}[\widehat{\tau_{-x}\phi}] = \tau_x u[\phi]$$

2.

$$\widehat{e_x u}[\phi] = e_x u[\hat{\phi}] = u[e_x \hat{\phi}] = u[\widehat{\tau_{-x}\phi}] = \hat{u}[\widehat{\tau_{-x}\phi}] = \tau_x u[\phi]$$

3.

$$\begin{aligned} \widehat{D^\alpha u}[\phi] &= D^\alpha u[\hat{\phi}] = (-1)^{|\alpha|} u[D^\alpha \hat{\phi}] = (-1)^{|\alpha|} u[(-i)^{|\alpha|} \widehat{X^\alpha \phi}] \\ &= i^{|\alpha|} u[\widehat{X^\alpha \phi}] = i^{|\alpha|} \hat{u}[X^\alpha \phi] = i^{|\alpha|} X^\alpha \hat{u}[\phi] \end{aligned}$$

4.

$$\begin{aligned} D^\alpha \hat{u}[\phi] &= (-1)^{|\alpha|} \hat{u}[D^\alpha \phi] = (-1)^{|\alpha|} u[\widehat{D^\alpha \phi}] \\ &= (-1)^{|\alpha|} u[i^{|\alpha|} X^\alpha \hat{\phi}] = (-i)^{|\alpha|} X^\alpha u[\hat{\phi}] = (-i)^{|\alpha|} \widehat{X^\alpha u}[\hat{\phi}] \end{aligned}$$

5.

$$\hat{\hat{u}}[\phi] = u[\hat{\hat{\phi}}] = (2\pi)^n u[\check{\phi}] = (2\pi)^n \check{u}[\phi]$$

Therefore, the map $\mathcal{F} : \mathcal{S}^* \rightarrow \mathcal{S}^*$ defined by $u \mapsto \hat{u}$ is a linear bijection.

Theorem. $\mathcal{F}$ is a Homeomorphism if $\mathcal{S}^*$ has the weak-star topology.

Proof. It suffices to prove that $\mathcal{F}$ is continuous since this is clearly equivalent to continuity of its inverse (note that if $\mathcal{F}$ is continuous then so is $\mathcal{F}^3$ which is a multiple of $\mathcal{F}$).

Note that the weak-star topology always induces a topological vector space (see linear analysis) so we have translation invariance. It therefore suffices to verify that the preimage of a basic open neighbourhood is open. A basic neighbourhood of the origin in $\mathcal{S}^*$ is of the form $V(\phi_1, \dots, \phi_k, \epsilon) = \{u \in \mathcal{S}^* : |u[\phi_j]| < \epsilon \text{ for all } j\}$, where $\phi_j \in \mathcal{S}$. The preimage under $\mathcal{F}$ is simply

$$\mathcal{F}^{-1}(V(\phi_1, \dots, \phi_k, \epsilon)) = \{u \in \mathcal{S}^* : |\hat{u}[\phi_j]| < \epsilon\} = \{u \in \mathcal{S}^* : |u[\hat{\phi}_j]| < \epsilon\} = V(\hat{\phi}_1, \dots, \hat{\phi}_k, \epsilon)$$

Since the Fourier transform maps $\mathcal{S}$ to $\mathcal{S}$, we know $\hat{\phi}_j \in \mathcal{S}$. Therefore, the preimage is exactly another basic open neighbourhood in the weak-* topology, which proves continuity. □

⁴From now on in ambiguous cases like this, I'll try to remember to reserve τ for translation and T for a function as a distribution. Also, I don't know why the minus sign convention exists here but it does so deal with it. Sorry, it only just occurred to me now I was using T_{stuff} to mean 2 different things.

4.3 Convolutions in the Schwartz space

Recall that if $u \in \mathcal{D}^*(\mathbb{R}^n)$ and $\phi \in \mathcal{D}(\mathbb{R}^n)$ then $u \star \phi(x) = u[\tau_x \check{\phi}]$. If $u \in \mathcal{S}^*$ and $\phi \in \mathcal{S}$ then this definition still makes sense, and has the following basic properties, where $u \in \mathcal{S}^*$ and $\phi \in \mathcal{S}$ are fixed and $u \star \phi$ is a function $\mathbb{R}^n \rightarrow \mathbb{C}$.

Proposition. $u \star \phi \in C^\infty(\mathbb{R}^n)$ with $D^\alpha(u \star \phi) = (D^\alpha u) \star \phi = u \star (D^\alpha \phi)$

Proof. We have previously shown that the analogous proposition holds for distributions in general, and this is almost established if you follow that proof. The only thing to check is that for $\phi \in \mathcal{S}$ we have $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{S}$ as $h \rightarrow 0$.

To justify that $\Delta_i^h \phi \rightarrow D_i \phi$ in $\mathcal{S}$ as $h \rightarrow 0$, we must show that convergence holds under all Schwartz seminorms. That is, for any multi-indices α and β :

$$\sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta(\Delta_i^h \phi(x) - D_i \phi(x))| \rightarrow 0 \text{ as } h \rightarrow 0.$$

Since the difference quotient commutes with partial derivatives and ϕ is smooth, we can let $\psi = D^\beta \phi$. Because $\phi \in \mathcal{S}$, we know $\psi \in \mathcal{S}$. Using the integral form of Taylor's theorem, we can express the error in the difference quotient exactly:

$$\begin{aligned} \Delta_i^h \psi(x) - D_i \psi(x) &= \frac{1}{h} [\psi(x+h) - \psi(x) - h D_i \psi(x)] \\ &= \frac{1}{h} \int_x^{x+he_i} (x+he_i-t) D_i^2(t) dt = \frac{1}{h} \int_0^h (h-s) D_i^2(x+se_i) ds \\ &= \frac{1}{h} \int_0^h \int_s^h D_i^2 \psi(x+se_i) dt ds = \frac{1}{h} \int_0^h \int_0^t D_i^2 \psi(x+se_i) ds dt \end{aligned}$$

Multiplying by x^α , we need to bound $x^\alpha D_i^2 \psi(x+se_i)$ uniformly. Because $\psi \in \mathcal{S}$, $D_i^2 \psi$ decays faster than the reciprocal of any polynomial. Assuming $|h| \leq 1$, we have $s \leq 1$, meaning $|x| \leq |x+se_i| + 1$. This guarantees that we can bound $|x^\alpha D_i^2 \psi(x+se_i)| \leq C$ for some uniform constant C that depends on α and ψ , but is independent of x and h . Therefore, we can bound the supremum:

$$\sup_{x \in \mathbb{R}^n} |x^\alpha (\Delta_i^h \psi(x) - D_i \psi(x))| \leq \frac{1}{|h|} \left| \int_0^h \int_0^t C ds dt \right| = \frac{C|h|}{2}$$

As $h \rightarrow 0$, this goes to 0, satisfying the condition for convergence in the Schwartz topology. □

Proposition. There exist constants $N \in \mathbb{N}$ and $K > 0$ depending on u and ϕ such that

$$|u \star \phi(x)| \leq K(1+|x|)^N$$

Proof. Note that $1+|x-y| \leq (1+|x|)(1+|y|)$ universally in $\mathbb{R}^n$.

Recall that a linear map $\mathcal{S} \rightarrow \mathbb{C}$ is continuous if and only if there exist $N, k \in \mathbb{N}$ and $C > 0$ such that

$$|u[\phi]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} \left| (1+|x|)^N D^\alpha \phi(x) \right| \text{ for all } \phi \in \mathcal{S}$$

Applying this inequality with $\psi = \tau_x \check{\phi}$ gives

$$|u \star \phi(x)| = |u[\tau_x \check{\phi}]| \leq C \sup_{y \in \mathbb{R}^n, |\alpha| \leq k} \left| (1+|y|)^N D^\alpha \phi(x-y) \right|$$

Set $z = x - y$ and note that $|y| = |-y|$.

$$= C \sup_{z \in \mathbb{R}^n, |\alpha| \leq k} \left| (1+|z-x|)^N D^\alpha \phi(z) \right| \leq \left[C \sup_{z \in \mathbb{R}^n, |\alpha| \leq k} \left| (1+|z|)^N D^\alpha \phi(z) \right| \right] (1+|x|)^N$$

Which gives the result on setting $K = C \sup_{z \in \mathbb{R}^n, |\alpha| \leq k} \left| (1+|z|)^N D^\alpha \phi(z) \right|$ □

Theorem. $\mathcal{D}(\mathbb{R}^n)$ (which is $C_c^\infty(\mathbb{R}^n)$) is dense in $\mathcal{S}(\mathbb{R}^n)$:

Proof. For any $f \in \mathcal{S}(\mathbb{R}^n)$, take a smooth cutoff function $\psi \in C_c^\infty(\mathbb{R}^n)$ such that $\psi(x) = 1$ for $|x| \leq 1$. Define the sequence $f_k(x) = f(x)\psi(x/k)$. By definition, each $f_k \in \mathcal{D}(\mathbb{R}^n)$.

To show $f_k \rightarrow f$ in $\mathcal{S}(\mathbb{R}^n)$, we must show that for any multi-indices α and β , $\sup_x |x^\alpha D^\beta [f(x)(1 - \psi(x/k))]| \rightarrow 0$ as $k \rightarrow \infty$. By the multivariate Leibniz rule:

$$D^\beta [f(x)(1 - \psi(x/k))] = D^\beta f(x)(1 - \psi(x/k)) - \sum_{0 < \gamma \leq \beta} \binom{\beta}{\gamma} D^{\beta-\gamma} f(x) k^{-|\gamma|} (D^\gamma \psi)(x/k)$$

Multiplying by x^α : The $\gamma = 0$ term is $x^\alpha D^\beta f(x)(1 - \psi(x/k))$. This is exactly 0 for $|x| \leq k$. Since $f \in \mathcal{S}$, $x^\alpha D^\beta f(x) \rightarrow 0$ as $|x| \rightarrow \infty$, so the supremum over the remaining region $|x| > k$ goes to 0 as $k \rightarrow \infty$. The $\gamma > 0$ terms each contain the factor $k^{-|\gamma|}$. Since $x^\alpha D^{\beta-\gamma} f$ is bounded globally (as $f \in \mathcal{S}$) and $D^\gamma \psi$ is bounded, each term is bounded by a constant times $k^{-|\gamma|}$, which clearly vanishes as $k \rightarrow \infty$.

□

Proposition. $T_{u \star \phi} \in \mathcal{S}^*$ and $\widehat{T_{u \star \phi}} = \hat{\phi} \hat{u}$

Proof. By the above 2 results we know that $T_{u \star \phi} \in \mathcal{S}^*$ because $u \star \phi$ is smooth and doesn't grow too fast. Therefore, it makes sense to consider the Fourier transform. Suppose that $\psi \in \mathcal{D}(\mathbb{R}^n)$. Then we have

$$\begin{aligned} \widehat{T_{u \star \phi}} [\hat{\psi}] &= T_{u \star \phi} [\hat{\psi}] = (2\pi)^n T_{u \star \phi} [\check{\psi}] = (2\pi)^n \int_{\mathbb{R}^n} u \star \phi(x) \psi(-x) dx \\ &= (2\pi)^n \int_{\mathbb{R}^n} u [\tau_x \check{\phi}] \psi(-x) dx = (2\pi)^n \int_{\mathbb{R}^n} u [\psi(-x) \tau_x \check{\phi}] dx \end{aligned}$$

We will now write that this is $= (2\pi)^n u [\int_{\mathbb{R}^n} \psi(-x) \tau_x \check{\phi} dx]$. The proof that this is valid is deferred to later in this document. Continuing, we get

$$= (2\pi)^n u [(\phi \star \check{\psi})] = u \left[\widehat{(\phi \star \check{\psi})} \right] = \hat{u} \left[\widehat{(\phi \star \check{\psi})} \right] = \hat{u} \left[(\hat{\phi} \hat{\psi}) \right] = (\hat{\phi} \hat{u}) [\hat{\psi}]$$

We conclude that $\widehat{T_{u \star \phi}} [\hat{\psi}] = (\hat{\phi} \hat{u}) [\hat{\psi}]$ is true for all $\psi \in \mathcal{D}(\mathbb{R}^n)$.

Since $\mathcal{D}(\mathbb{R}^n)$ is dense in $\mathcal{S}$ and $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a homeomorphism (We proved this above with $\mathcal{S}$ replaced with $\mathcal{S}^*$ and a bit above that we proved that $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is a continuous map, the inverse is continuous by Fourier inversion theorem), we have that the set $\mathcal{F}[\mathcal{D}(\mathbb{R}^n)] = \{\hat{\psi} : \psi \in \mathcal{D}(\mathbb{R}^n)\}$ is dense in $\mathcal{S}$. Thus, for any $\chi \in \mathcal{S}$ and by continuity and density we have that for all $\chi \in \mathcal{S}$ that

$$\widehat{T_{u \star \phi}} [\chi] = (\hat{\phi} \hat{u}) [\chi]$$

□

Lemma. Let $\phi \in \mathcal{S}$. Then the map $\mathcal{S} \rightarrow \mathcal{S}$ defined by $\psi \mapsto \phi\psi$ is continuous.

Proof. we must bound an arbitrary Schwartz seminorm of $\phi\psi$ by a finite linear combination of the seminorms of ψ . By definition, the seminorm is⁵:

$$|\phi\psi|_{\alpha,\beta} = \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta (\phi\psi)(x)|$$

Applying the general Leibniz rule:

$$D^\beta (\phi\psi) = \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} D^\gamma \phi(x) D^{\beta-\gamma} \psi(x)$$

⁵Well I've defined the seminorm slightly differently but it's just a finite linear combination of these so don't worry about it the proof still works

This gives:

$$|x^\alpha D^\beta(\phi\psi)(x)| \leq \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} |D^\gamma \phi(x)| |x^\alpha D^{\beta-\gamma} \psi(x)|$$

Since $\phi \in \mathcal{S}$, its derivatives are globally bounded. Let $C_\gamma = \sup_x |D^\gamma \phi(x)| < \infty$. Substituting this in:

$$|x^\alpha D^\beta(\phi\psi)(x)| \leq \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} C_\gamma |x^\alpha D^{\beta-\gamma} \psi(x)| \leq \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} C_\gamma |\psi|_{\alpha, \beta-\gamma}$$

Taking the supremum over all x yields:

$$|\phi\psi|_{\alpha, \beta} \leq \sum_{\gamma \leq \beta} \binom{\beta}{\gamma} C_\gamma |\psi|_{\alpha, \beta-\gamma}$$

Because $|\phi\psi|_{\alpha, \beta}$ is bounded by a finite sum of seminorms of ψ , the map $\psi \mapsto \phi\psi$ is continuous.

□

Theorem. Let $\phi \in \mathcal{S}$. Then the map $\mathcal{S} \rightarrow \mathcal{S}$ defined by $\psi \mapsto \phi \star \psi$ is continuous.

Proof. Taking the Fourier transform turns the convolution into pointwise multiplication: $\mathcal{F}(\phi \star \psi) = \hat{\phi}\hat{\psi}$. Since $\hat{\phi} \in \mathcal{S}$, pointwise multiplication by it is a continuous map from $\mathcal{S} \rightarrow \mathcal{S}$, and mapping back through the inverse Fourier transform preserves that continuity.

□

Proposition. For any $\psi \in \mathcal{S}$ we have $(u \star \phi) \star \psi = u \star (\phi \star \psi)$

Proof. In the proof of the previous proposition (except for one step to be justified later) we established that for any $\psi \in \mathcal{D}(\mathbb{R}^n)$ we have

$$\int_{\mathbb{R}^n} u \star \phi(x) \psi(-x) dx = u[(\phi \check{\star} \psi)]$$

which is equivalent to saying

$$(u \star \phi) \star \psi(0) = u \star (\phi \star \psi)(0)$$

It is clear from the definitions that

$$\phi \star \tau_y \psi = \tau_y(\phi \star \psi) \text{ and } u \star \tau_y \phi = \tau_y(u \star \phi)$$

To see why the second one is true, since it's difficult to keep all the signs straight,

Left side: $u \star (\tau_y \phi)$

Let $\psi(z) = \tau_y \phi(z) = \phi(z - y)$. Reflecting this gives $\check{\psi}(z) = \psi(-z) = \phi(-z - y)$. Translating by x gives the test function u acts on:

$$\tau_x \check{\psi}(z) = \check{\psi}(z - x) = \phi(-(z - x) - y) = \phi(-z + x - y)$$

So, $(u \star \tau_y \phi)(x) = u[\phi(-z + x - y)]$

Right side: $\tau_y(u \star \phi)$

Evaluating this at x gives $(u \star \phi)(x - y)$, which means u acts on $\tau_{x-y} \check{\phi}$. Reflecting ϕ gives $\check{\phi}(z) = \phi(-z)$. Translating by $x - y$ gives:

$$\tau_{x-y} \check{\phi}(z) = \check{\phi}(z - (x - y)) = \phi(-(z - x + y)) = \phi(-z + x - y)$$

So, $\tau_y(u \star \phi)(x) = u[\phi(-z + x - y)]$. Since the test functions are identical, the two expressions are equal.

Using this and the fact $(u \star \phi) \star \psi(0) = u \star (\phi \star \psi)(0)$ and replacing ψ with $\tau_y \psi$ we conclude that the formula holds at all y for all $\psi \in \mathcal{D}(\mathbb{R}^n)$ and this is a dense subset of $\mathcal{S}$. We would conclude, then, if we could verify continuity of the relevant maps. Specifically,

Left side: $\psi \mapsto ((u * \phi) * \psi)(y) = \int (u * \phi)(z) \psi(y - z) dz$. This is a continuous functional on $\mathcal{S}$ because $u * \phi$ is polynomially bounded (as proven in a proposition a bit ago).

Right side: $\psi \mapsto (u * (\phi * \psi))(y) = u [\tau_y (\phi \check{*} \psi)]$. This is continuous because convolution by a fixed $\phi \in \mathcal{S}$ is a continuous map from $\mathcal{S} \rightarrow \mathcal{S}$, and $u \in \mathcal{S}^*$ is continuous by definition.

□

Proposition. We have $T_{\hat{u} * \hat{\phi}} = (2\pi)^n \widehat{\phi u}$

Proof. Recall $\widehat{T_{u * \phi}} = \hat{\phi} \hat{u}$ (proved earlier). Using this, we have

$$\widehat{T_{\hat{u} * \hat{\phi}}} = \hat{\hat{\phi}} \hat{\hat{u}} = (2\pi)^{2n} \check{\phi} \check{u} = (2\pi)^{2n} (\check{\phi} u) = (2\pi)^n \left(\widehat{\widehat{\phi u}} \right)$$

Then done since the Fourier transform is a bijection on $\mathcal{S}^*$.

□

To complete the proof of the last few propositions it remains to justify the step where a tempered distribution and integration was interchanged.

Let $\Omega \subseteq \mathbb{R}^n$ be open and let $\Omega' \subseteq \mathbb{R}^m$ be open. Let $f \in C(\Omega \times \Omega')$ be uniformly continuous and suppose there is some $R > 0$ such that for each $y \in \Omega'$ we have

$$\text{supp } f(\cdot, y) \subset [-R, R]^n \subset \Omega$$

Next, define a partition of this cube into cubes as follows: Π_0 just consists of the rectangle and if Π_k is known define Π_{k+1} as Π_k with each cube split into 2^n equal sized half-open cubes. Then for each fixed k , $\bigcup \Pi_k = [-R, R]^n$. For each $\pi \in \Pi_k$ let x_π be the point at the center of that cube. Define

$$S_k(y) = \sum_{\pi \in \Pi_k} f(x_\pi, y) \mu(\pi)$$

Lemma. With the definitions as above

$$S_k(y) \rightarrow \int_{\Omega} f(x, y) dx$$

uniformly in $y \in \Omega'$.

Proof. Since $f \mapsto f(x, y)$ is continuous with compact support it is certainly Riemann integrable. Thus, for each fixed y we have

$$S_k(y) \rightarrow \int_{\Omega} f(x, y) dx$$

Now since f is uniformly continuous, the result follows easily: Specifically for each ε there is a δ that works for $\epsilon := \frac{\varepsilon}{(2R)^n}$ and if k is large enough the partition has any 2 points in each cube within a distance of this δ , where one point is at the center and the other is at, say, a point where the mean value is attained.

□

We will now prove the result we actually need.

Recall the result from Analysis that if f is Riemann integrable on $[a, b]$ then for all $\varepsilon > 0$ there exists a $\delta > 0$ such that the mesh being $< \delta$ implies the difference between the upper and lower sums is $< \varepsilon$. A similar proof strategy generalizes this to closed rectangles in $\mathbb{R}^n$ and integrating functions over those. However, I leave the details as an exercise.

Theorem. Let $u \in \mathcal{S}^*$ and $\phi \in \mathcal{S}$ and $\psi \in \mathcal{D}(\mathbb{R}^n)$. Then

$$u \left[\int_{\mathbb{R}^n} \psi(-x) \tau_x \check{\phi} dx \right] = \int_{\mathbb{R}^n} u [\psi(-x) \tau_x \check{\phi}] dx$$

Proof. Fix $\Omega, R > 0$ such that $\text{supp } \check{\psi} \subset [-R, R]^n \subset \Omega$. Define $f : \Omega \times \mathbb{R}^n \rightarrow \mathbb{C}$ by $(x, y) \mapsto \psi(-x) \phi(y - x)$.

Note that by a Leibniz rule trick and the fact that ϕ, ψ are in the Schwartz space and thus have uniformly continuous derivatives, we have that for any α, N that $(1 + |y|)^N D_y^\alpha f$ is uniformly continuous on $\Omega \times \mathbb{R}^n$, as the part of ϕ that the derivative 'sees' always decays faster than the $(1 + |y|)^N$ factor grows in y .

Now define $S_k(y)$ as in the lemma above. Then differentiating under the finite sum gives

$$(1 + |y|)^N D_y^\alpha S_k(y) = \sum_{\pi \in \Pi_k} \left[(1 + |y|)^N D_y^\alpha f(x_\pi, y) \right] \mu(\pi)$$

Given this, the previous lemma implies

$$S_k(y) \rightarrow \int_{\mathbb{R}^n} \psi(-x) \tau_x \check{\phi} dx \text{ in } \mathcal{S}$$

Continuity of u implies

$$u \left[\int_{\mathbb{R}^n} \psi(-x) \tau_x \check{\phi} dx \right] = u \left[\lim_{k \rightarrow \infty} S_k \right] = \lim_{k \rightarrow \infty} u[S_k]$$

By linearity of u we have

$$u[S_k] = u \left[\sum_{\pi \in \Pi_k} f(x_\pi, \cdot) \mu(\pi) \right] = \sum_{\pi \in \Pi_k} u[f(x_\pi, \cdot)] \mu(\pi) = \sum_{\pi \in \Pi_k} u[\psi(-x_\pi) \tau_{x_\pi} \check{\phi}] \mu(\pi)$$

Because $x \mapsto u[\psi(-x) \tau_x \check{\phi}]$ is smooth (since it is $\psi(-x) u[\tau_x \check{\phi}]$ which can be expressed as a convolution and thus is smooth) it is Riemann integrable and we have that

$$\lim_{k \rightarrow \infty} \sum_{\pi \in \Pi_k} u[\psi(-x_\pi) \tau_{x_\pi} \check{\phi}] \mu(\pi) = \int u[\psi(-x) \tau_x \check{\phi}] dx$$

□

4.4 The Fourier transform on $\mathcal{E}^*(\mathbb{R}^n)$

Recall that $\mathcal{E}^*(\mathbb{R}^n)$ is a continuously embedded subspace of $\mathcal{S}^*(\mathbb{R}^n)$ consisting of distributions of compact support. For these distributions, we have the following:

Theorem. Suppose that $u \in \mathcal{E}^*(\mathbb{R}^n)$. Then $\hat{u} = T_{\hat{v}}$ for some $\hat{v} \in C^\infty(\mathbb{R}^n)$ with $\hat{v}(\xi) = u[e_{-\xi}]$

Proof. Suppose that $\text{supp}(u) \subset B_R(0)$. Pick $\psi \in \mathcal{D}(\mathbb{R}^n)$ with $\psi = 1$ on $B_{R+1}(0)$ so that $\psi u = u$. Then we have (by properties from the previous subsection)

$$\hat{u} = \widehat{\psi u} = \frac{1}{(2\pi)^n} T_{\hat{u} \star \hat{\psi}}$$

Thus we have $\hat{u} = T_{\hat{v}}$ for $\hat{v} = \frac{1}{(2\pi)^n} \hat{u} \star \hat{\psi} \in C^\infty(\mathbb{R}^n)$. Now let $\phi \in \mathcal{S}$ be such that $\hat{\phi} = \psi$. We calculate

$$\begin{aligned} \hat{v}(\xi) &= \frac{1}{(2\pi)^n} \hat{u} \star \hat{\psi}(\xi) = \hat{u} \star \check{\phi}(\xi) = \hat{u}[\tau_\xi \phi] = u \left[\widehat{\tau_\xi \phi} \right] \\ &= u[e_{-\xi} \psi] = (\psi u)[e_{-\xi}] = u[e_{-\xi}] \end{aligned}$$

□

Therefore in practice we don't distinguish the distribution $\hat{u}$ and the function $\hat{v}$.

4.5 Periodic distributions and Poisson's summation formula

Note: In this section, sums over $\mathbb{Z}^n$ are understood to be the limit as sums over the cubes centered about the origin. In most cases they should be absolutely convergent and this distinction won't matter.

Definition. A distribution $u \in \mathcal{D}'(\mathbb{R}^n)$ is said to be periodic if for each $g \in \mathbb{Z}^n$ we have $\tau_g u = u$.

Example. The distribution $T_{e_{2\pi g}}$ is periodic for any $g \in \mathbb{Z}^n$.

Example. Suppose $v \in \mathcal{E}'(\mathbb{R}^n)$ and let $u = \sum_{g \in \mathbb{Z}^n} \tau_g v$. If we know that this defines a distribution then it is easy to see that it is periodic.

To see that it defines a distribution this is the sketch of the proof: note that for all $\phi \in \mathcal{D}(\mathbb{R}^n)$ its support is contained in a compact set which therefore must contain finitely many integer lattice points. Also, the support of v only has finitely many integer lattice points. All said and done it's easy to see that $u = \sum_{g \in \mathbb{Z}^n} \tau_g v$ effectively becomes a finite sum for any ϕ , and it is sequentially continuous and thus continuous as the effectively finite sum (any convergent sequence in $\mathcal{D}$ must be contained in a fixed compact set) of sequentially continuous stuff - only for finitely many g does the support of something in the sequence and the support of v overlap.

Define $q = \{x \in \mathbb{R}^n : -\frac{1}{2} \leq x_i < \frac{1}{2} \text{ for } i = 1, 2, \dots, n\}$. This is a fundamental cell of the periodic lattice. We define $M(f) = \int_q f(x) dx$. We would like to define $M(u) = u[1_q]$ but we can't since $1_q \notin \mathcal{D}(\mathbb{R}^n)$. However, we can solve this using the following proposition.

Proposition. Let $Q \subset \mathbb{R}^n$ be the open cube of diameter 2 centered at the origin. Then there exists a smooth non-negative function ψ supported in Q such that $\sum_{g \in \mathbb{Z}^n} \tau_g \psi = 1$. Also, if ψ and ψ' both satisfy this proposition and u is a periodic distribution then $u[\psi] = u[\psi']$. We then define $M(u) = u[\psi]$ satisfied that this is a well defined distribution.

Proof. Note that $\bar{q} = \{x \in \mathbb{R}^n : |x_i| \leq \frac{1}{2} \text{ for } i = 1, 2, \dots, n\}$. By earlier properties there is known to exist a non-negative function $\psi_0 \in C_c^\infty(Q)$ that is 1 on $\bar{q}$. Define $S(x) := \sum_{g \in \mathbb{Z}^n} \psi_0(x - g)$. Then this is a smooth function and it is ≥ 1 everywhere so we may take $\psi(x) = \frac{\psi_0(x)}{S(x)}$ and this does the job.

For the second part, we calculate

$$u[\psi] = u \left[\psi \sum_{g \in \mathbb{Z}^n} \tau_g \psi' \right] = \sum_{g \in \mathbb{Z}^n} u[\psi \tau_g \psi']$$

We can take the sum out as it is effectively finite because of the support being bounded. But then we can write

$$= \sum_{g \in \mathbb{Z}^n} \tau_{-g} u[\tau_{-g} \psi \psi'] = u \left[\psi' \sum_{g \in \mathbb{Z}^n} \tau_{-g} \psi \right] = u[\psi']$$

□

If $u = T_f$ for some locally integrable periodic function f then by choosing a bounded sequence of ψ_j 's that tends to $1_{\bar{q}}$ pointwise we have

$$M(T_f) = \int_{\bar{q}} f(x) dx$$

We call M the mean of the distribution.

Note that all periodic distributions are tempered and obtained by translating a compact distribution, as shown by the following proposition.

Lemma. $\sum_{g \in \mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}}$ converges.

Sketch of proof. Let g_M have largest absolute value coordinate M , then it is $O(M^{n-1})$ large but has values the size of $O(M^{-1-N})$ so the total sum is like summing $O(M^{-2})$ over M .

Proposition. Suppose $v \in \mathcal{E}^*(\mathbb{R}^n)$ is a compact distribution. Then $u = \sum_{g \in \mathbb{Z}^n} \tau_g v$ converges in $\mathcal{S}^*$. Conversely, suppose that $u \in \mathcal{D}^*(\mathbb{R}^n)$ is a periodic distribution. Then there exists $v \in \mathcal{E}^*(\mathbb{R}^n)$ such that $u = \sum_{g \in \mathbb{Z}^n} \tau_g v$ holds and u extends uniquely to a tempered distribution $u \in \mathcal{S}^*$.

Proof. By the characterization of continuity of linear maps $\mathcal{E} \rightarrow \mathbb{C}$ there exists $C > 0, N \in \mathbb{N}$ and a compact K such that for all $\phi \in \mathcal{E}(\mathbb{R}^n)$ we have

$$|v[\phi]| \leq C \sup_{x \in K, |\alpha| \leq N} |D^\alpha \phi(x)|$$

Now suppose that $\phi \in \mathcal{S}$. Then we have

$$|\tau_g v[\phi]| = |v[\tau_{-g} \phi]| \leq C \sup_{x \in K, |\alpha| \leq N} |D^\alpha \phi(x + g)|$$

Since K is bounded, we have that $K \subset B_R(0)$ for some $R > 0$. For all $x \in K$ we have

$$1 + |g| \leq 1 + |-x| + |x + g| \leq 1 + R + |x + g| \leq (1 + R)(1 + |x + g|)$$

Therefore for any $M \geq 1$ we have

$$\begin{aligned} |\tau_g v[\phi]| &\leq \frac{C(1+R)^M}{(1+|g|)^M} \sup_{x \in K, |\alpha| \leq N} (1+|x+g|)^M |D^\alpha \phi(x+g)| \\ &\leq \frac{C(1+R)^M}{(1+|g|)^M} \sup_{y \in \mathbb{R}^n, |\alpha| \leq N} (1+|y|)^M |D^\alpha \phi(y)| \end{aligned}$$

Since $\phi \in \mathcal{S}$ we have

$$\tau_g v[\phi] \leq \frac{C_{v,\phi}}{(1+|g|)^{n+1}}$$

Now it is easy to check that $\sum_{g \in \mathbb{Z}^n} \frac{1}{(1+|g|)^{n+1}}$ converges. Therefore for each $\phi \in \mathcal{S}$ the sum $\sum_{g \in \mathbb{Z}^n} \tau_g v[\phi]$ converges. This is precisely the statement that the sum converges in $\mathcal{S}^*$ with the weak-star topology.

Now let $u \in \mathcal{D}^*(\mathbb{R}^n)$ be periodic and take ψ as in the previous proposition. Let $\phi \in \mathcal{D}(\mathbb{R}^n)$ be arbitrary. Then we have

$$u[\phi] = \left(\sum_{g \in \mathbb{Z}^n} \tau_g \psi \right) u[\phi] = \left(\sum_{g \in \mathbb{Z}^n} u[\tau_g \psi \phi] \right)$$

Since u is periodic,

$$u[\tau_g \psi \phi] = u[\psi \tau_{-g} \phi] = \psi u[\tau_{-g} \phi] = \tau_g(\psi u)[\phi]$$

Now ψu has compact support so it extends uniquely (see the subsection on distributions with compact support) to a distribution $v \in \mathcal{E}^*(\mathbb{R}^n)$. Thus we have

$$u = \sum_{g \in \mathbb{Z}^n} \tau_g v$$

By the first part of the proof this sum converges in $\mathcal{S}^*$. By earlier in the proof if we set $M = n + 1$ then what happens is that the sum is bounded by a finite multiple of a seminorm in $\mathcal{S}$ so we indeed have continuity in $\mathcal{S}^*$. □

Since periodic distributions are tempered, it makes sense to talk about their Fourier transform. As it turns out, the result is a generalization of a Fourier series, which makes sense.

Proposition. Suppose that $u \in \mathcal{S}^*$ satisfies $(e_{-g'} - 1)u = 0$ for all $g' \in \mathbb{Z}^n$. Then

$$u = \sum_{g \in \mathbb{Z}^n} c_g \delta_{2\pi g}$$

where $c_g \in \mathbb{C}$ satisfy $|c_g| \leq K(1+|g|)^N$ for some $K > 0, N \in \mathbb{Z}$ and the sum converges in $\mathcal{S}^*$.

Proof. Define $\Lambda = \{2\pi g : g \in \mathbb{Z}^n\}$. Let $\phi \in \mathcal{D}(\mathbb{R}^n)$ with $\text{supp } \phi \subset \mathbb{R}^n \setminus \Lambda$.

Since $\text{supp}(\phi)$ is a compact set disjoint from Λ , for every $x \in \text{supp}(\phi)$ there exists at least one g' (in fact one g' that is a basis vector) such that $e_{-g'}(x) - 1 \neq 0$. By compactness, we can use a finite partition of unity (see level 9.5) to write $\phi = \sum \phi_j$, where each ϕ_j is supported on a small neighbourhood where a specific $e_{-g'_j} - 1$ has no zeros. We can then divide to get $(e_{-g'_j} - 1)^{-1} \phi_j \in \mathcal{D}$, allowing us to use the condition to conclude $u[\phi_j] = 0$, and therefore $u[\phi] = 0$. Specifically, we have

$$0 = (e_{-g'_j} - 1) u \left[(e_{-g'_j} - 1)^{-1} \phi_j \right] = u[\phi_j]$$

This implies that $\text{supp } u \subseteq \Lambda$.

Now take ψ as earlier, i.e. the 'partition of unity' building block bump function thingy and define $\tilde{\psi} = \psi\left(\frac{x}{2\pi}\right)$. Note that $\sum_{g \in \mathbb{Z}^n} \tau_{2\pi g} \tilde{\psi} = 1$ and $\text{supp } \tilde{\psi} \subset \{x \in \mathbb{R}^n : \forall i |x_i| < 2\pi\}$.

For each $g \in \mathbb{Z}^n$ define $v_g = (\tau_{2\pi g} \tilde{\psi}) u$. This distribution is supported at $2\pi g$ and we have $(e_{-g'} - 1) v_g = 0$ by the assumption on u .

Taking $g' = e_j$ where $\{e_j\}$ is the standard basis for $\mathbb{R}^n$ we have

$$(e^{-i(x_j - 2\pi g_j)} - 1) v_g = (e^{-i(x_j)} - 1) v_g = 0$$

Also, there exists a function $\kappa(x_j)$ that is non-zero on a neighbourhood of $2\pi g_j$ such that

$$e^{-i(x_j - 2\pi g_j)} - 1 = (x_j - 2\pi g_j) \kappa(x_j)$$

Thus, we conclude that $(x_j - 2\pi g_j) v_g = 0$ for the reason that v_g is entirely supported at $2\pi g$ so we only care about test functions in a neighbourhood about that point. As for why you can divide a distributional equation by a non-zero compact supported (after maybe multiplying κ by a cutoff function that won't affect it since the support is one point) function like that: Remember that multiplying a distribution by a function is just changing the function it acts on.

Suppose $\phi \in \mathcal{S}$ and $g \in \mathbb{Z}^n$. Then we can write $\phi(x)$ as $\phi(2\pi g)$ plus a smooth function that vanishes at $2\pi g$, which we will write as $\sum_{j=1}^n (x_j - 2\pi g_j) \phi_j(x)$ for some smooth functions ϕ_j .

Since v_g has compact support it extends to act on $\mathcal{E}(\mathbb{R}^n)$ and we calculate

$$v_g[\phi] = v_g[\phi(2\pi g)] + \sum_{j=1}^n (x_j - 2\pi g_j) v_g[\phi_j] = v_g[\phi(2\pi g)]$$

By definition of v_g we have

$$(\tau_{2\pi g} \tilde{\psi}) u[\phi] = (\tau_{2\pi g} \tilde{\psi}) u[\phi(2\pi g)] = u \left[(\tau_{2\pi g} \tilde{\psi}) \phi(2\pi g) \right] = u \left[\tau_{2\pi g} \tilde{\psi} \right] \delta_{2\pi g}[\phi]$$

so that $(\tau_{2\pi g} \tilde{\psi}) u = u \left[\tau_{2\pi g} \tilde{\psi} \right] \delta_{2\pi g}$. Then summing gives

$$\sum_{g \in \mathbb{Z}^n} \left((\tau_{2\pi g} \tilde{\psi}) u \right) = \left(\sum_{g \in \mathbb{Z}^n} (\tau_{2\pi g} \tilde{\psi}) \right) u$$

The reason we can do this is the partial sums $S_N = \sum_{|g| \leq N} \tau_{2\pi g} \tilde{\psi}$ have their derivatives of all orders are uniformly bounded independently of N . Because ϕ decays faster than any polynomial, the sequence $S_N \phi$ converges to $(\sum_{g \in \mathbb{Z}^n} \tau_{2\pi g} \tilde{\psi}) \phi = 1 \cdot \phi = \phi$ in the topology of $\mathcal{S}$ - this is a straight forward Leibniz rule application. By the linearity and continuity of u , the limit can be pulled outside: $u[\lim_{N \rightarrow \infty} S_N \phi] = \lim_{N \rightarrow \infty} u[S_N \phi]$.

To be really explicit (Since this is similar to the density proof above but without the extra $k^{-|\lambda|}$ factor, which causes it to look a bit suspicious):

By the definition of the partition of unity, the infinite sum converges to exactly 1 everywhere. For any finite N , $S_N(x) = 1$ identically on an expanding ball $B_{R_N}(0)$, where the radius $R_N \rightarrow \infty$ as $N \rightarrow \infty$.

Consequently, on the region $|x| \leq R_N$:

$$1 - S_N(x) = 0 \text{ and } D^\gamma(1 - S_N(x)) = 0$$

To show $S_N\phi \rightarrow \phi$ in $\mathcal{S}$, we evaluate the supremum of $x^\alpha D^\beta[(1 - S_N(x))\phi(x)]$. Using the Leibniz rule, this expands into a finite sum of terms of the form:

$$x^\alpha D^\gamma(1 - S_N(x))D^{\beta-\gamma}\phi(x)$$

We split the supremum into two regions. For $|x| \leq R_N$: Every term is exactly 0 because the derivatives of $1 - S_N(x)$ are 0. For $|x| > R_N$: The derivatives $D^\gamma(1 - S_N(x))$ do not vanish, but they are bounded by our uniform constant C_γ . Therefore, each term is bounded by:

$$C_\gamma \sup_{|x| > R_N} |x^\alpha D^{\beta-\gamma}\phi(x)|$$

Because $\phi \in \mathcal{S}$, the function $x^\alpha D^{\beta-\gamma}\phi(x)$ decays to 0 as $|x| \rightarrow \infty$. Since we are only taking the supremum over the region $|x| > R_N$, and $R_N \rightarrow \infty$, that supremum must go to 0 as a linear combination of things that go to 0.

Thus, we have

$$\sum_{g \in \mathbb{Z}^n} \left((\tau_{2\pi g} \tilde{\psi}) u \right) = \left(\sum_{g \in \mathbb{Z}^n} (\tau_{2\pi g} \tilde{\psi}) \right) u = u = \sum_{g \in \mathbb{Z}^n} c_g \delta_{2\pi g}$$

where $c_g = u \left[\tau_{2\pi g} \tilde{\psi} \right]$.

This proves the sum converges in $\mathcal{S}^*$. It remains to check the polynomial bound.

Recall that by the characterization of continuity for maps $u : \mathcal{S} \rightarrow \mathbb{C}$ there exist $N, k \in \mathbb{N}$ and $C > 0$ such that for all $\phi \in \mathcal{S}$ we have

$$|u[\phi]| \leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} \left| (1 + |x|)^N D^\alpha \phi(x) \right|$$

Applying this to $\tau_{2\pi g} \tilde{\psi}$ gives

$$\begin{aligned} |c_g| &\leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} \left| (1 + |x|)^N D^\alpha \tilde{\psi}(x - 2\pi g) \right| \\ &\leq C \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} \left| (1 + |x + 2\pi g|)^N D^\alpha \tilde{\psi}(x) \right| \\ &\leq C' \sup_{x \in \mathbb{R}^n, |\alpha| \leq k} \left| (1 + |x|)^N D^\alpha \tilde{\psi}(x) \right| \times (1 + |g|)^N \leq K (1 + |g|)^N \end{aligned}$$

□

Theorem. Suppose $u \in \mathcal{D}^*(\mathbb{R}^n)$ is a periodic distribution. Then there exist constants $c_g \in \mathbb{C}$ such that

$$u = \sum_{g \in \mathbb{Z}^n} c_g T_{e_{2\pi g}}$$

where $c_g = M(e_{-2\pi g} u)$ and we have $|c_g| \leq K(1 + |g|)^N$ for some $K > 0, N \in \mathbb{Z}$

Proof. Since u is periodic it is tempered so we may talk about its Fourier transform. For all $g' \in \mathbb{Z}^n$ we have $\tau_{g'} u = u$ which implies that $e_{-g'} \hat{u} = \hat{u}$ so $(e_{-g'} - 1) \hat{u} = 0$. Now we go like “ohhhh that’s why we set up the previous proposition that way!” so we can write $\hat{u} = (2\pi)^n \sum_{g \in \mathbb{Z}^n} c_g \delta_{2\pi g}$ for some c_g satisfying the required bound.

Since this sum converges in $\mathcal{S}^*$ and the Fourier transform is a homeomorphism, we may take the inverse Fourier transform to get that the sum $\sum_{g \in \mathbb{Z}^n} c_g T_{e_{2\pi g}}$ converges in $\mathcal{S}^*$. As for why $(2\pi)^n \delta_x$ is the Fourier transform of T_{e_x} , we conveniently did this in an earlier example.

To establish the formula for c_g we make use of the earlier observation that $M(T_f) = \int_{\bar{q}} f(x) dx$ to get that

$$M\left(e_{-2\pi g} T_{e_{2\pi g'}}\right) = \int_{\bar{q}} e^{2\pi i(g-g') \cdot x} dx = \delta_{gg'}$$

By definition, the mean of a periodic distribution is defined as $M(v) := v[\psi]$, where $\psi \in C_c^\infty(\mathbb{R}^n)$ is a fixed partition of unity function from earlier. Applying this to $u \mapsto M(e_{-2\pi g}u)$:

$$M(e_{-2\pi g}u) = (e_{-2\pi g}u)[\psi] = u[e_{-2\pi g}\psi]$$

Because ψ is smooth and compactly supported, the product $e_{-2\pi g}\psi$ is a fixed test function belonging to the Schwartz space $\mathcal{S}$. By the definition of the weak-* topology on $\mathcal{S}^*$, the evaluation functional $u \mapsto u[\phi]$ is continuous for any fixed test function $\phi \in \mathcal{S}$. Therefore, the map $u \mapsto u[e_{-2\pi g}\psi]$ is continuous. Therefore, since continuous functions commute with limits, we have

$$M(e_{-2\pi g}u) = \sum_{g' \in \mathbb{Z}^n} c_g M(e_{-2\pi g} T_{e_{2\pi g'}}) = c_g$$

To see why that last sum was justified: Convergence of the series $u = \sum c_{g'} T_{e_{2\pi g'}}$ in $\mathcal{S}^*$ means its sequence of partial sums S_N converges to u in the weak-* topology. By the definition of the weak-* topology, a sequence $S_N \rightarrow u$ in $\mathcal{S}^*$ if and only if $S_N[\phi] \rightarrow u[\phi]$ in $\mathbb{C}$ for every test function $\phi \in \mathcal{S}$. Since the map $u \mapsto M(e_{-2\pi g}u)$ is exactly the evaluation of u on the fixed test function $\phi = e_{-2\pi g}\psi$, it commutes with the limit of the partial sums by the very definition of convergence in $\mathcal{S}^*$. □

I just realized we're literally looking at functions on the set of linear maps on the space of functions. No wonder this is confusing.

We often write the series as $u = \sum_{g \in \mathbb{Z}^n} c_g e_{2\pi g}$ where convergence is understood to be in $\mathcal{S}^*$ and we may not always distinguish the function $e_{2\pi g}$ from the distribution it defines.

(Note: Convergence in $\mathcal{D}$ is stronger than convergence in $\mathcal{S}$ but once we take duals we are working with the weak-star topology so the order flips, so the limit in $\mathcal{S}^*$ defines a valid distribution)

Example. Consider $u = \sum_{g \in \mathbb{Z}^n} \delta_g$. Then since $\delta_0 \in \mathcal{E}^*(\mathbb{R}^n)$ an earlier proposition implies that this defines a periodic distribution. If ψ is a partition of unity function as earlier, then since this is supported inside the side length 2 cube about the origin we have that $\tau_g \psi(0) = 0$ for $g \in \mathbb{Z}^n$ with $g \neq 0$. Thus, since $\sum_{g \in \mathbb{Z}^n} \tau_g \psi = 1$ we must have $\psi(0) = 1$. We then calculate

$$c_g = M(e_{-2\pi g}u) = u[\psi e_{-2\pi g}] = \psi(0) e^{-2\pi i g \cdot 0} = 1$$

Thus, we get Poisson's formula:

$$\sum_{g \in \mathbb{Z}^n} \delta_g = \sum_{g \in \mathbb{Z}^n} T_{e_{2\pi g}}$$

This is sometimes written, using abuse of notation, as

$$\sum_{g \in \mathbb{Z}^n} \delta(x - g) = \sum_{g \in \mathbb{Z}^n} e^{2\pi i g \cdot x}$$

I'm willing to bet we'll see it written this way in the Methods course.

We can now deduce some properties of Fourier series.

Lemma. Differentiation is a continuous map from $\mathcal{S}^*$ to itself

Proof.

$$p_\phi(D_j u) = |u[D_j \phi]| = p_{D_j \phi}(u)$$

Since $D_j \phi \in \mathcal{S}$, the operator D_j is bounded by a defining seminorm and is therefore continuous - see the characterization of continuity of maps between topological vector spaces that was proved in the topological vector space chapter. Earlier work implies that weak-star topologies are topological vector spaces. □

Proposition. Let u be a periodic distribution and suppose c_g are its Fourier coefficients. Then $D_j u$ is a periodic distribution and has Fourier series

$$D_j u = \sum_{g \in \mathbb{Z}^n} (2\pi i g_j c_g) T_{e_{2\pi g}}$$

Proof. Since differentiation is continuous as established earlier, for a similar reason to the above with the other continuous map and swapping the sum, we may differentiate term by term and then the result follows because $D_j T_{e_{2\pi g}} = (2\pi i g_j) T_{e_{2\pi g}}$.

□

Lemma. Let $1 \leq p < \infty$ and let $f \in L^p_{loc}(\mathbb{R}^n)$ be periodic and let Q be the open cube centered at the origin of side length 1. Then for all $\varepsilon > 0$ there exists a smooth periodic function f_ε such that $\|f - f_\varepsilon\|_{L^p(Q)} < \varepsilon$

Proof. Let $Q = 2q$. Then by general analysis there exists $h_\varepsilon \in C^\infty(\mathbb{R}^n)$ such that $\text{supp } h_\varepsilon \subset Q$ and $\|f1_Q - h_\varepsilon\|_p < \varepsilon$, since compact supported smooth functions are dense in L^p .

Define $f_\varepsilon = \sum_{g \in \mathbb{Z}^n} \tau_g h_\varepsilon$, then this is clearly smooth and periodic.

Now consider $\|f - f_\varepsilon\|_p$, then this picks up ε from $\|f1_Q - h_\varepsilon\|_p < \varepsilon$ and at most $(3^n - 1)\varepsilon$ from the small extra bits from the bordering copies of h_ε , so that $\|f - f_\varepsilon\|_p < 3^n \varepsilon$. When n is fixed, this is just a constant, so done.

□

Proposition. Let $f \in L^1_{loc}(\mathbb{R}^n)$, then

$$|c_g| \leq \|f\|_{L^1(Q)}$$

and $c_g \rightarrow 0$ as $|g| \rightarrow \infty$

Proof. Note that

$$|c_g| = \left| \int_Q e^{-2\pi i g \cdot x} f(x) dx \right| \leq \int_Q |f(x)| dx = \|f\|_{L^1(Q)}$$

Now let f_ε be such that $\|f - f_\varepsilon\|_{L^p(Q)} < \varepsilon/2$ as in the lemma above. Suppose f_ε has Fourier coefficients c'_g . Since $D_j D_j f_\varepsilon$ is locally integrable we have that $|g|^2 |c'_g| < C$ by the Fourier coefficients of the derivatives and the fact that we know Fourier coefficients are bounded.

This implies that there exists $R > 0$ such that $|c'_g| < \frac{\varepsilon}{2}$ when $|g| > R$. We have

$$|c_g - c'_g| \leq \|f - f_\varepsilon\|_{L^1(Q)} < \frac{\varepsilon}{2}$$

so we conclude that for $|g| > R$

$$|c_g| = |c_g - c'_g + c'_g| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus $c_g \rightarrow 0$ as $|g| \rightarrow \infty$.

□

Proposition. Let $f \in C^{n+1}(\mathbb{R}^n)$ be periodic. Then

$$f(x) = \sum_{g \in \mathbb{Z}^n} c_g e^{2\pi i g \cdot x}$$

with the sum converging uniformly.

Proof. Have $D^\alpha f \in L^1_{loc}(\mathbb{R}^n)$ for $|\alpha| < n + 1$. By boundedness of the coefficients of $D^\alpha f$ and what happens when we differentiate we conclude $|c_g| \leq K(1 + |g|)^{-(n+1)}$ for some $K > 0$. Thus the partial sums converge uniformly by the Weierstrass M -test. Uniform convergence implies convergence in $\mathcal{S}^*$ and we conclude because the mapping between continuous functions and distributions is injective so we get convergence to the correct limit.

□

Example. In $\mathbb{R}$ we conclude that the Fourier series of any twice continuously differentiable periodic function converges uniformly to the function. That's quite nice.

Remark. Furthermore, we can conclude from the same method as the above proof that more differentiable functions have faster convergent Fourier series.

Proposition. To generalize the L^2 theorem in $\mathbb{R}$ from linear analysis to $\mathbb{R}^n$:

Let $f, h \in L^2_{loc}(\mathbb{R}^n)$ be periodic with Fourier coefficients f_g, h_g respectively. Then we have yet another version of Parseval's formula:

$$\int_q \overline{f(x)} h(x) dx = \sum_{g \in \mathbb{Z}^n} \overline{f_g} h_g$$

Also, the sum $f(x) = \sum_{g \in \mathbb{Z}^n} c_g e^{2\pi i g \cdot x}$ converges in $L^2(q)$.

Proof. First, suppose that $f, h \in C^\infty(\mathbb{R}^n)$. Then the previous theorem implies

$$f(x) = \sum_{g \in \mathbb{Z}^n} f_g e^{2\pi i g \cdot x} \text{ and } h(x) = \sum_{g \in \mathbb{Z}^n} h_g e^{2\pi i g \cdot x}$$

with $\sup_{g \in \mathbb{Z}^n} (1 + |g|)^N |f_g| < \infty$ for all $N \in \mathbb{N}$ and similarly for h_g . We have (dominated convergence for any dodgy steps)

$$\begin{aligned} \int_q \overline{f(x)} h(x) dx &= \int_q \left(\sum_{g \in \mathbb{Z}^n} \overline{f_g} e^{-2\pi i g \cdot x} \right) \left(\sum_{g' \in \mathbb{Z}^n} h_{g'} e^{2\pi i g' \cdot x} \right) dx \\ &= \sum_{g \in \mathbb{Z}^n} \sum_{g' \in \mathbb{Z}^n} \overline{f_g} h_{g'} \int_q e^{2\pi i (g' - g) \cdot x} dx \\ &= \sum_{g \in \mathbb{Z}^n} \sum_{g' \in \mathbb{Z}^n} \overline{f_g} h_{g'} \delta_{gg'} = \sum_{g \in \mathbb{Z}^n} \overline{f_g} h_g \end{aligned}$$

In particular, we have that

$$\|f\|_{L^2(q)} = \|f_g\|_{l^2(\mathbb{Z}^n)}$$

(Note: the $l^2(\mathbb{Z}^n)$ norm is such that $\|f_g\|_{l^2(\mathbb{Z}^n)} = \sqrt{\sum_{g \in \mathbb{Z}^n} |f_g|^2}$)

But this was under the assumption that f, h were smooth. If they are in L^2_{loc} we can find $f^{(k)} \in C^\infty(\mathbb{R}^n)$ with Fourier coefficients $f_g^{(k)}$ such that

$$\|f - f^{(k)}\|_2 < \frac{1}{k}$$

Note that since q is a finite measure space, $f \in L^1(q)$ as well. By Cauchy-Schwarz we have

$$\|f\|_{L^1(q)} = \int_q |f| \leq \sqrt{\int_q |f|^2} \sqrt{\int_q 1} = \|f\|_{L^2(q)}$$

Therefore

$$\|f_g - f_g^{(k)}\|_{l^\infty(\mathbb{Z}^n)} := \sup_{g \in \mathbb{Z}^n} |f_g - f_g^{(k)}| < \frac{1}{k}$$

Since $f^{(k)}$ is Cauchy in $L^2(q)$ we have that $\{f_g^{(k)}\}$ is a Cauchy sequence in $l^2(\mathbb{Z}^n)$ so it converges in the space $l^2(\mathbb{Z}^n)$ of $\mathbb{Z}^n$ -tuples of complex numbers that have a finite norm as defined above. In particular, its limit f_g is in l^2 by completeness of that space.

We also know that $f_g^{(k)} \rightarrow f_g$ in $l^\infty(\mathbb{Z}^n)$ and so f_g must be the identity of the limit of the Cauchy sequence in l^2 . Taking a similar sequence $h^{(k)}$ of smooth functions with Fourier coefficients $h_g^{(k)}$ and using the above identity for smooth functions

$$(f^{(k)}, h^{(k)})_{L^2(q)} = (f_g^{(k)}, h_g^{(k)})_{l^2(\mathbb{Z}^n)}$$

the result follows on letting $k \rightarrow \infty$, since inner products are continuous in Hilbert spaces. The limit actually exists because as discussed above f_g is actually in l^2 .

□

5 Sobolev spaces and Partial differential equations

5.1 Weak derivatives and the $W^{k,p}(\Omega)$ spaces

Definition. Let $\Omega \subseteq \mathbb{R}^n$ be an open set. For $k \in \mathbb{Z}_{\geq 0}$ and $1 \leq p \leq \infty$ we say that a function $f \in L^p$ belongs to the Sobolev space $W^{k,p}(\Omega)$ if for any $|\alpha| \leq k$ there exists $f^\alpha \in L^p(\Omega)$ with $D^\alpha T_f = T_{f^\alpha}$. We say f^α is the weak, or distributional derivative of f .

That was kind of two definitions (weak derivative and Sobolev space) in one so let's unpack it.

Example. If H is the Heaviside step function then $1+2H$ is a weak derivative of $|x|$. This means that $|x| \in W^{1,1}((-1,1))$ despite not having a classical derivative.

Proposition. Weak derivatives are unique up to a.e. equality

Proof. Suppose v and w in $L^p(\Omega)$ are both weak α -derivatives of $f \in L^p(\Omega)$. By the definition of the weak derivative, for any test function $\phi \in C_c^\infty(\Omega)$, we have:

$$\begin{aligned} \int_{\Omega} f D^\alpha \phi \, dx &= (-1)^{|\alpha|} \int_{\Omega} v \phi \, dx \\ \int_{\Omega} f D^\alpha \phi \, dx &= (-1)^{|\alpha|} \int_{\Omega} w \phi \, dx \end{aligned}$$

Subtracting the second equation from the first yields:

$$\int_{\Omega} (v - w) \phi \, dx = 0$$

Let $u = v - w \in L^p(\Omega)$. We just showed that $\int_{\Omega} u \phi \, dx = 0$ for all $\phi \in C_c^\infty(\Omega)$

If $1 < p \leq \infty$, the conjugate exponent q satisfies $1 \leq q < \infty$. The map $g \mapsto \int_{\Omega} u g \, dx$ is a bounded linear functional on $L^q(\Omega)$. Because $C_c^\infty(\Omega)$ is dense in $L^q(\Omega)$ for $q < \infty$, this continuous functional uniquely extends to 0 on all of $L^q(\Omega)$. By the duality $(L^q)^* \cong L^p$, this implies $u = 0$ as a vector in $L^p(\Omega)$, so $u = 0$ a.e. If $p = 1$, then $q = \infty$. The density argument directly fails here because $C_c^\infty(\Omega)$ is not dense in $L^\infty(\Omega)$ under the L^∞ norm.

To handle $p = 1$, we use mollification. Let ϕ_ε be a standard symmetric mollifier. Therefore, the convolution $u * \phi_\varepsilon(x) = \int_{\Omega} u(y) \phi_\varepsilon(x - y) dy$ is exactly 0 everywhere on this interior domain. Since $u \in L^p(\Omega) \subset L^1_{loc}(\Omega)$, standard mollifier properties ensure that $u * \phi_\varepsilon \rightarrow u$ in $L^1_{loc}(\Omega)$ (and pointwise a.e. by the Lebesgue Differentiation Theorem) as $\varepsilon \rightarrow 0$. This forces $u = 0$ a.e.

□

Example. I should actually show why $2H(x) - 1$ is the weak derivative of $|x|$ as an example.

$$f'[\phi] = -f[\phi'] = - \int_{-\infty}^{\infty} |x| \phi'(x) dx$$

Split the integral at $x = 0$ to substitute the piecewise definition of the absolute value:

$$= - \left(\int_{-\infty}^0 (-x) \phi'(x) dx + \int_0^{\infty} x \phi'(x) dx \right)$$

Apply integration by parts to each integral independently. For the first term on $(-\infty, 0)$:

$$\int_{-\infty}^0 (-x) \phi'(x) dx = [-x \phi(x)]_{-\infty}^0 - \int_{-\infty}^0 (-1) \phi(x) dx$$

Since ϕ has compact support, ϕ vanishes as $x \rightarrow -\infty$. At the upper limit $x = 0$, the term evaluates to $-0 \cdot \phi(0) = 0$. The boundary terms vanish entirely, leaving $\int_{-\infty}^0 \phi(x) dx$. For the second term, similar reasoning gives that it is $-\int_0^\infty \phi(x) dx$. Substituting these resolved integrals back into the original split equation yields:

This can be combined into a single integral over $\mathbb{R}$:

$$= \int_{-\infty}^{\infty} (2H(x) - 1)\phi(x) dx$$

Because this integral relation $\int f' \phi = \int (2H - 1)\phi$ holds for all test functions $\phi \in C_c^\infty(\mathbb{R})$, the weak derivative of $|x|$ is strictly $2H(x) - 1$ almost everywhere.

Similar reasoning works for any piecewise differentiable continuous function.

$W^{k,p}(\Omega)$ may be equipped with the norm

$$\|f\|_{W^{k,p}(\Omega)} := \sum_{|\alpha| \leq k} \|D^\alpha f\|_p$$

Proposition. With this norm, we have a complete metric space.

Proof. Let (f_m) be a Cauchy sequence in $W^{k,p}(\Omega)$. By the definition of the norm $\|f\|_{W^{k,p}(\Omega)} = \sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p(\Omega)}$, the sequence of weak derivatives $(D^\alpha f_m)$ must be a Cauchy sequence in $L^p(\Omega)$ for every multi-index α with $|\alpha| \leq k$. Because $L^p(\Omega)$ is a Banach space, there exist limit functions $g_\alpha \in L^p(\Omega)$ such that $D^\alpha f_m \rightarrow g_\alpha$ in $L^p(\Omega)$. Let $f := g_0$ be the limit of the functions themselves. To verify that g_α is actually the weak derivative $D^\alpha f$, take any test function $\phi \in C_c^\infty(\Omega)$. For each m , the definition of the weak derivative states:

$$\int_{\Omega} (D^\alpha f_m) \phi dx = (-1)^{|\alpha|} \int_{\Omega} f_m (D^\alpha \phi) dx$$

Take the limit as $m \rightarrow \infty$ on both sides. Because $D^\alpha f_m \rightarrow g_\alpha$ and $f_m \rightarrow f$ in $L^p(\Omega)$, and because ϕ and $D^\alpha \phi$ are bounded with compact support (thus belonging to the conjugate space $L^q(\Omega)$), Holder's inequality ensures the integrals converge to:

$$\int_{\Omega} g_\alpha \phi dx = (-1)^{|\alpha|} \int_{\Omega} f (D^\alpha \phi) dx$$

This identity holds for all $\phi \in C_c^\infty(\Omega)$, which is the exact definition of g_α being the weak derivative $D^\alpha f$. Since $D^\alpha f = g_\alpha \in L^p(\Omega)$ for all $|\alpha| \leq k$, we know $f \in W^{k,p}(\Omega)$. Furthermore, since $D^\alpha f_m \rightarrow D^\alpha f$ in $L^p(\Omega)$ for all these multi-indices, $f_m \rightarrow f$ in the $W^{k,p}(\Omega)$ norm, meaning every Cauchy sequence converges to a limit within the space. □

Crudely speaking, k tells us how differentiable our function is and p tells us how integrable our function is. If p is larger we have less spiky functions and if k is larger we have smoother functions.

5.2 The space $H^s(\mathbb{R}^n)$

We will now specialize to the case $p = 2$ and $\Omega = \mathbb{R}^n$, as properties of general Sobolev spaces are beyond the scope of this course. This is good because we now have a Hilbert space.

Also, if we let $f \in L^2(\mathbb{R}^n)$ and $\hat{f}$ be the Fourier-Plancherel transform.

Proposition. If $f \in L^2$ then $\widehat{T_f} = T_{\hat{f}}$

Proof. Let $f_j \in \mathcal{S}$ be a sequence approximating f in $L^2(\mathbb{R}^n)$. For any test function $\phi \in \mathcal{S}$: By definition of the distributional transform, $\widehat{T_f}[\phi] = \int f \hat{\phi}$. Since $\hat{\phi} \in \mathcal{S} \subset L^2(\mathbb{R}^n)$, the L^2 inner product is continuous, giving $\int f \hat{\phi} = \lim_{j \rightarrow \infty} \int f_j \hat{\phi}$. The L^2 Fourier transform $\hat{f}$ is defined such that $f_j \rightarrow \hat{f}$ in $L^2(\mathbb{R}^n)$ because it is an isometry. Because $\phi \in \mathcal{S} \subset L^2(\mathbb{R}^n)$, we have $T_{\hat{f}}[\phi] = \int \hat{f} \phi = \lim_{j \rightarrow \infty} \int \hat{f}_j \phi$. For Schwartz functions f_j and ϕ , Fubini's theorem gives $\int f_j \hat{\phi} = \int \hat{f}_j \phi$. Equating the limits proves $\widehat{T_f}[\phi] = T_{\hat{f}}[\phi]$, meaning $\widehat{T_f} = T_{\hat{f}}$.

□

Proposition. $f \in L^2(\mathbb{R}^n)$ belongs to $W^{k,2}(\mathbb{R}^n)$ if and only if

$$\int_{\mathbb{R}^n} \left(1 + |\xi|^2\right)^k \left|\hat{f}(\xi)\right|^2 d\xi < \infty$$

Proof. By definition, $f \in W^{k,2}(\mathbb{R}^n)$ if and only if its weak derivatives $D^\alpha f \in L^2(\mathbb{R}^n)$ for all $|\alpha| \leq k$. From basic properties earlier, taking the Fourier transform on tempered distributions yields $\widehat{D^\alpha f}(\xi) = (i\xi)^\alpha \hat{f}(\xi)$.

Since the Fourier transform is an isometry on $L^2(\mathbb{R}^n)$ (up to constants), $D^\alpha f \in L^2(\mathbb{R}^n)$ if and only if $\xi^\alpha \hat{f}(\xi) \in L^2(\mathbb{R}^n)$. For f to be in $W^{k,2}(\mathbb{R}^n)$, this must hold for all $|\alpha| \leq k$, meaning the integral $\int_{\mathbb{R}^n} \sum_{|\alpha| \leq k} |\xi^\alpha|^2 |\hat{f}(\xi)|^2 d\xi$ must be finite.

By expanding, $(1 + |\xi|^2)^k = (1 + \xi_1^2 + \dots + \xi_n^2)^k$ expands exactly into a linear combination of $\xi^{2\alpha}$ (which is $|\xi^\alpha|^2$) for all $|\alpha| \leq k$ with strictly positive coefficients. Because the weight $(1 + |\xi|^2)^k$ and the sum $\sum_{|\alpha| \leq k} |\xi^\alpha|^2$ bound each other above and below by strictly positive constants, integrating $|\hat{f}(\xi)|^2$ against either expression tests for the exact same L^2 convergence.

□

Given this definition, we can generalize to k a non-integer. We do this noting that $L^2 \subset \mathcal{S}^*$ so the following definition indeed agrees for integers.

Definition. f is said to belong to $H^s(\mathbb{R}^n)$ if and only if

$$\int_{\mathbb{R}^n} \left(1 + |\xi|^2\right)^s \left|\hat{f}(\xi)\right|^2 d\xi < \infty$$

The square root of the above quantity is the norm of f in H_s .

Note that f will be taken to mean both the function and the distribution from now on.

Proposition. These are actually Hilbert spaces

Proof. The inner product that makes it a Hilbert space is defined as

$$\langle f, g \rangle_{H^s} = \int_{\mathbb{R}^n} (1 + |\xi|^2)^s \overline{\hat{f}(\xi)} \hat{g}(\xi) d\xi$$

To show completeness, observe that the map $f \mapsto \hat{f}(\xi)(1 + |\xi|^2)^{s/2}$ forms an isometry between $H^s(\mathbb{R}^n)$ and $L^2(\mathbb{R}^n)$. Because $L^2(\mathbb{R}^n)$ is complete, any Cauchy sequence $\{f_m\}$ in H^s maps to a Cauchy sequence in L^2 . This converges to some limit $g \in L^2$, which then pulls back to a valid limit $f = \mathcal{F}^{-1}[g(\xi)(1 + |\xi|^2)^{-s/2}]$ in H^s , completing the proof.

□

Theorem. Fix $k \in \mathbb{Z}_{\geq 0}$. Suppose that $f \in H^s(\mathbb{R}^n)$ for some $s > k + \frac{n}{2}$, then f is equivalent (i.e. equal a.e.) to a function in $C^k(\mathbb{R}^n)$. That is, $H^s(\mathbb{R}^n) \subset C^k(\mathbb{R}^n)$.

Proof. First assume $f \in \mathcal{S}(\mathbb{R}^n)$. Then by Fourier inversion for $|\alpha| \leq k$ we have

$$D^\alpha f(x) = \frac{i^{|\alpha|}}{(2\pi)^n} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \xi^\alpha \hat{f}(\xi) d\xi$$

By Cauchy-Schwarz:

$$|D^\alpha f(x)| = \frac{1}{(2\pi)^n} \left| \int_{\mathbb{R}^n} e^{ix \cdot \xi} \xi^\alpha \hat{f}(\xi) d\xi \right|$$

$$\begin{aligned} &\leq \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} |\xi^\alpha \hat{f}(\xi)| d\xi \\ &\leq \frac{1}{(2\pi)^n} \sqrt{\left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi \right) \left(\int_{\mathbb{R}^n} \frac{|\xi^\alpha|^2}{(1 + |\xi|^2)^s} d\xi \right)} \end{aligned}$$

We'll prove the right integral converges in a second.

Since $|\xi^\alpha|^2 \leq (1 + |\xi|^2)^k$ by expanding, we have

$$\frac{1}{(2\pi)^n} \sqrt{\left(\int_{\mathbb{R}^n} \frac{|\xi^\alpha|^2}{(1 + |\xi|^2)^s} d\xi \right)} \leq \frac{1}{(2\pi)^n} \sqrt{\left(\int_{\mathbb{R}^n} \frac{1}{(1 + |\xi|^2)^{s-k}} d\xi \right)} =: C_{n,k,s}$$

It is here that we used $s > k + \frac{n}{2}$, as it guarantees the integral converges. We thus have that

$$\sup_{|\alpha| \leq k, x \in \mathbb{R}^n} |D^\alpha f(x)| \leq C_{n,k,s} \|f\|_{H^s(\mathbb{R}^n)}$$

Now suppose $f \in H^s(\mathbb{R}^n)$. Then by taking the Fourier transform and then approximating with functions in $\mathcal{S}$ and pulling back, we have that f can be approximated by a sequence with $f_m \rightarrow f$ in $H^s(\mathbb{R}^n)$ and $f_m \in \mathcal{S}$. By taking a subsequence we have that $f_m \rightarrow f$ pointwise a.e.

Since f_m is Cauchy in H^s , the display formula above applied to $f_m - f_n$ implies that there exists $f^* \in C^k(\mathbb{R}^n)$ such that $D^\alpha f_m \rightarrow D^\alpha f^*$ uniformly for all $|\alpha| \leq k$ (remember earlier about the result from analysis about uniform convergence of derivatives generalizing by induction on $|\alpha|$).

Thus f^* is the function we want.

□

Theorem. (Bounded Linear Transformation Theorem) Let X be a normed space, Y a Banach space, and D a dense linear subspace of X . If $T : D \rightarrow Y$ is a bounded linear operator, then there exists a unique bounded linear operator $\tilde{T} : X \rightarrow Y$ such that $\tilde{T}x = Tx$ for all $x \in D$. Furthermore, $|\tilde{T}|_{X \rightarrow Y} = |T|_{D \rightarrow Y}$.

Proof. **Step 1.** Construction of $\tilde{T}$

Let $x \in X$. Since D is dense in X , there exists a sequence (x_n) in D such that $x_n \rightarrow x$. Because (x_n) is convergent, it is a Cauchy sequence. Since T is a bounded linear operator on D , we have:

$$|Tx_n - Tx_m| = |T(x_n - x_m)| \leq |T||x_n - x_m|$$

This inequality shows that (Tx_n) is a Cauchy sequence in Y . Since Y is a Banach space, it is complete, so (Tx_n) converges to some $y \in Y$. We define $\tilde{T}x = y$.

Step 2. Well-definedness: We must show $\tilde{T}x$ is independent of the choice of sequence.

Let (z_n) be another sequence in D converging to x . Then $x_n - z_n \rightarrow 0$. By the boundedness of T :

$$|Tx_n - Tz_n| \leq |T||x_n - z_n| \rightarrow 0$$

Therefore, $\lim_{n \rightarrow \infty} Tx_n = \lim_{n \rightarrow \infty} Tz_n$, so $\tilde{T}$ is well-defined. If $x \in D$, we can take the constant sequence $x_n = x$, yielding $\tilde{T}x = Tx$.

Step 3. Linearity

Exercise

Step 4. Boundedness and Norm

For any $x \in X$ and $(x_n) \rightarrow x$ in D , the continuity of the norm implies:

$$|\tilde{T}x| = \lim_{n \rightarrow \infty} |Tx_n| \leq \lim_{n \rightarrow \infty} |T||x_n| = |T||x|$$

This shows $\tilde{T}$ is bounded and $|\tilde{T}| \leq |T|$. Conversely, since $\tilde{T}$ is an extension of T , its supremum over the unit ball in X must be at least the supremum over the unit ball in D , giving $|\tilde{T}| \geq |T|$. Therefore, $|\tilde{T}| = |T|$.

Step 5. Uniqueness

Suppose $S : X \rightarrow Y$ is another bounded linear operator such that $Sx = Tx$ for all $x \in D$. Since S and $\tilde{T}$ are both continuous mappings that agree on a dense subset D , they must agree on all of X . Explicitly, for any $x \in X$ and $(x_n) \rightarrow x$ in D :

$$Sx = S\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} Sx_n = \lim_{n \rightarrow \infty} Tx_n = \tilde{T}x$$

This completes the proof. □

Theorem. Let $s > \frac{1}{2}$. Then there is a bounded linear map $T : H^s(\mathbb{R}^n) \rightarrow H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})$ such that

$$Tf = f|_{\{x_n=0\}}$$

for all $f \in H^s(\mathbb{R}^n) \cap C^0(\mathbb{R}^n)$

This is a special case of the Trace theorem. Before we prove it, it helps to understand what this is actually saying.

Since $\mathbb{R}^{n-1}$ is a measure 0 subset of $\mathbb{R}^n$, the function could take completely different values on $\mathbb{R}^{n-1}$ and still be equivalent. However, this theorem basically tells us what the value of f on $\mathbb{R}^{n-1} \times 0$ is in what we will call the trace sense.

Proof. Step 1. We will suppose $u \in \mathcal{S}(\mathbb{R}^n)$ and define $Tu \in \mathcal{S}(\mathbb{R}^{n-1})$ by $Tu(x') = u(x', 0)$. We want to show that if $\xi' \in \mathbb{R}^{n-1}$ then $\widehat{Tu}(\xi') = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{u}(\xi', \xi_n) d\xi_n$

By the Fourier Inversion Theorem, since $u \in \mathcal{S}(\mathbb{R}^n)$, we can express $u(x)$ as:

$$u(x', x_n) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} \hat{u}(\xi', \xi_n) e^{i(x' \cdot \xi' + x_n \xi_n)} d\xi_n d\xi'$$

Evaluating this at $x_n = 0$ yields $Tu(x')$:

$$Tu(x') = u(x', 0) = \frac{1}{(2\pi)^{n-1}} \int_{\mathbb{R}^{n-1}} \left(\frac{1}{2\pi} \int_{\mathbb{R}} \hat{u}(\xi', \xi_n) d\xi_n \right) e^{ix' \cdot \xi'} d\xi'$$

Since $Tu \in \mathcal{S}(\mathbb{R}^{n-1})$, we can also write $Tu(x')$ using the $(n-1)$ -dimensional Fourier inversion formula:

$$Tu(x') = \frac{1}{(2\pi)^{n-1}} \int_{\mathbb{R}^{n-1}} \widehat{Tu}(\xi') e^{ix' \cdot \xi'} d\xi'$$

We have now expressed $Tu(x')$ as the inverse Fourier transform of 2 functions. By the injectivity of the Fourier transform on Schwartz spaces, we can equate the terms being inverted to conclude:

$$\widehat{Tu}(\xi') = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{u}(\xi', \xi_n) d\xi_n$$

From this it follows from Holder's inequality that we have the following and the second integral converges since $s > \frac{1}{2}$.

$$\left| \widehat{Tu}(\xi') \right|^2 \leq \frac{1}{(2\pi)^2} \left(\int_{\mathbb{R}} \left(1 + |\xi|^2 \right)^s |\hat{u}(\xi', \xi_n)|^2 d\xi_n \right) \left(\int_{\mathbb{R}} \frac{d\xi_n}{\left(1 + |\xi|^2 \right)^s} \right)$$

Step 2. We want to show that there exists a constant $C(s)$ such that

$$\|Tu\|_{H^{s-\frac{1}{2}}(\mathbb{R}^{n-1})} \leq C(s) \|u\|_{H^s(\mathbb{R}^n)}$$

Consider the integral $\int_{\mathbb{R}} \frac{d\xi_n}{(1+|\xi|^2)^s}$. Call this integral $I(\xi')$. Since $\xi = (\xi', \xi_n)$, we know $|\xi|^2 = |\xi'|^2 + \xi_n^2$. Making the substitution $\xi_n = t\sqrt{1+|\xi'|^2}$, we get $d\xi_n = \sqrt{1+|\xi'|^2} dt$.

$$\begin{aligned} I(\xi') &= \int_{\mathbb{R}} \frac{1}{(1+|\xi'|^2 + \xi_n^2)^s} d\xi_n \\ I(\xi') &= \int_{\mathbb{R}} \frac{\sqrt{1+|\xi'|^2}}{(1+|\xi'|^2 + t^2(1+|\xi'|^2))^s} dt \\ I(\xi') &= (1+|\xi'|^2)^{\frac{1}{2}-s} \int_{\mathbb{R}} \frac{1}{(1+t^2)^s} dt \end{aligned}$$

Since it is given that $s > \frac{1}{2}$, the integral over t converges to a finite constant that depends only on s . Let $K(s) = \int_{\mathbb{R}} \frac{1}{(1+t^2)^s} dt$. Substituting $I(\xi')$ back into the inequality from step 1 yields:

$$|\widehat{Tu}(\xi')|^2 \leq \frac{K(s)}{2\pi} (1+|\xi'|^2)^{\frac{1}{2}-s} \int_{\mathbb{R}} (1+|\xi|^2)^s |\hat{u}(\xi', \xi_n)|^2 d\xi_n$$

By definition,

$$\|Tu\|_{H^{s-1/2}(\mathbb{R}^{n-1})}^2 = \int_{\mathbb{R}^{n-1}} (1+|\xi'|^2)^{s-1/2} |\widehat{Tu}(\xi')|^2 d\xi'$$

Substituting the bounded inequality:

$$\begin{aligned} \|Tu\|_{H^{s-1/2}(\mathbb{R}^{n-1})}^2 &\leq \int_{\mathbb{R}^{n-1}} (1+|\xi'|^2)^{s-1/2} \left[\frac{K(s)}{2\pi} (1+|\xi'|^2)^{\frac{1}{2}-s} \int_{\mathbb{R}} (1+|\xi|^2)^s |\hat{u}(\xi', \xi_n)|^2 d\xi_n \right] d\xi' \\ &= \frac{K(s)}{(2\pi)^2} \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} (1+|\xi|^2)^s |\hat{u}(\xi', \xi_n)|^2 d\xi_n d\xi' \end{aligned}$$

The double integral on the right is exactly the squared $H^s(\mathbb{R}^n)$ norm of u :

$$\|Tu\|_{H^{s-1/2}(\mathbb{R}^{n-1})}^2 \leq \frac{K(s)}{2\pi} \|u\|_{H^s(\mathbb{R}^n)}^2$$

Taking the square root of both sides gives

$$\|Tu\|_{H^{s-1/2}(\mathbb{R}^{n-1})} \leq C(s) \|u\|_{H^s(\mathbb{R}^n)}$$

where $C(s) = \sqrt{\frac{K(s)}{2\pi}}$.

Step 3. We will conclude the theorem.

From step 2, we have established that $T : \mathcal{S}(\mathbb{R}^n) \rightarrow H^{s-1/2}(\mathbb{R}^{n-1})$ is a bounded linear operator with the bound $\|Tu\|_{H^{s-1/2}(\mathbb{R}^{n-1})} \leq C(s) \|u\|_{H^s(\mathbb{R}^n)}$. Since $\mathcal{S}$ is dense in H^s , $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$. Since $H^{s-1/2}(\mathbb{R}^{n-1})$ is a complete normed space, the Bounded Linear Transformation theorem guarantees that T extends uniquely to a continuous linear operator $T : H^s(\mathbb{R}^n) \rightarrow H^{s-1/2}(\mathbb{R}^{n-1})$.

Therefore T extends to a genuine bounded linear map. We must prove it satisfies the condition in the theorem.

We must show that for any $f \in H^s(\mathbb{R}^n) \cap C^0(\mathbb{R}^n)$, this extended operator yields $Tf = f|_{\{x_n=0\}}$. We cannot rely on an arbitrary sequence in $\mathcal{S}(\mathbb{R}^n)$ approximating f in H^s , because H^s -convergence does not guarantee pointwise convergence on sets of measure zero (such as the hyperplane $x_n = 0$). We must construct a sequence that converges in both topologies simultaneously. Pick a smooth cutoff function $\psi_R \in C_c^\infty(\mathbb{R}^n)$ such that $\psi_R \equiv 1$ on $B_R(0)$, and let $g = \psi_R f$. We have $g \in H^s(\mathbb{R}^n) \cap C_c^0(\mathbb{R}^n)$. Let ϕ_ε be a standard mollifier and define $g_\varepsilon = g \star \phi_\varepsilon$. Because g has compact support, $g_\varepsilon \in C_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$. By standard properties of mollifiers, $g_\varepsilon \rightarrow g$ in $H^s(\mathbb{R}^n)$ as $\varepsilon \rightarrow 0$ and $g_\varepsilon \rightarrow g$ uniformly on $\mathbb{R}^n$ (it is here we use that f and therefore g is continuous). By the definition of T on $\mathcal{S}(\mathbb{R}^n)$, $Tg_\varepsilon(x') = g_\varepsilon(x', 0)$. Since T is continuous on $H^s(\mathbb{R}^n)$ and $g_\varepsilon \rightarrow g$ in $H^s(\mathbb{R}^n)$, $Tg_\varepsilon \rightarrow Tg$ in $H^{s-1/2}(\mathbb{R}^{n-1})$. Convergence in $H^{s-1/2}(\mathbb{R}^{n-1})$

implies convergence in $L^2(\mathbb{R}^{n-1})$. Simultaneously, the uniform convergence $g_\varepsilon \rightarrow g$ on $\mathbb{R}^n$ implies $g_\varepsilon(x', 0) \rightarrow g(x', 0)$ uniformly on $\mathbb{R}^{n-1}$, which also implies convergence in $L^2(\mathbb{R}^{n-1})$ by compact support. Since limits in $L^2(\mathbb{R}^{n-1})$ are unique, $Tg(x') = g(x', 0)$ almost everywhere. Because $R > 0$ is arbitrary and $\psi_R \equiv 1$ on $B_R(0)$, we conclude $Tf(x') = f(x', 0)$ almost everywhere on $\mathbb{R}^{n-1}$. □

Ok now I will crash out.

Notes for the II analysis of functions course often say “oh it trivially generalizes to other smooth manifolds by change of coordinates” to do a claimed proof of the general trace theorem well I don’t think it’s trivial I have many questions about how it works and it ought to be made explicit. Therefore, I will stick with the half plane case proved above and if differential equation boundary value problems happen in applied courses that need Sobolev spaces to be rigorous then it better be on a half-plane or I will have to ignore it. If anyone can find an explicit treatment of this that has these notes and earlier levels as background I’m open to suggestions but will only look when I feel ready as right now I just want to be done with this course to be honest this is way too hard for me to do before even starting IB but I have to because IB methods exists. Perhaps I can use a heuristic ansatz to solve such PDEs when they come up in applied courses and just verify the solutions. The proofs in my reference notes have quietly devolved into sketches and exercises that I have to make LLMs complete around the time I got to the Fourier stuff and that’s the reason I’m sick of it, not because I inherently dislike the course.

5.3 The space $H_0^1(\Omega)$

Definition. Let $\Omega \subset \mathbb{R}^n$ be open. Then $H_0^1(\Omega)$ is the completion of $C_c^\infty(\Omega)$ with respect to the $H^1(\mathbb{R}^n)$ norm.

Then this is a Hilbert space equipped with the inner product

$$(u, v)_{H^1} = \int_{\Omega} \left(\overline{Du(x)} \cdot Dv(x) + \overline{u(x)}v(x) \right) dx$$

By Parseval’s formula and properties of derivatives of Fourier transforms this is equal to the inner product from the standard H^1 norm.

If $f \in H_0^1(\Omega)$ then there exists a sequence f_n with $f_n \in C_c^\infty(\Omega)$ and $f_n \rightarrow f$ in $H^1(\mathbb{R}^n)$. For any open $U \subseteq \mathbb{R}^n$ we have

$$\|f_n - f\|_{L^2(U)} \leq \|f_n - f\|_{L^2(\mathbb{R}^n)} \leq \|f_n - f\|_{H^1(\mathbb{R}^n)}$$

Therefore we have $f_n|_U \rightarrow f|_U$ in L^2 . This means that if $U = \Omega^c$ then if $f \in H_0^1(\Omega)$ then $f|_U = 0$ a.e.

Proposition. Suppose Ω is of the form $\{x : a \cdot x > b\}$ for fixed vector $a \in \mathbb{R}^n$ and scalar $b \in \mathbb{R}$, then every $f \in H_0^1(\Omega)$ vanishes in the trace sense on $\{x : a \cdot x = b\}$

Proof. By the definition of $H_0^1(\Omega)$, any $f \in H_0^1(\Omega)$ is the limit in $H^1(\mathbb{R}^n)$ of a sequence (ϕ_m) in $C_c^\infty(\Omega)$. Because each ϕ_m has compact support strictly inside Ω , ϕ_m vanishes identically on $\partial\Omega$. Therefore, its trace is $T\phi_m = 0$.

The Trace theorem (special case above, after allowing you to rotate and move your head) establishes that T is a bounded (and thus continuous) linear map. By continuity, since $\phi_m \rightarrow f$ in $H^1(\mathbb{R}^n)$, their traces must converge: $T\phi_m \rightarrow Tf$ in $H^{1/2}(\partial\Omega)$. Since $T\phi_m = 0$ for all m , the limit Tf is also 0, meaning f vanishes on $\partial\Omega$ in the trace sense. □

This means that if we have a PDE problem with the condition of vanishing on a plane then to get a solution we can seek a solution in $H_0^1(\Omega)$.

Proposition. Let $B = \{v \in H : |v| \leq K\}$, and suppose $w_j \in B$ converges weakly to u ($w_j \rightharpoonup u$). Then $u \in B$.

Proof. By definition of weak convergence, for any $y \in H$, $\langle w_j, y \rangle \rightarrow \langle u, y \rangle$. Choose $y = u$. Then:

$$|u|^2 = \langle u, u \rangle = \lim_{j \rightarrow \infty} \langle w_j, u \rangle \leq \lim_{j \rightarrow \infty} |w_j| |u| \leq K |u|$$

If $u = 0$ then trivial else divide by $|u|$ then done.

□

Theorem. (Rellich-Kondrachov) Let $\Omega \subset \mathbb{R}^n$ be an open set with finite measure and let u_i be a bounded sequence in $H_0^1(\Omega)$. Then there exists $u \in H_0^1(\Omega)$ and a subsequence u_{i_j} such that $u_{i_j} \rightharpoonup u$ in $H_0^1(\Omega)$ and $u_{i_j} \rightarrow u$ in $L^2(\Omega)$.

Proof. By assumption there exists K such that

$$\|u_i\|_{L^2(\Omega)} \leq \|u_i\|_{H_0^1(\Omega)} \leq K$$

$H_0^1(\Omega)$ and $L^2(\Omega)$ are Hilbert spaces, and therefore reflexive. Because a reflexive space is isomorphic to its dual, the weak topology on the space coincides with the weak-star topology. Banach-Alaoglu therefore implies that closed, bounded balls in these spaces are weakly compact.

Since the weak-star topology on the closed unit ball is metric we have sequential compactness so there is a subsequence that converges weakly in one space and a further subsequence converging weakly in the other. If this subsequence is (w_j) our goal is to prove that in fact w_j has strong L^2 convergence.

Fix $\varepsilon > 0$ and note that there is $R > 0$ to be chosen later. Parseval's formula implies

$$\|w_j - u\|_2^2 = \frac{1}{(2\pi)^n} \|\hat{w}_j - \hat{u}\|_2^2 = \frac{1}{(2\pi)^n} \int_{|\xi| < R} |\hat{w}_j(\xi) - \hat{u}(\xi)|^2 d\xi + \frac{1}{(2\pi)^n} \int_{|\xi| \geq R} |\hat{w}_j(\xi) - \hat{u}(\xi)|^2 d\xi$$

We start by dealing with the second term.

Observe that if R is sufficiently large then, making use of the simple inequality $(a - b) \cdot (a - b) \leq 2((a.a) + (b.b))$ which is because $-2(a.b) \leq 2\|a\| \|b\| \leq \|a\|^2 + \|b\|^2 \leq (a.a) + (b.b)$ we get (using the proposition above in the second step)

$$\int_{|\xi| \geq R} |\hat{w}_j(\xi) - \hat{u}(\xi)|^2 d\xi \leq \frac{2}{R^2} \int_{|\xi| \geq R} |\xi|^2 (|\hat{w}_j(\xi)|^2 + |\hat{u}(\xi)|^2) d\xi \leq \frac{4K^2}{R^2}$$

Therefore if R is sufficiently large the $\frac{1}{(2\pi)^n} \int_{|\xi| \geq R} |\hat{w}_j(\xi) - \hat{u}(\xi)|^2 d\xi$ term is $< \frac{\varepsilon}{2}$. First note that $\hat{w}_j(\xi) = (w_j, e_{-\xi})_{L^2(\Omega)}$. Note that here we use Ω having finite measure as $e_{-\xi}$ needs to be in $L^2(\Omega)$.

Since $w_j \rightharpoonup u$ in L^2 it means by definition of weak convergence and the linear map $x \mapsto (x, e_{-\xi})$ that $\hat{w}_j(\xi) \rightarrow \hat{u}(\xi)$ pointwise.

For $|\xi| < R$ we have

$$\begin{aligned} |\hat{w}_j(\xi) - \hat{u}(\xi)|^2 &\leq 2|\hat{w}_j(\xi)|^2 + 2|\hat{u}(\xi)|^2 \leq 2\left(\|\hat{w}_j\|_\infty^2 + \|\hat{u}\|_\infty^2\right) \\ &\leq 2\left(\|w_j\|_1^2 + \|u\|_1^2\right) \leq 2\mu(\Omega)\left(\|w_j\|_2^2 + \|u\|_2^2\right) \end{aligned}$$

last step by Holder's inequality with the constant function 1

$$\leq 4K^2\mu(\Omega) \in L^1(B^R(0))$$

So Dominated convergence implies

$$\frac{1}{(2\pi)^n} \int_{|\xi| < R} |\hat{w}_j(\xi) - \hat{u}(\xi)|^2 d\xi \rightarrow 0$$

Therefore for j large enough we established $\|w_j - u\|_2^2 < \varepsilon$

□

Corollary. Let Ω have finite measure. If $L : L^2(\Omega) \rightarrow H_0^1(\Omega)$ is a bounded linear operator then L viewed as a map $L : L^2(\Omega) \rightarrow L^2(\Omega)$ is compact.

Proof. Let (f_n) be a bounded sequence in $L^2(\Omega)$. Since $L : L^2(\Omega) \rightarrow H_0^1(\Omega)$ is a bounded linear operator, the image sequence $u_n = Lf_n$ is a bounded sequence in $H_0^1(\Omega)$. By the theorem above, the bounded sequence (u_n) admits a subsequence (u_{n_k}) that converges strongly in $L^2(\Omega)$. Because every bounded sequence (f_n) in $L^2(\Omega)$ has a subsequence such that Lf_{n_k} converges in $L^2(\Omega)$, the operator $L : L^2(\Omega) \rightarrow L^2(\Omega)$ is compact by definition. $\square$

5.4 Applications to PDEs

This example will consider eigenfunctions of the Laplacian operator as well as properties of the E function we will define below.

For this first example let $\Omega \subset \mathbb{R}^n$ be open and bounded, let $u \in H_0^1(\Omega)$ and define $E[u] := \int_{\Omega} |Du|^2 dx$.

We first want to prove the following statement: Let $u_i \in H_0^1(\Omega)$ be a sequence such that $u_i \rightharpoonup u$. Then $E[u] \leq \liminf_i E[u_i]$ i.e. the E operator is weakly lower semicontinuous.

Consider the gradient operator $D : H_0^1(\Omega) \rightarrow L^2(\Omega)^n$. For any vector-valued function $v \in L^2(\Omega)^n$, the map $w \mapsto \int_{\Omega} Dw \cdot \bar{v} dx$ is a bounded linear functional on $H_0^1(\Omega)$ because, by the Cauchy-Schwarz inequality and the definition of the H^1 norm:

$$\left| \int_{\Omega} Dw \cdot \bar{v} dx \right| \leq |Dw|_{L^2} |v|_{L^2} \leq |w|_{H^1} |v|_{L^2}$$

Since $u_i \rightharpoonup u$ weakly in $H_0^1(\Omega)$, applying this bounded linear functional yields:

$$\int_{\Omega} Du_i \cdot \bar{v} dx \rightarrow \int_{\Omega} Du \cdot \bar{v} dx$$

for all $v \in L^2(\Omega)^n$. This means that $Du_i \rightharpoonup Du$ weakly in $L^2(\Omega)^n$. Using a result from Linear analysis that the norm is weakly lower semi-continuous in the normed space $L^2(\Omega)^n$, we have:

$$|Du|_{L^2} \leq \liminf_{i \rightarrow \infty} |Du_i|_{L^2}$$

Since the E operator is exactly the squared L^2 norm of the gradient ($E[u] = |Du|_{L^2}^2$) and the square function is strictly monotonically increasing on $[0, \infty)$, this directly implies:

$$E[u] \leq \liminf_{i \rightarrow \infty} E[u_i]$$

Next, define $\mathcal{E}_1 = \{E[u] : u \in H_0^1(\Omega), \|u\|_2 = 1\}$ and let $\lambda_1 := \inf(\mathcal{E}_1)$. Then we will now prove that there exists $w_1 \in H_0^1(\Omega)$ with $\|w_1\|_2 = 1$ and $E[w_1] = \lambda_1$ and $\lambda_1 > 0$.

Let $(u_n)_{n=1}^{\infty}$ be a minimizing sequence for the set $\mathcal{E}_1$. By definition, $u_n \in H_0^1(\Omega)$, $\|u_n\|_{L^2} = 1$, and $\lim_{n \rightarrow \infty} E[u_n] = \lambda_1$. Since $E[u_n]$ converges, it is bounded, meaning there exists some constant C such that $E[u_n] \leq C$ for all n . The H^1 norm of u_n is given by:

$$\|u_n\|_{H^1}^2 = \|u_n\|_{L^2}^2 + \|Du_n\|_{L^2}^2 = 1 + E[u_n] \leq 1 + C$$

Therefore, (u_n) is a bounded sequence in $H_0^1(\Omega)$. By the Rellich-Kondrachov theorem, there exists a subsequence, which we will still denote as (u_n) , that converges weakly in $H_0^1(\Omega)$ and strongly in $L^2(\Omega)$ to some limit $w_1 \in H_0^1(\Omega)$. Because the convergence in $L^2(\Omega)$ is strong, the norm is preserved:

$$\|w_1\|_{L^2} = \lim_{n \rightarrow \infty} \|u_n\|_{L^2} = 1$$

This shows that w_1 satisfies the constraint to be in the set defining $\mathcal{E}_1$. Since $u_n \rightharpoonup w_1$ weakly in $H_0^1(\Omega)$, we have:

$$E[w_1] \leq \liminf_{n \rightarrow \infty} E[u_n] = \lambda_1$$

Hence, w_1 is the required minimizer. To deduce that $\lambda_1 > 0$, assume for a contradiction that $\lambda_1 = 0$. Since $E[w_1] = \lambda_1 = 0$, we have:

$$\int_{\Omega} |Dw_1|^2 dx = 0$$

This implies that $Dw_1 = 0$ almost everywhere in Ω . Since $w_1 \in H_0^1(\Omega)$, it is 0 outside Ω so we have a function in $H^1(\mathbb{R}^n)$ whose weak derivative is zero almost everywhere, except possibly $\partial\Omega$ but it is 0 there too as we will prove.

Since Ω is open, $\partial\Omega \subset \Omega^c$. The approximating functions $\phi_n \in C_c^\infty(\Omega)$ are identically 0 on Ω^c . Therefore, $D\phi_n = 0$ everywhere on Ω^c . Passing to the limit in H_0^1 and considering the norm, this implies $Dw_1 = 0$ almost everywhere on Ω^c .

A function in L^p with $1 \leq p < \infty$ with a zero weak derivative on $\mathbb{R}^n$ must be constant almost everywhere. The reason: Convolving w with a standard smooth mollifier ϕ_ε produces a smooth function $w_\varepsilon = w \star \eta_\varepsilon$. Its classical derivative is $\nabla w_\varepsilon = (\nabla w) \star \phi_\varepsilon = 0 \star \phi_\varepsilon = 0$. A smooth function with a zero classical gradient on $\mathbb{R}^n$ is a constant. Since $w_\varepsilon \rightarrow w$ in L^2 as $\varepsilon \rightarrow 0$ since w , w itself must be that constant a.e.

Because w_1 vanishes outside the bounded set Ω , this constant must be 0. Thus, $w_1 = 0$ almost everywhere, which implies $\|w_1\|_{L^2} = 0$. This directly contradicts the established fact that $\|w_1\|_{L^2} = 1$. Therefore, we conclude that $\lambda_1 > 0$.

We can now prove Poincare's inequality: $\lambda_1 \|u\|_2^2 \leq \int_\Omega |Du|^2 dx$ holds for all $u \in H_0^1(\Omega)$ with equality for $u = w_1$.

Let $u \in H_0^1(\Omega)$. If $u = 0$, then $\|u\|_{L^2} = 0$ and $\int_\Omega |Du|^2 dx = 0$, so the inequality $0 \leq 0$ holds trivially. Assume $u \neq 0$, which implies $\|u\|_{L^2} > 0$. Consider the normalized function:

$$v = \frac{u}{\|u\|_{L^2}}$$

Since $H_0^1(\Omega)$ is a vector space, $v \in H_0^1(\Omega)$. By construction, $\|v\|_{L^2} = 1$, which means $E[v] \in \mathcal{E}_1$. From earlier, we know $\lambda_1 = \inf \mathcal{E}_1$, so we must have:

$$\lambda_1 \leq E[v]$$

Using the definition of the E operator and the linearity of the weak derivative, we can expand $E[v]$:

$$E[v] = \int_\Omega \left| D \left(\frac{u}{\|u\|_{L^2}} \right) \right|^2 dx = \frac{1}{\|u\|_{L^2}^2} \int_\Omega |Du|^2 dx = \frac{E[u]}{\|u\|_{L^2}^2}$$

Substituting this back into the inequality and multiplying both sides by $\|u\|_{L^2}^2$ yields the desired Poincare inequality:

$$\lambda_1 \|u\|_{L^2}^2 \leq E[u] = \int_\Omega |Du|^2 dx$$

To verify the case for equality, recall that there exists a $w_1 \in H_0^1(\Omega)$ such that $\|w_1\|_{L^2} = 1$ and $E[w_1] = \lambda_1$. Substituting w_1 into the left-hand side of the inequality gives:

$$\lambda_1 \|w_1\|_{L^2}^2 = \lambda_1 (1)^2 = \lambda_1$$

Since $\lambda_1 = E[w_1]$, this is exactly the right-hand side, confirming that equality holds for $u = w_1$.

Our next goal is to prove that w_1 as defined above satisfies $-\nabla^2 w_1 = \lambda_1 w_1$ in $\mathcal{D}^*(\Omega)$.

From earlier, the Poincare inequality $E[u] \geq \lambda_1 \|u\|_{L^2}^2$ holds for all $u \in H_0^1(\Omega)$, with equality attained at $u = w_1$. Let $\phi \in \mathcal{D}(\Omega)$ and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by:

$$f(t) = E[w_1 + t\phi] - \lambda_1 \|w_1 + t\phi\|_{L^2}^2$$

Note that f is a quadratic:

$$f(t) = \int_\Omega |Dw_1 + tD\phi|^2 dx - \lambda_1 \int_\Omega |w_1 + t\phi|^2 dx$$

By expanding the squared magnitudes (using $|z|^2 = z\bar{z}$) and pulling the real variable t out of the integrals, you can group the terms by powers of t :

$$\begin{aligned} f(t) &= \left(\int_\Omega |Dw_1|^2 dx - \lambda_1 \int_\Omega |w_1|^2 dx \right) \\ &+ t \left(\int_\Omega (Dw_1 \cdot \overline{D\phi} + \overline{Dw_1} \cdot D\phi - \lambda_1 (w_1 \bar{\phi} + \overline{w_1} \phi)) dx \right) \\ &+ t^2 \left(\int_\Omega (|D\phi|^2 - \lambda_1 |\phi|^2) dx \right) \end{aligned}$$

Because $w_1 \in H_0^1(\Omega)$ and $\phi \in \mathcal{D}(\Omega)$, all of these integrals evaluate to finite constants. This means $f(t)$ is entirely of the form $f(t) = A + Bt + Ct^2$ for some constants $A, B, C \in \mathbb{R}$, which is smoothly differentiable everywhere.

Since $f(t) \geq 0$ for all $t \in \mathbb{R}$ and $f(0) = 0$, the function attains a global minimum at $t = 0$, meaning $f'(0) = 0$ and in fact $f(t) = At^2$ with $A \geq 0$. Expanding the function:

$$f(t) = \int_{\Omega} (|Dw_1 + tD\phi|^2 - \lambda_1|w_1 + t\phi|^2) dx$$

Differentiating with respect to t and evaluating at $t = 0$ yields:

$$f'(0) = \int_{\Omega} (Dw_1 \cdot \overline{D\phi} + \overline{Dw_1} \cdot D\phi - \lambda_1(w_1\overline{\phi} + \overline{w_1}\phi)) dx = 2 \operatorname{Re} \int_{\Omega} (Dw_1 \cdot \overline{D\phi} - \lambda_1 w_1 \overline{\phi}) dx = 0$$

Since $\phi \in \mathcal{D}(\Omega)$ is arbitrary, replacing ϕ with $i\phi$ shows that the imaginary part of the integral also vanishes. Therefore:

$$\int_{\Omega} (Dw_1 \cdot \overline{D\phi} - \lambda_1 w_1 \overline{\phi}) dx = 0$$

Because $\mathcal{D}(\Omega)$ is closed under complex conjugation, we can replace $\overline{\phi}$ with a general test function $\phi \in \mathcal{D}(\Omega)$ to obtain:

$$\int_{\Omega} (Dw_1 \cdot D\phi - \lambda_1 w_1 \phi) dx = 0$$

By the definition of the distributional derivative for functions in Sobolev spaces, $D_j w_1[\phi] = -w_1[D_j \phi]$. Applying this to the Laplacian:

$$(-\nabla^2 w_1)[\phi] = \sum_{j=1}^n \int_{\Omega} D_j w_1 D_j \phi dx = \int_{\Omega} Dw_1 \cdot D\phi dx$$

Substituting this back into our integral identity gives:

$$(-\nabla^2 w_1)[\phi] - (\lambda_1 w_1)[\phi] = 0$$

As this holds for all test functions $\phi \in \mathcal{D}(\Omega)$, it follows that $-\nabla^2 w_1 = \lambda_1 w_1$ in $\mathcal{D}'(\Omega)$.

Now suppose $\chi \in C_c^\infty(\Omega)$ and w_1 is defined as above and $v = \chi w_1$. Then we aim to prove that v satisfies $-\nabla^2 v + v = f$ where the equation holds in $\mathcal{S}'(\mathbb{R}^n)$ and $f \in L^2(\mathbb{R}^n)$ and that $v \in H^2(\mathbb{R}^n)$ and $w_1 \in H_0^1(\Omega) \cap C^\infty(\Omega)$.

To do this, we first need a version of the product rule for weak derivatives:

To prove the product rule for a weakly differentiable function u and a smooth function v , take a test function $\phi \in C_c^\infty(\Omega)$. We need to show that the weak derivative $\partial_i(uv)$ is $(\partial_i u)v + u(\partial_i v)$, which by definition means showing:

$$\int_{\Omega} (uv) \partial_i \phi dx = - \int_{\Omega} ((\partial_i u)v + u(\partial_i v)) \phi dx$$

Start with the left side, group v with ϕ , and apply the standard product rule for smooth functions, $v \partial_i \phi = \partial_i(v\phi) - \phi \partial_i v$:

$$\int_{\Omega} (uv) \partial_i \phi dx = \int_{\Omega} u(v \partial_i \phi) dx = \int_{\Omega} u(\partial_i(v\phi) - \phi \partial_i v) dx$$

Because v is smooth and ϕ has compact support, the product $v\phi$ is also in $C_c^\infty(\Omega)$. This makes $v\phi$ a valid test function for the weak derivative of u , allowing us to evaluate the first term:

$$\begin{aligned} \int_{\Omega} u \partial_i(v\phi) dx - \int_{\Omega} u \phi \partial_i v dx &= - \int_{\Omega} (\partial_i u)(v\phi) dx - \int_{\Omega} u(\partial_i v) \phi dx \\ &= - \int_{\Omega} ((\partial_i u)v + u(\partial_i v)) \phi dx \end{aligned}$$

This satisfies the exact definition of the weak derivative for uv .

Now we go back to the main example: Let $v = \chi w_1$. Using the product rule for weak derivatives in the sense of distributions in $\mathcal{S}'(\mathbb{R}^n)$, we have:

$$-\nabla^2 v = -(\nabla^2 \chi)w_1 - 2\nabla \chi \cdot \nabla w_1 - \chi \nabla^2 w_1$$

From earlier, w_1 satisfies $-\nabla^2 w_1 = \lambda_1 w_1$ in $\mathcal{D}^*(\Omega)$. Because $\chi \in C_c^\infty(\Omega)$, the identity $-\chi \nabla^2 w_1 = \lambda_1 \chi w_1$ holds globally in $\mathbb{R}^n$. Adding $v = \chi w_1$ to both sides yields:

$$-\nabla^2 v + v = -(\nabla^2 \chi) w_1 - 2\nabla \chi \cdot \nabla w_1 + (\lambda_1 + 1) \chi w_1$$

Let f denote the right-hand side. Because $w_1 \in H_0^1(\Omega)$, we know $w_1, \nabla w_1 \in L^2(\Omega)$. Since χ and its derivatives are smooth and compactly supported in Ω , each term in f belongs to $L^2(\mathbb{R}^n)$. Thus, $-\nabla^2 v + v = f$ where $f \in L^2(\mathbb{R}^n)$. Taking the Fourier transform of $-\nabla^2 v + v = f$ gives:

$$(|\xi|^2 + 1)\hat{v}(\xi) = \hat{f}(\xi)$$

(Note: The Fourier transform relation is allowed because $L^2 \subseteq L^1$ on a finite measure space)

Since $f \in L^2(\mathbb{R}^n)$, we have $\hat{f} \in L^2(\mathbb{R}^n)$. We evaluate the $H^2(\mathbb{R}^n)$ norm of v :⁶

$$|v|_{H^2}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^2 |\hat{v}(\xi)|^2 d\xi = \int_{\mathbb{R}^n} |\hat{f}(\xi)|^2 d\xi = |\hat{f}|_{L^2}^2 = J|f|_{L^2}^2 < \infty$$

Therefore, $v \in H^2(\mathbb{R}^n)$. Because $\chi \in C_c^\infty(\Omega)$ was arbitrary, this implies $w_1 \in H_{\text{loc}}^2(\Omega)$. We define H_{loc}^s for integer s as the space of distributions $u \in \mathcal{D}'(U)$ on an open set $U \subseteq \mathbb{R}^n$ such that $\psi u \in H^s(\mathbb{R}^n)$ for every compactly supported smooth test function $\psi \in C_c^\infty(U)$.

Now assume inductively that $w_1 \in H_{\text{loc}}^k(\Omega)$ for some integer $k \geq 1$. For any $\chi \in C_c^\infty(\Omega)$, the right-hand side f depends on w_1 and ∇w_1 . By the inductive hypothesis, $w_1 \in H_{\text{loc}}^k(\Omega)$ implies $D_i w_1 \in H_{\text{loc}}^{k-1}(\Omega)$ for each i . Multiplication by smooth, compactly supported functions preserves the property of being in these local Sobolev spaces, so $(\nabla^2 \chi) w_1$ and χw_1 are in $H^k(\mathbb{R}^n) \subset H^{k-1}(\mathbb{R}^n)$, and $\nabla \chi \cdot \nabla w_1 \in H^{k-1}(\mathbb{R}^n)$ by adding up each component. Thus, $f \in H^{k-1}(\mathbb{R}^n)$. Using the Fourier transform relation $(1 + |\xi|^2)\hat{v}(\xi) = \hat{f}(\xi)$, we find:

$$|v|_{H^{k+1}}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{k+1} |\hat{v}(\xi)|^2 d\xi = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{k-1} |\hat{f}(\xi)|^2 d\xi = |f|_{H^{k-1}}^2 < \infty$$

Hence, $v = \chi w_1 \in H^{k+1}(\mathbb{R}^n)$, which proves $w_1 \in H_{\text{loc}}^{k+1}(\Omega)$. By induction, $w_1 \in H_{\text{loc}}^m(\Omega)$ for all $m \geq 1$. By the Sobolev embedding theorem (the $H^s \subseteq C^k$ thing), $H_{\text{loc}}^m(\Omega) \subset C_{\text{loc}}^{m-\lfloor n/2 \rfloor - 1}(\Omega)$. Since this holds for all m , we conclude $w_1 \in C^\infty(\Omega)$. Combining this with the initial condition, we get $w_1 \in H_0^1(\Omega) \cap C^\infty(\Omega)$.

Finally, we will prove that there exists $\lambda_2 \geq \lambda_1$ and $w_2 \in H_0^1(\Omega) \cap C^\infty(\Omega)$ with $w_2 \neq w_1$ and $\|w_2\|_2 = 1$ and $-\nabla^2 w_2 = \lambda_2 w_2$.

Let

$$\mathcal{E}_2 := \{E[u] : u \in H_0^1(\Omega), |u|_{L^2} = 1, (u, w_1)_{L^2} = 0\}$$

so that we seek eigenfunctions perpendicular to w_1 . We now repeat the earlier argument.

Since the set of admissible functions for $\mathcal{E}_2$ is a subset of the admissible functions for $\mathcal{E}_1$, taking the infimum over a smaller set can only increase or maintain the value. Therefore, $\lambda_2 = \inf \mathcal{E}_2 \geq \inf \mathcal{E}_1 = \lambda_1$. Take a minimizing sequence $(u_i) \subset H_0^1(\Omega)$ such that $|u_i|_{L^2} = 1$, $(u_i, w_1)_{L^2} = 0$, and $E[u_i] \rightarrow \lambda_2$. Because $E[u_i]$ converges, the sequence of gradients Du_i is bounded in $L^2(\Omega)$. Since the L^2 norms are also fixed at 1, (u_i) is a bounded sequence in $H_0^1(\Omega)$. By the Rellich-Kondrachov theorem, there exists a subsequence (which we also denote as u_i) that converges weakly in $H_0^1(\Omega)$ and strongly in $L^2(\Omega)$ to some limit $w_2 \in H_0^1(\Omega)$. The strong convergence in $L^2(\Omega)$ implies: $|w_2|_{L^2} = \lim_{i \rightarrow \infty} |u_i|_{L^2} = 1$. $(w_2, w_1)_{L^2} = \lim_{i \rightarrow \infty} (u_i, w_1)_{L^2} = 0$. Since $|w_1|_{L^2} = |w_2|_{L^2} = 1$ and they are orthogonal, $w_2 \neq w_1$. Furthermore, w_2 satisfies the constraints of $\mathcal{E}_2$. By the weak lower semicontinuity of the E operator, we have $E[w_2] \leq \liminf_{i \rightarrow \infty} E[u_i] = \lambda_2$. Since w_2 is in the admissible set, its E cannot be strictly less than the infimum λ_2 . Thus, $E[w_2] = \lambda_2$. To show that $-\nabla^2 w_2 = \lambda_2 w_2$, we take variations of w_2 that remain orthogonal to w_1 . Let $\psi \in \mathcal{D}(\Omega)$ be an arbitrary smooth test function with compact support. We project ϕ onto the orthogonal complement of w_1 in L^2 by setting $\tilde{\psi} = \psi - (\psi, w_1)_{L^2} w_1$. This ensures $(\tilde{\psi}, w_1)_{L^2} = 0$.

We define a real-valued function of a real variable t similar to earlier:

$$g(t) = \int_{\Omega} (|D(w_2 + t\tilde{\psi})|^2 - \lambda_2 |w_2 + t\tilde{\psi}|^2) dx$$

⁶Where J is whatever the constant is in the Fourier-Plancherel isometry from the Fourier convention nowadays it doesn't matter as we just care about finiteness

Because w_2 is the minimizer in $\mathcal{E}_2$ with E value λ_2 , $g(t) \geq 0$ for all $t \in \mathbb{R}$ and $g(0) = 0$. Expanding the squared magnitudes using $|z|^2 = z\bar{z}$ and grouping by powers of t :

$$g(t) = g(0) + t \int_{\Omega} (Dw_2 \cdot \overline{D\tilde{\psi}} + \overline{Dw_2} \cdot D\tilde{\psi} - \lambda_2(w_2\tilde{\psi} + \overline{w_2}\tilde{\psi}))dx + t^2 \dots$$

Differentiating with respect to t at $t = 0$ isolates the linear coefficient:

$$g'(0) = 2\operatorname{Re} \int_{\Omega} (Dw_2 \cdot \overline{D\tilde{\psi}} - \lambda_2 w_2 \tilde{\psi})dx = 0$$

Because replacing ψ with $i\psi$ swaps $\tilde{\psi}$ for $i\tilde{\psi}$, the exact same logic from earlier shows the imaginary part also vanishes. This leaves:

$$\int_{\Omega} (Dw_2 \cdot \overline{D\tilde{\psi}} - \lambda_2 w_2 \tilde{\psi})dx = 0$$

Now, substitute $\tilde{\psi} = \bar{\psi} - \overline{(\psi, w_1)_{L^2} w_1}$ to expand the integral:

$$\int_{\Omega} (Dw_2 \cdot \overline{D\bar{\psi}} - \lambda_2 w_2 \bar{\psi})dx - \overline{(\psi, w_1)_{L^2}} \int_{\Omega} (Dw_2 \cdot \overline{Dw_1} - \lambda_2 w_2 \bar{w}_1)dx = 0$$

The second integral vanishes entirely. First, $\int_{\Omega} w_2 \bar{w}_1 dx = 0$ because w_2 is orthogonal to w_1 . Second, taking the complex conjugate of the distributional identity $-\nabla^2 w_1 = \lambda_1 w_1$ and testing it against w_2 gives $\int_{\Omega} Dw_2 \cdot \overline{Dw_1} dx = \lambda_1 \int_{\Omega} w_2 \bar{w}_1 dx = 0$. With the projection terms gone, we are left with:

$$\int_{\Omega} (Dw_2 \cdot \overline{D\bar{\psi}} - \lambda_2 w_2 \bar{\psi})dx = 0$$

Since $\mathcal{D}(\Omega)$ is closed under complex conjugation, replacing $\bar{\psi}$ with a general ϕ yields the identity $-\nabla^2 w_2 = \lambda_2 w_2$. This concludes our first example application.

Our next example will be the equation $-\nabla^2 u + k^2 u = f$ where f is given and we want to find u . We will prove that if $f \in H^s(\mathbb{R}^n)$ for some $s \in \mathbb{R}$ then there is a unique solution $u \in H^{s+2}(\mathbb{R}^n)$.

By assumptions on u, f we can take the Fourier transform of the equation and we get that $(|\xi|^2 + k^2) \hat{u}(\xi) = \hat{f}(\xi)$ holds pointwise a.e. We write this as $\hat{u}(\xi) = \frac{\hat{f}(\xi)}{|\xi|^2 + k^2}$. Since there exists a constant C such that $|\xi|^2 + k^2 \geq C(1 + |\xi|^2)$, it follows that

$$\|u\|_{H^{s+2}(\mathbb{R}^n)} \leq C \|f\|_{H^s(\mathbb{R}^n)}$$

Therefore we have $u \in H^{s+2}(\mathbb{R}^n)$ and uniqueness follows from injectivity of the Fourier transform. Note that if $s > \frac{n}{2}$ then $f \in C^0(\mathbb{R}^n)$ and $u \in C^2(\mathbb{R}^n)$ so this gives a classical solution.

Now suppose we modify this equation so that we have the following condition. Let Ω be an open set in $\mathbb{R}^n$ and because of the crashout from earlier assume that we are in the case where $\partial\Omega$ is a plane or hyperplane. The condition we want to consider is where $f : \Omega \rightarrow \mathbb{R}$ is given and

$$\begin{cases} -\nabla^2 u + u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

To incorporate the condition we will seek a solution $u \in H_0^1(\Omega)$ by the version of the trace theorem my reference notes actually prove.

We will now take this equation and conjugate it, and multiply by the test function $v \in C_c^\infty(\Omega)$:

$$-\nabla^2 \bar{u}v + \bar{u}v = \bar{f}v$$

Integrating both sides gives

$$\int_{\Omega} (-\nabla^2 \bar{u})v dx + \int_{\Omega} \bar{u}v dx = \int_{\Omega} \bar{f}v dx$$

These integrals converge because v is a test function. We will now assume u is C^2 .

Our goal is to simplify the integral $\int_{\Omega} (-\nabla^2 \bar{u})v dx$.

Assume Ω is the half-space $\{x_n > 0\}$. We can enclose the support K inside a finite box $B = [a_1, b_1] \times \cdots \times [a_n, b_n]$ strictly inside Ω (meaning $a_n > 0$). The integral over Ω is then exactly equal to the integral over B . Expanding the Laplacian and applying the 1D Fundamental Theorem of Calculus to the i -th coordinate over the box:

$$\int_{a_i}^{b_i} (-\partial_{ii}\bar{u})v dx_i = [-\partial_i\bar{u}v]_{a_i}^{b_i} + \int_{a_i}^{b_i} \partial_i\bar{u}\partial_i v dx_i$$

It is the Fubini theorem that lets you extend this for a 1D integral to the whole thing where for the i 'th term in the sum you integrate the i 'th coordinate first.

Because the support of v is strictly inside the box, $v = 0$ on all boundaries ($x_i = a_i$ and $x_i = b_i$). The boundary evaluation $[-\partial_i\bar{u}v]_{a_i}^{b_i}$ is therefore exactly zero. Summing this over all $i = 1, \dots, n$ and integrating over the remaining variables yields:

$$\int_B (-\nabla^2\bar{u})v dx = \int_B \sum_{i=1}^n (\partial_i\bar{u}\partial_i v) dx = \int_B \overline{Du} \cdot Dv dx$$

We therefore conclude that for all $v \in C_c^\infty(\Omega)$ we have

$$\int_\Omega \overline{Du} \cdot Dv + \bar{u}v dx = \int_\Omega \bar{f}v dx$$

Therefore if $f \in L^2(\Omega)$ we declare that $u \in H$ is a weak solution if it satisfies this integral equation, in other words we seek $u \in H_0^1(\Omega)$ such that

$$(u, v)_{H^1} = (\bar{f}, v)_{L^2}$$

Note that a classical solution is a weak solution.

If $f \in L^2$ then the map $F : H_0^1(\mathbb{C})$ sending $f \mapsto (\bar{f}, v)_{L^2}$ is a bounded linear operator. To apply the Riesz representation theorem, F must be shown to be bounded with respect to the $H_0^1(\Omega)$ norm. By the Cauchy-Schwarz inequality in L^2 :

$$|F(v)| = |(\bar{f}, v)_{L^2}| \leq \|f\|_{L^2} \|v\|_{L^2}$$

Use the definition of the H^1 norm, which is $\|v\|_{H^1}^2 = \|v\|_{L^2}^2 + \|Dv\|_{L^2}^2$. From this, it is trivially true that $\|v\|_{L^2} \leq \|v\|_{H^1}$. Combining these gives:

$$|F(v)| \leq \|f\|_{L^2} \|v\|_{H^1}$$

Since $f \in L^2(\Omega)$, $\|f\|_{L^2}$ is a finite constant. This proves F is a bounded linear functional on $H_0^1(\Omega)$, perfectly satisfying the requirements for the Riesz representation theorem.

Now the Riesz representation theorem implies that there is a unique $u \in H_0^1(\Omega)$ satisfying $(u, v)_{H^1} = (\bar{f}, v)_{L^2}$. Setting $u = v$ in $(u, v)_{H^1} = (\bar{f}, v)_{L^2}$ keeps the equation valid as $C_c^\infty(\Omega)$ is dense in $H_0^1(\Omega)$ by definition. Therefore, we have

$$\|u\|_{H^1}^2 = (\bar{f}, u)_{L^2} \leq \|f\|_2 \|u\|_2 \leq \|f\|_2 \|u\|_{H^1}$$

If $\|u\|_{H^1} = 0$ then the following inequality holds trivially so we may divide by it to get

$$\|u\|_{H^1} \leq \|f\|_2$$

Fix a compact $K \subset \Omega$ and suppose that the real function $\chi \in C_c^\infty$ satisfies $\chi(x) = 1$ for all $x \in K$. Let $\phi \in \mathcal{S}(\mathbb{R}^n)$, then since $\chi\phi \in C_c^\infty(\Omega)$ we can set $v = \chi\phi$ in $\int_\Omega \overline{Du} \cdot Dv + \bar{u}v dx = \int_\Omega \bar{f}v dx$ to get

$$\int_\Omega \overline{Du} \cdot D(\chi\phi) + \bar{u}\chi\phi dx = \int_\Omega \bar{f}\chi\phi dx$$

By the product rule, we have

$$\int_\Omega \overline{Du} \cdot \chi(D\phi) + \overline{Du} \cdot (D\chi)\phi + \bar{u}\chi\phi dx = \int_\Omega \bar{f}\chi\phi dx$$

Since χ is real it equals its own conjugate so we have

$$\int_\Omega (\overline{D(\chi u)}) \cdot (D\phi) + \overline{Du} \cdot (D\chi)\phi - D\phi \cdot (D\chi)\bar{u} + \bar{u}\chi\phi dx = \int_\Omega \bar{f}\chi\phi dx$$

Therefore, by the same integration by parts trick:

$$\int_{\Omega} -(\overline{\chi u}) \nabla^2 \phi + 2 \overline{Du} \cdot (D\chi) \phi + \phi (\nabla^2 \chi) \bar{u} + \bar{u} \chi \phi dx = \int_{\Omega} \bar{f} \chi \phi dx$$

Set $g := -2Du \cdot D\chi - u \nabla^2 \chi + f\chi \in L^2(\mathbb{R}^n)$. Then by reality of χ , $v = \chi u$ satisfies

$$\int_{\mathbb{R}^n} \bar{v} (-\nabla^2 \phi + \phi) dx = \int_{\mathbb{R}^n} \bar{g} \phi dx$$

since everything is supported in K . Therefore we have that in the distributional sense or the $\mathcal{S}'(\mathbb{R}^n)$ sense that

$$-\nabla^2 v + v = g$$

By the earlier work in here and the Fourier-Plancherel transform being an isometry up to whatever constant it is, $v \in H^2$ with

$$\|v\|_{H^2} \leq C \|g\|_{H^0} = C' \|g\|_{L^2}$$

Also, $v(x) = u(x)$ for all $x \in K$ so if we multiply u by a test function supported in K it is also in H^2 . If in reverse we multiply u by a test function, call its support K , let χ be another corresponding test function and then apply the above argument, it follows that $u \in H_{loc}^2(\mathbb{R}^n)$.

If $f \in H_{loc}^1$ then repeating the above argument g is in H^1 (the other terms in g are sufficiently regular by the fact $u \in H^2$) so then $u \in H_{loc}^3(\mathbb{R}^n)$. Therefore, by iterating this argument, if $f \in L^2 \cap C^\infty(\Omega)$ then $u \in H_0^1 \cap C^\infty(\Omega)$.

Crash out time episode 2

WHY DOES THIS TRIPOS COURSE CLAIM TO PROVE THE SPECTRAL THEOREM WHEN THE PROOF RELIES ON THE TRACE THEOREM WHICH IT LARPS AS PROVING BUT DOESN'T

I'm so sick of this bro I'm just gonna pretend like I never saw the spectral theorem existing in this course schedules I'm gonna pray that I can still do most of Methods.

Ok moving on. Next PDE example.

Definition. Given a Banach space X and an interval $I \subseteq \mathbb{R}$ the space $C^0(I; X)$ is the space of continuous functions $I \rightarrow X$. The space $C^k(I; X)$ can be defined inductively: $u \in C^{k-1}(I; X)$ belongs to $C^k(I; X)$ if there exists $u' \in C^{k-1}(I; X)$ such that for each $t \in I$ we have

$$\lim_{\varepsilon \rightarrow 0} \left\| \frac{u(t + \varepsilon) - u(t)}{\varepsilon} - u'(t) \right\| = 0$$

and $C^\infty(I; X)$ is the intersection of all $C^k(I; X)$.

We will now consider the problem of given $u_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ find a $u : \mathbb{R}^n \times [0, T] \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{\partial u}{\partial t} = \nabla^2 u & \text{in } (0, T) \times \mathbb{R}^n \\ u = u_0 & \text{on } \{0\} \times \mathbb{R}^n \end{cases}$$

By a convention that one is supposed to magically know that was never stated, the ∇^2 does not include the u_{tt} term. Don't ask me why it's suddenly this way, I don't make the rules.

This is called the heat equation because in physics u might represent the temperature at points in space and time.

We will suppose that for each t our function $\mathbf{u}(t) := u(t, \cdot)$ is an element of $H^2(\mathbb{R}^n)$ and that $\mathbf{u} \in C^0((0, T); H^2(\mathbb{R}^n))$. We will also assume that $\mathbf{u} \in C^1((0, T); L^2(\mathbb{R}^n))$ and $\mathbf{u} \in C^0([0, T]; L^2(\mathbb{R}^n))$.

We first aim to prove that if $u \in C^0((0, T); H^2(\mathbb{R}^n))$ and $u \in C^1((0, T); L^2(\mathbb{R}^n))$ and we define $\hat{u}(t, \xi)$ for each t to be the Fourier-Plancherel transform of what u would be if t is held constant then $\hat{u} \in C^0((0, T); L^2(\mathbb{R}^n)) \cap C^1((0, T); L^2(\mathbb{R}^n))$.

Since $H^2(\mathbb{R}^n)$ continuously embeds into $L^2(\mathbb{R}^n)$, the condition $u \in C^0((0, T); H^2(\mathbb{R}^n))$ implies $u \in C^0((0, T); L^2(\mathbb{R}^n))$. The Fourier-Plancherel transform $\mathcal{F}$ is a bounded linear operator on $L^2(\mathbb{R}^n)$ satisfying $\|\hat{f}\|_{L^2} = C \|f\|_{L^2}$ for whatever C

is. For continuity in t , take $t, s \in (0, T)$. By the linearity and boundedness of the transform, we have:

$$\|\hat{u}(t) - \hat{u}(s)\|_{L^2} = C\|u(t) - u(s)\|_{L^2}$$

Since $u \in C^0((0, T); L^2(\mathbb{R}^n))$, taking the limit as $s \rightarrow t$ gives 0, which establishes $\hat{u} \in C^0((0, T); L^2(\mathbb{R}^n))$. To prove continuous differentiability, consider the difference quotient evaluated in the L^2 norm:

$$\left\| \frac{\hat{u}(t+h) - \hat{u}(t)}{h} - \hat{u}_t(t) \right\|_2 = C \left\| \frac{u(t+h) - u(t)}{h} - u_t(t) \right\|_2$$

Since $u \in C^1((0, T); L^2(\mathbb{R}^n))$, the term inside the norm on the right-hand side converges to 0 as $h \rightarrow 0$. This shows that the strong derivative exists in L^2 and is given by $\partial_t \hat{u}(t) = \hat{u}_t(t)$. Because $u_t \in C^0((0, T); L^2(\mathbb{R}^n))$, the exact same continuity argument applied to u_t yields $\hat{u}_t \in C^0((0, T); L^2(\mathbb{R}^n))$. Therefore, $\hat{u} \in C^1((0, T); L^2(\mathbb{R}^n))$.

We will now seek a $\mathbf{u}$ satisfying the conditions above. Taking the Fourier transform of the equation gives the new equation

$$\begin{cases} \hat{u}_t(t, \xi) = -|\xi|^2 \hat{u}(t, \xi) & \text{in } (0, T) \times \mathbb{R}^n \\ \hat{u}(0, \xi) = u_0(\xi) & \xi \in \mathbb{R}^n \end{cases}$$

For each fixed ξ this is an ODE. The equation $\hat{u}_t(t, \xi) = -|\xi|^2 \hat{u}(t, \xi)$ is a standard separable first-order linear ODE of the form $y' = -ky$ (where $k = |\xi|^2$). Therefore its unique solution is given by $\hat{u}(t, \xi) = \hat{u}_0(\xi) e^{-t|\xi|^2}$.

If $u_0 \in L^2(\mathbb{R}^n)$ then $\hat{u}_0 \in L^2(\mathbb{R}^n)$ which implies that $\hat{u} \in C^0([0, T]; L^2(\mathbb{R}^n))$. For $t > 0$ we have that both $\hat{u}(t, \xi)$ and $\hat{u}_t(t, \xi)$ are rapidly decaying (in the sense they are literally L^2 functions times polynomial growing functions times gaussian decaying) functions of ξ so the functions u, u_t belong to every H^s so $u(t, x)$ is smooth in x . By induction, since mixed partial derivatives commute, you can just repeatedly substitute the original heat equation:

$$u_{tt} = \partial_t(\nabla^2 u) = \nabla^2(u_t) = \nabla^2(\nabla^2 u) = (\nabla^2)^2 u$$

and so on for higher powers of n so that $(\partial_t)^n u = (\nabla^2)^n u$. The reason we have smoothness for the partials to commute is: The explicit solution $\hat{u}(t, \xi) = \hat{u}_0(\xi) e^{-t|\xi|^2}$ is trivially infinitely differentiable with respect to t for any $t > 0$. Each t derivative brings down a factor of $-|\xi|^2$, which is easily absorbed by the rapid exponential decay of $e^{-t|\xi|^2}$. This keeps the function in H^s for all s and guarantees the spatial function has all t derivatives.

We can therefore recover $u(t, x)$ via the original Fourier transform

$$u(t, x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{u}_0(\xi) e^{-t|\xi|^2} e^{i\xi \cdot x} d\xi$$

Note that even if the initial condition in t is a rough function u_0 it 'instantaneously' smooths itself out.

Note that u is smooth in x, t but we need it to be jointly smooth. Here is how we prove this:

$$u(t, x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{u}_0(\xi) e^{-t|\xi|^2} e^{i\xi \cdot x} d\xi$$

Applying any mixed partial derivative $\partial_t^m \partial_x^\alpha$ to the integrand brings down a polynomial factor of $(-|\xi|^2)^m (i\xi)^\alpha$. For any $t > 0$, the rapid exponential decay of $e^{-t|\xi|^2}$ dominates this polynomial growth. This ensures that the differentiated integrand remains in $L^1(\mathbb{R}^n)$ and is uniformly bounded by an integrable function on regions like $[\varepsilon, T) \times \mathbb{R}^n$ for any $\varepsilon > 0$. By the Dominated Convergence Theorem, you can pass all mixed derivatives under the integral sign, and the resulting mixed partial derivatives are jointly continuous. Thus, u is jointly C^∞ on $(0, T) \times \mathbb{R}^n$. We have sequential continuity by dominated convergence since we have the Gaussian thing that makes it L^1 away from $t = 0$ and thus continuity of all derivatives.

As the final example we will consider the wave equation. Again we do that very silly thing where the Laplacian doesn't include the special variable because silly physicist convention hahaha. The equation says determine $u : \mathbb{R}^n \times (-T, T) \rightarrow \mathbb{R}$ such that

$$\begin{cases} u_{tt} = \nabla^2 u & \text{in } (-T, T) \times \mathbb{R}^n \\ u = u_0 \\ u_t = u_1 \end{cases} \quad \text{on } 0 \times \mathbb{R}^n$$

We will seek a solution in the space

$$X_s := C^0((-T, T), H^{s+2}(\mathbb{R}^n)) \cap C^2((-T, T) \times H^s(\mathbb{R}^n))$$

Taking the Fourier transform in the non- t variables as above the equation becomes

$$\begin{cases} \hat{u}_{tt}(t, \xi) = -|\xi|^2 \hat{u}(t, \xi) & (t, \xi) \in (-T, T) \times \mathbb{R}^n \\ \hat{u}(0, \xi) = \hat{u}_0(\xi) \\ \hat{u}_t(0, \xi) = \hat{u}_1(\xi) \end{cases} \quad \xi \in \mathbb{R}^{n-1}$$

Again this is an ODE for each fixed ξ and this time the solution is

$$\hat{u}(t, \xi) = \hat{u}_0(\xi) \cos(|\xi|t) + \hat{u}_1(\xi) \frac{\sin(|\xi|t)}{|\xi|}$$

Suppose $u_0 \in H^{s+2}(\mathbb{R}^n)$ and $u_1 \in H^{s+1}(\mathbb{R}^n)$. To show $u \in C^0((-T, T); H^{s+2}(\mathbb{R}^n))$, you bound the H^{s+2} norm using the Fourier characterisation:

$$|u(t)|_{H^{s+2}}^2 = \int_{\mathbb{R}^n} (1 + |\xi|^2)^{s+2} \left| \hat{u}_0(\xi) \cos(|\xi|t) + \hat{u}_1(\xi) \frac{\sin(|\xi|t)}{|\xi|} \right|^2 d\xi$$

Using $|a + b|^2 \leq 2|a|^2 + 2|b|^2$, the integrand is bounded by:

$$2(1 + |\xi|^2)^{s+2} |\hat{u}_0(\xi)|^2 \cos^2(|\xi|t) + 2(1 + |\xi|^2)^{s+2} |\hat{u}_1(\xi)|^2 \frac{\sin^2(|\xi|t)}{|\xi|^2}$$

Since $\cos^2(|\xi|t) \leq 1$ and $\frac{\sin^2(|\xi|t)}{|\xi|^2} \leq \frac{1+t^2}{1+|\xi|^2}$, this is bounded by:

$$2(1 + |\xi|^2)^{s+2} |\hat{u}_0(\xi)|^2 + 2(1 + t^2)(1 + |\xi|^2)^{s+1} |\hat{u}_1(\xi)|^2$$

These integrate to $2|u_0|_{H^{s+2}}^2 + 2(1 + t^2)|u_1|_{H^{s+1}}^2$, which is finite given the initial conditions $u_0 \in H^{s+2}$ and $u_1 \in H^{s+1}$. To show $u \in C^2((-T, T); H^s(\mathbb{R}^n))$, differentiate $\hat{u}$ twice with respect to t :

$$\partial_t^2 \hat{u}(t, \xi) = -|\xi|^2 \hat{u}_0(\xi) \cos(|\xi|t) - |\xi| \hat{u}_1(\xi) \sin(|\xi|t)$$

The H^s norm requires bounding $\int (1 + |\xi|^2)^s |\partial_t^2 \hat{u}(t, \xi)|^2 d\xi$. The squared term is bounded by:

$$2(1 + |\xi|^2)^s |\xi|^4 |\hat{u}_0(\xi)|^2 + 2(1 + |\xi|^2)^s |\xi|^2 |\hat{u}_1(\xi)|^2$$

Since $(1 + |\xi|^2)^s |\xi|^4 \leq (1 + |\xi|^2)^{s+2}$ and $(1 + |\xi|^2)^s |\xi|^2 \leq (1 + |\xi|^2)^{s+1}$, the integral is again bounded by $2|u_0|_{H^{s+2}}^2 + 2|u_1|_{H^{s+1}}^2$, which is finite.

By the bounds, continuity in t for both spaces follows directly from the Dominated Convergence Theorem. Because the Fourier transform is a linear isometry, taking the t derivative of the map $t \mapsto u(t)$ in the H^s norm is equivalent to taking the pointwise t derivative of $\hat{u}(t, \xi)$ in the weighted L^2 space.

Therefore if $u_0 \in H^{s+2}(\mathbb{R}^n)$ and $u_1 \in H^{s+1}(\mathbb{R}^n)$ then $u \in X^s$. By Fourier inversion we therefore have a unique solution in H^s .

Now we will specialize to $\mathbb{R}^3$ and assume u_0 and u_1 is in the Schwartz space $\mathcal{S}$.

For $\phi \in \mathcal{S}^*$ define the distribution

$$U_t[\phi] := \frac{1}{4\pi t} \int_{\partial B_t(0)} \phi(y) dA(y)$$

To prove this is actually a distribution: By the definition of the topology on the test space (whether the Schwartz space $\mathcal{S}$ or the space of compactly supported smooth functions $\mathcal{D}$), a sequence $\phi_j \rightarrow 0$ implies that ϕ_j converges to 0 uniformly on any compact set. To show continuity, we bound $U_t[\phi]$ using the supremum of ϕ on the compact sphere $K = \partial B_t(0)$:

$$|U_t[\phi]| \leq \frac{1}{4\pi t} \sup_{y \in K} |\phi(y)| \int_K dA(y)$$

Since the surface area of K in $\mathbb{R}^3$ is $4\pi t^2$, the integral evaluates exactly to this area, simplifying the bound to:

$$|U_t[\phi]| \leq t \sup_{y \in K} |\phi(y)|$$

Because $\sup_{y \in K} |\phi_j(y)| \rightarrow 0$ as $j \rightarrow \infty$ in the test space, it directly follows that $|U_t[\phi_j]| \rightarrow 0$. This bound by the L^∞ norm over a compact set is the exact condition required for a linear functional to be continuous on $\mathcal{S}$ or $\mathcal{D}$.

We define this as a tempered distribution for all $\phi \in \mathcal{S}^*$ and in fact it has compact support which implies by a theorem in the section on the Fourier transform in $\mathcal{E}^*$ that $\hat{U}_t = T_{\hat{v}_t}$ where $\hat{v}_t(\xi) = U_t[e_{-\xi}] = \frac{1}{4\pi t} \int_{\partial B_t(0)} e^{-i\xi \cdot y} dA(y)$. We can do this integral in spherical polar coordinates (with the axis chosen to be parallel to θ) as follows:

$$\begin{aligned} \hat{v}_t(\xi) &= \frac{1}{4\pi t} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} e^{-i|\xi|t \cos(\theta)} t^2 \sin(\theta) d\theta d\phi \\ &= \frac{t}{2} \int_{\theta=0}^{\pi} e^{-i|\xi|t \cos(\theta)} \sin(\theta) d\theta = \frac{t}{2} \int_{z=-1}^1 e^{-i|\xi|tz} dz \\ &= \frac{t}{2} \left(\frac{e^{-i|\xi|t}}{-i|\xi|t} - \frac{e^{i|\xi|t}}{-i|\xi|t} \right) = \frac{\sin(|\xi|t)}{|\xi|} \end{aligned}$$

Now we return to our expression for u :

$$\begin{aligned} \hat{u}(t, \xi) &= \hat{u}_0(\xi) \cos(|\xi|t) + \hat{u}_1(\xi) \frac{\sin(|\xi|t)}{|\xi|} \\ &= \frac{\partial}{\partial t} \left(\hat{u}_0(\xi) \frac{\sin(|\xi|t)}{|\xi|} \right) + \hat{u}_1(\xi) \frac{\sin(|\xi|t)}{|\xi|} \\ &= \frac{\partial}{\partial t} \left(\hat{u}_0(\xi) \widehat{U}_t \right) + \hat{u}_1(\xi) \widehat{U}_t \end{aligned}$$

By properties of convolutions in the Schwartz space we have

$$\begin{aligned} u(t, x) &= \text{Inverse Fourier} \left(\frac{\partial}{\partial t} \left(\hat{u}_0(\xi) \widehat{U}_t \right) + \hat{u}_1(\xi) \widehat{U}_t \right) \\ &= \frac{\partial}{\partial t} (U_t \star u_0(x)) + U_t \star u_1(x) \\ &= \frac{\partial}{\partial t} \left(\frac{1}{4\pi t} \int_{\partial B_t(0)} u_0(x-y) dA \right) + \frac{1}{4\pi t} \int_{\partial B_t(0)} u_1(x-y) dA \\ &= \frac{\partial}{\partial t} \left(\frac{1}{4\pi t} \int_{\partial B_t(x)} u_0(y) dA \right) + \frac{1}{4\pi t} \int_{\partial B_t(x)} u_1(y) dA \end{aligned}$$

Define $\oint f dA$ as the surface integral divided by the surface area. Then

$$= \frac{\partial}{\partial t} \left(t \oint_{\partial B_t(x)} u_0(y) dA \right) + t \oint_{\partial B_t(x)} u_1(y) dA$$

Interestingly what happens at t depends only on what happens at this sphere - intuitively this means information is propagated at finite speed in the wave equation.